

GUYANA | EXPLORATION TO PRODUCTION

GUYANA'S PETROLEUM TRANSFORMATION

The Guyana Sequence: exploration, contracts, first oil, sovereign wealth, environmental stewardship, national development and a transferable blueprint for emerging producers

Independent research by Ragunauth Ramsaroop

Edition 1.2 | Scrutiny-hardened publication candidate | Evidence cut-off: 27 September 2026

Independent research. No political or electoral ranking. Government, operator, multilateral, judicial and stakeholder positions are attributed. External peer review remains pending.

Contents

Contents	2
Executive brief	5
Audit hardening addendum	8
AUDIT HARDENING ADDENDUM	8
Legal transition, sovereign risk, latest fund evidence and contested climate-finance issues	8
Part I-X Integrated flagship manuscript	14
PART I	15
Before the Discovery: The Foundations of Guyana's Petroleum Era	15
Edition 2.0 deepening: the pre-discovery state was built through repeated cycles	25
PART II	32
Discovery, Contract Design, Cost Recovery and Revenue Assurance	32
Edition 2.0 deepening: from headline fiscal terms to an operating petroleum revenue system	48
PART III	58
From First Oil to Large-Scale Production	58
Edition 2.0 deepening: production scale-up as an integrated control system	70
Edition 2.0 project deepening: seven development decisions as one evolving system	76
PART IV	84
The Natural Resource Fund and the Conversion of Petroleum into National Wealth	84
Edition 2.0 deepening: the Natural Resource Fund as one account inside a national wealth system.	
Edition 2.0 evidence deepening: the year-by-year NRF record	104
PART V	110
Environmental Governance, Offshore Risk and Climate Coexistence	110
Edition 2.0 deepening: environmental governance as a lifecycle assurance system	127
PART VI	136
Local Content, Human Capital and the Distribution of Opportunity	136
Edition 2.0 deepening: local content as capability formation, not only compliance	152
Edition 2.0 evidence deepening: from reported local spend to verified economic retention	159
PART VII	167
Economic Transformation, Infrastructure, Energy and Diversification	167
Edition 2.0 deepening: turning a petroleum boom into a second growth engine	180
FLAGSHIP PART VIII	189

Accountability and institutional learning	189
39. Extractive disclosure as an operating system	189
40. Cost audits and revenue assurance	190
41. Procurement and spending oversight	191
42. Parliament, judiciary and institutional checks	191
43. Statistical institutions and public data	192
Institutional learning cycle	193
Part VIII conclusions	193
Edition 2.0 deepening: accountability as a system of reconciled public ledgers	193
FLAGSHIP PART IX	200
International comparators and transferability	200
44. Norway: converting petroleum cash into a financial asset	200
45. Ghana: stabilization and heritage within a producing developing economy	200
46. Timor-Leste: a small state's dependence risk	201
47. Suriname: the regional pre-first-oil comparison	202
48. Trinidad and Tobago: a mature Caribbean producer	202
49. Small islands and microstates	203
Comparative matrix	203
50. A method for responsible policy transfer	203
Part IX conclusions	204
Edition 2.0 deepening: a wider comparator universe	204
FLAGSHIP PART X	213
New-producer operating manual	213
50. Exploration-stage operating manual	213
51. Discovery-to-final-investment-decision manual	214
52. First-oil readiness manual	215
53. Years 1-5 scale-up manual	216
54. Adverse scenario playbooks	217
55. Cabinet and regulator checklists	218
56. The 24-month implementation sequence for a prospective state	220
57. Evidence standard for declaring readiness	221
Part X conclusion	221

Edition 2.0 deepening: the operating manual as a sequence of irreversible gates	221
Evidence dossiers and implementation workbooks	231
Technical annexes	418
Scenario tables Contract sensitivity	556
Scenario tables Production shocks	557
Scenario tables Sovereign-fund stress	558
Adversarial publication audit	559
High-risk claim audit	568
Publication review gates	572
Dated source register	575
Open evidence items	588

Executive brief

Read alongside MANUSCRIPT_WORKING_DRAFT.md, SOURCE_REGISTER.md, ADVERSARIAL_PUBLICATION_AUDIT.md and CLAIM_SOURCE_MATRIX.md. Edition 1.1 has completed an internal adversarial review cycle. Independent external expert review remains pending.

Audit status

Edition 1.1 is an internally audited publication candidate, not an externally peer-reviewed or independently validated publication. The internal audit identified and corrected publication-status, source-coverage, legal-scope, geopolitical-risk and PDF-design weaknesses. High-risk legal characterization remains deliberately narrow where an authenticated judgment, commencement instrument, insurance instrument or later dispute determination has not been located in the public sources reviewed. Guyana's 2026 EITI Validation remains ongoing at the evidence cut-off. The Bank of Guyana's audited 2025 NRF statements and current 2026 monthly reporting are now part of the evidence register. [S88-S89, S96]

Purpose

Reconstruct the institutional and commercial sequence behind Guyana's transition to a large offshore petroleum producer. Test where its history supplies documentary evidence for another country preparing for first oil, and where forest endowment, geological scale, fiscal terms and local political institutions limit direct transfer. The intended audience is senior officials, new petroleum regulators, central banks, finance ministries, environmental agencies, civil-society researchers and offshore investors.

Verified sequence

Guyana has decades of exploration history. Liza was announced as commercially discovered in May 2015. The Stabroek petroleum agreement was entered in June 2016 and the official block register shows an October 2016 effective date. The first Natural Resource Fund law received assent on 23 January 2019, before oil production commenced in December 2019. The 2021 statute replaced the first framework and 2024 legislation changed withdrawal limits. The fourth operating FPSO, Yellowtail, began production in August 2025. Four FPSOs were operating during the first half of 2026. This chronology separates pre-oil institutional decisions from subsequent modifications. [S01-S05, S11]

September 2026 evidence snapshot

The September 2026 Ministry of Finance mid-year report, describing activity through June, records 162 crude lifts, of which 21 represented government profit oil. During the half-year, government received USD 1,778.6 million in profit-oil proceeds and USD 218.4 million in royalties. The Natural Resource Fund stood at USD 4.2942 billion on June 30 after USD 1.02 billion of first-half withdrawals. The ministry revised its projected full-year 2026 petroleum deposits to USD 6.4976 billion. The latter number is a projection rather than revenue already received. Government estimated overall real GDP growth of 33.3% and non-oil growth of 10.1% for the first half; GDP estimates do not directly show changes in household poverty. [S07, S10]

The distinction most useful to a new producer

The 2016 Stabroek PSA and Guyana's later model PSA do not have the same fiscal terms. For 2016 Stabroek, the published administrative summary describes a 2% royalty, a cost-recovery ceiling of 75% and equal sharing of profit oil. Under the government revenue authority's simplified binding-cap example, 12.5% profit oil plus 2% royalty yields 14.5% of gross revenue. The figure changes as recoverable costs change. The later model for new licensing announced a 10% royalty, 65% maximum cost recovery and 10% corporate tax. The two regimes must be analysed separately, along with the information available at the time each was negotiated. [S02, S14-S15]

Institutional subjects for detailed scrutiny

1. Early-law timing. The NRF predates first oil, but the institution changed over time. Assess legislation, actual reporting, parliamentary authorization, investment stewardship and expense outcomes individually.
2. Fiscal and contract administration. Petroleum volumes, government lifts, sales, cost-bank reductions, royalties, cash deposits and budget transfers are different measures. Cost-oil dispute resolution and audit resources matter alongside contract percentages.
3. Environmental assurance. Project-specific EIAs and permits, monitoring, emergency assets, enforceable insurance, operator liability and cross-border response are separate protections. The 2025 pollution act was enacted, but the 2026 mid-year report described implementation regulations and commencement as ongoing work. The May 2026 Court of Appeal decision on financial guarantees requires analysis of the full judgment and underlying instruments.
4. Economic absorption. A high fiscal balance does not establish that roads, hospitals, schools, housing and flood defenses meet service and maintenance targets. IMF staff in July 2026 called for long-run fiscal guideposts, spending-outcome monitoring, procurement compliance and stronger cost-oil audits.
5. Social and community outcomes. Gross domestic supplier spend is not local value added. Old poverty-survey results cannot demonstrate current welfare. Indigenous village allocations require programme-level and locally grounded verification.
6. Transparency and institutional participation. The EITI Board launched Guyana's delayed second validation in May 2026 and recorded concerns about confidence in the independence of civil-society representative selection. The final paper will capture independent stakeholder responses and the final validation decision if published by the cut-off.
7. Parallel climate systems. Guyana's 2009 LCDS and Norway partnership preceded oil. Carbon-credit contracts, payments and village benefit-sharing require separate accounting from the NRF and domestic petroleum emissions. Forest results cannot be transferred mechanically to reef-based or low-forest island states.

Comparator lenses

Ghana's 2011 revenue law offers distinct stabilization and heritage funds and public publication requirements. Norway's fund documents describe transferring net state petroleum cash into a large investment fund and a long-run real-return-linked fiscal guideline. Timor-Leste's 2005 Petroleum Fund provides another small-state legal design. Suriname's GranMorgu was still under construction in September 2026, with first oil planned for 2028, making the country a relevant pre-production comparison rather than a second operating case. No design is treated as an automatic policy choice for another country. [S24-S27]

Implementation workbook for a newly discovering jurisdiction

Exploration: Record basin geology and uncertainty, licensing authority and contract disclosure, ownership, environmental baseline, marine response capabilities, available technical expertise and prospective contingent liabilities.

Commercial discovery: Build independent resource and development assessment, model fiscal terms on common assumptions, identify costs and the risk allocation actually negotiated, obtain project-specific environmental assessment, map nearby fisheries and communities, establish audit rights and publish authorized agreements subject to legitimate protections.

Before first oil: Test metering and valuation, recoverable-cost records, royalty invoicing, financial custody, parliamentary procedures, periodic public reporting, spill drills, grievance channels and beneficial-ownership records. Determine budget contingencies without booking forecast oil prices as guaranteed receipts.

First two years of production: Publish reconciled lifts, realized prices, royalty and profit-oil cash, fund balances, parliamentary withdrawals, government project execution and all material environmental incidents. Check whether procurement and supplier development result in verified domestic value.

Scaling: Stress-test price changes, new project cost-bank additions, maintenance obligations, delayed infrastructure and major spill financial exposure. Compare national growth with independently sourced real household outcomes.

Publication acceptance conditions

Complete the full multi-chapter manuscript and original, reproducible exhibits. Independently reconcile NRF period accounts, obtain the Court of Appeal judgment and original insurance and permitting instruments, resolve the legal commencement status of the 2025 spill act, incorporate final EITI 2026 status, document Indigenous stakeholder evidence, check page-specific citations and have relevant experts challenge fiscal, legal, environmental and social conclusions. Count only substantive PDF pages, inspect charts and citations, and describe independent external review as pending unless completed.

Primary citations are maintained in SOURCE_REGISTER.md and the high-risk claim crosswalk in CLAIM_SOURCE_MATRIX.md. The 500-plus-page document is a research compendium and publication candidate. External peer review, authenticated resolution of open legal evidence, and documented responses to specialist reviewers remain separate release gates.

Audit hardening addendum

AUDIT HARDENING ADDENDUM

Legal transition, sovereign risk, latest fund evidence and contested climate-finance issues

Evidence cut-off: 27 September 2026

This addendum closes material scope gaps identified in the adversarial audit of Edition 1.0. It does not alter the publication's neutral treatment of political choices or predict unresolved legal proceedings.

1. Petroleum law after the early Stabroek era

A complete exploration-to-production case study must distinguish the legal framework that existed when the Stabroek agreement was negotiated from the statutory framework later adopted for subsequent petroleum administration.

Guyana's Petroleum Activities Act 17 of 2023 provides the newer statutory framework for petroleum activities. It should be analysed separately from the 1986 legislation and separately from contractual rights created under earlier agreements. The Act is relevant to licensing, administration, information, safety, decommissioning and regulatory powers. The existence of a newer statute does not itself rewrite an executed production-sharing agreement. Transitional provisions, preserved rights, licence terms and contract clauses determine how older rights interact with later legislation. [S38, S87]

For a newcomer, the transfer lesson is institutional rather than political. A petroleum statute should be version-controlled. The public record should show:

- which law applied when acreage was awarded;
- which law applied when a development licence was granted;
- which later provisions apply prospectively;
- which obligations arise from the contract rather than statute;
- which regulator owns each decision;
- which rules govern decommissioning, data, safety, environmental coordination and transfer of interests.

This avoids a common analytical error: using today's legal framework to judge whether a historical decision complied with rules that did not yet exist.

2. Licensing evolution and contract generations

Guyana's petroleum framework now contains at least two analytically distinct contract generations: the 2016 Stabroek agreement and later model terms intended for subsequent licensing. Their differences reflect different information states, different geological knowledge and different policy choices. [S02, S14]

The 2023 Act adds a statutory layer around later petroleum administration. Edition 1.1 should therefore use a four-column legal crosswalk:

1. governing statute at the decision date;
2. executed agreement or model agreement;
3. production or exploration licence;
4. environmental and other sectoral authorisations.

No later model PSA should be applied retroactively when calculating Stabroek entitlement. No legacy contract term should automatically be treated as the rule for a later acreage award.

3. Territorial and maritime risk belongs in a petroleum blueprint

An offshore petroleum blueprint is incomplete if it treats geology and fiscal design without sovereign-boundary risk.

Guyana and Venezuela remain parties to proceedings before the International Court of Justice concerning the Arbitral Award of 3 October 1899. The ICJ case record includes jurisdictional decisions, a preliminary-objection judgment and provisional-measures orders. The Court's 1 May 2025 order is an interim judicial measure. The merits are not resolved by this research and no outcome is predicted here. [S90, S91]

CARICOM stated on 28 April 2026 that it continued to support Guyana's sovereignty and territorial integrity and the peaceful resolution of the controversy through the Court. That is a regional diplomatic position and should be identified as such, not converted into an independent legal merits finding. [S92]

For a new producer, the operational questions are:

- Are maritime boundaries settled?
- Are licence coordinates compatible with recognized maritime claims?
- What is the legal consequence of a disputed boundary?
- Which coast guard, navy or maritime authority protects offshore infrastructure?
- How would a security incident affect drilling, tanker movement, insurance and emergency response?
- Which regional organizations or neighbouring states have mutual-assistance arrangements?
- Does the operator's force-majeure and political-risk insurance align with the state's contingency planning?

These questions apply to many frontier basins even where no active territorial dispute exists.

4. The Natural Resource Fund should use the newest audited baseline

The Bank of Guyana now publishes audited financial statements for the Natural Resource Fund for the year ended 31 December 2025. This is a stronger annual baseline than relying only on 2024 audited statements plus later government summaries. [S44, S88]

Edition 1.1 should reconstruct the fund using audited annual statements through 2025 and then bridge into 2026 using monthly and mid-year reporting. The bridge should preserve:

- opening cash or financial assets;
- petroleum deposits;
- royalties;
- profit-oil receipts;
- investment income;
- lawful withdrawals;
- foreign-exchange translation where relevant;
- closing balance.

The Ministry of Finance publishes 2026 monthly NRF reports through July on its monthly-report portal. These reports should be reconciled to the September 2026 mid-year report before final numerical tables are locked. [S07, S96]

The audit principle is straightforward: a stock at 31 December 2025, a monthly cash movement in 2026 and a full-year 2026 projection are three different quantities.

5. Current EITI status must remain time-stamped

As of 27 September 2026, EITI's validation schedule still lists Guyana among ongoing Validations and notes commencement on 15 May 2026. The publication should therefore describe the 2026 Validation as ongoing at the evidence cut-off. It should not report a final score or outcome that EITI has not published. [S20, S89]

The EITI Board's concerns about stakeholder confidence in the civil-society selection process are attributed to EITI. The paper should also preserve the distinction between:

- disclosure of contracts and data;
- timeliness of reporting;
- beneficial-ownership information;
- stakeholder participation;
- the formal Validation outcome.

A state may perform strongly on one of these dimensions and still face questions on another.

6. Carbon markets: eligibility, accounting and contestation

Guyana's forest-carbon programme needs a more complete market-integrity section.

The LCDS programme reported issuance of 7.14 million 2021-vintage ART TREES credits in February 2024 and stated that the credits were eligible for use in the first phase of CORSIA, with corresponding-adjustment context. [S95]

This status is materially different from a voluntary-market sale. The final paper should keep at least five records separate:

- units issued;
- registry status;
- buyer contract;
- cash received;
- units retired or transferred for a specific compliance or voluntary purpose.

The accounting boundary matters. Forest-carbon credits do not mechanically offset all emissions associated with exported petroleum. Domestic operational emissions, land-use accounting, buyer retirement claims and downstream combustion belong to different accounting systems.

7. Indigenous safeguards require competing evidence

The paper should not rely solely on programme descriptions when evaluating Indigenous participation.

Official LCDS sources describe the village allocation mechanism and consultation process. [S62, S63] Independent critical analysis has raised questions about ART certification, complaint mechanisms and Indigenous safeguards. [S93]

These sources should be presented as different evidence types. Government and ART documentation establishes programme rules and official processes. Advocacy and community-focused sources document criticisms and disputed interpretations. Neither should be treated as a substitute for direct community-level outcome evidence.

A rigorous village-level audit would ask:

- Was the village eligible and did it participate?
- What consultation record exists?
- What amount was allocated and received?
- Which village plan authorized spending?
- What procurement occurred?
- What asset or service was delivered?
- What grievance route existed?
- What independent or community evidence confirms the outcome?

Where such evidence is unavailable, the paper should say so.

8. Strategic environmental assessment belongs above project EIAs

Project EIAs answer project-specific questions. A rapidly expanding offshore basin also requires cumulative and strategic analysis.

Terms of reference under the Guyana Oil & Gas Capacity Building Project identified strategic environmental and social issues including seismic effects, drilling waste, fisheries, marine habitats, coastal support facilities, accidental spills and community impacts. [S94]

A newcomer blueprint should therefore distinguish:

- project EIA;
- cumulative-impact assessment;
- strategic environmental and social assessment;
- national oil-spill planning;
- climate and energy-system analysis.

Approval of one project does not answer the cumulative question created by several FPSOs, support bases, tanker movements and onshore infrastructure.

9. Audit-grade treatment of unresolved law

The 2025 Oil Pollution Prevention, Preparedness, Response and Responsibility Act is enacted and published. [S53, S54] This research has not established, from an authoritative commencement instrument located during the audit, that every provision and implementing regulation was operational by 27 September 2026.

The correct wording is therefore: "Act 6 of 2025 was enacted and published. Full commencement and implementing-regulation status should be verified from authoritative instruments before relying on a specific provision as operational law."

That wording is narrower and more defensible than either assuming the whole statute was already operational or asserting that it had not commenced.

10. Court of Appeal financial-assurance case

Government and contemporaneous press reporting describe the May 2026 Court of Appeal decision concerning the environmental permit and financial assurance. [S20] The publication should not go beyond the proposition supported by the sources reviewed.

Until an authenticated judgment is reviewed directly:

- do not paraphrase detailed ratio decidendi beyond verified summaries;
- distinguish operator liability from the amount and form of financial security;
- avoid treating a press statement as the court's own language;
- preserve later appeal status if any.

11. New audit controls for Edition 1.1

Edition 1.1 should carry, for every high-risk claim:

- claim ID;
- claim text;
- source ID;
- pinpoint page/section;
- observation date;
- evidence class;
- contradictory source if any;
- confidence level;
- unresolved question;
- revision date.

High-risk classes are:

1. legal interpretation;
2. petroleum fiscal arithmetic;
3. fund balances and withdrawals;
4. production and capacity;
5. environmental compliance;
6. financial assurance;
7. Indigenous safeguards;
8. social outcomes;
9. local value added;
10. cross-country comparisons.

Addendum conclusion

Guyana remains a highly informative modern case because exploration, commercial discovery, first oil, multiple FPSO developments, a sovereign resource fund, local-content legislation, forest-carbon finance and major public-investment expansion occurred within a compressed period.

That density of change is also the reason a serious case study must resist a single-label conclusion. The transferable value lies in dated institutional decisions, measurable results, unresolved risks, correction mechanisms and transparent limits on what the evidence establishes.

The blueprint should help another jurisdiction ask better questions before discovery, before first oil and before the first large public withdrawal. It should not imply that Guyana's precise fiscal terms, political choices, forest endowment or institutional pathway are universally transferable.

Part I-X | Integrated flagship manuscript

PART I

Before the Discovery: The Foundations of Guyana's Petroleum Era

Analytical section 1. The correct place to begin

A petroleum case study becomes misleading when its history starts with the commercial discovery. The discovery is the point at which petroleum moves from geological possibility toward a bankable development proposition. It is not the point at which a country begins to accumulate law, institutions, technical knowledge, environmental responsibilities, infrastructure constraints or experience with resource governance.

Guyana's modern offshore petroleum story therefore needs two starting dates. One is May 2015, when the Liza discovery changed the commercial outlook for the Stabroek Block. The other lies decades earlier, when successive exploration campaigns, legal reforms, geological surveys and maritime disputes shaped the conditions inherited by the state in 2015. The official petroleum history records exploration activity extending across much of the twentieth century, while the statutory framework for petroleum exploration and production dates to 1986. Environmental law also predates the oil boom. The Environmental Protection Act was enacted in 1996. [S35, S38, S39]

This longer chronology matters for a newcomer country. The central lesson is not that a government should build a large petroleum bureaucracy before finding commercial oil. Such a course risks creating fixed public costs against uncertain resources. The more defensible lesson is narrower. A prospective producer benefits from a scalable legal and technical foundation before the negotiating stakes rise. It benefits from knowing who owns geological information, which agency grants rights, how environmental assessment works, which maritime boundaries remain disputed, who assesses tax and royalty, and how expert advice is procured.

A country that enters a commercial discovery with those questions unresolved faces a compressed learning problem. Geological uncertainty starts to fall while commercial pressure rises. The investor wants to appraise quickly, define a development concept, secure approvals and move toward a final investment decision. Government needs time to test reservoir assumptions, fiscal terms, environmental impacts, infrastructure consequences and long-term liabilities. The negotiation therefore involves both economic value and institutional time.

Guyana's experience is useful because the transition happened quickly enough to expose this time problem. It is also useful because important pre-oil policies existed outside petroleum. The Low Carbon Development Strategy began in 2009. Guyana's forest partnership with Norway also predates commercial offshore oil. These facts make Guyana a more complex case than a simple resource-boom narrative. [S22]

The final case study should therefore resist two stories. The first says oil created every modern institution. The second says pre-existing institutions were fully prepared for the later scale of deepwater production. The evidence supports neither proposition. The research question is how inherited systems interacted with rapid post-discovery adaptation.

Analytical section 2. Economic conditions before commercial oil

A national petroleum strategy is constrained by the economy receiving the resource. Before 2015, Guyana relied on a mix of mining, agriculture, forestry, services and remittance-supported household activity. Gold, bauxite, rice and sugar were important sources of foreign exchange and employment. Mining had already familiarized public institutions with licences, royalties, environmental oversight and commodity price cycles. Yet deepwater oil introduced a different order of capital intensity, technical complexity, marine risk and potential revenue concentration.

The pre-oil economic baseline should be used for comparison rather than nostalgia. It identifies the problems petroleum revenue was expected to help address. Those problems included infrastructure gaps, coastal vulnerability, energy cost and reliability, uneven access to services, limited fiscal space and the need for stronger technical capacity. A credible assessment of post-oil development must later ask whether those constraints improved, shifted or became more expensive under rapid growth.

The baseline also establishes the denominator for institutional absorption. A small economy that receives a sudden increase in foreign exchange and public revenue faces a different challenge from a large diversified economy. Construction demand may rise faster than the supply of engineers, contractors and imported materials. Land prices and rents may respond quickly. Public agencies may lose technical staff to higher-paying private employers. New projects increase future maintenance obligations. These effects do not prove a resource curse. They describe transmission channels that need measurement.

For this reason, the flagship should present several pre-oil baselines rather than one GDP number. The economic baseline needs national output, sector composition, public debt, government revenue, capital spending, external balances, inflation, household conditions and labour-force indicators. The infrastructure baseline needs power generation, outages, roads, bridges, ports, drainage, sea defence, water, hospitals and schools. The institutional baseline needs staffing, budgets and statutory mandates for the agencies later expected to regulate oil.

The historical comparison should preserve data revisions. A national account series revised after 2015 should not be silently mixed with an earlier series using another base year. Household evidence also needs an observation date. A 2018 or 2019 household survey cannot be reported as a 2026 welfare measure merely because it is the latest complete historical dataset in a source. [S61]

The same principle applies to poverty and inequality. Where current representative evidence is absent, the correct statement is that the current outcome is not established by the available survey. Strong GDP growth and large public investment do not logically reveal the distribution of real household consumption. The absence of a current household statistic also does not prove a negative outcome. The paper should keep the gap visible.

This discipline matters for the newcomer blueprint. Before a discovery, a government should freeze a baseline dataset. It should record what electricity costs, how long outages last, where poverty is measured, how much public debt exists, what infrastructure is deficient, what coastal assets are exposed and what technical staff are available. Later claims of transformation then have a documented starting point.

Analytical section 3. Petroleum exploration as a long-cycle process

The official exploration history shows that Guyana did not move directly from no petroleum activity to commercial oil in 2015. Exploration stretched across decades. That record included onshore work, offshore seismic acquisition, exploratory wells, changes in investors and periods of reduced activity. [S35]

The important analytical point is uncertainty. An exploration licence grants a right to search under specified conditions. It does not create a commercial reserve. A discovery proves hydrocarbons in a geological structure. It does not automatically establish an economically recoverable field. Appraisal work reduces uncertainty about reservoir size, continuity, pressure, fluid properties and expected recovery. Only after technical and commercial review does a field move toward development.

The distinction has fiscal consequences. Governments are often tempted to discuss future revenue immediately after a discovery. At that stage, several variables remain uncertain. The recoverable volume is not final. The development design is not final. Capital cost is not final. Oil prices over the production life are not known. The timing of first production remains subject to engineering, permitting, procurement and financing. A newcomer country should therefore separate a geological success announcement from the public budget.

Guyana's long exploration history also shows why historical data rights matter. Seismic data, well logs, cores and geological interpretations accumulate across licence generations. If a state does not preserve those records or the legal right to use them after confidentiality periods expire, later negotiators start with less information than private operators. A national data policy is thus a governance instrument.

A robust data policy identifies who owns raw seismic, processed seismic, well logs, samples, test results and interpretation reports. It identifies confidentiality periods and the conditions under which data move into a national repository. It identifies technical formats and metadata standards. It also protects legitimate commercial confidentiality while ensuring the state retains the information needed to regulate future development.

The same logic applies to unsuccessful wells. Dry holes are information. They reduce uncertainty about where petroleum is not present in commercial quantities. A national geological archive therefore has value even if commercial oil is never produced.

This is a practical lesson for small states. Permanent in-house teams need not replicate the technical depth of a global operator. Government still needs enough internal competence to frame questions, preserve data, supervise external advisers and understand the consequences of a technical recommendation. External advisers should add depth without becoming the only holders of institutional memory.

Analytical section 4. The 1986 legal foundation

The Petroleum (Exploration and Production) Act 1986 forms part of the legal inheritance that predates Liza. [S38] Its historical importance lies in establishing a statutory framework for exploration and production rights. The final legal analysis needs the exact version of the law that governed each historical decision, together with later amendments, regulations and any successor legislation.

A petroleum act performs several distinct functions. It defines state ownership or control over petroleum. It authorizes the granting of exploration and production rights. It establishes the relationship between licences and agreements. It provides powers for work programmes, information, inspection, conservation and development. It may address assignments, relinquishment, unitization, surrender and termination.

The presence of a statute does not settle whether every operational detail is sufficiently developed. Regulations, licence terms and individual petroleum agreements often carry much of the commercial and technical detail. The state also needs administrative capability to enforce them.

This distinction is useful in comparing legal readiness. A newcomer country should not ask only whether a petroleum act exists. It should ask whether the act and associated regulations define a coherent sequence from acreage award through exploration, discovery, development, production, abandonment and environmental restoration.

The legal review should also test interaction with other law. Petroleum operations involve environmental law, tax, customs, labour, local content, maritime law, public procurement, occupational safety, corporate law and sometimes Indigenous or customary rights. Conflicting or unclear statutes increase administrative discretion and litigation risk.

A practical pre-discovery legal audit should therefore map:

1. petroleum title;
2. licensing power;
3. contract approval;
4. environmental assessment;
5. tax and royalty authority;
6. beneficial ownership;
7. audit rights;
8. emergency response;
9. decommissioning;
10. public disclosure;
11. judicial review;
12. maritime boundaries.

The objective is not to eliminate discretion. Some discretion is unavoidable in complex technical regulation. The objective is to know where discretion sits, who exercises it, what evidence supports a decision and which review mechanism applies.

Analytical section 5. Maritime boundaries and petroleum rights

Offshore petroleum policy rests on geography and international law. A state cannot treat a disputed maritime area as if the legal status were irrelevant to investment risk.

The Permanent Court of Arbitration's *Guyana v. Suriname* case record documents an offshore dispute connected to petroleum activity. It records a 2000 incident involving an oil rig and drill ship in a disputed area and Guyana's commencement of arbitration in 2004. [S40] The final legal chapter should rely on the authenticated award for detailed propositions about the boundary and the parties' conduct.

The significance for a petroleum blueprint is broader than the result of one case. Maritime boundary uncertainty affects acreage promotion, seismic acquisition, investor risk, security arrangements and the potential for neighboring-state claims. An operator may hesitate to commit large capital to an unresolved area. A government may face diplomatic pressure if commercial activity starts before a dispute is managed.

A newcomer country therefore needs a maritime boundary register alongside its petroleum block map. The register should show agreed boundaries, pending negotiations, overlapping claims, treaties, provisional arrangements and dispute mechanisms.

Where a boundary is unresolved, options include negotiation, provisional joint arrangements or international adjudication under applicable law. The choice is sovereign and fact specific. The administrative lesson is to make the legal risk visible before a commercial award.

Maritime boundaries also matter after a spill. Oil, response vessels and environmental effects do not remain inside block coordinates. A transboundary preparedness plan should identify notification channels, evidence-sharing, joint response arrangements and compensation issues before an emergency.

Analytical section 6. Environmental law before the petroleum boom

The Environmental Protection Act 1996 is another important pre-oil institution. [S39] Its existence shows that environmental impact assessment and pollution control were part of Guyana's legal framework long before large-scale offshore production.

The analytical question is how a general environmental framework adapts to a new industry with deepwater wells, subsea systems, FPSOs, marine transport, flaring, produced water and low-probability high-consequence spill risk.

A general environmental law establishes authority. Project permits translate that authority into operating conditions. Monitoring and enforcement reveal how those conditions function in practice. These stages should be analyzed separately.

The flagship should therefore reconstruct the environmental system at three dates. The first is before Liza. The second is during approval of Liza Phase 1. The third is after several large offshore developments are operating.

The comparison should ask whether permit conditions become more detailed as regulators gain experience. It should ask whether monitoring frequency grows with the number of projects. It should test whether regulator staffing and technical support keep pace with the industry.

This is a better measure of institutional learning than counting amendments alone.

Analytical section 7. Low-carbon policy before first oil

Guyana's Low Carbon Development Strategy adds another pre-oil dimension. The strategy began in 2009, before the 2015 commercial discovery and before first oil. [S22]

This chronology is important because later discussions sometimes place petroleum and forest finance into one environmental narrative. The two systems have different origins, accounting rules and environmental effects.

The early LCDS sought to link forest conservation with national development and results-based climate finance. The Norway partnership provided one source of external payments connected to forest outcomes. Later versions of the strategy expanded into jurisdictional carbon markets and other national development priorities.

A petroleum case study should preserve that institutional history. Guyana did not first acquire a forest monitoring system as a reaction to offshore oil. Its forest policy developed under an earlier development and climate framework.

The transferable lesson is not that every producer should sell forest carbon. A small island may have limited forest area and a larger share of national natural capital in reefs, fisheries, mangroves or tourism assets. The stronger lesson concerns natural-capital accounting. A country entering petroleum production benefits from knowing which ecological assets already support livelihoods, public safety and export earnings.

The national balance sheet should therefore treat petroleum as one asset category among several. Extraction increases cash while reducing underground reserves. Public investment may increase produced capital. Forest conservation may protect ecosystem value. Pollution may reduce natural capital. The accounting need not force every asset into a single speculative monetary value. It needs to prevent the analysis from treating petroleum cash as the only form of wealth.

Analytical section 8. Coastal geography and development constraints

Guyana's coastal geography shapes the way petroleum revenue interacts with development.

Population, economic activity and public infrastructure are concentrated heavily along a low coastal plain. Flood protection, drainage and sea defence therefore remain central public investment functions. This makes the country different from many petroleum states and different again from small islands where storm surge, reef degradation and concentrated tourism infrastructure shape risk.

The flagship should establish a coastal baseline before measuring post-oil investment. Key variables include sea-defence condition, drainage capacity, pump infrastructure, historical flood

losses, exposed housing, transport corridors and the location of hospitals, schools, utilities and industrial facilities.

This baseline supports a more rigorous evaluation of petroleum-financed infrastructure. A drainage project should not be measured only by contract value. A stronger outcome measure asks whether flood duration falls, whether critical infrastructure remains accessible, whether maintenance is funded and whether the system performs under severe rainfall and tide conditions.

The same approach applies to sea defence. Construction length and expenditure are useful inputs. Outcome assessment needs condition, design standard, maintenance and residual risk.

For a newcomer country, the lesson is to identify national bottlenecks before revenue arrives. Otherwise, the political pressure created by a windfall may produce a long project list without a clear ranking of economic and resilience constraints.

Analytical section 9. Institutional capacity as a network

Petroleum governance is often discussed as if one ministry manages the sector. In practice, the state is a network.

The petroleum ministry or regulator handles licences, development plans and operator oversight. The environmental agency handles environmental assessment and permit conditions. The revenue authority handles tax and royalty functions. The finance ministry handles fiscal policy and the budget. The central bank may hold sovereign petroleum assets. The supreme audit institution reviews public accounts. Parliament authorizes spending and scrutinizes reports. Courts review legal disputes. Maritime and emergency agencies handle safety and response.

The quality of the system depends on information moving between these institutions.

A production number recorded by an operator matters to the revenue authority because it affects royalty and entitlement. A cost statement matters to petroleum auditors. A cargo sale matters to the sovereign fund. An environmental incident matters to the EPA and emergency authorities. A public withdrawal matters to parliament and the auditor.

The institutional baseline should therefore map not only organizations but data flows.

For 2014, the map should record which petroleum-specific functions existed, which were located inside broader natural-resource institutions and which depended on external support. For 2019, it should show the systems needed at first oil. For 2026, it should show the scale required by multiple producing FPSOs.

The comparison should avoid an impressionistic score. It should use measurable indicators where records permit. Examples include staffing, budget, audit completion time, permit workload, production reporting frequency, unresolved disputes and publication lag.

Analytical section 10. The pre-discovery skills problem

A prospective producer faces a difficult human-capital choice. Deepwater petroleum demands specialized skills that have little domestic market value if no commercial development occurs. Building every capability in-house before discovery is therefore expensive. Waiting until after discovery leaves government under time pressure.

The solution is a layered capacity model.

The first layer is a small permanent public team with strong general competence in law, geology, finance, environmental assessment and public administration. This team owns data, defines questions and supervises advisers.

The second layer is domestic technical institutions, including universities, professional bodies, geological agencies and environmental laboratories. Their capability supports both petroleum and other sectors.

The third layer is external specialist support used for high-complexity tasks such as reservoir simulation, petroleum fiscal modelling, international tax, offshore engineering or catastrophe insurance.

The fourth layer is peer and multilateral support. International institutions and other regulators offer comparative experience, training and technical assistance.

The key governance rule is knowledge transfer. An adviser report that government cannot later reproduce, challenge or update creates dependence rather than capacity.

Contracts for advisers should therefore state what data government receives, what models remain accessible, how staff are trained, what conflicts are prohibited and what intellectual property remains available after the engagement.

Analytical section 11. What changed after Liza

The Liza discovery altered the state capacity problem because the probability of large-scale petroleum development rose sharply.

The government then faced parallel workstreams:

- evaluate the discovery and development plans;
- negotiate or administer contractual arrangements;
- review environmental applications;
- prepare production licensing;
- build cost-audit capability;
- develop sovereign revenue management;
- plan local-content policy;
- strengthen public transparency.

The time horizon compressed.

This period is important to the final case because several institutions often described as part of today's petroleum system did not all exist in final form at discovery. Some came before first oil. Others followed first oil. Still others changed again as production expanded.

A rigorous chronology should therefore mark each institution as:

- inherited before 2015;
- created 2015-2019;

- created after first oil;
- later amended.

This avoids retrospective simplification.

Analytical section 12. A baseline for evaluating later reform

The value of Part I is that it creates controls for the rest of the publication.

When the paper examines the NRF, Part I identifies what public finance looked like before the windfall.

When the paper examines local content, Part I identifies the skills and supplier base before large-scale petroleum demand.

When the paper examines environmental regulation, Part I identifies the environmental statute and natural-capital policy that already existed.

When the paper examines infrastructure, Part I identifies the earlier gaps and coastal constraints.

When the paper examines institutional capacity, Part I identifies which agencies and mandates predate oil.

The case study therefore avoids attributing every observed change to petroleum. Economic development is influenced by global commodity prices, domestic policy, demographic change, private investment and other sectors.

The final empirical sections should use petroleum timing as one explanatory factor, not as a universal causal claim.

Analytical section 13. Transferable lesson for an island or small country

A small state considering petroleum exploration should start with a readiness inventory rather than a sovereign fund law copied from another country.

The readiness inventory should contain:

- maritime boundaries;
- block map;
- petroleum law;
- environmental law;
- tax treatment;
- geological data policy;
- regulator map;
- conflict rules;
- emergency response;
- social and ecological baseline.

If commercial discovery has not occurred, public spending assumptions should exclude future oil revenue.

If a discovery occurs, the inventory becomes a gap assessment. The country identifies what needs to be strengthened before development approval and first production.

An island should add special controls for:

- reefs;
- coastal tourism;
- desalination;
- hurricane disruption;
- single-port dependence;
- limited waste disposal;
- small labour market.

Guyana's forest-finance experience does not translate automatically to an island without comparable forests. The transferable principle is to identify natural capital before extraction changes the economy.

Analytical section 14. Counterfactual tests

A strong case study should ask what would weaken its own interpretation.

The proposition that early legal preparation matters would be weakened if a country with no pre-discovery framework consistently achieved equal or stronger revenue, environmental and transparency results without greater transition risk.

The proposition that geological data ownership matters would be weakened if government could reproduce independent technical decisions without retained data.

The proposition that baseline household and environmental information matters would be weakened if credible post-development impact attribution were possible without a starting measure.

The proposition that maritime boundary clarity matters would be weakened if disputed acreage attracted equivalent investment on equivalent terms without legal risk.

These tests keep the analysis falsifiable.

Analytical section 15. Part I findings

The documentary record supports several limited findings.

First, Guyana possessed petroleum, environmental and public institutions before the 2015 commercial discovery. [S38, S39]

Second, petroleum exploration stretched across decades rather than beginning with Liza. [S35]

Third, the country entered the oil era with a pre-existing low-carbon and forest-finance strategy. [S22]

Fourth, maritime boundary management formed part of the longer offshore petroleum context. [S40]

Fifth, the commercial discovery created a new scale and speed of institutional demand. The later chapters test the adequacy and adaptation of those systems rather than assuming either success or failure.

For a newcomer state, the strongest lesson from this opening period is chronological discipline. Build what remains useful even if no oil is found. Preserve national data. Clarify legal authority. Measure the environment and economy before the shock. Scale specialist capacity as commercial certainty rises. Do not spend revenue before it exists.

These are administrative design principles. They do not determine a country's political choices about ownership, taxation, extraction pace or the use of future public revenue.

Edition 2.0 deepening: the pre-discovery state was built through repeated cycles

Analytical section 15A. The 1980s petroleum project as institutional prehistory

The modern petroleum story should not begin with the 2015 Liza discovery. It should begin much earlier, when the state was trying to decide whether an uncertain petroleum endowment justified investment in law, data and specialist capacity.

The World Bank's Project Completion Report for the Petroleum Exploration Promotion Project gives unusually useful evidence because it records both the intended design and the shortcomings of an early petroleum-capacity programme. Credit 1208-GUA was approved in 1982. The project sought to attract foreign exploration, examine heavy-oil potential and strengthen the government's ability to administer the petroleum sector. The historical value of this record is not that the project produced oil. It did not. Its importance is that the state had already begun to assemble the institutional components that later became relevant when commercial discovery finally occurred. [S107]

The project supported a Petroleum Unit, petroleum-potential studies, promotional activity, training and the legal framework needed to negotiate with international companies. The completion report records four exploration contracts negotiated with international firms and describes petroleum legislation being enacted under separate financing. It also records implementation problems, including delays, incomplete components and a capacity-building approach that the Bank considered too weighted toward academic training. [S107]

This makes the project a useful case in institutional sequencing. Capacity created before a discovery can appear underutilized for years. Yet a frontier state that waits for a commercial discovery before building even basic legal and geological capability will enter negotiations at the exact moment when bargaining pressure and information asymmetry become greatest.

The lesson is not that every frontier state should create a large petroleum bureaucracy before hydrocarbons are proven. The lesson is to build scalable capability whose value survives a dry-hole outcome. A small geological data unit, a clear licensing law, a model data-room protocol,

trained public lawyers and a documented regulator map can be justified even if exploration never becomes commercial.

Analytical section 15B. Exploration promotion and the problem of credible commitment

A frontier petroleum state faces a strategic contradiction. It wants to attract high-risk capital, but it also wants to preserve the ability to regulate, tax and supervise a discovery if one occurs.

The World Bank completion report shows that this problem was already visible in Guyana's early petroleum programme. The project sought a legal and fiscal structure capable of attracting foreign exploration while preserving a framework for government administration. The report records debate over complex sliding-scale approaches and the need for a structure flexible enough to accommodate different agreement types. [S107]

For a new producer, the deeper issue is credible commitment. Before discovery, the state needs investors to believe that exploration rights, data obligations and development rights are sufficiently clear for capital to be risked. After discovery, citizens need confidence that the same legal system can protect public revenue, environmental obligations and national interests.

These are not opposing objectives if the framework is designed well. A credible petroleum regime should specify what is stable and what is subject to ordinary law. It should define the process for licence renewal, discovery declaration, appraisal, development approval, assignment, unitisation, decommissioning and termination. It should also distinguish fiscal commitments from the state's continuing authority over safety, environment, reporting and emergency response.

The pre-discovery period is therefore the best time to decide which issues belong in statute, which belong in regulation, which belong in the contract and which require case-specific permit conditions.

Analytical section 15C. Data sovereignty is a national asset

Geological and geophysical data can be more durable than any single licence.

The historical petroleum-promotion programme recognized data as an economic and institutional asset. The state sought to synthesize exploration information, use it to promote acreage and retain the ability to support later licensing decisions. [S35, S107]

This principle becomes more important after repeated exploration cycles. A company may leave a basin after drilling one unsuccessful well. The well still produces national information. Seismic processing may be superseded by later technology, but the underlying acquisition remains useful. Core samples, logs, mud reports, pressure data, geochemical results and well-location records can all reduce uncertainty for later explorers and improve government negotiating capacity.

A national petroleum data regime should therefore answer six questions.

First, who owns raw exploration data?

Second, when must contractors submit data to government?

Third, how long may data remain confidential?

Fourth, who is responsible for long-term preservation and metadata quality?

Fifth, under what conditions can the state re-license or commercialize historical data?

Sixth, how are consultants prevented from becoming the only holders of institutional memory?

The state does not need to perform every interpretation itself. It does need to preserve the inputs required for independent interpretation.

Analytical section 15D. Failed exploration is part of the institutional record

A serious case study should account for failed and interrupted exploration rather than tell the story backward from success.

Guyana's record included early wells, periods of inactivity, investor withdrawals, financing constraints and border-related interruptions before Liza. The official exploration history records repeated pauses and restarts across decades. [S35]

This matters because a state can learn from dry holes in at least four ways.

A geological lesson changes basin understanding.

A contractual lesson changes acreage design or work commitments.

A regulatory lesson changes the way licences and reporting obligations are administered.

An institutional lesson changes how much permanent capacity is justified.

A newcomer country should therefore maintain a "failed exploration archive". It should record why each licence ended, which work obligations were completed, what data was produced, whether environmental obligations were closed, and whether any contingent liabilities remain.

That archive prevents a future licensing round from treating the basin as if institutional history began with the current investor.

Analytical section 15E. Maritime boundaries are economic infrastructure

Maritime boundaries are often discussed as questions of sovereignty and international law. For offshore petroleum they are also a form of investment infrastructure.

A block cannot be assessed only by geology. Investors need confidence that the state granting the licence has jurisdiction over the acreage and that drilling can proceed without interruption. Guyana's history includes the 2000 CGX incident with Suriname and the subsequent arbitration process, which became part of the operating context for offshore exploration. [S35, S40]

The transferable lesson is not that all boundary uncertainty must be eliminated before a country conducts any offshore work. Some states explore while boundaries remain contested. The lesson is that boundary risk should be explicit in the petroleum-risk register.

A frontier government should map:

- undisputed acreage;

- overlapping claims;
- areas subject to judicial or arbitral process;
- navigation and fisheries implications;
- coast-guard and maritime-security capability;
- force-majeure implications for contracts;
- insurance implications for operators and contractors.

If acreage is disputed, the economic model should include the possibility of delay, suspension or exclusion.

Analytical section 15F. The institutional minimum viable state

The pre-discovery experience suggests that a new producer does not need every institution at full scale before exploration. It needs a minimum viable petroleum state.

That minimum should include:

- a legally identified licensing authority;
- a geological data custodian;
- public-sector access to petroleum legal expertise;
- a tax authority capable of identifying petroleum-specific risks;
- an environmental authority with offshore jurisdiction;
- a finance ministry able to model contingent future revenue rather than assume it;
- a maritime authority able to coordinate offshore safety and incident response;
- an audit trail for licences, amendments and work obligations.

The institution can be small. What matters is that mandates are clear.

A common failure mode in small states is to compensate for limited staffing by creating committees without assigning ownership. A committee can coordinate work. It cannot substitute for a named institution responsible for a statutory decision.

Analytical section 15G. Capacity should scale with decision risk

Petroleum staffing should not be sized only to production volume.

Before discovery, a handful of specialists may be adequate if the country has limited active acreage.

After a commercial discovery, the state faces a step-change in decision risk. It must review appraisal plans, development economics, field-development concepts, petroleum metering, environmental studies, cost-recovery systems, decommissioning assumptions, marketing arrangements and sovereign-revenue preparation.

This is why discovery should trigger a pre-defined capacity escalation plan.

The plan should identify:

- positions that must become permanent;
- functions that can be outsourced temporarily;
- conflicts that prevent one adviser from serving multiple roles;
- knowledge-transfer requirements;
- records that remain state property;
- the date by which external advisers must be replaced or shadowed by national staff.

The World Bank's earlier finding that academic training alone was insufficient remains relevant. Capacity is not the number of degrees funded. Capacity is the ability to make, document, challenge and repeat a decision under real operating conditions. [S107]

Analytical section 15H. Environmental baselines should precede the investment shock

Environmental baseline work becomes harder after commercial discovery because development pressure accelerates.

The best baseline is created before the location of production infrastructure, support bases, pipelines and tanker routes becomes politically and commercially salient.

For an offshore frontier, baseline ownership should include:

- marine mammals;
- fish and spawning areas;
- seabed conditions;
- water and sediment quality;
- coastal habitats;
- fisheries effort;
- port traffic;
- shoreline sensitivity;
- protected areas;
- disaster-response resources.

The state does not need to measure every variable continuously before a discovery. It should know what data exists, who owns it, whether methods are reproducible and which gaps would become material if development moves forward.

Guyana's pre-existing environmental statute and later project-specific EIA framework make it possible to distinguish law that existed before the petroleum boom from systems created in response to the scale of development. [S39, S48-S56]

Analytical section 15I. Natural capital should be on the same balance sheet as petroleum

A resource-discovery narrative can become distorted if petroleum is treated as the only national asset.

Guyana entered the oil era with forests, biodiversity, freshwater, agriculture, mineral resources and a pre-existing low-carbon strategy. The LCDS and forest-finance architecture therefore belong in the petroleum case study because they demonstrate that the country was already attempting to assign economic value to natural capital before first oil. [S22]

The accounting principle is broader than any specific carbon programme.

A country converts underground petroleum into financial and physical assets. It should simultaneously track whether extraction, infrastructure and associated growth reduce other forms of wealth.

A complete national balance sheet should therefore ask:

- what petroleum stock is being depleted;
- what financial assets are accumulating;
- what infrastructure is being created;
- what human capital is improving;
- what ecosystems are being degraded or protected;
- what future liabilities are being created.

This is the bridge between resource governance and sustainable development.

Analytical section 15J. The discovery shock should trigger a formal readiness review

The moment of discovery is often treated as a celebration or a geological milestone. Administratively, it should trigger a structured readiness review.

Within ninety days of a commercial discovery, a small state should reassess:

- petroleum law and regulations;
- contract administration;
- environmental authority;
- fiscal modelling;
- tax administration;
- sovereign-revenue custody;
- procurement rules;
- beneficial ownership;
- emergency response;
- labour and immigration;
- infrastructure demand;
- public communication.

The output should be a gap register with owners, deadlines and budget.

This is preferable to creating institutions reactively as each new problem appears.

Analytical section 15K. The strongest pre-discovery lesson is optionality

Good pre-discovery institutions preserve national options.

A state with organized data can run a better licensing round.

A state with clear law can choose among fiscal structures with less legal uncertainty.

A state with environmental baselines can compare development options.

A state with trained negotiators can decide when independent advice is necessary.

A state with a maritime-risk map can sequence acreage.

A state with conservative fiscal rules can avoid spending against unproven revenue.

Optionality matters because the state does not know whether the basin will remain marginal, produce one discovery or become a major province.

The best early institutions are therefore those that remain useful across all three outcomes.

Analytical section 15L. Expanded Part I conclusion

The deeper historical record changes the interpretation of Guyana's petroleum rise.

Commercial success arrived quickly after Liza, but the institutional prehistory was long. Petroleum promotion, legislation, geological data management, exploration contracting and public-sector training had been attempted decades earlier. The World Bank project of the 1980s did not solve every institutional problem and did not create a complete petroleum state. It did create a record of how a small frontier country tried to build petroleum capability before commercial certainty. [S107]

That earlier period also reveals a recurring constraint that matters today: institutional capacity is not static. The skills needed to attract an explorer are different from those needed to audit a multibillion-dollar development. The staff required to manage one exploration licence are different from those required to oversee several FPSOs, large petroleum receipts and complex environmental exposure.

For a newcomer, the practical standard is therefore staged readiness.

Before discovery, build the legal map, preserve data and establish environmental and maritime baselines.

After discovery, expand fiscal, legal, engineering and environmental capability before irreversible approvals.

Before first oil, test revenue custody, metering, auditing and emergency response.

After production begins, scale institutions with the number and complexity of projects rather than with the age of the petroleum sector.

Guyana's pre-discovery history is valuable not because it offers a perfect model. It is valuable because it shows that petroleum governance begins long before oil reaches a production vessel.

PART II

Discovery, Contract Design, Cost Recovery and Revenue Assurance

Analytical section 16. Why contract analysis requires historical context

A petroleum contract should be read twice. The first reading asks what the clauses mean. The second asks what information, risks and alternatives existed when the parties agreed to them.

This distinction matters in Guyana because commercial knowledge of the Stabroek Block changed sharply after the 2015 Liza discovery and continued to change as appraisal wells and later discoveries expanded the known resource base. A fiscal arrangement signed under frontier uncertainty and a later model agreement developed after multiple successful discoveries do not arise under the same information set. [S36, S41]

Historical context does not remove accountability. A government remains responsible for understanding the powers it exercises and the agreements it signs. Historical context instead prevents a different error, which is to judge an earlier negotiation solely with geological and commercial information learned years later.

The correct analysis therefore has three layers.

The first layer is the executed legal text. It states the rights and obligations.

The second layer is the information environment at the relevant date. It identifies reserve uncertainty, development cost, oil price, market interest and the state's technical capacity.

The third layer is observed performance. It records development, cost recovery, production, government receipts, audits and disputes.

The flagship uses all three.

Analytical section 17. From Liza discovery to the 2016 agreement

The Liza-1 record places the discovery well in May 2015. The official Stabroek Block register records the petroleum agreement date as 27 June 2016 and an effective/licence date of 7 October 2016. [S36, S41]

These dates should not be collapsed.

Between discovery and agreement, the parties were still learning about the field. Appraisal reduced geological uncertainty. Project concepts were being evaluated. Commercial expectations changed as evidence accumulated.

A historical fiscal assessment therefore needs a dated resource-estimate series. It should identify the resource view at:

- discovery;

- first appraisal;
- agreement signature;
- production licence;
- final investment decision;
- first oil.

The same time series should contain oil price conditions and estimated project costs.

Without this chronology, a contract comparison risks hindsight bias.

Analytical section 18. What a production-sharing agreement does

A production-sharing agreement allocates petroleum and costs through a contractual formula.

At a simplified level, gross petroleum production is valued. Royalty is calculated according to the agreement. A portion of production or value is allocated to recover eligible costs, subject to the contractual ceiling. The remaining profit petroleum is divided between the state and contractor under the agreed share. Taxes and other obligations then follow the applicable agreement and law.

The simplified arithmetic is useful for teaching.

It is not enough for an audit.

Actual implementation needs:

- measured production;
- crude valuation;
- lifting schedules;
- eligible cost;
- cost allocation;
- currency treatment;
- affiliate transactions;
- tax;
- year-end receivables;
- decommissioning.

The flagship therefore distinguishes a teaching waterfall from the audited account.

Analytical section 19. The 2 percent royalty

The public explanation of the 2016 Stabroek fiscal arrangement identifies a 2 percent royalty.
[S15]

A royalty is attractive from a revenue-administration perspective because it is calculated before profit and does not depend on whether the contractor has recovered all development cost.

That does not make royalty design simple.

The government still needs:

- verified production;
- verified sales volume;
- a valuation rule;
- due dates;
- currency;
- audit access.

A royalty rate also affects project economics. A higher royalty takes revenue even when costs are high and profit is low. A lower royalty leaves more downside risk with the state in exchange for greater investor cash during cost-intensive periods.

The economic effect should therefore be tested across field quality, capital cost, price and production decline.

A policy argument about a royalty rate should not be reduced to whether the number appears large or small.

Analytical section 20. The cost-recovery ceiling

The Stabroek arrangement permits cost recovery subject to a ceiling. Public explanations commonly use 75 percent as the maximum cost-recovery share. [S02, S15]

The word maximum is essential.

If recoverable costs exceed the ceiling in a period, some costs remain unrecovered and move forward according to the contract.

If recoverable costs are below the ceiling, less petroleum is allocated to cost recovery and more remains in the profit pool.

This creates time variation in government receipts even when the formal contract does not change.

The cost bank is therefore a central variable.

Analytical section 21. How the cost bank changes government cash

Consider a simplified field at constant production and price.

During early development, large capital costs enter the recoverable pool. If costs exceed the monthly ceiling, the ceiling binds. The residual profit pool is relatively small.

As older development costs are recovered, the actual cost share can fall below the ceiling. The residual profit pool grows and government profit oil rises.

A new sanctioned development can then add a large new set of recoverable costs.

Government cash therefore depends on:

- oil price;
- production;
- recoverable cost;
- timing;
- new investment;
- prior unrecovered balance.

A forecast that assumes a permanently low cost-recovery share after one high-revenue period risks overstating future receipts.

The 2026 government reporting discussed this dynamic when revising expected petroleum deposits and noting that new project development can affect the cost bank. [S07]

Analytical section 22. A 100-unit teaching example

Normalize gross petroleum value to 100.

Under a simplified case where cost recovery uses the full 75 percent ceiling:

- 75 units go to cost recovery;
- 25 units remain as profit petroleum;
- a 50 percent government share of profit petroleum equals 12.5 units;
- a 2 percent royalty equals 2 units.

Under the public teaching convention, the combined royalty plus state profit share equals 14.5 units. [S15]

Now reduce cost recovery to 50 units:

- 50 remain as profit petroleum;
- government's half is 25;
- royalty remains 2 under the simplified example.

Combined units become 27.

At 20 units of cost recovery:

- 80 remain as profit petroleum;
- government profit share is 40;
- royalty is 2.

Combined units become 42.

These are teaching cases.

They are not a claim that the state received 14.5, 27 or 42 percent in any specific calendar period.

Analytical section 23. Why government take is not one number

Petroleum fiscal analysis often produces a single government-take percentage.

That number depends on assumptions.

A full economic model needs:

- project capex;
- opex;
- production curve;
- oil price;
- royalty;
- cost recovery;
- profit share;
- corporate tax;
- state participation if any;
- bonuses;
- abandonment;
- discount rate.

Government take also has different definitions.

A cash-flow government take is not the same as a share of gross revenue.

A discounted government take is not the same as an undiscounted lifetime share.

A marginal effective tax rate is another concept again.

The flagship should therefore label every ratio with its denominator and model assumptions.

Analytical section 24. Cost recovery turns the contract into an audit system

A production-sharing agreement does not regulate revenue only through percentages.

It also regulates which costs are recoverable.

This creates a large administrative task.

The state needs to test:

- whether an expenditure occurred;
- whether the contractor paid it;
- whether it relates to the contract area;
- whether it is permitted;
- whether it is allocated correctly;
- whether it is duplicated;

- whether it is reasonable under the agreement;
- whether affiliate transactions follow the required basis;
- whether timing is correct.

The government-published cost audits provide direct evidence that these questions arise in practice. [S42, S71]

Analytical section 25. The 1999-2017 audit as a governance case

The government-published IHS Markit audit covered cost-recovery submissions for 1999-2017. The publication states that the contractor submitted about USD 1.678 billion for review and that the audit identified about USD 214.4 million, plus specified overhead adjustments, as having grounds for dispute. [S71]

Those figures must be described with precise verbs.

An auditor identified or recommended an adjustment.

Government disputed an amount.

A contractor responded.

A sole expert or arbitration process may later determine treatment.

A final accounting credit changes the cost bank.

Cash revenue follows only through the resulting petroleum allocation and sales process.

The word recovered should therefore be reserved for a completed financial outcome.

Analytical section 26. The 2018-2020 audit and category-level controls

The 2018-2020 audit offers more detailed examples of cost classification and allocation. [S42]

The audit material contains issues associated with:

- allocation between blocks;
- shared costs;
- office-related construction;
- emergency-response studies;
- pre-effective-date insurance;
- duplicate charges.

The significance is systemic.

Each type of issue suggests a control.

Block-allocation issues require an allocation methodology.

Shared services require cost drivers.

Pre-effective-date costs require contractual date controls.

Duplicate charges require invoice-level testing.

Affiliate charges require transfer-pricing and contractual review.

The paper should use the examples to explain audit design, not to imply that every contractor cost is disputed.

Analytical section 27. Effective date as an accounting boundary

The official block register gives 7 October 2016 as an effective/licence date. The 2018-2020 audit also uses the effective date in its treatment of specified costs. [S41, S42]

This shows why contract chronology belongs inside accounting.

A cost incurred before signature may have different treatment from a cost incurred after the effective date.

Some agreements expressly recognize pre-contract costs.

Others limit or condition them.

An audit model must therefore tag every material cost with:

- invoice date;
- service period;
- payment date;
- applicable contractual period.

The final cost-bank dataset should also preserve the audit period in which an adjustment is recognized.

Analytical section 28. Affiliate transactions

Large offshore developments rely on specialized global supply chains.

Contractors often procure services from affiliated companies or global framework arrangements.

The issue is not whether affiliate transactions exist.

The issue is whether the agreement's pricing and allocation rules are followed.

Audit controls should identify:

- affiliate;
- service;
- cost basis;
- markup;
- allocation;
- market comparison where required;
- invoice support.

A newcomer country needs access to international tax and petroleum accounting expertise for this work.

Analytical section 29. Allocation across multiple blocks and projects

A company operating several blocks or projects may incur costs that benefit more than one area.

Examples include:

- shared office;
- aviation;
- marine support;
- emergency planning;
- technical studies;
- management systems.

The accounting question is how much belongs to each contract area.

An arbitrary allocation can shift recoverable cost into a producing block.

A defensible allocation uses a stated driver such as:

- activity;
- headcount;
- vessel days;
- well count;
- direct usage.

The audit should test the driver and consistency over time.

Analytical section 30. Petroleum valuation

Cost oil and profit oil are calculated from physical petroleum and value.

The valuation rule therefore matters.

A fiscal model needs:

- crude grade;
- quality adjustment;
- benchmark;
- sale transaction;
- affiliate/non-affiliate status;
- timing;
- freight treatment.

A state should reconcile contract valuation to actual cargo sales where the agreement permits.

This is distinct from production measurement.

One system answers how many barrels existed.

Another answers what those barrels were worth under the contract.

Analytical section 31. Cargo timing and government receipts

Government profit-oil entitlement is often delivered in cargoes.

A cargo is not a monthly production statistic.

Crude produced over several days enters FPSO storage. A tanker lifts a cargo when scheduled. Settlement occurs later.

The 2020 NRF audit record demonstrates the effect. A late-year government lift was recognized as revenue but remained receivable at year end. [S45]

A contract-accounting system should therefore include:

- production period;
- entitlement;
- lift date;
- invoice;
- settlement;
- NRF deposit.

The same cargo should have one persistent identifier across all systems.

Analytical section 32. Tax treatment

Petroleum fiscal analysis also needs tax law and contractual tax provisions.

The model should identify:

- taxpayer;
- taxable base;
- rate;
- deductions;
- payment mechanism;
- credits;
- interaction with production-sharing terms.

Tax treatment should not be inferred from royalty and profit oil.

A later model PSA with an explicit corporate tax rate belongs in a different agreement generation. [S14]

Analytical section 33. Decommissioning and end-of-life cost

Every offshore project ends.

Wells need abandonment.

Subsea equipment may require removal or safe disposition.

The FPSO leaves.

Environmental restoration and monitoring may continue.

The agreement and production licence should therefore identify:

- who pays;
- when funding begins;
- how security is held;
- what happens if ownership changes;
- what standards apply at abandonment.

A fiscal model that excludes decommissioning overstates project cash available for distribution.

Analytical section 34. Later model terms

Guyana later announced model fiscal terms for new offshore licensing that differ materially from the Stabroek arrangement. The official announcement describes a higher royalty, a lower cost-recovery ceiling and a corporate tax for future arrangements. [S14]

The later model shows institutional learning and a changed bargaining environment.

It does not amend Stabroek.

The final comparison should include:

- royalty;
- cost ceiling;
- profit share;
- tax;
- relinquishment;
- work commitments;
- acreage;
- audit;
- local content;
- environmental terms.

Each column should also include the date and information environment.

Analytical section 35. Why later terms do not prove earlier terms were available

A later government may offer different terms because:

- geological risk is lower;
- more companies are interested;
- infrastructure exists;
- the regulator has more experience;
- comparable developments reduce technical uncertainty.

The market itself has changed.

A strong historical analysis therefore asks a counterfactual.

What terms would a competitive investor have accepted at the earlier date, given the information then available?

This is an economic question, not a slogan.

It requires contemporaneous frontier-basin agreements, project economics and investor interest.

Analytical section 36. Why frontier risk does not eliminate scrutiny

The opposite error also needs attention.

A high-risk frontier does not mean any fiscal term was unavoidable.

Government still has options:

- commission independent advice;
- compare jurisdictions;
- stage commitments;
- use work obligations;
- retain acreage;
- publish contracts after award;
- preserve audit rights.

The final paper should therefore evaluate process quality independently from hindsight.

A process can be strong even if a later price or geological outcome looks unfavorable.

A process can be weak even if good luck produces high revenue.

Analytical section 37. Contract transparency

Contract publication allows researchers and citizens to test fiscal claims against the legal text.

Publication also reduces confusion between:

- executed agreements;
- draft model agreements;
- production licences;
- public summaries.

The Petroleum Management Programme publishes petroleum documents, including the Stabroek materials and later model agreements. [S02, S43]

A contract portal should preserve:

- version;
- signature date;
- effective date;
- amendments;
- annexes.

A missing annex can make an apparently transparent contract incomplete for audit.

Analytical section 38. Confidentiality and legitimate commercial information

Transparency has limits grounded in law and legitimate commercial protection.

Seismic interpretation, proprietary technology, current bid strategy and some pricing information may require confidentiality for a period.

A strong system defines the exception rather than labeling the entire petroleum relationship confidential.

The public register should state:

- legal basis;
- category;
- duration;
- authority;
- review.

When confidentiality expires, the record should move into the public archive under the relevant law.

Analytical section 39. The negotiating team

A newcomer state needs a negotiating team whose composition reflects the contract.

Core disciplines include:

- petroleum law;
- geology/reservoir engineering;
- petroleum economics;

- tax;
- accounting;
- environmental law;
- decommissioning;
- insurance.

The team should have a written mandate.

Advisers should disclose conflicts.

Technical models should remain accessible to the state after the engagement.

This avoids a situation in which a government receives a recommendation without retaining the ability to test it.

Analytical section 40. Negotiation decision record

A useful negotiation record is not a transcript.

It is a structured decision log.

For each material clause:

- objective;
- operator position;
- government position;
- alternatives;
- fiscal impact;
- legal risk;
- final outcome.

The log provides institutional memory when officials change.

It also helps later auditors distinguish a negotiated policy choice from an implementation error.

Analytical section 41. Contract administration after signature

A signed agreement creates an administrative calendar.

The state must track:

- work programme;
- budget;
- discovery notice;
- appraisal;
- development plan;
- production licence;
- relinquishment;

- audit window;
- royalties;
- tax;
- local content;
- training;
- environmental obligations;
- insurance;
- decommissioning.

A calendar-based contract management system turns clauses into deadlines and evidence.

Analytical section 42. Audit capacity as a long-term asset

Petroleum audit should not be treated as an occasional consulting exercise.

The state benefits when each audit improves its permanent ability to:

- understand cost categories;
- query operator systems;
- test allocation;
- reconstruct cargo valuation;
- manage disputes.

External firms remain useful for specialist capacity.

Government still needs an internal team that owns the record.

Analytical section 43. Dispute resolution

Large petroleum contracts include mechanisms for technical and legal disputes.

The correct mechanism depends on the issue.

An accounting disagreement may move through audit responses and expert determination.

A legal interpretation dispute may move to arbitration or court, depending on the agreement and law.

The flagship should track disputes by procedural stage.

It should avoid describing an unresolved government claim as a final entitlement.

Analytical section 44. Revenue assurance as a system

Revenue assurance links five controls:

1. production measurement;
2. valuation;
3. cost audit;

4. fiscal calculation;
5. cash reconciliation.

Weakness in any one control affects the final public receipt.

A strong dashboard should therefore show more than money deposited.

It should show whether the underlying physical and contractual calculations reconcile.

Analytical section 45. Stress test one: low price

Take the simplified model.

A lower price reduces gross sales.

Royalty falls in proportion to sales.

Profit oil also falls.

If recoverable costs remain high, the state receives a smaller residual profit pool.

The budget impact depends on:

- cash already held in the NRF;
- lawful withdrawal;
- non-oil revenue;
- debt;
- project commitments.

This demonstrates why high-price revenue should not be treated as a permanent baseline.

Analytical section 46. Stress test two: new project cost

Suppose a mature field has recovered much of its historical cost and government profit oil rises.

A new development is sanctioned.

New recoverable cost enters the system.

The cost bank increases.

Government profit oil can fall relative to the mature-field period even if production later rises.

This is an important forecasting control.

Analytical section 47. Stress test three: production interruption

A production outage reduces physical barrels.

The impact on government cash may arrive later because cargoes and settlements lag production.

A robust fiscal model therefore needs timing rather than annual averages alone.

The new production-shock dataset in the repository combines lower volume, price and cost-recovery assumptions to demonstrate this interaction.

Analytical section 48. What another country should copy

The strongest transferable elements are procedural:

- preserve data;
- date every agreement version;
- obtain independent modelling;
- keep an audit calendar;
- reconcile physical and financial records;
- retain dispute status;
- publish executed terms under law;
- maintain institutional memory.

The exact royalty, cost ceiling or profit share is not a template.

Those terms depend on geology, market conditions, state capacity, capital cost and sovereign policy.

Analytical section 49. What another country should avoid

A newcomer should avoid:

- budgeting a discovery as proven revenue;
- copying fiscal terms from a mature basin without modelling;
- using a headline government-take number without assumptions;
- treating audit findings as recovered money;
- losing geological data to advisers;
- allowing contract versions to become unclear;
- permitting audit windows to expire through administrative delay.

Analytical section 50. Part II findings

The Guyana contract case is valuable because it displays the entire lifecycle of a production-sharing system.

The 2016 Stabroek terms define the baseline. [S02]

The cost audits show the agreement becoming an accounting system. [S42, S71]

The later model terms show a changed fiscal approach under a different information environment. [S14]

The NRF and cargo records show how contract entitlement eventually becomes public cash. [S45, S46]

The case does not yield one political verdict.

It yields a set of technical questions that every new producer should answer before signing, before first oil and throughout production.

Edition 2.0 deepening: from headline fiscal terms to an operating petroleum revenue system

Analytical section 50A. A production-sharing agreement is an operating constitution for the field

A petroleum agreement is often reduced in public discussion to a royalty percentage, a cost-recovery ceiling and a profit split.

Those numbers matter, but they do not describe the whole economic system.

A production-sharing agreement also governs:

- when costs become recoverable;
- which costs are excluded;
- how affiliate charges are tested;
- when audit rights expire;
- how petroleum is valued;
- how production is measured;
- which currency is used;
- how tax obligations are treated;
- how assets are transferred or decommissioned;
- how disputes are escalated;
- how assignments are approved;
- how stabilization and change-of-law provisions interact with later legislation.

The fiscal result is therefore produced by a chain of clauses rather than one percentage.

For audit purposes, every contract analysis should begin with a clause map. The clause map identifies the economic effect of each provision, the institution responsible for administering it, the data needed to test compliance and the time limit for challenge.

The executed 2016 Stabroek agreement remains the controlling baseline for that contract. Later model terms provide evidence of later policy design, not a substitute for the executed agreement. [S02, S112]

Analytical section 50B. Contract economics should be modelled as a waterfall

The most reliable way to explain a production-sharing agreement is to model it as a waterfall.

Start with gross production.

Subtract volumes that do not reach sale because of operational use, loss or other contractually recognized treatment.

Apply the valuation rule to saleable petroleum.

Calculate royalty on the correct base.

Determine the amount of recoverable cost subject to the contractual ceiling.

Calculate the residual profit petroleum.

Apply the profit split.

Add tax treatment.

Then bridge the state's physical entitlement to the timing of actual sale proceeds and cash receipts.

This sequence should be performed for each accounting period.

A single annual average can hide timing effects. A high-cost month, a delayed cargo or a large development expenditure can change the period result even when annual production remains strong.

The teaching value of a waterfall is that it makes every assumption visible.

Analytical section 50C. The cost bank is a balance-sheet item, not a rhetorical category

Cost recovery is frequently discussed as if recovered cost disappears.

In accounting terms, unrecovered cost is better understood as a stock carried forward subject to the agreement.

The opening cost bank increases with eligible current-period cost and may also reflect allowed carry-forward treatment.

It decreases when cost oil is allocated.

The closing cost bank becomes an input into future periods.

This creates an intertemporal link between today's investment and tomorrow's government profit petroleum.

A major new development can therefore increase production capacity while simultaneously increasing the stock of recoverable cost.

The state needs to forecast both.

A petroleum revenue model should report:

- opening recoverable-cost balance;
- current eligible cost;
- adjustments and audit disallowances;
- amount recovered in period;
- closing recoverable-cost balance.

Without this table, a public forecast can make government revenue appear to move mysteriously when the real driver is the cost bank.

Analytical section 50D. Audit rights are perishable assets

An audit right that expires unused has no economic value.

The GRA account of the inaugural cost audit is important because it shows how audit administration actually works. The audit covered pre-contract and exploration/development costs through 31 December 2017. Guyanese officials from several agencies worked with IHS Markit, and the process included draft findings, government review, auditee response and later dispute over the quality and finality of the audit product. [S111]

The administrative lesson is that an audit is not a single event.

It has at least eight stages:

1. define the audit population;
2. preserve records;
3. test invoices and allocation;
4. identify exceptions;
5. obtain auditee response;
6. evaluate supporting evidence;
7. settle or dispute findings;
8. update the recoverable-cost ledger.

Every stage requires ownership.

A country can have a strong audit clause and still lose value through late mobilisation, incomplete records, weak specialist support or poor dispute tracking.

Analytical section 50E. Audit scope should follow risk rather than invoice volume

Petroleum auditing should not treat every dollar as equally risky.

A risk-based audit plan should prioritize:

- affiliate charges;
- shared-service allocations;

- head-office overhead;
- procurement from related entities;
- financing costs if recoverable;
- inventory and asset transfers;
- pre-contract cost;
- abandonment and decommissioning accruals;
- costs allocated across projects or blocks;
- costs incurred outside the country but charged to the operation;
- abnormal or non-routine events.

The state should maintain a petroleum audit risk register updated after each audit.

Recurring findings should change future audit procedures.

If the same category of cost repeatedly produces disputes, the response should not be another isolated exception. The state should consider clearer guidance, standardized supporting schedules or contract amendments where legally available.

Analytical section 50F. Affiliate transactions require an arm's-length control system

Integrated petroleum groups purchase significant services from affiliated entities.

An affiliate transaction is not automatically improper.

The audit question is whether the charge is permitted, correctly allocated and consistent with the contract's accounting rules.

A robust system should test:

- service description;
- actual service delivery;
- allocation key;
- markup;
- transfer-pricing basis;
- duplication;
- relationship between payer and supplier;
- whether the same cost appears in another block or project.

The state's petroleum audit team and tax authority should coordinate without conflating their legal mandates.

A cost can be deductible for one purpose and non-recoverable for another, depending on the governing rules.

Analytical section 50G. Petroleum valuation deserves its own control framework

Government entitlement is sensitive not only to volume but also to value.

A valuation system should identify:

- crude grade;
- quality adjustment;
- benchmark;
- differential;
- freight;
- timing;
- related-party sales;
- destination;
- sales-agent fee where applicable.

If the government takes physical profit oil and sells its own cargoes, the state also needs a transparent sales protocol.

The sales record should preserve bids, counterparty selection, pricing formula, settlement terms and realized price.

This creates a second audit chain alongside cost recovery.

One chain protects the cost base.

The other protects the value of petroleum sold.

Analytical section 50H. Government take should be reported as a range with assumptions

There is no single government-take number that describes a petroleum agreement under all conditions.

Government take varies with:

- oil price;
- recoverable cost;
- production;
- tax treatment;
- project timing;
- discount rate;
- financing structure;
- decommissioning;
- treatment of bonuses and rentals;

- state participation where applicable.

A defensible fiscal comparison should therefore publish assumptions before the result.

At minimum, a comparative table should use a common oil price path, cost schedule, production profile and discount rate.

The same model should then run both contract systems.

This is especially important when comparing the 2016 Stabroek agreement with the later model PSA. The model agreement includes different fiscal and contractual architecture, including later-generation provisions on financial guarantees, insurance, decommissioning, audit and other matters. [S112]

A difference between the contracts is evidence of policy evolution. It does not establish that both sets of terms were available under the same geological information and market conditions.

Analytical section 50I. Stabilization clauses should be interpreted narrowly and precisely

Stabilization is often discussed in absolute language.

The real legal effect depends on the wording of the clause, the governing law, the type of government action and the available remedy.

A responsible case study should therefore avoid statements such as "the contract freezes all law" unless the text actually supports that conclusion.

The analytical questions are:

- which laws are covered;
- what counts as an adverse change;
- whether environmental and safety measures are treated differently;
- whether the remedy is compensation, renegotiation or another mechanism;
- whether international law limits the clause;
- how the clause interacts with later statutes.

The same discipline applies to tax provisions.

Legal characterization should begin with the executed text.

Analytical section 50J. Decommissioning is a present fiscal issue even when closure is decades away

Decommissioning is often deferred in public debate because production is still expanding.

Financially, it begins much earlier.

The state should understand:

- estimated abandonment cost;
- timing of contributions to any decommissioning fund;

- whether contributions are recoverable;
- asset-transfer obligations;
- residual environmental liability;
- security if a contractor exits or becomes insolvent;
- treatment of unused funds.

A late decommissioning framework creates intergenerational risk.

The public may receive revenue for years and then inherit a liability after production declines.

The later model PSA contains explicit decommissioning architecture that can be compared with the executed Stabroek framework, but the comparison must preserve contract version. [S112]

Analytical section 50K. Negotiation files are national institutional memory

A petroleum agreement records the final bargain.

It does not record all options considered.

For future accountability and learning, the state should retain a negotiation file containing:

- government objectives;
- fiscal model versions;
- adviser reports;
- risk assumptions;
- marked-up drafts;
- legal issues;
- decision memoranda;
- approvals;
- conflict declarations.

Some documents may require lawful confidentiality.

That does not justify institutional amnesia.

A later government should be able to reconstruct why a clause was accepted and what information was available at the time.

This is essential for distinguishing hindsight from genuine process failure.

Analytical section 50L. Contract administration needs a calendar, not just a filing cabinet

The contract should be converted into a compliance calendar.

Every recurring obligation should have:

- responsible institution;

- due date;
- data input;
- review procedure;
- escalation path;
- retention rule.

The calendar should cover:

- work programmes;
- budgets;
- cost statements;
- production reports;
- royalty;
- audit windows;
- training obligations;
- local content;
- insurance;
- guarantees;
- decommissioning updates;
- assignment approvals.

A state that administers the contract only when a dispute arises has already lost leverage.

Analytical section 50M. A petroleum contract should be tested against an adverse case before signature

The fiscal model should not use only a base case.

Before signature, the state should test:

- low oil price;
- high development cost;
- two-year delay;
- dry appraisal;
- rapid cost escalation;
- major outage;
- high decommissioning cost;
- contractor insolvency;
- disputed affiliate charges;
- major environmental event.

The purpose is not to predict these events.

The purpose is to identify which risks remain with the contractor, which move to the state and which become shared.

A contract that looks strong in a single central scenario can behave very differently under stress.

Analytical section 50N. Later model terms should be used as a reform crosswalk

The 2023 draft model PSA is valuable because it shows a later policy architecture in a basin with more geological information and a different institutional setting. It includes explicit articles on financial guarantees, liability and insurance, local content, cost recovery, taxation and royalty, decommissioning, audits, dispute resolution and stabilization. [S112]

A rigorous crosswalk should classify each change into one of five categories:

1. fiscal;
2. administrative;
3. risk allocation;
4. transparency and reporting;
5. end-of-life responsibility.

The crosswalk should then ask whether the change arose from:

- stronger geological knowledge;
- lessons from contract administration;
- changed market conditions;
- public-policy reform;
- a different licensing strategy.

This produces more insight than a simple "old versus new" percentage table.

Analytical section 50O. Expanded Part II conclusion

The contract chapter should leave the reader with one central idea: petroleum revenue is administered, not merely received.

The agreement creates rights, but government value depends on institutions converting those rights into measured production, validated costs, priced cargoes, settled disputes and reconciled cash.

Guyana's cost-audit history illustrates both the importance and difficulty of this work. The GRA account shows that government agencies had to coordinate with external specialists, challenge audit methodology, provide the contractor a response process and continue working through unresolved issues. [S111]

The later model PSA shows how contract architecture can evolve as a petroleum province matures. [S112]

For a newcomer, the practical blueprint is therefore to build four systems before first oil:

- a contract register;
- a cost-recovery ledger;
- a valuation and cargo-sales ledger;
- an audit and dispute calendar.

The state should then reconcile those systems to the sovereign revenue account.

If that chain cannot be reproduced, headline fiscal terms provide less protection than they appear to offer.

PART III

From First Oil to Large-Scale Production

Analytical section 51. First oil is a chain, not a single event

The phrase first oil is useful shorthand. It also hides several different milestones.

A development moves through project sanction, engineering, construction, drilling, installation, commissioning and production startup. Once crude begins flowing, it still needs processing, storage, lifting, sale and settlement before public petroleum cash appears in the sovereign account.

For a case study, each milestone should have its own date.

Guyana's Liza Phase 1 began production in December 2019. Liza Phase 2 started production on 11 February 2022. Payara started in November 2023. Yellowtail began production on 8 August 2025. [S81, S82, S83, S84]

This sequence demonstrates the speed at which one offshore development became a multi-FPSO production system.

The sequence also created a regulatory scaling problem.

Every new FPSO increases:

- production measurement;
- cargo scheduling;
- cost submissions;
- environmental monitoring;
- inspections;
- local-content reporting;
- emergency exposure;
- future decommissioning obligations.

The number of barrels is therefore also a measure of administrative workload.

Analytical section 52. Liza Phase 1 as the first operating test

The first producing project is the point at which petroleum governance moves from plans to live transactions.

For Liza Phase 1, the state needed to supervise:

- the production licence;
- environmental permit;
- production reporting;
- cargo entitlement;

- royalty;
- profit oil;
- cost recovery;
- cash receipt;
- NRF accounting.

The first project therefore functions as an institutional test bench.

It reveals whether laws written for exploration translate into operating controls.

It also exposes gaps that are hard to identify on paper.

Examples include:

- mismatch between production and sales periods;
- accounting receivables at year end;
- operating flaring;
- timing of cost audits;
- public reporting definitions.

The later permit modification associated with Liza Phase 1 flaring provides one example of regulation responding to observed operations rather than remaining fixed at approval. [S55]

Analytical section 53. The production licence

A petroleum agreement establishes the contractual relationship.

A production licence authorizes development and production under law.

The distinction matters.

The production licence identifies:

- area;
- project;
- development plan;
- term;
- reporting;
- operational obligations;
- environmental requirements;
- local-content requirements;
- abandonment.

The official petroleum portal publishes the Liza production licence and later project licences. [S86]

The flagship should build a production-licence register for every development and preserve amendments separately.

A newcomer country benefits from the same discipline. Contract rights and project approval should not be treated as one undifferentiated document.

Analytical section 54. Liza Unity and the move to two FPSOs

Liza Phase 2 began production on 11 February 2022. ExxonMobil stated that the Liza Unity FPSO was expected to ramp toward 220,000 barrels per day. [S82]

The startup doubled the complexity of several systems.

Production reports now needed project identification.

Environmental monitoring needed to separate FPSOs.

Cargoes needed entitlement tracking across two sources.

Cost accounting needed project-level detail.

Emergency planning needed to consider concurrent operations.

This is an important governance threshold.

A regulator that works comfortably with one producing asset does not automatically scale to two, four or six.

Analytical section 55. Payara and Prosperity

The government announced first oil at the Prosperity FPSO in November 2023. [S83]

By this stage, the production system contained:

- Liza Destiny;
- Liza Unity;
- Prosperity.

The third project added another production licence, environmental permit, cost stream, local-content plan and operating history.

The flagship should use Payara to test institutional standardization.

Questions include:

- Did production data use the same definitions?
- Were environmental reporting formats comparable?
- Were audit procedures consistent?
- Were public project documents easier to access?
- Did lessons from earlier permits appear in later conditions?

This is how regulatory learning is measured.

Analytical section 56. Yellowtail and the fourth operating development

Yellowtail began production on 8 August 2025. The operator stated that the ONE GUYANA FPSO increased combined installed capacity above 900,000 barrels per day. [S84]

In November 2025, the operator announced a daily production milestone of 900,000 barrels. [S85]

These two facts should remain separate.

Installed capacity is an engineering measure.

A daily milestone is a point observation.

Neither is the same as annual average production.

The final production chapter should display all three on separate lines.

Analytical section 57. Capacity versus realized production

A recurring error in petroleum reporting is to multiply nameplate capacity by 365 and treat the result as actual production.

Realized output is affected by:

- ramp-up;
- maintenance;
- well availability;
- reservoir performance;
- operating constraints;
- unplanned shutdowns.

The official production dashboard extends from December 2019 through 31 July 2026. [S81]

The flagship should calculate:

- monthly barrels;
- monthly daily average;
- annual barrels;
- annual daily average;
- utilization relative to stated capacity.

The utilization ratio should be treated cautiously because nameplate capacity can change after optimization.

Analytical section 58. Production optimization

Offshore facilities sometimes operate above original nameplate assumptions after engineering review and optimization.

This does not mean the original capacity figure was false.

It means facility constraints, well performance or operating procedures changed.

The data series should therefore preserve:

- original design;
- approved optimization;
- revised operating capacity;
- realized output.

A newcomer regulator should require technical support for material capacity increases, including environmental and safety review where required.

Analytical section 59. Production measurement

Petroleum revenue begins with physical measurement.

A credible measurement system requires:

- calibrated meters;
- sampling;
- temperature and pressure correction;
- volume reconciliation;
- tank measurement;
- cargo measurement.

The government needs independent access to the measurement chain.

A contract formula is only as accurate as the physical data entering it.

The flagship should identify which standards govern custody transfer and which agency verifies metering.

Analytical section 60. Production data architecture

The public production database should use persistent identifiers.

Each record should include:

- project;
- FPSO;
- field;
- date;
- oil;
- gas;
- water;
- flaring;

- reinjection;
- downtime.

Production data should link to:

- environmental data;
- cargo data;
- government entitlement.

This makes cross-checking possible.

Analytical section 61. Cargo lifting

FPSOs store produced crude until a tanker lifts a cargo.

The cargo date therefore follows production.

A cargo register should include:

- vessel;
- load date;
- barrels;
- crude grade;
- entitlement owner;
- pricing method;
- sale date;
- settlement date.

For government cargoes, the record should link to the NRF cash deposit.

The same cargo identifier should appear in both petroleum and financial records.

Analytical section 62. Government entitlement

Under the production-sharing arrangement, government receives a share of profit oil and royalty.

Government entitlement is influenced by:

- total production;
- valuation;
- recoverable cost;
- profit pool;
- contractual share.

Entitlement does not guarantee cash on the same day.

The government may lift a cargo and receive sale proceeds later.

This creates timing differences across production, fiscal and NRF reports.

Analytical section 63. The first-year receivable lesson

The 2020 audited NRF statements illustrate timing directly.

Four government profit-oil lifts were recognized for 2020, while the last lift remained a receivable at year end. [S45]

This is a small accounting detail with major analytical importance.

A calendar-year chart using cash deposits would show one number.

An accrual financial statement would show another.

Both can be correct.

The paper should never reconcile them by forcing one definition onto the other.

Analytical section 64. Royalty timing

Royalty is calculated from petroleum revenue under the applicable agreement.

Payment timing may differ from cargo sale timing.

The annual audited statements and petroleum reporting should be reconciled to the same transaction register.

Fields should include:

- royalty period;
- amount;
- due date;
- payment date;
- exchange rate;
- fund deposit.

This supports audit and public understanding.

Analytical section 65. The 2026 scale

By the first half of 2026, four FPSOs were operating.

The Ministry of Finance reported substantial profit-oil receipts and royalties during the period and a large NRF balance at end-June. [S07]

These figures establish scale.

They do not settle sustainability.

A high-revenue period can reflect:

- strong production;
- oil price;
- lower cost recovery;

- cargo timing.

The fiscal chapter therefore needs the physical production chapter underneath it.

Analytical section 66. Regulatory staffing versus asset count

A useful performance indicator is the ratio between regulator capacity and operating complexity.

Possible measures include:

- petroleum engineers per FPSO;
- environmental specialists per active permit;
- auditors per annual contractor cost submission;
- inspection days per project;
- unresolved audit value;
- publication delay.

The objective is not to establish an arbitrary international benchmark.

The objective is to detect whether workload grows faster than institutional resources.

Analytical section 67. Production and environmental monitoring

Production expansion changes environmental exposure.

Each project adds:

- emissions;
- flaring risk;
- produced water;
- vessel traffic;
- waste;
- subsea infrastructure.

Monitoring should therefore be both project-specific and cumulative.

A regulator needs to answer two questions.

Is each permit being followed?

What is the basin-wide effect of several projects operating together?

The second question becomes more important as the fleet expands.

Analytical section 68. Production and local content

More FPSOs increase procurement and employment demand.

This can widen local opportunities.

It can also strain:

- skilled labour;
- port capacity;
- housing;
- transport.

The local-content chapter should therefore align procurement data with project startup dates.

A rise in supplier spending immediately before first oil may reflect construction and commissioning rather than steady-state operations.

Analytical section 69. Production and infrastructure

Offshore operations rely on shore-based systems.

These include:

- port services;
- aviation;
- warehousing;
- waste handling;
- fuel;
- roads;
- telecommunications.

The final case study should distinguish privately financed oil-support infrastructure from public national infrastructure.

This avoids attributing every new facility to public petroleum spending.

Analytical section 70. Production and emergency readiness

Each additional offshore asset increases the number of possible simultaneous incidents.

Emergency planning needs to test concurrency.

Examples:

- a medical evacuation during severe weather;
- an operational incident on one FPSO while another is in maintenance;
- spill-response equipment already deployed elsewhere.

The response plan should state which resources are dedicated and which are shared.

Analytical section 71. Production data as a transparency tool

Public production data let citizens and analysts verify several other claims.

They support:

- royalty estimation;

- export analysis;
- emissions intensity;
- production forecasts;
- fund receipt expectations.

A public dashboard has greater value when it provides downloadable data rather than only a chart.

The dataset should retain historical revisions.

Analytical section 72. Forecasting production

A production forecast should show:

- producing projects;
- sanctioned projects;
- unsanctioned prospects.

Each category has different confidence.

Producing projects use observed decline and operating data.

Sanctioned projects use construction and startup assumptions.

Unsanctioned prospects should remain scenario cases.

A national budget should not treat all three categories as equivalent.

Analytical section 73. Startup risk

A project scheduled for first oil in a given quarter may face:

- fabrication delay;
- subsea delay;
- drilling delay;
- weather;
- commissioning issue.

A prudent fiscal forecast should include schedule sensitivity.

The public document should show which production year depends on a new project starting on time.

Analytical section 74. Downtime risk

A large FPSO contributes a material share of national output.

An outage therefore affects:

- exports;
- government entitlement;

- royalties;
- GDP;
- foreign exchange.

The model suite now includes lower-volume stress cases to demonstrate the arithmetic effect.

A national fiscal framework should test multi-month interruption, not only daily variation.

Analytical section 75. Price and production together

Price and volume are not independent risk controls.

A lower oil price can coincide with production interruption.

The combined effect on gross petroleum value is multiplicative.

Government cash then depends on cost recovery and fiscal terms.

This is why the stress tests use both price and volume changes.

Analytical section 76. First-oil readiness for another country

A newcomer should rehearse the first cargo before it happens.

The exercise should assign:

- producer;
- petroleum regulator;
- revenue authority;
- sales agent;
- central bank;
- finance ministry;
- auditor.

A fictional cargo should include one deliberate error.

The system passes only when institutions identify and correct it.

Analytical section 77. A production data room for government

The government should maintain a permanent production data room containing:

- licence;
- development plan;
- meter calibration;
- daily production;
- cargo;
- price;
- royalty;

- cost statement;
- entitlement;
- permit monitoring.

Access controls should protect commercial data.

Institutional memory should survive personnel changes.

Analytical section 78. A public petroleum ledger

Not every field in the government data room belongs in public.

The public ledger should still allow reconciliation of:

- production;
- government cargoes;
- public petroleum receipts;
- NRF deposits.

It should define every term.

Publication should follow a fixed calendar.

Analytical section 79. What rapid scale-up demonstrates

Guyana's production sequence demonstrates that institutional preparation is not a one-time first-oil task.

The state needs to scale with the sector.

A system adequate for one project may become inadequate at four.

Laws need implementation.

Audits need completion.

Environmental monitoring needs expansion.

Data systems need standardization.

Analytical section 80. Part III findings

The production record establishes a fast move from first oil in December 2019 to four operating FPSOs by August 2025. [S81-S85]

The main transferable lesson is not the speed itself.

It is the need to separate every physical and financial stage.

Discovery is not production.

Capacity is not output.

Production is not cargo.

Cargo is not cash.

Cash is not budget spending.

A newcomer country that preserves these distinctions gains a traceable petroleum system.

That traceability becomes the foundation for sovereign wealth management, environmental oversight and public accountability in the later parts of this study.

Edition 2.0 deepening: production scale-up as an integrated control system

Analytical section 80A. First oil should be decomposed into a readiness sequence

"First oil" compresses many separate events into one phrase.

Before production begins, a state has already had to approve or monitor:

- appraisal;
- commerciality;
- field-development design;
- production licence;
- environmental authorization;
- FPSO construction and commissioning;
- subsea installation;
- metering;
- lifting procedures;
- customs and import treatment;
- emergency arrangements;
- revenue custody.

The correct readiness test is therefore not whether the FPSO can produce.

It is whether the state can independently account for the first barrel from reservoir to public receipt.

A new producer should conduct a full simulated production month before startup. The exercise should create synthetic daily production, meter data, cargo nominations, lifting statements, benchmark pricing, royalty calculations, cost-recovery entries, government entitlement and a mock NRF deposit.

Every responsible institution should sign off on the same transaction.

Analytical section 80B. FPSO-by-FPSO expansion changes regulatory complexity

Guyana moved from the Liza Destiny to additional FPSOs in a short period. The production dashboard and project records allow each startup to be dated separately. [S81-S85]

The policy significance is not only higher capacity.

Each additional FPSO creates another operating system with:

- production meters;
- maintenance schedules;
- flare and emissions records;
- tanker traffic;
- helicopter and marine support;
- environmental monitoring;
- emergency-response requirements;
- cargo schedules;
- project-specific cost accounts.

A regulator that was adequate for one project may not remain adequate for four or more.

Capacity planning should therefore use project count and control points, not only barrels per day.

Analytical section 80C. Installed capacity, observed output and annual production are different measures

An FPSO's nameplate capacity is an engineering design parameter.

Observed daily production is a point-in-time operating result.

Monthly production is an aggregation of realized output.

Annual production incorporates ramp-up, downtime, maintenance and reservoir performance.

These measures should never be substituted for one another.

The distinction becomes especially important during rapid scale-up. A new vessel may add large installed capacity but contribute only part of a year's production because of startup timing.

Forecasting should therefore be done monthly, project by project.

The public dashboard should display capacity and realized production on separate axes.

Analytical section 80D. Metering is the first line of revenue assurance

Revenue assurance begins before price and cost recovery.

It begins with volume.

The state needs confidence in:

- meter specification;
- calibration;
- proving;
- failure procedures;
- correction factors;
- custody-transfer points;
- data retention.

If a production meter fails, the agreement and licence should define how production is reconstructed.

The regulator should preserve calibration certificates and exception logs.

This is not an engineering detail separate from fiscal governance.

A systematic measurement error becomes a revenue error.

Analytical section 80E. Cargo accounting is a bridge between reservoir and treasury

The public revenue chain should be reproducible at cargo level.

For each cargo, government should be able to connect:

- production period;
- entitlement calculation;
- lifting date;
- volume;
- quality;
- buyer;
- pricing formula;
- realized price;
- settlement date;
- cash receipt;
- NRF deposit.

This table should not expose legitimate commercial secrets unnecessarily.

It should exist inside government.

A summarized public version can then reconcile total government cargoes to reported petroleum receipts.

The advantage of cargo-level accounting is that timing differences become visible. Production in one month may generate a sale in another and a cash receipt later still.

Analytical section 80F. Production forecasting should be probabilistic

A single production forecast hides operational risk.

A more useful framework separates:

- base production;
- planned maintenance;
- startup ramp;
- reservoir decline;
- unplanned downtime;
- new project contribution.

Government revenue forecasting should then run at least three production cases.

The low case should not be an arbitrary percentage reduction. It should represent plausible operational conditions such as delayed startup or extended maintenance.

This produces a more credible fiscal contingency range.

Analytical section 80G. Maintenance is part of sovereign revenue risk

As an offshore province matures, maintenance becomes economically material.

Shutdowns affect:

- production;
- cargo timing;
- cost recovery;
- government revenue;
- service demand;
- safety risk.

The state should therefore distinguish planned from unplanned downtime.

A public production series can show the output effect.

The regulator's internal record should also preserve the reason for material outages where legally reportable.

This matters because repeated operational interruptions can signal a different risk from one scheduled maintenance campaign.

Analytical section 80H. Production data should be governed as public infrastructure

A production dashboard is not merely a communications product.

It is part of national statistical infrastructure.

The series should have:

- documented units;
- project identifiers;
- revision policy;
- observation frequency;
- publication lag;
- metadata;
- archival versions.

If historical values are revised, the state should preserve the earlier release and explain the revision.

This makes the dataset usable for fiscal analysis, academic work and public accountability.

The official Guyana production series through July 2026 provides a foundation for this approach.
[S81]

Analytical section 80I. Production and cost data should meet at project level

A national production total is not enough for petroleum economics.

The state should be able to associate each project with:

- cumulative production;
- development cost;
- operating cost;
- recoverable-cost balance;
- government entitlement;
- environmental obligations;
- decommissioning estimate.

This creates a project economics ledger.

The ledger helps distinguish whether lower government profit petroleum results from price, production, cost recovery or project mix.

Analytical section 80J. Rapid scale creates a regulator financing question

A producing state should ask how regulatory capacity is funded.

If the regulator depends entirely on annual general-budget appropriations, staffing may lag sector growth.

Alternative mechanisms can include lawful licence fees, regulatory charges or dedicated appropriations.

The design must preserve independence and public accountability.

The key principle is that the state should not be forced to supervise a multibillion-dollar industry with a staffing model designed for a pre-production frontier.

Analytical section 80K. Production expansion should trigger emergency-response scaling

Every additional offshore project changes the response problem.

Response planning should account for:

- simultaneous operations;
- distance between installations;
- capping-resource availability;
- vessel capacity;
- aviation capacity;
- medical evacuation;
- shoreline sensitivity;
- regional assistance.

The emergency plan should test concurrent incidents rather than assume only one project can fail at a time.

This links Part III directly to the environmental analysis in Part V.

Analytical section 80L. Mature production requires decline planning

A petroleum strategy should not focus only on startup.

Every field eventually declines.

The state should monitor:

- reservoir performance;
- water cut;
- gas handling;
- infill drilling;
- enhanced-recovery proposals;
- operating-cost changes;
- cessation-of-production planning.

A late-stage field may produce fewer barrels while carrying higher operating cost.

That can affect both cost recovery and government cash.

A country should therefore begin decline and decommissioning analysis while production is still growing.

Analytical section 80M. Expanded Part III conclusion

Guyana's rapid production scale-up makes the country especially useful for a newcomer blueprint because institutional stress becomes visible quickly.

The transition from one FPSO to several projects compressed regulatory learning into a few years. [S81-S85]

The central control principle is traceability.

Every barrel should pass through a chain of measurement, entitlement, sale, receipt and public accounting.

Every project should have a corresponding chain of licence, environmental authorization, operating data, cost account and decommissioning estimate.

The production chapter should therefore be read as a data-governance chapter as much as an engineering chapter.

A newcomer that builds these controls before first oil creates the evidence base needed for every later question in the book: fiscal performance, environmental impact, local content, public investment and intergenerational wealth.

Edition 2.0 project deepening: seven development decisions as one evolving system

Analytical section 80N. Liza Phase 1 as the institutional first test

Liza Phase 1 should be treated as more than the first production project.

It was the first time the state had to apply the new offshore petroleum system to a commercial deep-water development.

The project created simultaneous demands for:

- development-plan review;
- production licensing;
- environmental authorization;
- metering;
- cargo accounting;
- royalty administration;
- cost recovery;

- emergency response;
- public revenue custody.

First oil on 20 December 2019 therefore marks the beginning of a new operating regime, not simply a production milestone. [S01, S81, S86]

The project is useful for audit because it establishes the baseline against which later changes can be measured.

For every major rule or institutional improvement introduced later, the book should ask whether the control existed at Liza Phase 1 startup or was added after operating experience.

This produces an institutional learning timeline.

Analytical section 80O. Liza Phase 2 as the first scale replication test

Liza Phase 2 reached first oil on 11 February 2022. [S82]

The significance of the second development is that a country no longer has the excuse of first-project novelty.

The regulator should be able to compare:

- permit conditions;
- development cost;
- production capacity;
- project schedule;
- environmental monitoring;
- metering architecture;
- local content;
- startup ramp;
- operating incidents.

A second project creates the first true benchmark.

If a requirement changed between Phase 1 and Phase 2, the state should be able to explain whether the change resulted from new evidence, different project design or policy reform.

Analytical section 80P. Payara as the multi-project transition

Payara's first oil in November 2023 marked the transition from a two-project system to a multi-project basin. [S83]

This changes the regulator's workload non-linearly.

Several projects can now:

- submit cost statements simultaneously;
- schedule maintenance at overlapping times;

- compete for response resources;
- use the same supply base;
- employ the same contractors;
- affect the same fisheries and marine environment.

The state should therefore stop treating projects as isolated administrative files.

It needs portfolio management.

A portfolio dashboard should show the cumulative effect of all approved developments on:

- production;
- cost bank;
- regulator staffing;
- marine traffic;
- emergency resources;
- local labour;
- public revenue.

Analytical section 80Q. Yellowtail as the rapid-capacity step

Yellowtail began production on 8 August 2025, becoming the fourth producing project. [S84]

The later November 2025 operator report that production reached a daily milestone of 900,000 barrels should be treated as a point observation. [S85]

This is a useful example of the distinction between capacity and realized output.

The correct time series should preserve:

- startup date;
- monthly output;
- daily maximum where relevant;
- maintenance periods;
- national total.

A newcomer should resist the temptation to convert a production milestone into an annual revenue assumption without accounting for ramp-up and downtime.

Analytical section 80R. Uaru as the cutoff-date discipline test

Uaru is particularly important for the current edition because it demonstrates why an evidence cutoff matters.

The Errea Wittu FPSO arrived in Guyana in August 2026. Government and contractor sources described commissioning work ahead of first oil and identified a design capacity around 250,000 barrels per day. [S124, S125]

At the 27 September 2026 evidence cutoff used by this publication, the source set reviewed here does not establish an observed first-oil event.

The book should therefore say "commissioning ahead of first oil" rather than convert an expected September or fourth-quarter startup into an observed event.

This may become outdated immediately after publication.

That is acceptable.

A dated publication should prefer a correct cutoff over a premature milestone.

The corrections register can record the eventual startup in a later edition.

Analytical section 80S. Whiptail as an under-construction comparison

Whiptail provides a useful case because the state can observe development while another project is still being commissioned.

The 2026 maritime record confirms continuing offshore development activity, while project reporting identifies the Jaguar FPSO and a planned capacity around 250,000 barrels per day. [S126]

The correct analytical treatment is prospective.

The book can examine:

- approved project design;
- construction;
- subsea installation;
- expected capacity;
- stated startup target.

It should not include future Whiptail barrels in observed production.

This distinction becomes crucial in fiscal forecasting.

Analytical section 80T. Hammerhead as the seventh-development governance test

Hammerhead received development approval and a production licence in September 2025. Government and operator sources describe an estimated US\$6.8 billion project, planned production capacity of approximately 150,000 barrels per day and an anticipated 2029 startup. [S127, S128]

The project's analytical significance is that the state was approving a seventh development roughly a decade after the original Liza discovery.

This raises a cumulative governance question.

How should fiscal terms, environmental controls, financial assurance, local content and data requirements evolve as geological uncertainty declines and institutional experience increases?

The answer should be based on executed legal instruments and project-specific approvals.

It should not assume that later policy preferences automatically alter earlier contractual rights.

Analytical section 80U. Project approval should create a national commitments register

Each approved project creates future commitments for both operator and state.

The register should include:

- approved capital estimate;
- well count;
- FPSO;
- design capacity;
- startup target;
- production licence conditions;
- environmental permit;
- financial assurance;
- local-content plan;
- decommissioning plan;
- expected regulator staffing.

This allows government to see the full forward workload created by cumulative project approvals.

Analytical section 80V. Project cost should be normalized before comparison

Headline project costs cannot be compared without context.

A useful comparison should consider:

- design capacity;
- number of wells;
- water depth;
- subsea complexity;
- inflation;
- supply-chain conditions;
- construction timing;
- storage capacity;
- gas handling;
- local-content requirements.

The purpose is not to determine automatically whether one project is expensive.

It is to identify the drivers of cost.

This also improves cost-recovery forecasting.

Analytical section 80W. Production capacity should be connected to environmental capacity

Every new FPSO adds production capacity.

The environmental system must also add:

- monitoring capacity;
- inspection capacity;
- incident response;
- financial assurance;
- data review.

A country should therefore maintain an "environmental regulatory capacity per active project" indicator.

Possible measures include:

- inspectors per producing project;
- permit reports reviewed on time;
- drills completed;
- independent samples;
- unresolved permit findings.

This does not create a universal staffing ratio.

It makes regulator scaling visible.

Analytical section 80X. Marine traffic should be part of the project register

Offshore development changes tanker, supply-vessel, helicopter and support traffic.

The project register should therefore connect each development to:

- cargo frequency;
- support-vessel requirements;
- aviation demand;
- shore-base demand;
- navigational notices.

This helps quantify cumulative maritime exposure.

Analytical section 80Y. Development plans should include data-delivery obligations

The development approval should specify the data government expects during construction and operation.

At minimum:

- well data;
- reservoir model updates;
- cost forecasts;
- procurement;
- schedule;
- installation progress;
- commissioning;
- production;
- maintenance;
- incidents;
- decommissioning estimate.

A government that receives only final summary reports cannot independently update its own national model.

Analytical section 80Z. Startup readiness should be project-specific

A national first-oil system can be reused.

Each project still needs its own startup certification.

The checklist should confirm:

- permit conditions satisfied;
- metering commissioned;
- emergency plans tested;
- financial assurance valid;
- cargo and valuation protocols ready;
- government entitlement system configured;
- reporting templates active.

This avoids assuming that readiness for Liza automatically proves readiness for Uaru or Whiptail.

Analytical section 80AA. The project portfolio should be stress-tested as a system

A multi-project basin can experience common shocks.

Examples include:

- hurricane or severe weather affecting several facilities;
- helicopter-service constraint;
- port disruption;
- contractor failure;
- common equipment defect;
- cyber incident;
- shortage of specialist personnel.

The regulator should therefore run portfolio-level stress scenarios in addition to project-specific ones.

Analytical section 80AB. The offshore project register is now a core publication dataset

Edition 2.0 adds `OFFSHORE\_PROJECT\_REGISTER.csv` as a machine-readable control.

The register classifies each major project by observed or planned status.

Its most important rule is simple:

A planned startup remains planned until a dated source establishes production.

This rule should govern every chart, narrative and fiscal scenario in the final publication.

Analytical section 80AC. Project-level conclusion

The seven-development sequence shows why Guyana cannot be described only as a story of rapid production growth.

It is also a story of repeated development approvals, each adding another layer of fiscal, environmental, operational and administrative exposure.

For a newcomer, the key lesson is to build a project portfolio system before the second development is approved.

The state should be able to compare projects on common fields, identify cumulative burdens and preserve a dated record of what was forecast versus what actually occurred.

That is how rapid development becomes an auditable institutional learning process rather than a succession of isolated announcements.

PART IV

The Natural Resource Fund and the Conversion of Petroleum into National Wealth

Analytical section 81. The fund begins before the money

A sovereign resource fund is most useful when its rules exist before large recurring resource receipts create pressure for immediate spending.

Guyana's first Natural Resource Fund law was enacted in January 2019, before first offshore production in December 2019. [S37]

This timing is one of the most important documented features in the Guyana case.

The timing does not prove that every later rule was optimal.

The fund framework changed.

The 2019 law was replaced in 2021.

Withdrawal provisions changed again in 2024. [S04, S05]

The best-practice question is therefore not whether Guyana established one perfect rule before first oil.

The better question is whether the country created a formal petroleum-revenue institution before recurring production, published financial records, subjected the fund to audit and later revised the framework through law.

Analytical section 82. Why a fund exists

A petroleum fund solves several administrative problems.

It gives resource revenue a defined legal destination.

It separates receipt of petroleum cash from automatic public spending.

It creates a custodian.

It defines investment authority.

It creates reporting and audit obligations.

It supports a visible bridge from resource extraction to the national budget.

A fund does not decide whether money is spent well after transfer.

That is the job of public investment, procurement, service delivery and audit systems.

The distinction is central to this paper.

Analytical section 83. The legal objective

A resource-fund law should state its purpose clearly.

Possible objectives include:

- stabilization;
- intergenerational savings;
- budget support;
- macroeconomic management;
- preservation of national wealth.

One fund can combine several objectives.

Combining objectives creates trade-offs.

A highly liquid stabilization portfolio is designed differently from a long-term growth portfolio.

A rule designed to support major development spending differs from one designed to preserve capital indefinitely.

The legal purpose should therefore connect to the investment and withdrawal rules.

Analytical section 84. The custodian

Guyana's NRF uses the Bank of Guyana as a central operational institution.

The audited statements and quarterly reports provide the financial record. [S44-S47]

A custodian performs functions different from the political budget authority.

The custodian records:

- deposits;
- investments;
- income;
- expenses;
- transfers;
- closing assets.

The budget authority determines lawful public spending under the national fiscal framework.

Keeping the two roles clear improves accountability.

Analytical section 85. Cash versus accrual

The 2020 audited NRF statements show why fund analysis needs accounting precision. [S45]

Petroleum revenue recognized in the financial statements included a year-end receivable.

Cash at year end was therefore lower than the accumulated recognized income.

A researcher who reports the cash balance and a researcher who reports accrual revenue are not necessarily contradicting each other.

They are answering different questions.

A national dashboard should state the accounting basis beside every number.

Analytical section 86. The 2020 opening record

The first audited period establishes a reference point.

The 2020 statement records royalties, profit oil, limited investment income and no government withdrawal. [S45]

The account also identifies the receivable tied to a late-year government lift.

This gives the flagship a complete example of petroleum moving through three stages:

1. cargo entitlement;
2. accounting recognition;
3. later cash settlement.

The example belongs in the executive training module because the timing issue appears in many resource states.

Analytical section 87. The 2021 record

The 2021 audited statements show a larger fund and still no government withdrawal during the year. [S46]

The statement also carries restated comparative data.

Restatement is not an error to hide.

It is part of the financial reporting process.

The flagship should use the latest audited comparative where a later report formally restates an earlier period.

A public data system should preserve both the original publication and the restatement history.

Analytical section 88. The first withdrawals

Withdrawals began after the early accumulation period.

The Bank of Guyana reporting provides a dated series of transfers to the Consolidated Fund. [S47]

The analysis should separate:

- legal maximum;
- parliamentary appropriation;
- actual transfer;

- ultimate public expenditure.

A legal maximum is not an instruction to withdraw the full amount.

An appropriation is not a completed project.

A transfer is not an outcome.

Analytical section 89. The consolidated budget boundary

Once NRF money enters the Consolidated Fund, it enters the broader public budget.

At this point, the petroleum origin of the cash is economically important but accounting attribution to a specific project depends on budget structure.

If petroleum revenue is pooled with taxes and borrowing, a claim that one named road was paid for by one specific oil cargo requires a traceable appropriation mechanism.

The flagship should therefore distinguish macro-fiscal financing from project tracing.

Analytical section 90. The withdrawal rule

A withdrawal rule transforms fund assets into authorized budget space.

A rule can use:

- recent petroleum revenue;
- fund balance;
- expected long-run return;
- a fixed formula;
- a multi-tier formula;
- a fiscal-balance target.

Every formula creates different sensitivity to oil prices and asset values.

Guyana's legal rule changed over time.

The final crosswalk should show each version and apply it only to the periods in which it governed.

Analytical section 91. Rules and discretion

A mechanical formula reduces some forms of discretion.

It does not remove political choice.

Parliament still chooses appropriations within the legal framework.

Government still chooses projects.

A formula also depends on data definitions.

If the rule references revenue, the law must define which revenue and which period.

If it references assets, valuation matters.

A strong rule is therefore both mathematical and legal.

Analytical section 92. Stabilization

Petroleum prices move sharply.

Production also moves.

A stabilization objective seeks to prevent the public budget from rising and falling one-for-one with resource receipts.

The fund can support this by accumulating assets during high-revenue periods and providing lawful budget support during weaker periods.

The size of the buffer depends on:

- revenue volatility;
- budget rigidity;
- debt;
- import dependence;
- disaster exposure.

An island exposed to hurricanes may need a different liquidity profile from a continental state.

Analytical section 93. Intergenerational equity

Oil left underground has option value.

Oil produced today becomes cash.

If all public cash is consumed, future generations inherit less of the original asset unless current spending creates durable capital.

Intergenerational equity therefore does not necessarily mean saving all petroleum revenue as financial assets.

A country can convert resource wealth into:

- education;
- health;
- infrastructure;
- productive capacity;
- resilience.

The challenge is measuring whether the converted asset lasts.

Analytical section 94. Financial saving versus domestic investment

External financial saving has advantages.

It diversifies national wealth away from the domestic economy.

It creates liquid assets.

It reduces pressure on local construction markets.

Domestic investment also has advantages when high-return public projects remove constraints.

The comparison depends on project quality.

A poorly designed road is not automatically better than a global bond portfolio because it is domestic.

A high-return drainage, power or education investment is not automatically worse than financial saving because it is spent.

The case study should compare expected public value, risk and maintenance.

Analytical section 95. Absorptive capacity

A government can have the legal authority and cash to invest faster than the economy can efficiently deliver.

Absorptive limits include:

- engineering staff;
- contractor capacity;
- imported materials;
- land;
- procurement;
- project management.

When demand exceeds supply, project costs rise.

This is why the IMF's 2026 mission discussed public investment monitoring and absorptive capacity. [S64]

The statement should remain attributed to IMF staff.

The flagship should then test the proposition with independent data.

Analytical section 96. Construction inflation

Rapid public construction can increase:

- cement cost;
- aggregate cost;

- equipment rental;
- wages;
- land prices.

The national project register should record original and final contract cost.

Cost growth needs a cause classification.

Possible causes include:

- design change;
- inflation;
- scope expansion;
- delay;
- unforeseen conditions.

This gives a stronger basis for fiscal analysis than the size of the capital budget alone.

Analytical section 97. Project sequencing

A large revenue envelope creates a temptation to launch many projects simultaneously.

Project sequencing asks which constraints should be removed first.

For example, a new industrial facility may deliver less value if power and transport remain unreliable.

A new hospital may struggle if staff housing and training are not available.

The public investment system should therefore identify dependencies.

A project portfolio should be reviewed as a network.

Analytical section 98. Public investment appraisal

Before procurement, a major project should have an appraisal.

The appraisal should state:

- problem;
- alternatives;
- capital cost;
- operating cost;
- demand;
- economic benefit;
- environmental effect;
- climate resilience;
- implementation risk.

The strongest option may not be the largest project.

A transparent appraisal allows later comparison between forecast and outcome.

Analytical section 99. Procurement

A resource windfall increases procurement volume.

This increases the importance of:

- competition;
- qualification;
- bid evaluation;
- contract publication;
- variation control.

Procurement reform is therefore part of resource governance even when it sits outside petroleum law.

The supreme audit institution and parliamentary oversight become part of the petroleum wealth-conversion chain.

Analytical section 100. Completion is not operation

A physical asset produces value only after commissioning and operation.

A hospital building without staff is not full health capacity.

A power plant without dependable fuel or transmission is not reliable electricity.

A road without maintenance loses value.

The project register should therefore have separate fields for:

- physical completion;
- commissioning;
- service start;
- service performance.

Analytical section 101. Maintenance as a hidden liability

Capital expenditure creates future recurrent spending.

Roads need repair.

Drainage pumps need fuel and maintenance.

Hospitals need staff, medicines and power.

Power systems need overhaul.

A public balance sheet should estimate lifecycle cost.

This prevents the country from building a portfolio whose future maintenance burden exceeds stable revenue after oil growth slows.

Analytical section 102. Human capital

Education and health spending create assets that do not appear in conventional government balance sheets.

A petroleum wealth strategy should still measure them.

Indicators include:

- learning;
- certification;
- labour productivity;
- life expectancy;
- disease burden;
- access to care.

Spending alone is insufficient.

The national wealth framework should connect expenditure to outcome.

Analytical section 103. Natural capital

Petroleum is part of natural capital.

So are forests, fisheries, water and ecological systems.

Resource extraction reduces one natural asset.

Pollution can reduce others.

Conservation can preserve them.

A complete wealth framework should therefore report natural assets beside financial and produced capital.

This is particularly relevant in Guyana because forest finance predates the oil boom.

Analytical section 104. The non-oil primary balance

The non-oil primary balance measures the fiscal position excluding petroleum revenue and interest.

It helps show how dependent public expenditure has become on resource income.

A large non-oil deficit is not automatically unsustainable.

A developing country may choose temporary resource-financed investment.

The indicator still reveals the scale of the transition.

The IMF suggested consideration of a longer-run guide based on the non-oil primary balance.
[S64]

The paper should present that as an external proposal rather than its own political recommendation.

Analytical section 105. Public debt

A country with petroleum assets may still borrow.

Debt can finance projects before petroleum cash arrives, smooth project delivery or support capital markets.

Debt also creates fixed future payments.

A sovereign balance sheet should therefore report both fund assets and public debt.

Gross fund size alone can overstate net financial wealth.

Analytical section 106. Exchange-rate and import effects

Large public spending often increases imports.

Construction equipment, machinery and specialized services may come from abroad.

This creates foreign-exchange demand.

A petroleum fund invested abroad partially offsets domestic liquidity pressure until money is withdrawn.

The macroeconomic section should link withdrawals to imports, exchange rates and inflation.

Analytical section 107. Fund investment policy

A sovereign fund needs a risk policy.

Key dimensions include:

- liquidity;
- currency;
- duration;
- credit risk;
- equities;
- governance;
- benchmark.

A young fund expecting high withdrawals has different liquidity needs from a mature fund with stable fiscal use.

The investment policy should match the withdrawal profile.

Analytical section 108. Investment return

Investment return should be separated into:

- interest;
- dividends;
- realized gains;
- unrealized gains.

Market value can fall in a year even if petroleum deposits are strong.

The synthetic stress model includes a negative-return year followed by recovery to illustrate this point.

It is not a forecast.

Analytical section 109. Fund expenses

Professional management, custody and administration cost money.

A full return measure should be net of fund expenses.

The audit should identify:

- investment fees;
- custody;
- administration;
- external managers.

The materiality of these costs grows with the fund.

Analytical section 110. Transparency

A useful public fund report should provide:

- opening balance;
- deposits;
- investment return;
- withdrawals;
- expenses;
- closing balance.

It should state currency and accounting basis.

Annual audit provides assurance over the financial statements.

Quarterly reporting improves timeliness.

The Bank of Guyana publishes both annual and quarterly material. [S44, S47]

Analytical section 111. Audit

An external financial audit answers whether statements are fairly presented under the identified reporting framework.

It does not answer whether withdrawal policy was economically optimal.

It does not answer whether projects delivered value.

The paper should preserve these boundaries.

Different reviewers answer different questions.

Analytical section 112. Stress test one: low petroleum revenue

The fund-stress model includes a synthetic case where deposits fall for several years.

The purpose is to test:

- whether withdrawals remain affordable;
- whether liquid assets are sufficient;
- whether the budget needs adjustment.

The case does not predict Guyana's future receipts.

Analytical section 113. Stress test two: high spending

A second synthetic case increases withdrawals while deposits remain high.

The fund still grows in the example, but more slowly.

This demonstrates that a rising balance does not reveal the opportunity cost of spending.

Project quality remains the key question.

Analytical section 114. Stress test three: market loss

A fund invested in markets can experience negative return.

The synthetic model starts with a market loss and later positive returns.

The case tests whether spending rules force asset sales after a loss.

A liquidity reserve can reduce that risk.

Analytical section 115. Stress test four: combined disaster

A small producer should test a natural disaster occurring during weaker petroleum revenue.

Emergency spending rises while revenue falls.

Infrastructure damage may also interrupt production.

This combined stress case belongs in the island-state blueprint.

Analytical section 116. Reporting public wealth

A national petroleum dashboard should therefore have four panels.

Panel one shows petroleum production and government entitlement.

Panel two shows the NRF.

Panel three shows the national budget and public investment.

Panel four shows outcomes and national assets.

This structure prevents a large fund balance from becoming a substitute for development evidence.

Analytical section 117. What Guyana established early

The strongest factual statement is about timing.

A statutory resource fund existed before first oil. [S37]

The country also established an operational relationship with the central bank before significant production revenue accumulated.

This sequence gives new producers a concrete institutional milestone to examine.

Analytical section 118. What changed later

The framework did not remain fixed.

Later legislation changed governance and withdrawal rules.

This shows that pre-first-oil preparation is not the end of policy design.

A resource framework needs lawful revision as evidence changes.

The revision process itself should remain transparent.

Analytical section 119. Transferability

A newcomer country should decide first what problem its fund is intended to solve.

It should then align:

- deposit rules;
- investment;
- withdrawal;
- budget integration;
- audit.

It should not copy a withdrawal percentage without modelling:

- reserves;
- revenue;

- debt;
- development gap;
- disaster risk.

Analytical section 120. Part IV findings

Guyana's NRF case offers a sequence rather than a fixed formula.

The fund was established before first oil.

Audited financial statements provide a public account.

Withdrawals later moved revenue into the budget.

The law changed over time. [S37, S44-S47]

The blueprint lesson is to build the petroleum-to-budget accounting chain early, preserve independent financial reporting, and evaluate public investment separately from fund accounting.

A country ultimately needs both.

A clean fund without effective public investment leaves development opportunities unrealized.

Large public investment without financial control risks converting a finite resource into weak assets.

The purpose of the system is to convert depleting petroleum into a durable national balance sheet.

Edition 2.0 deepening: the Natural Resource Fund as one account inside a national wealth system

Analytical section 120A. A sovereign resource fund should solve a defined problem

A resource fund can serve several purposes, but those purposes should not be blurred.

It can:

- stabilize the budget;
- save for future generations;
- separate volatile petroleum receipts from annual spending decisions;
- preserve foreign assets;
- improve transparency;
- create a rules-based bridge into the national budget.

These objectives can conflict.

A rule optimized for stabilization may not maximize long-run saving.

A rule designed to build financial assets may constrain near-term development spending.

A state with large infrastructure gaps may decide to spend more than a mature high-income producer.

The design question is therefore not whether a country should "save" or "spend".

The question is how much can be spent productively without creating macroeconomic instability, weak projects or permanent expenditure commitments that outlive volatile petroleum revenue.

Analytical section 120B. The fund should be reconstructed as a stock-flow identity

The cleanest audit of a sovereign fund uses a simple identity:

opening balance plus petroleum inflows plus investment income plus or minus valuation and currency effects minus lawful withdrawals equals closing balance.

Every annual and quarterly report should reconcile to this identity.

The Bank of Guyana reporting architecture makes it possible to reconstruct the NRF across audited annual statements and quarterly or monthly reports. [S44-S47, S88, S109-S110]

The value of the reconstruction is not only arithmetic.

It separates four concepts that are often confused:

- petroleum revenue;
- investment return;
- withdrawal;
- fund balance.

A large withdrawal can occur in a year when petroleum receipts are also large.

A lower closing balance does not automatically mean the fund "lost" money.

The explanation may be lawful transfers to the Consolidated Fund.

Analytical section 120C. Cash and accrual accounting should not be mixed casually

Petroleum cash flows can straddle reporting periods.

A cargo lifted in December can be paid in January.

A quarterly report may recognize an accrual before the cash reaches the bank account.

A government budget may refer to expected receipts for a full year.

These are different accounting boundaries.

The Q4 2025 NRF report explicitly describes accrual accounting and notes petroleum payments due after year-end for some December lifts. [S109]

This is a valuable teaching example.

A publication should never add an accrual figure to a later cash receipt without checking whether they refer to the same transaction.

The NRF reconciliation workbook should therefore include:

- transaction date;
- recognition date;
- cash receipt date;
- accounting basis;
- source report.

Analytical section 120D. The custody chain is part of sovereign risk management

The resource fund is not simply a line in the budget.

It is a custody arrangement.

The state should know:

- where the assets are held;
- who can authorize transfers;
- who reconciles the account;
- who audits the financial statements;
- what investment mandate applies;
- what operational controls prevent unauthorized movement.

Bank of Guyana reporting identifies the central bank's operational role and the investment mandate. [S109]

A new producer should test the entire custody chain before the first material petroleum receipt.

A simulated transfer should verify bank instructions, dual authorization, accounting entries, reconciliation and audit evidence.

Analytical section 120E. Investment policy should match the purpose and horizon of the fund

A fund that may be used heavily for development spending has a different liquidity need from a mature intergenerational fund.

Investment policy should therefore distinguish:

- liquidity reserve;
- near-term withdrawal needs;
- long-duration savings;

- currency exposure;
- market risk;
- operational risk.

A conservative early-stage mandate can be rational when withdrawal needs are uncertain and institutions are still maturing.

Over time, the state can decide whether a diversified portfolio is appropriate.

The decision should be based on explicit risk tolerance, not a desire to imitate another sovereign fund.

Analytical section 120F. Withdrawal rules should be tested against revenue volatility

A withdrawal formula can look prudent under a high-price base case and become procyclical under stress.

The rule should therefore be tested under:

- oil price decline;
- delayed startup;
- major outage;
- cost-recovery increase;
- global recession;
- domestic disaster;
- investment-market loss.

The question is not only whether the formula is legal.

The question is whether the resulting budget path remains sustainable.

A country should know how quickly public spending would need to adjust if petroleum receipts fall.

Analytical section 120G. The non-oil primary balance is the key macro bridge

The fund balance alone does not show the fiscal impulse delivered to the domestic economy.

The non-oil primary balance is useful because it asks how large the fiscal position would be without petroleum revenue and interest.

A rapidly widening non-oil deficit can be sustainable for a period if the state is deliberately converting resource wealth into high-return public assets.

It becomes risky when spending growth exceeds implementation capacity or creates permanent recurrent obligations without future financing.

The indicator should therefore be read with public-investment quality, not in isolation.

Analytical section 120H. Absorptive capacity is measurable

"Absorptive capacity" should not remain a vague warning.

A government can track it through:

- project completion delays;
- procurement lead times;
- construction-cost inflation;
- skilled-labour vacancies;
- imported construction inputs;
- contractor concentration;
- project redesign;
- maintenance backlogs;
- land and housing prices.

If these indicators deteriorate as public capital spending rises, the marginal return on additional spending may be falling.

The response is not necessarily to stop investment.

It may be to sequence projects, expand engineering capacity, improve procurement and prioritize maintenance.

Analytical section 120I. Public investment should be tracked through an asset register

A budget appropriation is not a national asset.

A completed structure is not necessarily an operating public service.

The public-investment chain should therefore record:

1. project concept;
2. appraisal;
3. budget;
4. procurement;
5. contract;
6. construction progress;
7. commissioning;
8. operating agency;
9. maintenance plan;
10. service outcome.

For a resource-rich state, this asset register is the other half of the sovereign fund.

The NRF records financial wealth.

The public asset register records what withdrawals became.

Analytical section 120J. Maintenance converts construction into durable wealth

Resource booms create incentives to build visible capital projects.

The less visible question is whether those assets can be operated and maintained.

Every major public project should therefore have a lifecycle-cost estimate before construction begins.

The estimate should include:

- staffing;
- utilities;
- spare parts;
- periodic rehabilitation;
- digital systems;
- insurance;
- replacement cycle.

A hospital without recurrent staffing or a road without maintenance is not equivalent to a durable increase in national wealth.

Analytical section 120K. Human capital belongs in the wealth-conversion ledger

Petroleum revenue can finance education and health, but expenditure is not the same as human-capital formation.

The state should track:

- learning outcomes;
- tertiary completion;
- technical certification;
- health access;
- mortality and morbidity indicators;
- labour-force participation;
- occupational shortages.

The public wealth framework should therefore include financial, physical and human capital.

This makes it possible to compare competing uses of a finite resource.

Analytical section 120L. Natural capital prevents false accounting

A state can increase financial assets while degrading ecosystems.

A comprehensive wealth framework should record major natural-capital changes alongside fiscal assets.

For Guyana this is especially relevant because petroleum development coexists with forest-carbon policy, biodiversity assets, freshwater systems and coastal vulnerability.

The objective is not to reduce every ecological value to one monetary number.

The objective is to prevent the public balance sheet from treating natural assets as if they were free.

Analytical section 120M. A national wealth statement would improve accountability

An annual national wealth statement could present five accounts:

- petroleum depletion;
- NRF financial assets;
- public physical assets;
- human-capital indicators;
- natural-capital indicators.

The statement would not need to claim perfect valuation.

Its value would be conceptual discipline.

Citizens could see whether petroleum extraction is being matched by growth in other durable assets.

This is a stronger intergenerational test than the size of the NRF alone.

Analytical section 120N. The 2025 NRF record shows why reconciliation matters

The Bank of Guyana's Q4 2025 report provides a useful case because it records the year-end market value, quarterly inflows, transfers, investment return and the investment mandate in one report. [S109]

A reader can therefore observe that fund performance is affected simultaneously by new petroleum revenue, investment income and withdrawals.

This is exactly why political or media discussion should not infer fund performance from one number.

The same discipline should be carried into 2026 using the monthly reports and Ministry of Finance mid-year reporting. [S07, S110]

Analytical section 120O. Expanded Part IV conclusion

The Natural Resource Fund should be treated as a control account in a much larger national wealth system.

Its first job is to record and safeguard petroleum receipts under law.

Its second job is to create a disciplined interface with the budget.

Its third job is to preserve evidence about how much of a finite resource has been converted into financial assets.

But the development question begins after the withdrawal.

A state ultimately succeeds only if withdrawals become productive, maintained and socially valuable assets without destabilizing the economy.

For a newcomer, the operating model should therefore link four ledgers:

- petroleum entitlement;
- sovereign fund;
- national budget;
- public asset and outcome register.

The country should be able to trace one petroleum dollar through all four.

That is the standard required for true resource-to-wealth accountability.

Edition 2.0 evidence deepening: the year-by-year NRF record

Analytical section 120P. The NRF should be read as a continuous ledger

Edition 2.0 adds `NRF\_ANNUAL\_LEDGER.csv` because the sovereign-fund story is best understood as a continuous stock-flow account rather than a sequence of press releases.

The table separates:

- opening balance;
- petroleum and other eligible inflows;
- investment return;
- withdrawals;
- closing balance;
- accounting basis;
- reconciliation status.

This immediately reveals whether an apparent change in the fund is driven by petroleum revenue, financial return, withdrawal or accounting timing.

Analytical section 120Q. 2020: the first accounting year

The early NRF record contains an important accounting lesson.

The 2020 Q4 reporting shows both cash and a profit-oil receivable. That means a cargo-related amount could belong to the economic activity of one period while cash settlement occurred in the next. [S45]

The final historical ledger should therefore preserve receivables rather than treating every year-end balance as cash physically received by 31 December.

This is why the 2020 and 2021 rows remain explicitly marked as provisional early-year bridges until the audited annual statements are transcribed line by line into the master dataset.

The choice to label an early row provisional is a strength, not a weakness.

It prevents false precision.

Analytical section 120R. 2021: accumulation before withdrawals

The 2021 year-end fund balance increased materially while the fund had not yet entered the later period of large budget transfers.

This provides a useful baseline for evaluating the transition from accumulation to withdrawal.

The institutional question is not whether withdrawal is inherently better or worse than accumulation.

It is whether the transition occurs through the legal budget process and whether the assets financed by withdrawals can be traced.

Analytical section 120S. 2022: the first full withdrawal cycle

The 2022 record is the first clear example in the ledger of large petroleum inflows occurring alongside material withdrawals.

Official reporting records approximately US\$607.65 million transferred to the Consolidated Fund during the year and a year-end fund value around US\$1.43 billion. [S44, S47]

This is the point at which sovereign-fund governance and public-investment governance become inseparable.

The fund can remain correctly accounted for while the development value of the withdrawal depends on a separate budget and project-delivery system.

Analytical section 120T. 2023: the importance of accounting basis

The 2023 annual reporting states petroleum inflows around US\$1.608 billion, withdrawals around US\$1.002 billion and investment return around US\$86.84 million, with a year-end market value around US\$2.122 billion. [S44]

A separate public oversight source reported a lower closing figure in a different context.

This discrepancy is analytically useful.

A responsible publication should not silently choose the number that best fits the narrative.

It should first determine whether the difference reflects:

- cash versus accrual;
- receivables;
- reporting date;
- market value versus cash;
- transcription.

Edition 2.0 therefore treats the Bank of Guyana fund statement as the controlling financial source and records the alternate figure as a reconciliation note until the basis is fully matched.

Analytical section 120U. 2024: simultaneous high inflow and high withdrawal

The 2024 year illustrates why a closing balance cannot be read without the flow statement.

Official reporting records annual petroleum inflows of approximately US\$2.568 billion and withdrawals of approximately US\$1.586 billion. The year-end market value was approximately US\$3.246 billion. [S44]

Using the official GYD stock-flow figures, the implied annual investment result is approximately US\$141 million, consistent with public reporting on interest income.

The publication labels that annual return as derived from the official identity unless and until the exact annual line is transcribed from the final annual statement.

That disclosure matters because a derived figure should not be represented as a direct quote.

Analytical section 120V. 2025: withdrawals approach annual inflows

The 2025 record is particularly important for long-run fiscal analysis.

Annual reporting records inflows of approximately US\$2.510 billion, investment income around US\$142.13 million and withdrawals around US\$2.463 billion, with a year-end fund value around US\$3.435 billion. [S88, S109]

The key observation is not that the fund balance still rose.

It is that annual withdrawals were close in magnitude to annual petroleum inflows.

A fiscal sustainability analysis should therefore ask:

- Is this level of withdrawal temporary or structural?
- How much recurrent spending does it support?
- What happens if oil price or production falls?
- How much is being converted into productive assets?

- How quickly is the fund expected to grow under stress?

Those are forward-looking policy questions.

The ledger supplies the historical evidence needed to model them.

Analytical section 120W. H1 2026: partial-year data must stay partial

The H1 2026 official mid-year reporting records government petroleum receipts and a June 30 fund balance after first-half withdrawals. [S07]

The book should resist a common analytical shortcut: doubling the first-half number and calling it an annual result.

Second-half production, cargo timing, prices, startup schedules and withdrawals can differ materially.

The row is therefore labelled H1 2026 and remains visibly separate from audited full-year rows.

Analytical section 120X. The withdrawal-to-inflow ratio is a useful but incomplete indicator

One useful annual indicator is:

withdrawals / petroleum inflows.

A rising ratio shows that a larger share of current-year petroleum revenue is being transferred to the budget.

The indicator does not answer whether the withdrawal is prudent.

It must be read with:

- fund stock;
- expected future revenue;
- public debt;
- non-oil primary balance;
- investment quality;
- recurrent spending.

It is a diagnostic, not a policy verdict.

Analytical section 120Y. The fund-to-budget ratio should be paired with capital execution

A second indicator is petroleum-funded budget transfer relative to total central-government expenditure or capital expenditure.

This shows the fiscal importance of the fund.

Again, the ratio is not sufficient.

If capital execution is weak or projects are delayed, a high transfer may increase domestic liquidity faster than productive capacity.

The fiscal dashboard should therefore pair:

- withdrawal;
- capital appropriation;
- capital execution;
- commissioned assets.

Analytical section 120Z. The NRF ledger should reconcile to petroleum entitlement

The fund ledger begins at the point of revenue deposit.

The petroleum revenue chain begins earlier.

Edition 2.0 should ultimately connect:

- government cargo entitlement;
- sale;
- royalty invoice;
- settlement;
- NRF deposit.

If a government cargo is lifted in December and paid in January, the bridge should show that timing.

This creates one continuous chain from physical petroleum to sovereign financial asset.

Analytical section 120AA. The ledger should be reproducible by an outside reviewer

The final dataset should provide:

- source document;
- table/page;
- GYD figure;
- USD figure;
- exchange rate where relevant;
- accounting basis;
- whether direct or derived;
- formula.

An outside reviewer should be able to reconstruct the annual identity without asking the author for a private spreadsheet.

Analytical section 120AB. A fund can grow while fiscal dependence also grows

This is one of the most important conceptual points in resource governance.

A larger year-end fund balance can coexist with:

- higher annual withdrawals;
- a wider non-oil deficit;
- higher recurrent expenditure;
- larger public investment;
- rising dependence on petroleum transfers.

The fund stock and the fiscal flow answer different questions.

A country can be accumulating financial assets and increasing budget dependence at the same time.

Analytical section 120AC. The long-run stress model should start from the reconciled history

Illustrative future scenarios should not begin from an arbitrary opening balance.

They should begin from the most recent verified closing fund balance.

They should then model:

- petroleum receipts;
- investment return;
- withdrawal rule;
- expenditure adjustment;
- production delay;
- price shock.

This makes the scenario model auditable.

The synthetic stress tables already in the repository should therefore be linked to the NRF annual ledger in the final Edition 2.0 methods appendix.

Analytical section 120AD. Evidence conclusion

The year-by-year ledger changes Part IV from an institutional description into a financial history.

It shows exactly why fund governance should be judged through both stock and flow.

The final book should continue this record annually.

A sovereign-fund system becomes more credible when every new year can be added to the same reconciliation without changing definitions.

PART V

Environmental Governance, Offshore Risk and Climate Coexistence

Analytical section 121. Environmental governance is a chain

Environmental performance should not be inferred from the existence of an environmental law.

A functioning offshore environmental system contains several distinct steps:

- baseline science;
- impact assessment;
- public participation;
- authorization;
- operating conditions;
- monitoring;
- inspection;
- incident reporting;
- enforcement;
- remediation;
- financial responsibility.

Each step produces its own evidence.

A permit proves that an authority granted approval under specified conditions.

A monitoring report shows what the operator measured.

An inspection shows what the regulator observed.

A laboratory result provides a measurement.

An enforcement notice shows how a deviation was addressed.

A restoration assessment shows whether ecological conditions recovered.

The Guyana case needs all of these layers.

Analytical section 122. The legal foundation

Guyana's Environmental Protection Act predates commercial offshore oil. [S39]

This gives the state a general environmental framework before the petroleum boom.

The petroleum-specific question is how that framework is applied to deepwater development.

A general environmental statute needs project-level implementation through:

- EIA requirements;
- permits;
- standards;
- plans;
- audits;
- enforcement.

The final case should therefore move from statute to project.

Analytical section 123. Liza Phase 1 environmental sequence

The official permit record gives a clear chronology.

The project application was submitted on 5 July 2016.

The approved EIA and Environmental and Socioeconomic Management Plan are dated 1 June 2017.

The permit was issued on 1 June 2017.

The permit record also references revised oil-spill and wildlife-response plans.

The permit was later modified on 12 May 2021. [S48]

This is important because it shows environmental authorization occurring before first production.

It also shows the permit changing after operating experience.

Analytical section 124. Environmental impact assessment

An EIA is a forecast and decision-support document.

It identifies:

- baseline conditions;
- project activities;
- impacts;
- alternatives;
- mitigation;
- residual effects;
- monitoring.

An EIA is strongest when assumptions are explicit.

For an offshore project, important assumptions include:

- currents;
- spill rate;
- spill duration;

- oil weathering;
- drilling waste;
- produced water;
- flaring;
- vessel traffic;
- marine species.

The final paper should compare predicted conditions with observed operations where later data exist.

Analytical section 125. Baseline ownership

Environmental baseline data have public value beyond one project.

The government should know:

- who collected the data;
- who paid;
- who owns the raw data;
- whether methods are reproducible;
- whether independent researchers have access.

A baseline locked inside a project consultant's report is less useful for cumulative basin management.

A newcomer country should negotiate data rights before approvals.

Analytical section 126. Public participation

Public participation should be measured by more than a notice.

The Yellowtail public record provides an example of an EIA requirement and public submission process. [S52]

The flagship should ask:

- who received notice;
- how long the comment period lasted;
- where documents were available;
- which groups participated;
- what issues were raised;
- what the authority changed;
- what remained unresolved.

This produces a participation audit trail.

Analytical section 127. Permit conditions

A permit converts environmental findings into enforceable conditions.

Important offshore conditions include:

- flaring;
- emissions reporting;
- discharge;
- waste;
- monitoring;
- spill plans;
- wildlife response;
- audits;
- reporting deadlines.

The permit should be treated as a versioned document.

Amendments should remain visible.

Analytical section 128. Liza Phase 1 permit modification

The EPA's public notice on the Liza Phase 1 modification states that the revised permit added emissions reporting and flaring-related requirements, including specified time constraints and a payment associated with flaring beyond those timelines. [S55]

This is an example of adaptive regulation.

The operating issue did not remain solely inside the original EIA.

The regulator changed permit conditions.

The next question is whether the later operating record shows compliance.

That requires actual flaring data.

Analytical section 129. Flaring

Flaring has several categories.

Routine flaring is different from:

- commissioning flaring;
- emergency flaring;
- maintenance flaring;
- equipment-failure flaring.

The environmental impact depends on volume, composition and duration.

The final project table should record:

- gas flared;
- reason;
- start;
- end;
- permit condition;
- emissions;
- regulator response.

A company statement that flaring returned to background level is useful evidence of its position, not a substitute for the regulator's measured record.

Analytical section 130. Gas reinjection

Offshore FPSOs often reinject produced gas to support reservoir pressure and reduce routine flaring.

The system therefore connects production optimization and environmental performance.

A compression failure can affect both.

The environmental chapter should explain:

- design;
- redundancy;
- downtime;
- restart;
- flare impact.

This helps readers understand why flaring may occur during commissioning or equipment failure.

Analytical section 131. Produced water

Produced water is another operating stream.

The environmental assessment should state:

- expected volume;
- treatment;
- discharge standard;
- monitoring;
- chemical content.

The final case needs project-level observed data before drawing conclusions.

A permit limit is not an observed discharge.

Analytical section 132. Drilling waste

Development drilling produces cuttings, fluids and other waste.

The EIA and permit should identify:

- treatment;
- disposal;
- discharge;
- transport;
- hazardous classification.

A basin with many wells accumulates waste across projects.

Cumulative management matters.

Analytical section 133. Marine traffic

Offshore petroleum increases:

- supply vessels;
- tankers;
- helicopters;
- support traffic.

These create:

- collision risk;
- noise;
- emissions;
- wildlife interaction.

A cumulative assessment should include marine traffic density rather than analyzing each FPSO as if it operates alone.

Analytical section 134. Underwater noise

Seismic acquisition, drilling and vessel operations create underwater sound.

Potential receptors include:

- marine mammals;
- fish;
- turtles.

A strong monitoring framework records:

- source level;
- duration;
- season;

- species observations;
- mitigation.

The final paper should state where direct Guyana-specific data are absent.

Analytical section 135. Fisheries

Fisheries matter economically and socially.

A fisheries baseline should include:

- catch;
- landed value;
- employment;
- export;
- subsistence;
- season;
- location.

Without a baseline, compensation after an incident becomes more difficult to evaluate.

An island producer with high tourism and fishery dependence faces especially concentrated marine exposure.

Analytical section 136. Biodiversity

Guyana's offshore environment sits beside a country with extensive terrestrial biodiversity and forests.

The petroleum case should not allow the visibility of forests to obscure marine biodiversity.

The environmental asset register should contain both.

The national wealth chapter should avoid subtracting one environmental gain from another loss without a defensible common accounting method.

Analytical section 137. Oil spill risk

Oil spill risk is a combination of:

- probability;
- release size;
- duration;
- trajectory;
- receptor vulnerability;
- response.

A worst-case scenario does not state the probability of that event.

It tests whether institutions and resources are sufficient if it occurs.

This distinction should remain explicit.

Analytical section 138. Prevention

Spill governance begins before response.

Prevention includes:

- well design;
- barriers;
- blowout preventers;
- maintenance;
- operating procedures;
- training;
- inspection.

A response plan does not substitute for prevention.

The final case should separate prevention requirements from emergency assets.

Analytical section 139. Detection and notification

A response begins with detecting an event and notifying the right authorities.

The operational standard should define:

- detection method;
- notification deadline;
- regulator;
- maritime authority;
- emergency command;
- neighboring states.

A tabletop exercise should measure actual notification time.

Analytical section 140. Incident command

An offshore spill involves more than the operator.

Government responsibilities can include:

- environmental decisions;
- maritime safety;
- fisheries;
- health;
- coast guard;
- civil defence;

- public communication.

The incident command structure should identify who leads each function.

Ambiguity during an emergency increases delay.

Analytical section 141. Capping stack

A subsea loss of well control may require capping equipment.

The Uaru production licence contains spill-response requirements including capping capability.
[S56]

The final review should identify:

- equipment source;
- location;
- mobilization time;
- compatibility;
- transport;
- installation.

A subscription agreement is not the same as equipment located at the well.

Both facts matter.

Analytical section 142. Dispersants

Dispersant use involves environmental trade-offs.

A response plan should define:

- approval authority;
- product;
- stock;
- toxicity data;
- conditions for use;
- monitoring.

The presence of dispersant inventory does not establish that use is appropriate in every spill.

Analytical section 143. Mechanical recovery

Booms and skimmers work differently under different sea conditions.

The response capability assessment should record:

- equipment;
- capacity;
- sea-state limits;

- storage;
- waste disposal.

Nominal equipment capacity should be distinguished from realistic offshore recovery.

Analytical section 144. Shoreline protection

If a spill threatens the coast, priorities need to be established before impact.

Sensitive sites include:

- mangroves;
- fisheries;
- communities;
- industrial intakes.

A shoreline-response atlas should rank protection actions based on ecological and social evidence.

Analytical section 145. Transboundary response

Ocean currents do not follow national boundaries.

A serious offshore incident can affect neighboring states.

A transboundary plan needs:

- notification;
- joint surveillance;
- evidence;
- responder access;
- claims;
- diplomacy.

Maritime boundary agreements do not automatically create environmental response arrangements.

Analytical section 146. The 2025 oil-pollution law

Parliament records the Oil Pollution Prevention, Preparedness, Response and Responsibility Act 2025. [S53]

The Official Gazette records publication. [S54]

At the research cut-off, the project has not treated every provision as fully operational because commencement and implementing regulations still require verification.

This is a critical legal-writing rule.

Enacted does not necessarily mean commenced.

Commenced does not mean fully implemented.

Analytical section 147. Financial responsibility

Spill response depends on money.

Financial responsibility is not one instrument.

It can include:

- operator assets;
- insurance;
- parent guarantee;
- affiliate support;
- response contracts;
- public emergency spending.

Each layer has different triggers and limits.

The paper should not combine them into one headline figure without reading the instruments.

Analytical section 148. Legal liability versus security

Legal liability describes what a party owes under law.

Financial security describes funds or guarantees available to satisfy obligations.

An operator might have broad legal liability while a specific guarantee has a stated limit.

The two concepts should remain separate.

This distinction is central to the financial-assurance litigation.

Analytical section 149. Court review

A 2026 appellate dispute concerning financial assurance is material to the environmental case.

The final manuscript must use the authenticated judgment before making a legal interpretation.

Government summaries and stakeholder commentary should be attributed.

The case illustrates why environmental governance includes courts alongside regulators.

Analytical section 150. Insurance

A useful insurance review asks:

- insured party;
- insurer;
- policy period;
- limit;

- deductible;
- exclusions;
- territory;
- pollution coverage;
- defense cost;
- claims trigger.

An insurance certificate alone provides less information than the full policy.

Where the full instrument is not public, the evidence gap should be stated.

Analytical section 151. Parent guarantees

A parent guarantee can add financial strength.

Its value depends on:

- guarantor;
- amount;
- trigger;
- law;
- duration;
- termination;
- enforcement.

A guarantee should be evaluated on its legal terms and guarantor capacity.

Analytical section 152. Cleanup cost and ecological damage

A serious spill creates several cost categories:

- source control;
- offshore response;
- shoreline cleanup;
- waste;
- fishery loss;
- tourism loss;
- ecological restoration;
- monitoring;
- legal claims.

A financial assurance assessment should test all material categories.

One cleanup-cost estimate should not be treated as a complete loss estimate.

Analytical section 153. Long-term monitoring

Environmental damage may persist after visible cleanup ends.

Monitoring can include:

- sediment;
- water;
- fisheries;
- habitat;
- species;
- health.

The responsible party and funding duration should be established in law or settlement.

Analytical section 154. Cumulative impact

The cumulative environmental question becomes more important as the number of projects grows.

Each project may satisfy its permit and still contribute to a larger cumulative load.

Relevant basin-level measures include:

- total flaring;
- total emissions;
- total produced water;
- vessel traffic;
- underwater noise;
- waste;
- incident frequency.

The regulator needs project and basin views.

Analytical section 155. Regulatory workload

Environmental oversight has a capacity dimension.

Metrics include:

- permits per specialist;
- inspections;
- audit reviews;
- data volume;
- unresolved notices.

As production expands, staffing and data systems should scale.

A permit system that works for one project should not be assumed adequate for six.

Analytical section 156. Independent monitoring

Independent monitoring can strengthen confidence where:

- methods are public;
- laboratories are accredited;
- raw data are accessible;
- conflicts are controlled.

Independence should be demonstrated by governance and method rather than the label of a contractor.

Analytical section 157. Environmental data portal

A national portal should publish:

- permit;
- EIA;
- monitoring summary;
- incident;
- enforcement;
- permit amendment.

Sensitive commercial or security information can be protected.

The public should still be able to trace material environmental obligations.

Analytical section 158. LCDS as a separate system

Guyana's Low Carbon Development Strategy began before oil. [S22]

The strategy therefore has an institutional history distinct from petroleum.

The later carbon-credit programme should be analyzed under its own accounting.

The petroleum system should not absorb or redefine it.

Analytical section 159. Forest-carbon accounting

A forest-credit programme has several stages:

- measured forest outcome;
- verification;
- issuance;
- sale;
- registry transfer;
- cash receipt;
- allocation.

These are separate.

A contract value is not a cash receipt.

A credit issued is not necessarily sold.

A payment allocated is not necessarily a completed project.

The same traceability principles used for oil should apply.

Analytical section 160. Community allocation

LCDS materials describe a share of forest-carbon revenues directed to participating Indigenous villages under village-led planning. [S62, S63]

The final social outcome review should trace:

- eligibility;
- allocation;
- receipt;
- plan;
- procurement;
- completion;
- outcome.

Government programme records are one source.

Village and independent evidence are also needed.

Analytical section 161. Petroleum emissions

Offshore petroleum has direct operational emissions.

Sources include:

- turbines;
- flaring;
- vessels;
- helicopters;
- shore facilities.

The inventory should state scope and method.

An emissions-intensity measure needs consistent production units.

Analytical section 162. Downstream combustion

Most emissions associated with crude occur when fuels produced from oil are later combusted.

Those emissions often occur outside Guyana.

They remain relevant to global climate analysis even if they sit outside the national territorial inventory.

The paper should state the accounting boundary.

Analytical section 163. Forest credits do not automatically cancel petroleum emissions

A forest credit represents an environmental claim under a specified methodology.

Petroleum emissions arise under a different accounting system.

The paper should not simply subtract one from the other and call the result national neutrality.

Any net claim needs a recognized accounting basis and complete boundaries.

Analytical section 164. Domestic energy transition

Petroleum development also interacts with Guyana's domestic power system.

Gas-to-energy, solar, hydro and efficiency policies affect domestic emissions, cost and reliability.

These policies should be measured on their own outcomes.

Lower domestic power emissions do not change the physical combustion of exported oil.

Analytical section 165. Climate resilience

Petroleum wealth can finance adaptation.

Guyana's coastal exposure gives this issue particular importance.

Adaptation investment should be measured by reduced risk.

Examples include:

- flood depth;
- outage duration;
- protected population;
- drainage performance.

Spending volume alone is not resilience.

Analytical section 166. Island-state adaptation

For a small island producer, the environmental chapter should put added weight on:

- reefs;
- beach tourism;
- desalination;
- small fisheries;
- hurricanes;

- one or two major ports.

A spill and a storm could occur together.

The response plan should test that combined case.

Analytical section 167. Stress test one: blowout during poor weather

Assume a subsea loss of well control during adverse weather.

Ask:

- Is detection immediate?
- Is capping equipment available?
- Can aircraft operate?
- Can vessels mobilize?
- Who approves dispersant?
- Are neighboring states notified?

The scenario tests readiness, not event probability.

Analytical section 168. Stress test two: major spill with insurance dispute

Assume physical response begins while an insurer disputes part of coverage.

The state still needs cash for immediate actions.

The financial assurance structure should identify who pays before the dispute is resolved.

This is why liquidity and legal liability are different questions.

Analytical section 169. Stress test three: simultaneous incidents

Assume response assets are already deployed to a smaller event when a second event occurs.

The plan should state:

- reserve capacity;
- external support;
- prioritization.

A multi-project basin needs concurrency planning.

Analytical section 170. What another country should adopt

The transferable environmental principles are procedural:

- baseline before development;
- public EIA;

- versioned permit;
- independent monitoring;
- incident transparency;
- tested emergency command;
- enforceable financial assurance;
- cumulative assessment.

The exact permit conditions depend on ecosystem and project.

Analytical section 171. Part V findings

Guyana's environmental case is not one decision.

It is a sequence from the 1996 statutory framework through project EIAs and permits, later permit modification, production-licence obligations, a dedicated 2025 pollution statute and judicial review. [S39, S48, S53-S56]

The existence of this architecture is a documented fact.

Its operating performance requires project-level evidence.

The blueprint lesson is equally precise.

A newcomer should not wait until a spill to define response authority, and it should not treat a permit as the end of environmental governance.

The environmental system should be designed to learn from operations, publish evidence, enforce conditions and finance the consequences of failure.

Edition 2.0 deepening: environmental governance as a lifecycle assurance system

Analytical section 171A. Environmental governance should begin before project approval

An offshore environmental regime should be designed as a lifecycle system.

The sequence begins with baseline data and strategic assessment.

It continues through project EIA, permit conditions, monitoring, incident reporting, enforcement, financial assurance, emergency response and decommissioning.

Each stage should answer a different question.

The EIA asks what impacts are reasonably foreseeable before approval.

The permit converts selected mitigation and monitoring requirements into enforceable obligations.

Operational monitoring asks what is actually occurring.

Incident reporting captures deviations and failures.

Financial assurance asks whether the operator can finance the consequences of non-compliance or an accident.

Decommissioning asks what remains after production ends.

Treating any one of these as the complete environmental system creates a blind spot.

Analytical section 171B. Baseline quality determines the credibility of later impact claims

A post-development environmental measurement has limited interpretive value if the pre-development condition is unknown.

This is particularly important in marine systems because fish abundance, currents, water quality and species distribution can vary naturally.

The baseline should therefore record:

- methodology;
- sampling locations;
- season;
- equipment;
- detection limits;
- uncertainty;
- raw data ownership;
- laboratory quality controls.

A baseline is not credible merely because it exists in an EIA.

It should be reproducible.

Where monitoring methods change, the record should explain whether new and old series remain comparable.

Analytical section 171C. Strategic assessment is different from project assessment

An EIA for one FPSO cannot answer every cumulative question created by a basin with several producing projects.

Strategic environmental and social assessment asks system-level questions:

- total tanker traffic;
- aggregate flaring and emissions;
- cumulative noise;
- fisheries interaction;
- waste handling;

- shared support bases;
- response capacity across simultaneous operations;
- onshore infrastructure effects.

The Guyana Oil and Gas Capacity Building Project's strategic-assessment terms identify many of these categories. [S94]

The transferable lesson is that project-by-project authorization should be supplemented by basin-wide assessment as development scales.

Analytical section 171D. Permit conditions should be transformed into a compliance matrix

A long environmental permit is difficult to administer if obligations remain embedded in prose.

Each permit should be converted into a compliance matrix containing:

- condition number;
- obligation;
- frequency;
- responsible party;
- evidence required;
- regulator reviewer;
- public disclosure status;
- breach consequence.

This makes compliance auditable.

The matrix should distinguish one-time conditions from recurring conditions.

A permit that requires annual reporting should generate an expected evidence item every year.

If the evidence is missing, the regulator should know immediately.

Analytical section 171E. Financial assurance has multiple layers

Financial assurance should not be reduced to a single headline guarantee.

The Uaru permit provides a useful structure because conditions 14.3 to 14.5 distinguish credible-cost estimation, liability insurance and parent or affiliate guarantee obligations. [S99]

These instruments serve different purposes.

Insurance transfers defined risks to an insurer subject to coverage terms, exclusions and limits.

A parent guarantee depends on the guarantor's obligation and financial capacity.

Other financial-assurance instruments may address specific regulatory obligations.

The audit questions are therefore:

- what loss is covered;

- what amount is available;
- when can it be called;
- who can draw;
- what exclusions apply;
- what happens if the instrument expires;
- what happens if the obligor becomes insolvent.

A stated liability principle is not the same thing as immediately available cash.

Analytical section 171F. The distinction between liability and financial security is fundamental

An operator may have legal responsibility for environmental damage while the state still faces a separate question about how quickly money can be accessed after a major event.

This is why legal liability and financial assurance should be reported separately.

The 2026 appellate litigation illustrates the importance of this distinction. Government and contemporaneous reporting describe the Court of Appeal as overturning the earlier unlimited-guarantee order and treating financial assurance as distinct from unlimited liability. [S101, S105]

The publication should remain narrow until authenticated full reasons are available.

The transferable point does not depend on the outcome of one case.

Every producer should distinguish:

- liability;
- insurance;
- guarantee;
- liquid emergency funding;
- ultimate recovery.

Analytical section 171G. A national spill plan should be built around decision rights

Oil-spill plans often contain equipment lists.

The more important question is who decides.

The plan should identify authority for:

- incident command;
- dispersant approval;
- fishing closure;
- shoreline protection;
- public health advice;

- international notification;
- wildlife response;
- claims administration;
- termination of emergency status.

These decisions should be exercised in drills.

A plan that has never been tested across agencies is not equivalent to operational readiness.

Analytical section 171H. Response equipment should be assessed by deployment time

An inventory can create false confidence.

A capping stack or response vessel has little operational value if it cannot arrive within the scenario window assumed by the response plan.

The state should therefore measure:

- location;
- mobilization time;
- customs and border requirements;
- vessel availability;
- weather limits;
- compatible connectors;
- trained personnel;
- maintenance status.

For a small state, regional agreements may be essential.

The response plan should show which resources are national, which are operator-owned and which depend on external assistance.

Analytical section 171I. Fisheries require both ecological and economic baselines

A fisheries impact programme should not rely only on biological sampling.

It should also document:

- fishing grounds;
- vessel types;
- seasonality;
- landing sites;
- value chain;
- household dependence;

- exclusion zones;
- navigation changes;
- complaints.

This creates the evidence needed to distinguish ecological change from access or economic disruption.

If compensation is contemplated, the baseline should define eligibility before a dispute arises.

Analytical section 171J. Flaring should be treated as an operating-performance indicator

Flaring has climate, air-quality and operational dimensions.

The regulator should distinguish:

- commissioning flaring;
- routine flaring;
- safety flaring;
- equipment failure;
- gas-injection constraints.

A single annual flare total does not explain the cause.

The operating record should therefore include duration, volume, reason and corrective action for material events.

This helps distinguish a temporary startup condition from persistent system performance.

Analytical section 171K. Produced water and drilling waste need a mass-balance approach

Environmental reporting becomes stronger when it moves from narrative statements to mass balance.

For produced water, the record should track:

- volume generated;
- treatment;
- discharge;
- quality;
- monitoring location.

For drilling waste, it should track:

- material type;
- quantity;
- treatment;
- discharge or disposal;

- destination.

This creates an auditable environmental flow account.

Analytical section 171L. Incident transparency should use severity tiers

Not every incident warrants the same reporting protocol.

A national system can define severity tiers based on:

- quantity released;
- toxicity;
- injury;
- operational shutdown;
- protected-area impact;
- shoreline impact;
- cross-border significance.

Higher tiers should trigger faster public reporting and more detailed investigation.

The system should preserve minor incidents as data rather than treating them as irrelevant.

Patterns can matter even when individual events are small.

Analytical section 171M. Independent verification is most valuable where information asymmetry is highest

Operator monitoring will always be a major source of environmental data.

Independence matters most where:

- the impact is difficult to observe;
- the operator controls the measurement system;
- consequences are large;
- public trust is low;
- disputes are likely.

Independent sampling does not require the regulator to duplicate every operator measurement.

It requires a risk-based verification programme.

Analytical section 171N. The 2025 pollution statute should be tracked from enactment to operation

The Official Gazette confirms publication of Act 6 of 2025. [S98]

That is a legal milestone.

Operational readiness requires more.

A complete implementation tracker should identify:

- commencement;
- regulations;
- designated authorities;
- national contingency arrangements;
- claims procedures;
- financial-assurance rules;
- offences and penalties;
- cross-border coordination.

The document should use "published" or "enacted" where those facts are established and reserve "in force" or "operational" for provisions supported by authoritative commencement evidence.

Analytical section 171O. Decommissioning is an environmental project

End-of-life planning should begin before facilities approach closure.

The environmental decommissioning plan should address:

- well abandonment;
- subsea infrastructure;
- residual contamination;
- waste;
- seabed recovery;
- monitoring period;
- liability after surrender.

The financial model should connect decommissioning obligations to secured funding.

A state should not assume that decades of production guarantee the operator or parent company will still have the same financial capacity at closure.

Analytical section 171P. Climate accounting should use explicit boundaries

Guyana's forest-carbon programmes and petroleum production belong in the same national development story, but they use different accounting systems.

Forest-carbon credits should not be presented as automatically cancelling petroleum lifecycle emissions.

The paper should distinguish:

- domestic operational emissions;
- exported-fuel combustion;

- national greenhouse-gas inventory treatment;
- forest-carbon accounting;
- buyer retirement claims;
- corresponding adjustments where applicable.

This avoids double counting and misleading equivalence. [S22, S95]

Analytical section 171Q. Expanded Part V conclusion

A strong environmental regime is not defined by the number of permits issued.

It is defined by the evidence chain from baseline to outcome.

Guyana has statutory, project-level, spill-response, financial-assurance and later legislative components that can be documented. [S39, S48-S56, S98-S100]

The next level of scrutiny is performance evidence.

For each material permit condition, the state should be able to show what was required, what evidence was submitted, what the regulator concluded and what corrective action followed where necessary.

For a newcomer, the blueprint is therefore to build five linked registers:

- baseline register;
- permit compliance matrix;
- incident register;
- response-capability register;
- financial-assurance register.

Together they turn environmental governance from a set of documents into an operating assurance system.

PART VI

Local Content, Human Capital and the Distribution of Opportunity

Analytical section 172. Why local content requires more than a procurement number

A petroleum industry creates demand for goods, services and labour.

A local-content policy attempts to increase the share of that demand supplied by nationals and nationally owned businesses.

The resulting headline number is often gross procurement.

Gross procurement is useful.

It is not the same as domestic value added.

A locally registered company may import a finished product and resell it.

Another local company may fabricate, engineer, test and maintain a product inside the country.

Both transactions can appear in local procurement.

Their contribution to national capability differs.

The flagship therefore separates:

- supplier registration;
- beneficial ownership;
- procurement;
- imported content;
- domestic wages;
- retained profit;
- training;
- capability.

Analytical section 173. The 2021 statutory framework

Guyana's Local Content Act was passed at the end of 2021 and published in early 2022. [S57]

This means the formal statutory framework arrived after first oil.

The chronology is important.

The industry had already begun hiring and procurement before the law.

The final chapter should therefore divide local-content history into:

- pre-law industry activity;
- statutory implementation;
- later reporting and target adjustments.

This gives a more accurate view of policy development.

Analytical section 174. Definitions

The legal definition of a Guyanese company matters.

A meaningful local-content rule needs to address:

- ownership;
- voting control;
- management;
- beneficial ownership;
- residency;
- workforce.

A paper company with nominal domestic ownership but external control presents a different economic case from a business whose management, assets and profits are genuinely domestic.

The final legal crosswalk should quote the operative definitions from the Act.

Analytical section 175. The supplier register

The official register provides a structured route for company certification. [S58]

The register supports:

- identification;
- procurement;
- compliance.

Certification should not be interpreted as an independent measure of economic value.

A verification sample should test:

- ownership;
- employees;
- domestic operations;
- imports;
- related parties.

The purpose is measurement, not suspicion.

Analytical section 176. Scheduled targets

A local-content schedule identifies services where the law expects a stated level of Guyanese participation.

A target should reflect actual national capacity.

A target set far above domestic capability can lead to:

- inflated cost;
- nominal compliance;
- subcontracting;
- supply bottlenecks.

A target set too low can miss realistic opportunities.

The policy should therefore be revised using evidence.

Analytical section 177. Capacity assessment

Before changing a target, government should know:

- number of qualified suppliers;
- capacity;
- safety record;
- price;
- certification;
- equipment;
- workforce.

The 2025 government reporting described ongoing capacity assessment and wider reporting. [S60]

The final paper should compare target changes with documented supplier capability.

Analytical section 178. Domestic value-added accounting

A practical domestic value formula starts with contract value.

Subtract:

- imported goods;
- imported subcontracted services;
- foreign intellectual property;
- external management charges directly tied to delivery.

The residual approximates domestic gross value added.

That residual can be divided into:

- wages;
- domestic intermediate purchases;
- taxes;
- operating surplus.

This produces a clearer picture than procurement alone.

Analytical section 179. Customs data

Customs records help estimate imported content.

A supplier reporting USD 10 million in local procurement may have imported most of the underlying equipment.

This does not make the transaction valueless.

The local contribution may include logistics, inventory, financing, installation and maintenance.

The research should quantify the difference.

Analytical section 180. Payroll data

Payroll provides evidence of:

- local employment;
- wage;
- occupation.

A local supplier with high domestic payroll generates a different value profile from a reseller with minimal employment.

Payroll analysis should protect personal data.

Aggregate occupational statistics are sufficient for the public paper.

Analytical section 181. Retained profit

Domestic ownership matters partly because profit can remain in the national economy.

Retained profit can support:

- reinvestment;
- equipment;
- expansion;
- tax;
- wealth formation.

The research should avoid assuming that every locally owned firm's profit remains inside the country.

Actual company data would be needed.

Analytical section 182. Technology and know-how

Local value also includes knowledge.

A training programme that qualifies technicians to work across mining, power and marine sectors creates value beyond one petroleum contract.

The flagship should therefore identify transferable skills.

Examples:

- welding;
- nondestructive testing;
- electrical;
- mechanical maintenance;
- marine safety;
- logistics;
- environmental monitoring;
- project controls.

Analytical section 183. Direct employment

Direct employment is the most visible local-content metric.

The 2024 government mid-year report included employment information from reporting petroleum companies and a broader estimate of Guyanese industry employment. [S59]

The paper should preserve the reporting population.

A number from 40 reporting companies should not be presented as a census unless the source says so.

Analytical section 184. Occupation matters

Ten jobs do not have equal capability effect.

The workforce dataset should distinguish:

- entry level;
- skilled trade;
- technician;
- engineer;
- supervisor;
- manager;
- professional services.

Progression into higher-responsibility roles is a better capability measure than headcount alone.

Analytical section 185. Expatriate succession

A petroleum industry often relies on expatriate specialists in early years.

A local-content strategy can define succession pathways.

A credible pathway states:

- role;
- competency;
- training;
- mentor;
- target date;
- assessment.

A quota without competency development can create safety and performance risk.

Analytical section 186. Safety

Employment quantity should never displace occupational safety.

The local-value dashboard should include:

- lost-time incidents;
- training;
- certification;
- offshore medical fitness;
- safety competence.

A workforce programme that expands rapidly needs a matching safety system.

Analytical section 187. Wage pressure

Petroleum demand can raise wages for scarce occupations.

Higher wages benefit workers.

They can also increase costs for:

- construction;
- utilities;
- mining;
- public agencies.

The national labour analysis should compare petroleum wages with other sectors and vacancy rates.

This is part of the absorption-capacity question.

Analytical section 188. Public-sector capacity

The government competes for some of the same specialists it regulates.

Engineers, accountants, lawyers and environmental professionals can move to private employers.

A resource strategy should therefore include public-sector retention.

Options include:

- career progression;
- training;
- specialist pay structures;
- secondments;
- targeted external expertise.

The policy choice belongs to government.

The analytical requirement is to measure vacancy and turnover.

Analytical section 189. Education pipeline

Long-run localization begins before recruitment.

The skills pipeline includes:

- secondary science;
- vocational education;
- university;
- apprenticeship;
- professional certification.

The flagship should map annual graduates against projected occupational demand.

This reduces the temptation to set targets based only on aspiration.

Analytical section 190. Training expenditure versus training outcome

Training expenditure is an input.

The outcome is:

- completion;
- certification;
- employment;
- promotion;
- retention;

- wage progression.

The local-content report should publish both.

A scholarship without completion data provides an incomplete picture.

Analytical section 191. Supplier finance

Local firms often face working-capital constraints.

Large petroleum contracts may require:

- equipment;
- inventory;
- insurance;
- payroll before payment.

The supplier-development system should examine:

- payment terms;
- collateral;
- banking;
- invoice financing.

Access to a contract does not mean a firm has the balance sheet to deliver it.

Analytical section 192. Standards and certification

Petroleum supply chains use demanding standards.

Local companies need:

- quality systems;
- safety certification;
- testing;
- traceability.

Supplier development is stronger when these capabilities also support exports and other sectors.

Analytical section 193. Procurement transparency

Local suppliers need visibility into opportunities.

A transparent procurement system should state:

- expected tender;
- qualification;
- schedule;
- contact.

Post-award reporting should state aggregate domestic participation.

Commercially sensitive bid information can remain protected under law.

Analytical section 194. Measuring capability after the contract

The best long-run local-content question is what remains after a petroleum contract ends.

Possible durable assets include:

- certified technicians;
- equipment;
- workshops;
- management systems;
- export clients;
- retained earnings.

This measure connects local content to diversification.

Analytical section 195. Household welfare

Petroleum employment reaches only part of the population directly.

National development therefore needs household evidence.

The Bureau of Statistics states that the major Household Budget and Living Condition Survey was fielded from late 2018 through late 2019. [S61]

That observation period is close to first oil.

It is not a 2026 household measurement.

The research should make the date visible beside every indicator.

Analytical section 196. Why GDP and household welfare differ

GDP records production.

Household welfare depends on:

- income;
- prices;
- taxes;
- transfers;
- public services;
- housing;
- environment.

A rise in petroleum GDP can coincide with a smaller change in median household consumption.

It can also support large public investments that improve welfare later.

Only household and service data resolve the question.

Analytical section 197. Real income

Nominal wage growth is not enough.

The household analysis should adjust for:

- food;
- rent;
- transport;
- utilities.

A household whose wage rises 10 percent while housing cost rises 20 percent experiences a different result from the nominal wage statistic.

Analytical section 198. Housing

Rapid economic growth can increase demand for:

- rentals;
- construction land;
- mortgages.

Housing policy can expand supply.

The final chapter should use:

- rent;
- homeownership;
- construction;
- household size;
- housing quality.

These measures show whether the housing response keeps pace.

Analytical section 199. Health

Petroleum-financed public investment can expand health infrastructure.

The outcome test should include:

- staffing;
- service availability;
- waiting time;
- referral;
- health outcomes.

A completed hospital is an important capital asset.

Its social return depends on operation.

Analytical section 200. Education

Education investment should be measured through:

- enrollment;
- attendance;
- completion;
- learning;
- teacher availability;
- technical certification.

A resource windfall creates an opportunity to invest in capabilities that remain after oil.

The evidence should test the outcome.

Analytical section 201. Water and sanitation

Access to reliable water and sanitation is a direct development measure.

The chapter should report:

- connection;
- quality;
- continuity;
- regional gaps.

Large national GDP growth does not reveal these household conditions.

Analytical section 202. Regional distribution

Guyana's ten administrative regions differ in geography and economic structure.

National averages can hide:

- coastal/hinterland differences;
- urban/rural differences;
- infrastructure access.

The final social dashboard should therefore report regional measures.

Analytical section 203. Amerindian and hinterland development

Indigenous communities have distinct governance, land and cultural contexts.

The paper should not reduce their development experience to petroleum transfers.

Relevant systems include:

- land titling;
- public services;

- connectivity;
- renewable energy;
- forest-carbon benefit sharing.

LCDS materials describe a community allocation mechanism and village-led plans. [S62, S63]

Independent community evidence is still required.

Analytical section 204. Village plans

A village-led plan should be traced from:

- consultation;
- approval;
- funding;
- procurement;
- completion;
- outcome.

This creates the same accountability structure used for national public investment.

The scale is different.

The principle is the same.

Analytical section 205. Consent and participation

A public programme should record more than the number of consultations.

The paper should seek:

- topics raised;
- dissent;
- revisions;
- grievance.

Government programme descriptions should be paired with community and independent sources.

Where positions conflict, the paper should state the disagreement.

Analytical section 206. Land rights

Resource development and forest finance interact with land rights.

The final chapter should use legal and community sources to distinguish:

- titled land;
- pending title;
- extension;

- customary claim.

No national development claim should imply settlement of a land dispute that remains open.

Analytical section 207. Gender

Local-content analysis should include gender.

Indicators include:

- employment;
- occupation;
- wage;
- training;
- management;
- supplier ownership.

A petroleum industry with high male participation may require deliberate skills pathways for broader access.

The paper should measure rather than assume.

Analytical section 208. Youth

Young people experience the resource transition through:

- education;
- jobs;
- housing;
- migration;
- expectations.

A boom can create pressure to leave education early for short-term work.

It can also finance technical education.

Longitudinal data are needed.

Analytical section 209. Migration

High wages can attract workers from abroad and encourage return migration.

Migration expands labour supply.

It also increases housing and service demand.

The labour-market chapter should include legal migration data where available.

Analytical section 210. Cost of living

A credible social dashboard should track:

- CPI;
- food;
- rent;
- transport;
- construction.

Average CPI can hide pressure in categories that dominate lower-income budgets.

The household analysis should identify expenditure shares.

Analytical section 211. Social transfers

Government can use petroleum-supported fiscal space for transfers or subsidies.

The distributional effect depends on:

- eligibility;
- amount;
- coverage;
- inflation;
- duration.

The paper should separate transfer spending from infrastructure investment.

Both affect welfare through different channels.

Analytical section 212. Grievance mechanisms

Stakeholders need routes to challenge:

- environmental permits;
- labour practices;
- local-content certification;
- public procurement;
- community programmes.

Each mechanism should state:

- who receives complaints;
- deadline;
- decision;
- appeal;
- publication.

A process that receives complaints but never records outcomes is incomplete.

Analytical section 213. Social licence

Social licence is not a legal licence.

It describes public acceptance and trust.

The concept should be treated carefully.

Polling or consultation provides evidence of attitudes.

It does not replace legal rights or environmental standards.

The paper should not infer consent from the absence of protest.

Analytical section 214. Measuring distribution

A petroleum transition can be evaluated across four distribution channels:

1. labour income;
2. business income;
3. public services;
4. direct transfers.

The final paper should map which groups receive each channel.

This gives a clearer picture than asking whether "the country" benefited in the abstract.

Analytical section 215. A local-content adverse case

Assume procurement rises rapidly while most high-value inputs remain imported.

Gross local procurement rises.

Domestic value added rises much less.

The policy response should depend on the cause.

If local manufacturing is uneconomic at national scale, forcing it may increase cost.

If capability exists but firms lack certification or finance, targeted support may have a higher return.

The audit should identify the bottleneck.

Analytical section 216. A labour adverse case

Assume petroleum wages attract skilled electricians from the power utility.

Oil localization improves.

Grid maintenance suffers.

This is a national allocation problem.

The workforce strategy should therefore monitor vacancies outside petroleum.

Analytical section 217. A household adverse case

Assume nominal wages rise but rent and food costs increase faster for lower-income households.

GDP and employment improve while real household conditions diverge by income group.

A current household survey is required to detect the effect.

Analytical section 218. New-producer checklist

Before first oil, a new producer should have:

- household baseline;
- labour baseline;
- supplier register;
- beneficial ownership;
- realistic local targets;
- training pathway;
- grievance process.

After first oil it should update these on a fixed schedule.

A country should not wait a decade for the next household benchmark.

Analytical section 219. Part VI findings

Guyana's local-content framework formalized after first oil and now provides a statutory platform for procurement, employment and supplier participation. [S57-S60]

The economic question goes beyond compliance.

The strongest future evidence will show how much domestic value, skill and business capability remains after contracts are delivered.

The social question is broader again.

Petroleum revenue, jobs and public investment need to be connected to current household and regional outcomes.

Where current representative data are absent, the white paper should preserve the gap instead of substituting an old figure.

A newcomer country should build those measurement systems before the resource transition makes the baseline harder to reconstruct.

Edition 2.0 deepening: local content as capability formation, not only compliance

Analytical section 219A. Local content should be measured through retained domestic value

A procurement value can overstate local economic contribution.

A company registered in Guyana may import most of the goods it supplies.

A contract awarded to a local firm may pass a large share of the revenue to foreign manufacturers, expatriate labour or imported equipment.

The stronger metric is retained domestic value.

For each procurement category, government should estimate:

- local wages;
- local ownership;
- imported inputs;
- domestic operating expenditure;
- retained profit;
- local tax contribution;
- training expenditure;
- technology or process capability retained after contract completion.

This produces a domestic-value bridge rather than a gross-spend headline.

The Local Content Secretariat's reporting framework already requires detailed information on expenditure, employment, salary scales, supplier participation, training and capacity development. [S108]

The next analytical step is to convert those reporting fields into value-retention metrics.

Analytical section 219B. Local ownership should be verified economically, not only legally

A company can meet formal registration requirements while exercising limited local economic control.

A credible ownership test should therefore examine:

- beneficial ownership;
- voting rights;
- management control;
- financing;
- profit distribution;

- related-party arrangements;
- operational capability.

This does not imply that joint ventures or foreign capital are undesirable.

It means that "local" should describe the economic relationship being measured.

A transparent beneficial-ownership regime is therefore part of local-content governance.

Analytical section 219C. Category targets should reflect capability pathways

Not every petroleum good or service should have the same localization strategy.

Some categories can be localized quickly because the required capability already exists.

Others require years of training, certification, capital or international partnership.

A rational local-content plan should classify categories into:

1. immediately competitive;
2. competitive with certification or finance;
3. competitive after workforce development;
4. unlikely to be efficient at domestic scale;
5. strategic capability worth developing despite a longer horizon.

This avoids treating every target as a static percentage.

The policy objective becomes capability progression.

Analytical section 219D. Supplier development should be tracked after the petroleum contract ends

A supplier programme succeeds when firms become more capable, not merely when they win one petroleum contract.

The audit should ask whether participating firms later:

- won larger contracts;
- exported services;
- entered other sectors;
- improved safety performance;
- obtained international certification;
- invested in equipment;
- hired and trained permanent staff.

This creates a longitudinal supplier dataset.

The strongest local-content outcome is a firm that becomes competitive beyond the protected or targeted procurement category.

Analytical section 219E. Workforce localization should be occupation-specific

A national employment percentage can hide skill concentration.

The workforce dataset should distinguish:

- executive and managerial;
- petroleum engineering;
- geoscience;
- subsea;
- marine;
- process operations;
- maintenance;
- safety;
- finance;
- legal;
- logistics;
- administrative support.

A country may have a high national employment share while specialist decision-making remains concentrated among expatriates.

The succession plan should therefore be occupation-specific.

Analytical section 219F. Training should be evaluated by competence, not attendance

Training expenditure and participant count are inputs.

The output is demonstrated competence.

A training programme should identify:

- role;
- required competency;
- training method;
- assessment;
- supervised practice;
- certification;
- date of independent responsibility.

This is especially important for safety-critical and technical roles.

A scholarship can contribute to long-term capability.

It is not the same as operational qualification.

Analytical section 219G. The public sector competes for the same scarce skills

A petroleum boom can improve private wages while weakening regulators, utilities and ministries if scarce specialists leave.

The workforce strategy should therefore track vacancies and turnover in:

- environmental regulation;
- tax administration;
- public engineering;
- power utilities;
- maritime authorities;
- statistics;
- public procurement.

If public institutions cannot retain critical staff, localization inside the private petroleum sector may coexist with weaker state capacity.

Compensation reform, specialist career tracks and targeted secondments may be required.

Analytical section 219H. Migration is part of the labour-market response

A small economy cannot always train workers fast enough to meet a sudden investment boom.

Migration can relieve labour shortages.

It can also increase demand for housing, transport, health care and schools.

The labour-market chapter should therefore track:

- work permits;
- occupation;
- duration;
- housing pressure;
- remittance flows;
- transition to local workers.

The objective is not to treat migration as success or failure.

It is to include it in capacity planning.

Analytical section 219I. Wage growth should be interpreted with prices

Nominal wage growth does not establish improved household welfare.

Real household conditions depend on:

- rent;
- food;
- transport;
- utilities;
- health costs;
- household size;
- employment stability.

A petroleum boom can increase average wages while widening differences between households connected to the new economy and those facing higher prices without equivalent income gains.

This is why current household data are essential.

Analytical section 219J. A rolling household survey system is part of resource governance

A country experiencing rapid structural change should not wait many years between household surveys.

A rolling system can combine:

- annual labour-force data;
- periodic household expenditure modules;
- price surveys;
- administrative social-protection data;
- regional service indicators.

The purpose is not to publish one poverty rate more frequently.

It is to understand how the transition is distributed across income, geography and household type.

The 2018/2019 household survey remains valuable as a baseline, but it should not be described as a current 2026 outcome. [S61]

Analytical section 219K. Regional equity should be measured through access and outcomes

National averages can obscure geographic gaps.

A regional dashboard should include:

- household income or consumption;
- electricity;
- water;
- internet;
- travel time to health care;

- school outcomes;
- housing;
- roads;
- employment;
- business registration.

For hinterland and riverine areas, physical distance can make the same budget allocation produce different service access.

This is especially relevant when national resource revenue expands rapidly.

Analytical section 219L. Indigenous and community benefit systems require transaction-level evidence

Community allocations should be evaluated through a chain similar to public investment.

The chain should record:

- eligibility;
- consultation;
- decision;
- allocation;
- receipt;
- procurement;
- asset or service;
- maintenance;
- grievance;
- outcome.

Official programme documents and critical stakeholder accounts should remain distinguishable.
[S62-S63, S93]

The strongest evidence is village-level implementation that can be verified by the community and an independent reviewer.

Analytical section 219M. Grievance systems should generate data

A grievance mechanism should not be a mailbox.

It should produce a dataset containing:

- category;
- location;
- date;
- institution responsible;

- resolution time;
- outcome;
- appeal;
- recurrence.

The data can reveal systemic issues in land access, employment, procurement, environmental impact or public services.

Recurring grievances should trigger policy review.

Analytical section 219N. Gender analysis should move beyond participation rates

A petroleum transition can affect women and men differently through:

- occupational segregation;
- unpaid care;
- migration;
- safety;
- access to technical training;
- entrepreneurship;
- land ownership;
- household price exposure.

The analysis should therefore track not only labour-force participation but also occupational level, pay, training completion, business ownership and regional access.

Analytical section 219O. Local content should have an exit test

A local-content rule should not be judged only by whether the target was met.

It should also ask whether protection or preference remains necessary.

An exit test can assess:

- number of capable suppliers;
- competitive pricing;
- quality;
- export success;
- safety performance;
- financing access.

A mature category may eventually need less preference.

A weak category may need a different intervention.

This keeps local content focused on capability rather than permanent insulation from competition.

Analytical section 219P. Expanded Part VI conclusion

Local content is one of the clearest places where petroleum policy can create durable capability, but only if the measurement moves beyond gross spend.

The reporting architecture already collects many of the fields needed for a more rigorous system. [S108]

The next-generation framework should reconcile procurement, employment, ownership, imports, wages, retained profit, training and supplier progression.

The social side requires equally disciplined evidence.

GDP growth, public expenditure and petroleum employment are not substitutes for current household and regional data.

For a newcomer, the practical blueprint is to build three linked systems before first oil:

- supplier and beneficial-ownership register;
- occupation and skills ledger;
- household and regional outcomes baseline.

These systems make it possible to distinguish temporary boom participation from lasting national capability.

Edition 2.0 evidence deepening: from reported local spend to verified economic retention

Analytical section 219Q. The H1 2026 procurement figure is an input, not the conclusion

The 2026 mid-year reporting provides a useful contemporary observation because it places a number on procurement within the categories prioritized for Guyanese participation. The reported first-half amount was approximately US\$466.2 million. The comparable first-half 2025 figure cited in the same reporting cycle was lower. [S07]

The correct analytical treatment is deliberately narrow.

The figure demonstrates that substantial procurement activity is being routed through categories subject to local-content policy.

It does not, by itself, establish:

- how much of the contract value remained in Guyana;
- how much represented imported equipment or material;
- how much became Guyanese wages;
- how much profit accrued to verified Guyanese beneficial owners;
- how much capability remained after the contract ended.

The final Edition 2.0 tables should therefore place reported procurement beside a second column labelled "retained domestic value: not yet independently reconciled" unless the underlying supplier data permit the calculation.

This is an example of how the book should distinguish a valid government statistic from a broader economic inference that the statistic alone cannot support.

Analytical section 219R. Cumulative procurement needs consistent start dates

Cumulative industry-spend figures are sensitive to the date from which the calculation begins.

One source may measure spending since 2015.

Another may measure spending since first oil in 2019.

Another may measure spending since the Local Content Act came into force.

These totals are not directly comparable.

Every cumulative local-content statistic should therefore include:

- start date;
- end date;
- included companies;
- whether goods and services are gross or net of imports;
- whether wages are included;
- currency conversion method.

A cumulative figure without those fields should be treated as a communications statistic rather than a full economic account.

Analytical section 219S. The supplier registry should be connected to contract outcomes

The number of registered suppliers measures access to the system.

It does not measure whether those suppliers received work.

A stronger registry should connect each supplier to:

- beneficial ownership;
- capability category;
- certifications;
- bids submitted;
- contracts won;
- contract value;
- contract performance;

- training received;
- later non-petroleum or export contracts.

This would allow the state to distinguish registration growth from supplier development.

It would also show whether procurement is concentrated among a small number of firms.

Analytical section 219T. Concentration is a local-content risk

A local-content regime can meet aggregate targets while opportunities remain concentrated.

The analysis should therefore calculate:

- top-five supplier share;
- top-ten supplier share;
- median contract value;
- number of first-time suppliers;
- number of repeat suppliers;
- region of supplier ownership;
- sector of supplier capability.

High concentration is not automatically evidence of failure.

Some petroleum services require scale and certification.

It is evidence that the policy should examine whether barriers to entry are technical, financial or informational.

Analytical section 219U. Procurement categories should be reviewed against cost and quality

Local-content targets can create value when they accelerate learning and market formation.

They can create cost when supply is constrained or firms exist only nominally.

For each reserved or targeted category, government should periodically compare:

- local price;
- international benchmark;
- quality;
- delivery time;
- safety performance;
- number of qualified suppliers.

The objective is to understand whether the category is moving toward competitive domestic capability.

A category that remains structurally expensive after years of support may need a different strategy.

Analytical section 219V. Offshore catering is a useful micro-case

Catering illustrates how a seemingly simple service can create a wider local value chain.

The economic chain can include:

- agriculture;
- food processing;
- cold storage;
- transport;
- hygiene standards;
- offshore logistics;
- workforce training.

A procurement contract therefore has greater developmental value when local firms progressively capture more of the upstream chain rather than only the final service.

The audit should ask what share of food was grown or processed domestically, which certifications were required and whether the supplier later gained access to other industrial clients.

Analytical section 219W. Fabrication illustrates the difference between assembly and capability

Fabrication can range from basic metal work to high-specification components requiring welding procedures, traceability, non-destructive testing and international certification.

A "local fabrication" statistic should therefore identify the technical level of the work.

Capability indicators could include:

- coded welders;
- certified procedures;
- quality-management systems;
- pressure or structural classes;
- client acceptance;
- export qualification.

This makes the local-content discussion relevant to industrial policy.

Analytical section 219X. Finance is a hidden constraint on supplier development

A local company can have technical capability and still fail to win large petroleum contracts because it cannot finance:

- inventory;
- equipment;

- insurance;
- performance bonds;
- long payment cycles.

The policy toolkit should therefore distinguish a capability gap from a financing gap.

Potential responses include:

- receivables finance;
- guarantee schemes;
- supplier-development lending;
- transparent contract pipelines;
- prompt-payment standards.

The risk is that public support socializes weak commercial decisions.

Financial support should therefore be linked to verified contracts and capability.

Analytical section 219Y. Safety performance is part of local value

A supplier that wins work but cannot meet petroleum safety standards is not durable local capacity.

Supplier-development reporting should include:

- safety training;
- incident rates;
- audit findings;
- quality non-conformances;
- corrective actions.

This prevents local content from being treated as a separate objective from operational integrity.

Analytical section 219Z. Local content should be connected to the education system

The workforce forecast should be shared with:

- technical institutes;
- universities;
- apprenticeship programmes;
- scholarship administrators.

This does not mean training institutions should produce workers for only one sector.

It means that labour-demand information should help shape course capacity and technical equipment.

The strongest skills investment creates portable competencies useful in mining, power, construction, manufacturing and maritime services as well as petroleum.

Analytical section 219AA. Skills portability is the post-petroleum test

A petroleum-specific skill can be valuable during production but vulnerable to sector decline.

A portable skill creates a stronger national asset.

The workforce strategy should therefore favour competencies such as:

- electrical systems;
- mechanical maintenance;
- process control;
- welding;
- instrumentation;
- marine operations;
- data analysis;
- safety management;
- project management.

The policy should measure where trained workers move over time.

Analytical section 219AB. Housing pressure belongs in the distribution analysis

A resource boom can raise rents rapidly near employment centres.

For households not receiving petroleum-linked wage gains, higher housing cost can reduce real welfare even while national income rises.

The social dashboard should therefore include:

- rent index;
- house-price index;
- household overcrowding;
- commute time;
- new housing supply.

These indicators help explain why GDP and household experience can diverge.

Analytical section 219AC. Food-price exposure matters for lower-income households

Imported demand, wage growth and logistics constraints can increase food prices.

The distributional effect is larger for households that spend a high share of income on food.

The paper should therefore present inflation by major household category where data permit rather than only the headline consumer-price index.

Analytical section 219AD. Regional investment should be tracked to service outcomes

Resource revenue may finance roads, schools, health facilities or digital infrastructure outside the main economic corridor.

The audit question is not only whether money was allocated.

It is whether access improved.

For each region, the public-investment register should track:

- facility or infrastructure delivered;
- operational status;
- staffing;
- utilization;
- maintenance;
- service outcome.

This is the social equivalent of the petroleum transaction ledger.

Analytical section 219AE. A local-content annual account

The final book should propose an annual local-content account with five statements.

Statement 1: Procurement. Gross local procurement by category and contractor.

Statement 2: Domestic value. Imported inputs, Guyanese wages, local profit, taxes and retained value.

Statement 3: Employment. Guyanese employment by occupation and seniority.

Statement 4: Capability. Training completion, certification, supplier progression and exports.

Statement 5: Distribution. Regional supplier participation, household indicators and community outcomes.

This would convert local content from a target-compliance exercise into a national capability account.

Analytical section 219AF. Evidence conclusion

The existence of detailed reporting requirements is a strength because it means many of the required data fields already exist in the regulatory system. [S108]

The analytical challenge is aggregation and verification.

Edition 2.0 should therefore avoid both extremes.

It should not dismiss official procurement numbers because they are not perfect measures of value added.

It should not present them as proof that the entire amount remained in-country.

The defensible position is to use the reported figure for what it measures and build the additional reconciliation needed to answer the deeper question.

PART VII

Economic Transformation, Infrastructure, Energy and Diversification

Analytical section 220. The development question after the windfall

Petroleum changes the scale of national income and public finance.

It does not automatically answer the development question.

A resource-rich state still needs to decide how to convert temporary extraction income into assets and capabilities that continue producing value after the fastest period of oil growth ends.

The framework used in this study therefore separates:

- petroleum output;
- non-oil output;
- financial saving;
- public capital;
- human capital;
- private productive capacity;
- natural capital.

The objective is national wealth conversion.

Analytical section 221. GDP as a production measure

GDP measures economic production within the country.

Guyana's petroleum expansion has produced exceptional headline growth.

IMF staff reported real GDP growth above 19 percent in 2025 after average growth near 40 percent during 2023 and 2024. [S64]

These figures establish the scale of the macroeconomic transition.

They do not show how income is distributed.

They do not show service quality.

They do not show the durability of infrastructure.

The flagship should therefore use GDP as the first line of analysis, not the final line.

Analytical section 222. Oil and non-oil GDP

A resource economy needs both total and non-resource measures.

Oil GDP responds to:

- production;
- price-volume accounting;
- new FPSO startup.

Non-oil GDP captures activity in:

- construction;
- agriculture;
- manufacturing;
- services;
- mining outside petroleum.

A strong non-oil growth rate can reflect genuine diversification.

It can also reflect oil-financed construction and services.

Sector detail is needed.

Analytical section 223. Construction

Construction is often one of the fastest channels through which a resource boom enters the domestic economy.

Government builds infrastructure.

Private firms build offices, housing and industrial facilities.

Households increase demand.

Construction growth creates jobs.

It also competes for:

- labour;
- cement;
- aggregate;
- equipment;
- land.

The final chapter should track output and cost together.

Analytical section 224. Agriculture

A petroleum boom can support agriculture through:

- roads;
- drainage;
- electricity;

- cold storage;
- finance.

It can also challenge agriculture through:

- higher wages;
- imported food;
- land prices.

The diversification analysis should use:

- production;
- yield;
- exports;
- imports;
- farm income.

A policy announcement about food security should be tested against those indicators.

Analytical section 225. Manufacturing

Manufacturing benefits when energy and logistics become more reliable.

It struggles when wages and imported input costs rise.

The paper should measure:

- manufacturing value added;
- employment;
- exports;
- electricity cost;
- industrial capacity.

Petroleum-related fabrication should be identified separately from broader manufacturing.

Analytical section 226. Mining

Gold and other mining sectors formed part of Guyana's economy before oil.

The petroleum era changes their relative share of GDP.

A declining share does not necessarily mean lower mining output.

The denominator has expanded.

The analysis should therefore report both absolute output and share.

Analytical section 227. Services

Oil creates demand for:

- finance;
- legal work;
- accounting;
- transport;
- logistics;
- hospitality;
- engineering.

Some services build exportable capability.

Others depend almost entirely on the oil investment cycle.

The diversification chapter should identify which have clients outside petroleum.

Analytical section 228. Public investment as a transmission channel

Petroleum revenue influences the non-oil economy strongly through public investment.

The 2026 Budget Speech records a large programme of capital and social spending. [S65]

A budget is an authorization and policy statement.

The final case needs a project outcome register.

For each major project it should show:

- appropriation;
- award;
- construction;
- completion;
- operation.

Analytical section 229. Roads

Road investment can reduce:

- travel time;
- vehicle cost;
- geographic isolation.

A road also creates:

- maintenance;
- land-use change;
- environmental effect.

The project outcome should therefore include:

- traffic;
- time saved;
- condition;
- maintenance cost.

Analytical section 230. Bridges

A major bridge can change national transport patterns.

The appraisal should compare:

- existing crossing;
- traffic;
- travel time;
- capacity;
- resilience;
- toll or operating model.

The post-completion review should compare actual use with forecast.

Analytical section 231. Ports

Petroleum and broader trade both depend on ports.

Port investment should be analyzed through:

- depth;
- berth;
- cargo throughput;
- turnaround;
- cost.

A petroleum shorebase is not automatically a national public port asset.

Ownership and financing should be recorded.

Analytical section 232. Airports

Economic growth increases passenger and cargo traffic.

Airport investment can support:

- tourism;
- business;
- emergency response.

The infrastructure ledger should distinguish capital expansion from operating performance.

Analytical section 233. Drainage and sea defence

For Guyana, climate resilience is economic infrastructure.

A flood-protection project should be assessed through:

- protected area;
- pump capacity;
- drainage time;
- failure frequency;
- maintenance.

This allows petroleum-financed resilience to enter the national wealth framework.

Analytical section 234. Housing

Housing investment affects:

- supply;
- rent;
- household wealth;
- construction demand.

A national housing programme should be evaluated through occupied units and infrastructure access, not allocated lots alone.

Analytical section 235. Hospitals

Health capital creates value only when combined with:

- staff;
- equipment;
- supplies;
- operating budget.

The public-investment register should include those recurrent requirements.

Analytical section 236. Schools and universities

Education infrastructure should support a broader human-capital strategy.

The petroleum transition increases demand for:

- engineering;
- technology;
- finance;
- project management.

A building without qualified teachers does not close the skills gap.

Analytical section 237. Electricity as a binding constraint

Power cost and reliability affect households and firms.

Guyana's energy strategy includes gas-to-energy and renewable resources. [S67]

The correct performance measures include:

- dependable capacity;
- generation;
- outages;
- losses;
- tariff;
- fuel cost;
- emissions.

Installed megawatts alone are not enough.

Analytical section 238. Gas-to-energy

A domestic gas project can change the power system through:

- fuel cost;
- generation efficiency;
- supply diversity.

A full project assessment needs:

- pipeline;
- processing;
- power plant;
- grid;
- financing;
- operating cost;
- gas terms.

The outcome should be tested through observed generation, reliability and tariff.

Analytical section 239. Project delay risk

Large integrated energy projects contain several dependencies.

A delay in one component can delay system benefits.

The project register should therefore use:

- original schedule;
- revised schedule;

- commissioning.

A policy target should not be described as an achieved outcome.

Analytical section 240. Demand growth

Economic growth increases power demand.

New housing, industry, commercial buildings and data infrastructure add load.

A generation plan must therefore compare new supply with demand.

The paper should not report a new power plant as excess capacity without a demand forecast.

Analytical section 241. Energy efficiency

The Guyana Energy Agency's 2026 demand-side management plan targets significant savings by 2030. [S66]

The final assessment should distinguish target from verified saving.

Efficiency programmes should report:

- participants;
- measure;
- cost;
- verified kWh saved.

Demand reduction is a resource.

Analytical section 242. Renewable energy

Solar, hydro and other renewable resources diversify the electricity mix.

A national energy comparison should use:

- installed capacity;
- annual generation;
- capacity factor;
- storage;
- grid integration;
- cost.

A renewable project should not be measured only by nameplate MW.

Analytical section 243. Reliability

Reliability has direct economic value.

Common measures include:

- outage frequency;

- outage duration;
- unserved energy.

The final paper should establish a pre-oil baseline and post-investment series.

This tests whether capital spending improves service.

Analytical section 244. Energy affordability

The household and business impact depends on the final tariff and quality of service.

A tariff reduction achieved through hidden subsidy has a different fiscal effect from one achieved through lower generation cost.

The paper should report:

- cost;
- tariff;
- subsidy.

Analytical section 245. Export concentration

Petroleum rapidly changes export composition.

A diversification strategy should therefore monitor concentration.

The Herfindahl index offers one measure.

The paper should also show non-oil exports in absolute terms.

A country can become more petroleum-concentrated even while non-oil exports grow.

Analytical section 246. Dutch-disease channels

The classic resource-boom concern operates through several channels.

Higher resource income increases spending.

Demand for non-tradable goods rises.

Wages and prices can increase.

Labour shifts toward resource-linked sectors.

The real exchange rate appreciates through nominal exchange rate movement, domestic prices or both.

Tradable non-resource firms become less competitive.

These channels should be tested, not assumed.

Analytical section 247. IMF 2026 assessment

IMF staff in July 2026 stated that available indicators did not show clear signs of overheating or resource-driven competitiveness pressure at that time, while identifying wage and foreign-exchange indicators for continued monitoring. [S64]

The paper should preserve the date.

A 2026 assessment is not a permanent conclusion.

The independent dashboard should update each indicator.

Analytical section 248. Wage-based indicators

Wage growth can reflect:

- productivity;
- labour shortage;
- inflation;
- sector demand.

The paper should compare wages with productivity.

A wage increase supported by productivity has different competitiveness implications from one without matching output.

Analytical section 249. Real exchange rate

The real exchange rate compares domestic costs with trading partners.

In a relatively stable nominal exchange-rate system, adjustment may appear through domestic inflation and wages.

The paper should use more than one indicator.

Analytical section 250. Imports

Large investment raises imports of:

- machinery;
- vehicles;
- materials;
- services.

Import growth is not automatically a weakness.

Capital imports can expand future productive capacity.

The analysis should distinguish capital, intermediate and consumption imports.

Analytical section 251. Non-oil exports

The most direct diversification test is whether sectors earn foreign exchange outside petroleum.

The paper should track:

- gold;
- agriculture;
- manufactured goods;
- services;
- tourism.

A stronger export base reduces long-run oil dependence.

Analytical section 252. Private investment

Government spending is only one development engine.

The private sector needs:

- reliable power;
- logistics;
- rule clarity;
- finance;
- skills.

The flagship should track non-oil private investment and firm formation.

This provides evidence about whether public infrastructure crowds in private activity.

Analytical section 253. Financial sector

Rapid growth affects banks through:

- deposits;
- credit;
- mortgages;
- corporate lending.

A resource boom can increase asset prices and concentration.

The financial stability chapter should monitor:

- credit growth;
- nonperforming loans;
- sector exposure;
- foreign currency.

Analytical section 254. Land prices

Infrastructure and income growth can increase land values.

This creates wealth for owners and affordability pressure for buyers.

A land-price series would strengthen the distribution analysis.

Where data are absent, the gap should be noted.

Analytical section 255. Public-sector wages

Government competes with private employers.

Public wage policy affects:

- staffing;
- inflation;
- service delivery.

The paper should analyze technical occupations separately from broad wage averages.

Analytical section 256. National productivity

The most durable economic test is productivity.

Infrastructure and human capital should allow workers and firms to produce more value per unit of input.

The flagship should calculate productivity cautiously because petroleum output dominates aggregate measures.

Non-oil productivity is more informative for diversification.

Analytical section 257. National wealth accounting

The economic transformation should ultimately appear in a balance-sheet view.

Petroleum reserves fall.

Financial assets rise.

Public infrastructure rises.

Human capital may rise.

Natural capital can rise or fall.

Debt and maintenance liabilities change.

The country should track these categories separately.

Analytical section 258. Conversion efficiency

A conceptual wealth-conversion ratio can compare the present value of durable assets created with petroleum revenue used.

The paper should not publish a precise ratio without defensible valuation.

It can still use a dashboard.

For example:

- financial assets;
- infrastructure condition;
- educational outcomes;
- health outcomes;
- natural capital.

Analytical section 259. A low-price development test

Assume petroleum receipts fall for three years.

Which public investments continue?

Which recurrent programmes depend on oil?

Which projects can be delayed without large economic loss?

A resilient development plan should identify priorities before the shock.

Analytical section 260. A capacity test

Assume capital budgets rise 30 percent but contractor capacity does not.

Expected effects include:

- fewer bids;
- higher price;
- delay.

The response options include sequencing, imports, training and procurement changes.

This is the practical meaning of absorptive capacity.

Analytical section 261. A maintenance test

Assume oil revenue growth slows after a large infrastructure build.

Can stable non-oil revenue cover:

- maintenance;
- staffing;
- operating cost?

If not, the asset portfolio creates future fiscal pressure.

This test should be part of appraisal.

Analytical section 262. New-producer blueprint

A newcomer should establish before first revenue:

- national infrastructure baseline;
- project appraisal;
- procurement control;
- maintenance accounting;
- non-oil GDP series;
- price and wage monitoring;
- export concentration;
- household baseline.

The country should then update them annually.

Analytical section 263. Part VII findings

Guyana's growth record demonstrates the scale of a petroleum-led macroeconomic transition.
[S64]

The strongest development case will rest on what the petroleum period leaves behind.

That includes:

- financial assets;
- infrastructure that works;
- educated and healthy people;
- competitive firms;
- resilient natural systems.

Budget size and GDP growth are important evidence.

They are not sufficient evidence of durable national wealth.

The blueprint lesson is to measure the conversion, not only the windfall.

Edition 2.0 deepening: turning a petroleum boom into a second growth engine

Analytical section 263A. The development test is productivity outside petroleum

Non-oil GDP growth is important, but it does not by itself establish diversification.

Construction, retail, transport and professional services can expand because petroleum investment and public spending are increasing.

A stronger diversification test asks whether the economy is building activities that can remain competitive when petroleum investment and revenue growth slow.

The core indicators should therefore include:

- non-oil exports;
- labour productivity;
- firm survival;
- private investment outside petroleum;
- energy intensity;
- logistics cost;
- export-market concentration;
- technological capability.

This distinguishes demand spillovers from structural transformation.

Analytical section 263B. Export diversification should be measured in foreign-exchange terms

An economy can produce more goods and services domestically while remaining dependent on petroleum for foreign exchange.

The diversification dashboard should therefore track:

- petroleum export share;
- gold and other mineral share;
- agricultural exports;
- manufactured exports;
- tourism receipts;
- business-service exports;
- regional trade.

The objective is not to reduce petroleum exports artificially.

It is to create additional sources of foreign exchange before the petroleum cycle matures.

Analytical section 263C. Public infrastructure should reduce economy-wide transaction cost

The development value of a road, bridge, port or airport is not the construction expenditure.

It is the reduction in time, cost, risk or congestion experienced by users.

Project evaluation should therefore estimate:

- travel time;
- freight cost;
- reliability;
- accident cost;
- market access;
- maintenance cost;
- induced demand;
- climate exposure.

A project can be strategically valuable even when it does not generate direct cash revenue.

The appraisal should still define the service outcome.

Analytical section 263D. Electricity is a cross-sector productivity input

Reliable and affordable electricity affects:

- manufacturing;
- cold storage;
- digital services;
- mining;
- hospitality;
- household welfare;
- public services.

The energy transition should therefore be assessed through delivered system performance rather than installed megawatts alone.

A national power dashboard should track:

- demand;
- available capacity;
- outage frequency;
- outage duration;
- system loss;
- fuel cost;
- generation mix;
- reserve margin;
- customer tariff.

Official Guyana energy planning and demand-side work provide inputs for this analysis. [S66-S67]

Analytical section 263E. Gas-to-energy should be evaluated as an integrated system

A gas-to-energy project is not one asset.

It can include:

- offshore gas infrastructure;
- pipeline;
- processing;
- power generation;
- transmission;
- distribution;
- backup supply.

The economic case depends on the performance of the full chain.

The risk model should therefore test:

- pipeline interruption;
- plant outage;
- demand forecast error;
- fuel-price assumptions;
- construction delay;
- transmission constraints.

A lower expected generation cost becomes a development benefit only when reliable power reaches users.

Analytical section 263F. Renewable energy and gas should be analysed through system reliability

Energy policy should not reduce to a contest between technologies.

A small system needs a portfolio that manages:

- variable demand;
- renewable intermittency;
- fuel risk;
- storage;
- transmission;
- reserve capacity;
- climate resilience.

The correct comparison is delivered reliability and lifecycle cost.

Analytical section 263G. Agriculture needs productivity, storage and market access

Oil revenue can support agricultural infrastructure, but long-run competitiveness requires more than subsidy.

The agricultural transformation chain includes:

- drainage and irrigation;
- seed and genetics;
- extension;
- mechanization;
- storage;
- processing;
- standards;
- logistics;
- export market access.

A country that expands primary production without storage or market access may create price volatility rather than durable value.

Analytical section 263H. Manufacturing requires scale and capability, not only cheaper power

Lower electricity cost can improve manufacturing economics.

It does not guarantee competitive industry.

Manufacturing strategy should also assess:

- domestic market size;
- regional market access;
- freight;
- labour skill;
- finance;
- quality standards;
- supplier networks;
- imported input dependence.

The most promising sectors may be those linked to existing resource and agricultural strengths rather than arbitrary import substitution.

Analytical section 263I. Tourism should be assessed against petroleum-era cost pressure

A petroleum boom can raise wages, land prices and construction cost.

Tourism competes for many of the same inputs.

A tourism strategy should therefore model:

- air access;
- room inventory;
- labour cost;
- service quality;
- nature assets;
- safety;
- seasonality;
- marketing;
- carrying capacity.

High national income does not automatically create a competitive tourism sector.

Analytical section 263J. Digital services offer a different diversification pathway

Digital exports are less dependent on physical freight.

Their constraints are different:

- connectivity;
- power reliability;
- skills;
- data protection;
- payment systems;
- market access.

A small state can therefore use digital services to diversify even when manufacturing scale is limited.

The petroleum transition can finance the infrastructure and education required, but the sector still needs firms capable of competing internationally.

Analytical section 263K. Urban growth should be treated as infrastructure demand

Rapid income and population changes affect:

- housing;
- land;
- transport;
- drainage;
- waste;

- water;
- schools;
- health care.

Urban planning should therefore be integrated into the resource-conversion model.

If housing and transport capacity lag, petroleum-driven income growth can create congestion and price pressure that reduce real welfare.

Analytical section 263L. Dutch-disease analysis should use evidence rather than labels

The phrase "Dutch disease" is often used too loosely.

A serious test should examine:

- real exchange-rate pressure;
- wages;
- tradable-sector employment;
- export performance;
- import penetration;
- productivity;
- government spending;
- migration.

A decline in one sector is not sufficient by itself.

The question is whether resource inflows are systematically reducing the competitiveness of non-resource tradables.

IMF analysis has treated this as a risk requiring monitoring rather than a conclusion that can be inferred from GDP growth alone. [S68]

Analytical section 263M. Public investment can crowd in or crowd out private investment

Infrastructure can crowd in private activity by lowering cost and risk.

Large state spending can also crowd out firms through:

- labour competition;
- construction capacity;
- credit;
- land;
- procurement concentration.

The development model should track both effects.

A public project that improves logistics may increase private investment.

A poorly sequenced capital boom may make private construction prohibitively expensive.

Analytical section 263N. Debt policy should be integrated with the sovereign fund

A country can simultaneously hold a large resource fund and borrow.

That is not automatically contradictory.

The correct comparison considers:

- borrowing cost;
- fund return;
- liquidity;
- project return;
- currency;
- maturity;
- risk.

Borrowing to preserve fund assets can be expensive if debt costs exceed the value of liquidity or the project return.

Using the fund excessively can weaken future buffers.

The optimal balance requires a consolidated public balance sheet.

Analytical section 263O. The second-engine test

By the time petroleum production matures, the economy should be able to point to at least one additional growth engine capable of earning foreign exchange and raising productivity without relying primarily on new petroleum investment.

That engine may be a portfolio rather than one sector.

The test should ask:

- Is it export capable?
- Is productivity rising?
- Can it attract private capital?
- Does it employ skills developed during the petroleum era?
- Does it remain viable under lower public spending?

This is a more demanding standard than non-oil GDP growth.

Analytical section 263P. Expanded Part VII conclusion

The economic-transformation chapter should connect petroleum wealth to the productive system that will remain after the boom.

The key sequence is: resource revenue to infrastructure and human capital to lower transaction cost to private investment and productivity to competitive exports to a broader tax base.

If one link is weak, large public spending can still produce visible growth without equivalent long-run capability.

For a newcomer, the blueprint is therefore to define the post-petroleum economy before peak revenue arrives.

Every major infrastructure project should be able to explain which productivity constraint it removes.

Every diversification programme should identify the market in which firms are expected to compete.

Every permanent expenditure commitment should be tested against a lower petroleum-revenue future.

The objective is not to predict the exact structure of the economy decades ahead.

It is to ensure that petroleum finance expands national options rather than narrowing them.

FLAGSHIP PART VIII

Accountability and institutional learning

Working manuscript for the 520-560-page flagship edition. Evidence cut-off: 27 September 2026. Source IDs refer to SOURCE_REGISTER.md. Political and institutional disagreements are attributed to documented sources. This section does not rank parties, governments or public officials.

39. Extractive disclosure as an operating system

Transparency in a petroleum state has little value if disclosure arrives too late, uses incompatible definitions or cannot be reconciled across agencies. The strongest disclosure architecture therefore behaves as an operating system rather than a periodic communications exercise. It links contracts, production, government entitlement, royalties, audited recoverable costs, fund deposits, investment income, parliamentary withdrawals and public expenditure through shared dates and definitions.

Guyana's current public record contains several layers. The Petroleum Management Programme publishes petroleum agreements, block information, production licences, cost-audit documents, environmental records and a production dashboard. The Bank of Guyana publishes Natural Resource Fund reports and audited statements. The Ministry of Finance publishes budget and mid-year fiscal reports. Guyana EITI publishes national reports and participates in the international EITI validation process. These sources create a substantial documentary trail, but the trail still requires reconciliation by date and transaction. [S18, S33-S34, S41-S47, S69-S72, S81-S86]

The EITI process deserves separate treatment because it tests more than numerical publication. In May 2026, the EITI Board launched a full validation of Guyana from 15 May and recorded concern about continuing stakeholder confidence in the selection of civil-society representatives. The International Secretariat also sought views on stakeholder engagement and barriers affecting participation. At the evidence cut-off used for this manuscript, a final 2026 validation result had not been established. The proper research treatment is therefore to describe the process as ongoing, record the Board's stated concern and avoid substituting an author's political conclusion for an unfinished institutional review. [S16-S18, S33-S34, S69-S70]

The design lesson for an emerging producer is to separate disclosure into at least six ledgers. The first is a legal ledger containing executed contracts, licences, permits and amendments. The second is a production ledger containing measured output, cargoes and entitlement. The third is a revenue ledger containing royalties, profit oil, taxes, penalties and settlement dates. The fourth is a cost ledger containing recoverable expenditures, audit adjustments and dispute status. The fifth is a sovereign-fund ledger containing opening cash, deposits, investment returns, withdrawals and closing cash. The sixth is a public-investment ledger containing appropriations, tenders, awards, variations, completion, operation and maintenance. A seventh social and

environmental ledger is warranted when the country has major local-content, community, biodiversity or climate-finance obligations.

A high-quality transparency platform should not force users to infer equivalence among these ledgers. For example, a monthly production volume does not imply a same-month government cash receipt. A disputed cost item does not imply a final cash recovery. An NRF withdrawal does not imply a completed road or hospital. A carbon-credit agreement does not imply a cash receipt in the petroleum fund. These distinctions protect public understanding and improve external audit.

40. Cost audits and revenue assurance

Cost recovery is one of the least intuitive components of a production-sharing system. Under the Stabroek framework, the economic effect of recoverable expenditure depends on the cost ceiling, the stock of eligible costs and the timing of production. The state's ability to verify cost claims therefore affects the timing and amount of profit oil even when headline contract terms remain unchanged.

Guyana's public petroleum portal includes an IHS Markit audit covering the 1999-2017 period and a VHE audit record for 2018-2020. The IHS Markit final report states that the audited cost-recovery submission totalled USD 1.6778 billion and identified USD 214.4 million plus overhead adjustments as having grounds for dispute. The Ministry of Natural Resources later stated that aspects of the earlier audit had moved into the production-sharing agreement's sole-expert procedure and described the status of several audit exercises. An audit exception, a government challenge, an expert determination and cash ultimately recovered are four different events. The flagship paper will preserve those statuses separately. [S42-S43, S71-S72]

For a new producer, the audit function should begin before first revenue. The state needs contract accountants, petroleum engineers, reservoir specialists, transfer-pricing expertise, legal counsel and access to source documentation. Cost categories should be mapped to contractual clauses before operators begin submitting large development claims. Data retention must be long enough to support later disputes. Audit timetables should account for the fact that offshore development projects spend billions before production begins, creating a large initial cost bank.

The revenue-assurance model should contain a claim register with fields for the cost category, project, invoice date, related party, contractual basis, amount claimed, amount accepted, amount challenged, audit reason, operator response, dispute stage, final determination and cash effect. Without these fields, public reporting risks compressing a multi-year accounting process into one headline number.

Audit independence also requires procurement safeguards. If the government hires external auditors, the public should know how the firm was selected, whether conflicts were screened, which years and entities fall within scope, and whether the contract permits access to underlying technical records. The state should maintain its own institutional memory rather than outsourcing the entire audit function to successive consulting teams.

41. Procurement and spending oversight

A petroleum windfall expands the public capital programme faster than administrative capacity may expand. Procurement oversight therefore becomes part of petroleum governance even when procurement law sits outside the petroleum sector. A country may account perfectly for oil revenue entering a sovereign fund and still convert the revenue poorly into durable assets.

Guyana's Ministry of Finance has published large public-investment allocations in successive budgets. IMF staff in 2026 highlighted absorption, project monitoring, procurement and public-enterprise oversight as policy priorities. The Audit Office publishes public-accounts and performance reports, including the 2024 public-accounts report presented to the National Assembly in September 2025. These records create a basis for project-level testing rather than broad claims about public-sector performance. [S64-S65, S73]

The flagship methodology therefore applies an outcome chain to major capital projects. The first stage is policy justification. The second is feasibility and lifecycle costing. The third is procurement and contractor selection. The fourth is construction and contract variation. The fifth is commissioning. The sixth is operation and maintenance. The seventh is measured service outcome. A project should not move to the final category because a contract was signed or because budgeted expenditure was executed.

For an emerging producer, the public-investment register should identify the original cost estimate, financing source, procurement route, evaluation criteria, winning bidder, beneficial owner, variation orders, completion date, maintenance provision, responsible operating agency and service indicators. Where climate exposure matters, the register should record flood, heat, hurricane, coastal or seismic design assumptions.

This framework also helps limit procyclical spending. If capital projects are approved faster than engineers, construction firms, ports, aggregate supply or procurement teams can deliver them, the result may be higher costs rather than proportionally higher public-service output. The correct test is not whether the government has the cash. The test is whether the economy and institutions can convert that cash into durable capacity at acceptable cost.

42. Parliament, judiciary and institutional checks

A resource-governance system is more resilient when multiple institutions can test decisions using different legal mandates. Parliament authorizes laws and public expenditure. Courts interpret legal obligations and review administrative action. Auditors examine financial records. Regulators issue permits and licences. Central banks hold or manage financial assets. EITI processes add a multi-stakeholder disclosure layer. None of these institutions performs the same function.

Guyana's petroleum framework provides several examples. Parliament enacted the 2019 Natural Resource Fund Act before first oil, later replaced it with the 2021 framework and amended withdrawal provisions in 2024. Parliament also enacted the Local Content Act 2021 and the Oil Pollution Prevention, Preparedness, Response and Responsibility Act 2025. Each statute needs a versioned legal crosswalk because enactment, assent, publication, commencement and

implementing regulations do not necessarily occur on the same date. [S05, S19, S37, S53-S54, S57]

The judiciary has also become part of petroleum governance. The May 2026 Court of Appeal dispute over financial assurance illustrates why the final publication must use authenticated judgments rather than relying on a press release or one party's summary. The government stated that the appeal court set aside the earlier High Court requirement for an unlimited guarantee and confirmed regulatory discretion over the amount of financial assurance. This work treats that statement as an attributed description pending full judgment review. It does not transform a contested legal question into an authorial verdict. [S20]

The lesson for a new producer is procedural. Every major decision should identify the statutory authority, decision-maker, evidence considered, public notice requirement, reasons, appeal route, judicial-review route and record-retention requirement. Where a regulator possesses discretion, the file should preserve the reasons for exercising that discretion. This improves the ability of courts and auditors to review the decision later without requiring them to re-create the technical record from fragments.

Institutional checks work best when their boundaries are explicit. A court does not run an environmental agency. An environmental regulator does not set the annual budget. A finance ministry should not certify its own external audit. A sovereign-fund board should not replace parliament's appropriation role. The operating model should therefore map both authority and limits.

43. Statistical institutions and public data

A resource boom changes the statistical system as much as it changes the fiscal system. Production, exports, government revenue, construction, wages, migration, prices and household consumption can move quickly. If statistical releases lag, public debate relies on nominal spending, operator announcements or national GDP, none of which answers the full distributional question.

Guyana's Bureau of Statistics completed the 2018/2019 Household Budget Survey and Living Conditions work across all ten regions. The World Bank and IMF have both noted limits in the availability of more recent household information. IMF reports in 2025 and 2026 also discuss statistical modernization and data gaps affecting analysis of the non-oil economy and household conditions. The responsible approach is to date every welfare statistic and avoid describing a 2019 poverty observation as a 2026 outcome. [S10, S61, S64, S68]

A flagship resource-governance dashboard should therefore carry three dates for each indicator: observation period, release date and last revision. It should also identify geographic coverage, sample size where relevant, price base, currency, nominal or real status and source owner. Petroleum data need equally careful definitions. Monthly barrels, daily peaks, nameplate capacity, government lifts and cash receipts should never occupy one undifferentiated column.

The public-data system should include a corrections policy. Resource states experience frequent revisions because cargo prices settle after lifting, cost claims change through audit, national accounts undergo rebasing, and surveys improve. Publishing a revised series with a version history is better than defending an outdated number for reputational reasons.

The most useful indicator architecture for the final paper will include petroleum output, non-oil output, government revenue, NRF flows, inflation, employment, real wages, household consumption, poverty or deprivation, school and health outcomes, regional service access, public debt, construction costs, exchange-rate conditions, imports, exports outside petroleum and environmental performance. The point is not to produce one composite score. The point is to prevent a high petroleum-growth number from substituting for evidence on other outcomes.

Institutional learning cycle

The final accountability model uses a six-stage loop.

1. **Decide.** A legally authorized institution makes a decision using a dated record.
2. **Disclose.** The underlying rule, assumptions and non-confidential evidence are published.
3. **Measure.** Actual production, cash, cost, environmental or service outcomes are recorded.
4. **Challenge.** Auditors, parliament, courts, stakeholders and independent experts test the evidence.
5. **Correct.** Errors, weak controls or outdated rules are revised with reasons.
6. **Archive.** The version history remains accessible so later analysts can distinguish what was known at each date.

This cycle is more transferable than any single petroleum contract or fund-withdrawal rule. It allows a new producer to learn without assuming that the first institutional design is permanent.

Part VIII conclusions

Guyana's public record shows a growing network of petroleum disclosure, fund reporting, audit material, parliamentary lawmaking, court review and EITI engagement. The same record also contains unresolved audit disputes, an ongoing EITI validation at the research cut-off, legal questions requiring primary judgment review and statistical gaps affecting social analysis. These facts belong together.

The blueprint value lies in the structure of verification. A new producer should design for reconciliation, version control, challenge and correction from the start. Institutional maturity is not the absence of disputes. It is the ability to record them, resolve them through lawful procedures and preserve evidence of what changed.

Edition 2.0 deepening: accountability as a system of reconciled public ledgers

Analytical section 43A. Transparency should be designed for reconciliation

Publication is necessary but not sufficient.

A resource state can publish contracts, production data, fund reports and budgets while still making it difficult for an outside reviewer to connect one record to another.

The stronger standard is reconcilability.

A reviewer should be able to move from: licence to production to state entitlement to sale to cash receipt to NRF deposit to budget withdrawal to procurement to completed asset.

Each transition should have a dated identifier.

This is the difference between disclosure and an auditable public operating system.

Analytical section 43B. Beneficial ownership belongs in both licensing and procurement

Beneficial ownership is often treated as an extractive-industry disclosure issue.

Its relevance is wider.

Government should understand who ultimately controls:

- licence holders;
- joint-venture partners;
- petroleum suppliers;
- public contractors;
- companies receiving local-content preferences.

The purpose is not to stigmatize complex ownership.

It is to identify conflicts, related-party transactions, concentration and hidden control.

The same ownership identifier should be usable across petroleum, company-registration and procurement systems where law permits.

Analytical section 43C. Conflict-of-interest controls should attach to decisions

A generic ethics code is not enough for high-value petroleum decisions.

Each major decision file should record:

- decision-maker;
- declared interests;
- recusal if any;
- adviser conflicts;
- contractor relationships;
- cooling-off restrictions where applicable.

This should apply to:

- licence awards;

- contract negotiation;
- audit procurement;
- cargo marketing;
- environmental permitting;
- major public-investment awards.

The public disclosure level may vary.

The internal control should not.

Analytical section 43D. EITI is most valuable when it becomes part of routine government data

EITI reporting can operate as a periodic reconciliation exercise.

The higher institutional standard is systematic disclosure.

Production, licence, ownership and revenue data should increasingly come from regular government systems rather than be recreated only for an EITI cycle.

At the September 2026 evidence cut-off, Guyana's Validation remained ongoing, and EITI committee records show continuing process review. [S89, S104]

The correct lesson for a newcomer is not to design transparency around one external assessment date.

It is to make the underlying data routine enough that an external assessment can test an already functioning system.

Analytical section 43E. Audit findings need a status taxonomy

Public debate often compresses all audit numbers into "money owed".

A better taxonomy distinguishes:

1. potential exception;
2. formal audit finding;
3. contractor response;
4. government position;
5. dispute;
6. expert or judicial determination;
7. accounting adjustment;
8. cash recovery.

This taxonomy should be used in every public update.

It prevents a disputed cost from being represented as recovered public revenue before the contractual process is complete.

Analytical section 43F. Parliamentary oversight needs transaction-level support

Parliamentary debate is stronger when members receive reconciled data rather than headline totals.

An annual petroleum accountability package could include:

- production by project;
- government lifts;
- realized public petroleum receipts;
- NRF opening and closing balance;
- withdrawals;
- outstanding cost-audit disputes;
- public investment funded from withdrawals;
- major environmental incidents;
- local-content outcomes.

The package should use definitions consistent with audited financial statements and regulator data.

Analytical section 43G. Judicial review works best when administrative records are complete

Courts should not have to reconstruct the factual record from press statements.

A regulator's decision file should preserve:

- legal authority;
- technical evidence;
- consultation;
- reasons;
- conditions;
- financial-assurance analysis;
- internal review.

This improves both accountability and institutional defensibility.

The environmental financial-assurance litigation demonstrates why the quality of the administrative record matters even when the ultimate legal interpretation remains contested.
[S101, S103, S105]

Analytical section 43H. Freedom-of-information capability should include data architecture

A transparency regime can be overwhelmed if records are poorly indexed.

Petroleum records should therefore have:

- document type;
- project;
- date;
- version;
- responsible agency;
- confidentiality status;
- retention period.

Good records management reduces the cost of both public disclosure and internal decision-making.

Analytical section 43I. Statistical independence depends on revision discipline

Fast-growing resource economies experience frequent revisions.

The credibility question is not whether revisions occur.

It is whether the revision process is transparent.

Every major statistical series should retain:

- previous release;
- revised value;
- reason;
- methodology;
- effective date.

This applies to national accounts, production, trade and public finance.

Analytical section 43J. Public dashboards should publish definitions before graphics

A dashboard can increase confusion if users do not know what a number means.

Every dashboard should define:

- unit;
- time period;
- source;
- gross or net;
- nominal or real;
- cash or accrual;
- provisional or final.

The definition should be visible without downloading a technical appendix.

Analytical section 43K. Independent scrutiny needs access to machine-readable data

PDF reports are valuable for narrative and legal record.

Auditors and researchers also need structured data.

A mature disclosure system should publish:

- CSV or API production series;
- machine-readable budget data;
- contract metadata;
- permit metadata;
- beneficial-ownership fields where lawful;
- procurement records.

This allows external parties to reproduce calculations rather than manually transcribe tables.

Analytical section 43L. A corrections register is a governance instrument

Institutions sometimes resist public correction because correction is perceived as failure.

The stronger practice is to make correction routine.

A correction register should show:

- original release;
- corrected value or wording;
- reason;
- date;
- effect on prior conclusions.

The publication itself now uses this principle.

The same logic should apply to government data.

Analytical section 43M. Institutional memory should survive political transition

Petroleum systems operate across administrations.

A licence can outlast a government.

A cost audit can span several years.

A decommissioning obligation can persist for decades.

Institutional records should therefore be designed to survive political turnover.

This means:

- permanent file structures;
- documented delegations;
- version-controlled models;
- handover protocols;
- protected archives.

The objective is not to eliminate policy change.

It is to ensure that later decision-makers inherit the evidence needed to change policy intelligently.

Analytical section 43N. Expanded Part VIII conclusion

Accountability should not be evaluated through the number of reports published.

The stronger test is whether the public record can be reconciled across institutions and across time.

Guyana's petroleum, fund, EITI, budget, audit and judicial records provide a substantial documentary base. [S18, S41-S47, S69-S73, S81-S89, S104]

The next step is integration.

For a newcomer, the blueprint is to build one common data architecture across legal, production, revenue, cost, fund, procurement, social and environmental systems.

Every material figure should have a source owner, observation date, revision history and transaction boundary.

That architecture makes scrutiny cheaper, dispute resolution faster and institutional learning more durable.

FLAGSHIP PART IX

International comparators and transferability

Working manuscript for the 520-560-page flagship edition. Comparisons below describe documented mechanisms. They do not rank countries, governments or political choices.

44. Norway: converting petroleum cash into a financial asset

Norway is frequently invoked in resource-governance debates because its Government Pension Fund Global sits within a long-established fiscal framework. The current official framework directs the state's net petroleum cash flow to the fund and uses a fiscal guideline linked to the fund's expected long-term real return, commonly expressed around 3%. The 2026 white paper continues to describe the fund as part of a broader framework for managing petroleum wealth over time. [S25, S74]

The relevant lesson is not the 3% number in isolation. Norway built the fund after developing strong tax administration, petroleum expertise, state ownership capacity, democratic institutions, mature capital markets and a large non-petroleum tax base. Its financial asset stock is the product of decades of accumulation. A new producer with severe infrastructure deficits and a small domestic tax base faces different intertemporal trade-offs.

The transferable mechanism is the separation of the petroleum cash flow from ordinary current revenue, the integration of fund withdrawals with the national budget, the preference for diversified external financial investment, transparent reporting and a fiscal rule intended to protect the real value of the fund. The receiving country must still determine its own acceptable spending path, debt strategy and public-investment capacity.

A useful comparison with Guyana will therefore examine five variables rather than headline fund size: annual petroleum revenue as a share of non-oil GDP, fund assets per capita, non-oil primary balance, domestic investment need, and institutional capacity to deliver public projects. The same nominal withdrawal ratio means different things under different values for those variables.

45. Ghana: stabilization and heritage within a producing developing economy

Ghana's Petroleum Revenue Management Act 2011 established a statutory framework that includes a Stabilisation Fund and a Heritage Fund. The Bank of Ghana publishes reports on the petroleum funds and describes their distinct purposes. Ghana therefore offers a more developmentally proximate comparison than Norway for the architecture of separating short-term stabilization from intergenerational saving. [S24, S75-S76]

The framework also provides a useful publication model. Semi-annual reporting, annual financial statements and central-bank administration create repeated disclosure points. The research comparison should examine whether the fund rules remained stable, how excess petroleum

revenue was allocated, and how withdrawals interacted with the national budget and debt over time. Those empirical questions are separate from the statutory design.

For Guyana, the comparison is useful because both states use petroleum revenue within economies that also require large infrastructure and social investment. Yet reserve size, production trajectory, population, exchange-rate system and political economy differ. A direct transfer of Ghana's allocation percentages would therefore be methodologically weak.

The adaptable lesson is to define the separate purposes of stabilization and intergenerational saving before money arrives, publish the rule, and specify when parliament or the executive may alter the allocation. A single undifferentiated petroleum fund can still achieve those functions, but the internal accounting should make them visible.

46. Timor-Leste: a small state's dependence risk

Timor-Leste established its Petroleum Fund by law in 2005 and later amended the framework. The central bank publishes monthly, quarterly and annual reports and describes the portfolio structure, including Liquidity and Growth components. This creates a strong public record for examining how a small state manages a large financial asset relative to the domestic economy. [S26, S77-S78]

Timor-Leste is analytically valuable because the central challenge is not only investment performance. It is the relationship among petroleum depletion, government withdrawals, domestic budget dependence and the future tax base. A fund can remain professionally managed while the broader fiscal system becomes increasingly dependent on withdrawals. The final comparator chapter will therefore use a depletion-adjusted fiscal lens rather than fund-return data alone.

For a new island producer, this case demonstrates the importance of modelling the post-field budget before first oil. If petroleum production has a limited life and the non-oil tax base is narrow, recurrent spending promises made during peak revenue years may become difficult to sustain. The operating manual should therefore require a 20-year expenditure path under low production and low return assumptions before permanent salary, subsidy or entitlement increases are adopted.

The comparison with Guyana also highlights geological uncertainty. Guyana's discovered resource base expanded rapidly after Liza, while a smaller state may face one dominant field. A rule that appears conservative under a long reserve life may be insufficient under a short one. Reserve duration belongs inside fiscal design rather than in a geological appendix.

47. Suriname: the regional pre-first-oil comparison

Suriname provides the closest regional comparison for a country moving from discovery and final investment decision toward first oil. TotalEnergies reported in September 2026 that the GranMorgu project, sanctioned in 2024, remained under construction with first oil targeted for 2028. Because production had not begun by the evidence cut-off, the proper comparison is institutional preparation rather than operating outcome. [S27, S79]

The research questions for Suriname include the role of Staatsolie, revenue-management arrangements, environmental regulation, supplier development, port capacity, fiscal exposure and public communication before first oil. The existence of an established national oil company creates a different information and capacity structure from Guyana's. That difference matters when comparing negotiating expertise and project monitoring.

For Guyana, Suriname also provides a test of regional spill preparedness. Offshore development in neighbouring jurisdictions increases the value of interoperable maritime response, shared training, compatible data standards and transboundary communication. Environmental risk does not stop at a licence boundary.

A useful regional blueprint would therefore include a shared minimum incident-data protocol, emergency contact architecture, mutual assistance terms, rules for independent sampling and compatible claims documentation. Regional cooperation is a practical extension of national regulation rather than a substitute for it.

48. Trinidad and Tobago: a mature Caribbean producer

Trinidad and Tobago offers a different Caribbean case because it has decades of petroleum and gas experience and a Heritage and Stabilisation Fund. The Ministry of Finance publishes fund reports through 2026. The country also has industrial, petrochemical and energy-sector capabilities that most new island producers lack. [S80]

Its relevance lies in three areas. First, it demonstrates the challenge of managing resource revenue across commodity cycles rather than only during a first production boom. Second, it shows how petroleum can shape industrial capability and the wider labour market. Third, it offers a regional fund structure explicitly combining heritage and stabilization purposes.

The comparison should not treat Trinidad's mature gas-based industrial system as a natural next stage for every Caribbean producer. Market size, pipeline infrastructure, feedstock availability, electricity systems and export logistics differ. The correct transfer question is which capabilities were needed before value-added industry became viable.

A small island considering domestic processing should therefore compare the full delivered cost of feedstock, plant scale, port infrastructure, electricity, water, carbon exposure, skilled labour and export contracts. Petroleum production alone does not establish a commercial downstream industry.

49. Small islands and microstates

The smallest prospective producers face a distinctive concentration problem. A single offshore project can dwarf the domestic economy, public budget, labour force and banking system. Imported investment may raise GDP while creating limited domestic value added. One accident may affect a large share of fisheries, beaches or tourism assets. A few specialist employees leaving the public service for contractors can materially weaken regulation.

The first adaptation is fiscal. The fund architecture should model revenue as volatile and temporary. A permanent expenditure increase should be tested against a lower production path and delayed projects. External financial investment may reduce domestic overheating, but a state with severe infrastructure deficits also faces a credible case for higher public investment. The decision therefore needs an absorption model rather than a slogan about saving or spending.

The second adaptation is environmental. Island economies often have concentrated coastal tourism, reefs, fisheries, desalination intakes and limited emergency ports. Spill plans should include hurricane disruption, unavailable airports, damaged telecommunications and cross-border assistance. The cost of response equipment relative to population may argue for regional pooling, but legal liability and immediate response authority must remain clear.

The third adaptation is local content. Setting high local percentages in every category can produce front companies, inflated costs or shortages when the domestic workforce is small. A better model may prioritize skills transfer, verified ownership, apprenticeship, safety competence, data services and a narrower group of tradable capabilities with export potential.

The fourth adaptation is sovereign administration. A microstate should resist creating a large permanent bureaucracy during uncertain exploration. It may instead maintain a smaller core regulator with transparent access to independent external expertise, regional technical arrangements and rigorous conflict-of-interest controls.

Comparative matrix

The matrix is descriptive. It does not assign a comparative score.

50. A method for responsible policy transfer

A policy should pass five tests before another state adopts it.

Context test. Was the originating country at exploration, discovery, early production, mature production or depletion when the policy was used?

Capacity test. Which institutions, professions, databases, courts, regulators and financial systems were needed for the rule to function?

Endowment test. Did the policy depend on a large forest estate, a long reserve life, an established national oil company, a mature capital market or a diversified tax base?

Stress test. What happens under lower prices, project delay, cost overruns, a spill, an audit dispute or political turnover?

Revision test. Does the receiving state have a lawful route to revise the instrument without undermining contract certainty or institutional credibility?

This framework converts comparison into a disciplined method. A newcomer studies the mechanism, measures the prerequisites, models failure and records the reasons for adaptation. It does not copy a headline percentage or institutional name.

Part IX conclusions

No comparator provides a universal template. Norway illustrates long-run financial saving and budget integration. Ghana separates stabilization and heritage functions. Timor-Leste shows the importance of depletion and withdrawal dependence for a small state. Suriname offers a live regional pre-first-oil comparison. Trinidad and Tobago provides a mature Caribbean petroleum and fund case. Guyana contributes a rapid discovery-to-scale sequence combined with a pre-existing forest-finance framework.

The transferable value lies in the questions these cases force a newcomer to answer. How long might the resource last? How volatile is revenue? How strong is the non-resource tax base? Which institutions must function before first cash? How much public investment can the economy absorb? How concentrated is environmental exposure? How quickly can the state correct a weak rule? Those questions are more durable than any copied numerical formula.

Edition 2.0 deepening: a wider comparator universe

Analytical section 50A. Why the comparator set must extend beyond famous success cases

Resource-governance comparison becomes misleading when every country is compared only with Norway.

A useful comparator set should include:

- a mature high-capacity producer;
- a developing producer with petroleum funds;
- a small state facing depletion;
- a neighbouring pre-first-oil state;
- a mature Caribbean producer;
- recent African first-oil or first-gas states;
- a pre-production state with a long preparation period;
- non-petroleum commodity states with fiscal stabilization experience.

The objective is mechanism comparison.

It is not a country ranking.

Each comparator should answer one defined question.

Analytical section 50B. Senegal: transparency after first oil and gas

Senegal is useful because the transition from preparation to production occurred recently.

EITI records the updated petroleum legal framework, oil and gas revenue-management law, PETROSEN's role, contract publication and local-content architecture. It also records the entry of Sangomar oil and Greater Tortue Ahmeyim gas into production. [S113-S114]

The comparator value lies in the overlap between:

- new hydrocarbon revenue;
- state participation;
- debt and financing questions;
- contract review;
- local content;
- EITI-based public disclosure.

The 2026 EITI Validation report highlights the importance of tracing state-participation financing and reconciling production-share revenues and related loans. [S114]

For Guyana, this provides a useful external test of the principle that state participation creates an additional accounting chain.

A national oil company or state equity stake does not replace the need for transparent budget and revenue accounting.

It adds:

- shareholder financing;
- dividends;
- debt service;
- production entitlement;
- related-party transfers.

A newcomer considering state participation should therefore model both commercial upside and contingent public liabilities.

Analytical section 50C. Mauritania: building revenue-administration capacity around first gas

Mauritania offers a different lesson because the Greater Tortue Ahmeyim gas project required revenue-administration capacity in a cross-border LNG development.

IMF technical assistance in 2025 focused directly on hydrocarbon tax administration and a medium-term capacity-development plan. [S115]

EITI's 2024 Validation also emphasized contract publication, national oil-company transparency, beneficial ownership and the need for stronger reporting around the hydrocarbon revenue fund. [S116]

The transferable lesson is that first gas can be institutionally different from first oil.

Gas projects often involve:

- processing;
- LNG pricing;
- long-term sales contracts;
- transportation;
- shared cross-border infrastructure;
- domestic gas allocation.

A newcomer should therefore design tax and transfer-pricing capability for the actual commercial chain.

A crude-oil audit model is not automatically sufficient for LNG.

Analytical section 50D. Uganda: a long pre-first-oil preparation period

Uganda provides a useful contrast to Guyana's compressed discovery-to-production timeline.

The country discovered commercial oil years before production, allowing a longer period for legal, institutional, infrastructure and local-content preparation.

Uganda EITI documents a petroleum authority, national oil company, petroleum fund, cost-recovery framework, beneficial-ownership reforms, environmental-management systems and local-content arrangements while the sector remained pre-production. [S117-S118]

This makes Uganda a valuable case for one question:

What should a state do when it has time?

A longer pre-first-oil window can support:

- institutional staffing;
- petroleum fund design;
- contract-administration systems;
- beneficial ownership;
- procurement preparation;
- environmental baseline work;
- public communication.

Long preparation does not guarantee strong operating outcomes.

It does reduce the argument that institutions could not be built because production arrived too quickly.

Analytical section 50E. Mozambique: statutory sovereign-fund design before full revenue scale

Mozambique's 2024 Sovereign Wealth Fund law provides a current example of formal rules designed around future petroleum and gas revenue.

The statute establishes development, stabilization and intergenerational objectives, specifies eligible revenue, defines transfer rules and assigns roles to the budget and fund. [S119]

Banco de Moçambique later documented operational management and initial capitalization. [S120]

This comparator is useful because the law explicitly tries to combine near-term development finance with saving.

For a newcomer, the mechanism questions are:

- How is the revenue projection calculated?
- How much goes to the budget?
- How much remains in the fund?
- What happens when actual revenue is below projection?
- What is the role of parliament?
- Which assets can the fund hold?

These questions are transferable even when the numerical percentages are not.

Analytical section 50F. Chile: the structural-balance principle

Chile is not a petroleum comparator.

It is a resource-revenue stabilization comparator.

Its fiscal framework uses a structural-balance concept to reduce the effect of cyclical commodity revenue on annual spending, supported by sovereign financial buffers. IMF analysis describes the interaction between the structural balance, stabilization fund and debt anchor. [S121]

The transferable mechanism is the separation of permanent or structural fiscal capacity from temporary commodity conditions.

For Guyana or another producer, the analogous question is whether annual expenditure is being set using a temporary petroleum-price and production peak or a more conservative medium-term resource envelope.

The exact Chilean formula should not be transplanted.

The principle of smoothing volatile revenue can be.

Analytical section 50G. Botswana: the risk of buffer depletion even after decades of resource income

Botswana provides a useful non-petroleum case because the Pula Fund and long diamond history demonstrate that resource wealth can support financial buffers for decades. [S122]

IMF analysis in 2024 also documents a different side of the story: mineral-revenue weakness and spending can reduce financial buffers even when public debt remains relatively low. [S123]

The lesson is not that the earlier model failed.

The lesson is that resource-management systems need to survive the mature and declining phases of the commodity cycle.

A fund should not be judged only by how well it accumulates during high-revenue years.

It should also be judged by how the fiscal system behaves when revenue weakens.

Analytical section 50H. Cross-border projects create a governance layer above the nation

The Senegal-Mauritania GTA project demonstrates a governance issue that is increasingly relevant to small states.

A petroleum or gas field can cross a maritime boundary or depend on shared infrastructure.

This creates questions about:

- unitization;
- revenue allocation;
- metering;
- tax jurisdiction;
- emergency response;
- environmental liability;
- domestic supply;
- dispute resolution.

A newcomer with a boundary-adjacent discovery should therefore prepare for treaty-level or intergovernmental arrangements in addition to domestic law.

Analytical section 50I. State participation should be treated as a financing decision

Comparators differ in the role of national oil companies and state equity.

The decision to take an equity share should be modelled like any other investment.

The state should estimate:

- capital contribution;
- financing source;
- cost of debt;
- downside exposure;
- dividend timing;

- governance;
- technical learning;
- opportunity cost.

Equity participation can create knowledge and revenue.

It can also create contingent liabilities or obscure the boundary between commercial and sovereign functions if poorly governed.

Analytical section 50J. Sovereign funds should be compared by cash-flow function

A fund comparison should not start with the fund's name.

It should start with what cash flow the fund is designed to manage.

A stabilization fund absorbs volatility.

A savings fund transfers wealth across generations.

A financing fund integrates resource revenue with the budget.

A pension reserve has a different liability structure.

Some funds combine several functions.

The comparator table should therefore show:

- deposit rule;
- withdrawal rule;
- investment objective;
- budget integration;
- audit;
- governance;
- escape clauses.

This is more useful than comparing asset size.

Analytical section 50K. New producers should be compared by institutional timing

The most important comparator question may be when an institution was created.

For each country, the timeline should record:

- discovery;
- petroleum law;
- model contract;
- environmental law;
- revenue fund;

- beneficial-ownership rules;
- local-content law;
- first production;
- first withdrawal.

A country that creates a revenue framework before first oil is solving a different problem from one that reforms after years of production.

The timing itself is a policy variable.

Analytical section 50L. Comparator failures are as valuable as successful mechanisms

A responsible comparative chapter should not search only for practices to copy.

It should identify failure modes.

Common failure modes include:

- rigid fund rules that are bypassed;
- excessive withdrawal;
- underinvestment in regulator capacity;
- state-company debt opacity;
- weak beneficial-ownership data;
- delayed cost auditing;
- infrastructure booms that exceed capacity;
- environmental response designed too late.

The point is not to label a country a failure.

It is to isolate a mechanism that can break.

Analytical section 50M. Small-state concentration should be its own comparative variable

A policy that works in a large diversified economy may behave differently in a country where one project dominates exports and public revenue.

The comparator table should therefore include:

- population;
- non-resource GDP;
- export concentration;
- public-sector size;
- labour force;
- number of major ports;

- disaster exposure;
- reserve life.

This makes the comparison more relevant to islands and small coastal states.

Analytical section 50N. A transferability scorecard without country rankings

The publication should not rank political systems or governments.

It can still assess whether a mechanism is transferable.

Each mechanism can be tested against:

- institutional complexity;
- data requirement;
- staffing requirement;
- fiscal volatility;
- market-size dependence;
- ecological dependence;
- legal dependence;
- reversibility.

The result is a mechanism profile.

For example, a simple monthly petroleum ledger may have high transferability.

A forest-carbon system requiring a large standing tropical forest has low transferability to a coral island.

Analytical section 50O. Expanded Part IX conclusion

The wider comparator set changes the purpose of the book.

Guyana is no longer being compared only with established petroleum producers.

It is placed in a sequence that includes recent first-oil states, first-gas states, long pre-production preparations and non-petroleum commodity fiscal frameworks.

Senegal shows the accounting complexity created by state participation and new production. [S113-S114]

Mauritania shows the need to build revenue administration around a complex LNG chain. [S115-S116]

Uganda shows what can be prepared during a long pre-first-oil window. [S117-S118]

Mozambique shows a new statutory sovereign-fund architecture. [S119-S120]

Chile and Botswana show that stabilization and buffer management can be studied outside petroleum. [S121-S123]

The transferable lesson is methodological.

A newcomer should not ask "Which country should we copy?"

It should ask "Which mechanism solves our specific problem, what prerequisites does it require, and how does it fail under stress?"

FLAGSHIP PART X

New-producer operating manual

Working manuscript for the 520-560-page flagship edition. This manual translates the research into decision stages for a prospective producer. It does not prescribe a political choice, ownership structure, contract type or spending ratio.

50. Exploration-stage operating manual

The exploration stage should be governed as if no commercial discovery will occur. This prevents the state from building expenditure promises around uncertain geology.

Step 1: establish the legal map

Publish the ministry responsible for petroleum policy, licensing authority, environmental regulator, revenue authority, central bank, auditor, procurement authority, maritime safety agency, emergency command, local-government role and parliamentary oversight bodies. For each institution, record the statute, decision rights, staffing, conflict-of-interest rules and publication duties.

The petroleum law should define ownership of subsurface resources, acreage allocation, data rights, work obligations, relinquishment, environmental duties, transfer of interests, decommissioning and dispute procedures. Existing environmental law should be cross-referenced rather than assumed to apply automatically in the same way offshore.

Step 2: establish the geological-data regime

Create an acreage map, licence register, seismic-data inventory, well register, data-room procedure, confidentiality schedule and beneficial-ownership register. State clearly which data become public and when. Do not use undiscovered resource estimates as budget forecasts.

The regulator should retain original or verified copies of all geological and engineering submissions required by law. Data storage should be resilient and independent of an operator's continued presence in the country.

Step 3: establish the environmental baseline

Before major drilling expands, compile baseline information on currents, fisheries, protected habitats, reefs, mangroves, wetlands, coastal water intakes, seabirds, marine mammals, shoreline communities and economic uses. Record seasonality. Baseline science commissioned only after a spill weakens the ability to establish causation.

The environmental agency should define when an EIA is required, which alternatives must be considered, who reviews the work, how the public comments, and how permit conditions are monitored. Permit issuance and permit compliance belong in different databases.

Step 4: build compact specialist capacity

A new state rarely needs a large petroleum bureaucracy before a commercial discovery. It needs a competent core able to procure and challenge external expertise. The minimum specialist pool includes petroleum law, fiscal modelling, reservoir engineering, environmental science, accounting, transfer pricing, maritime safety and emergency management.

External advisers should undergo conflict checks. Their terms of reference, procurement route and deliverables should be recorded. The state should retain working papers and models so knowledge does not leave when a consultancy ends.

Step 5: run the dry-hole fiscal test

Every proposed fiscal regime should be tested under exploration failure, marginal discovery, low price and high development cost. The objective is to understand incentives before a successful well changes bargaining conditions.

The model should separate royalty, cost recovery, profit share, corporate income tax, bonuses, duties, state participation, ring-fencing and decommissioning. It should disclose the discount rate and price path. A single government-take percentage is insufficient.

51. Discovery-to-final-investment-decision manual

A commercial discovery compresses time. Operators seek appraisal and project approval while public expectations rise. The state's task is to improve information quality before committing the country to long-lived rights and infrastructure.

First 90 days after commercial declaration

1. Obtain independent review of discovered resources, recovery assumptions and development concepts.
2. Freeze a versioned record of the contract and licence in force.
3. Identify every stabilization, tax, valuation, cost-recovery, transfer, decommissioning and dispute clause.
4. Publish a non-confidential summary of the state's fiscal exposure and environmental decision process.
5. Map expected staffing and training needs.
6. Begin a sovereign-revenue custody design, but do not authorize spending based on unproduced oil.

Months 3-9

Commission a full lifecycle environmental assessment with alternatives. Require a spill scenario and transboundary analysis. Define financial assurance and verify the legal trigger for drawing on it. Consult affected communities and document agency responses.

Build a project cost model with independent ranges for capital expenditure, operating expenditure, first oil, plateau output, decline and decommissioning. The model should show how each variable changes state revenue under the executed fiscal terms.

Months 9-18

Set up the production-and-revenue accounting chain. Define meter verification, cargo measurement, price valuation, royalty invoicing, cost submission, audit rights, profit-oil entitlement, sale settlement and deposit into the sovereign account.

Pass the sovereign revenue law before first cash. The law should define objectives, eligible receipts, deposit rules, withdrawal authority, investment policy, oversight, audit, publication and relationship to the annual budget. The numerical withdrawal formula should follow country-specific modelling.

Final investment decision gate

The government file should demonstrate that it understands the project economics, legal obligations, environmental risks, decommissioning exposure and public fiscal path. A decision should not rest solely on the operator's commercial model.

52. First-oil readiness manual

The first cargo is a live systems test.

Technical readiness

Verify production meters, cargo measurement, crude quality, custody-transfer procedures and regulator access. Confirm that the government can independently reproduce the state's entitlement from production and contract terms.

Financial readiness

Trace the first cargo from entitlement through sale, invoice, settlement and deposit. Reconcile the sovereign account to the operator's statement and government invoice. Publish the deposit in the legally required period.

Cost-accounting readiness

Test a sample of recoverable costs before the cost bank becomes too large for the administration to understand. Ensure the state possesses the underlying records and knows the deadline for challenging claims.

Environmental readiness

Conduct a full emergency exercise involving the operator, regulator, maritime authority, military or coast guard where applicable, health services, fisheries authority, local government and neighbouring states if transboundary effects are possible. Test communications loss and port disruption.

Parliamentary readiness

The first withdrawal from the resource fund should follow the same public budget procedures intended for later years. Emergency or exceptional clauses should be narrowly defined and publicly auditable.

Public communication

Publish a first-oil dashboard explaining production, entitlement, royalty, tax, fund deposits and withdrawals. Use plain definitions so citizens do not confuse gross export value with public revenue.

53. Years 1-5 scale-up manual

Year 1: prove the accounting

The first year should prioritize reconciliation over ambitious policy expansion. Complete audited fund statements, contract-cost audits, environmental compliance reports and a public record linking cargoes to government cash.

Commission a baseline household and labour survey if recent comparable data do not exist. This gives the country a reference point before the resource boom fully affects wages, migration and prices.

Year 2: prove domestic value

Audit local-content claims. Separate registered local suppliers from beneficial local ownership and domestic value added. Compare payroll, customs and procurement evidence. Identify skills where local firms can become internationally competitive.

Review public construction costs against pre-boom benchmarks. Rising unit costs may signal capacity constraints even when revenue is abundant.

Year 3: prove public-investment delivery

Select a sample of major projects and trace them from business case to operation. Record time, cost, procurement, variation, maintenance and service indicators. If the economy cannot deliver the planned programme efficiently, slow approvals rather than equating available cash with usable capacity.

Update the petroleum-revenue model with actual reservoir performance and audited cost data. Re-run low-price and delay scenarios.

Year 4: prove resilience

Stress-test the budget under a two- or three-year revenue decline. Test public-sector wage commitments, transfers, debt service and maintenance obligations. Conduct a national emergency exercise simulating a serious offshore incident during severe weather or infrastructure disruption.

Review the sovereign fund's asset allocation and liquidity needs. Do not increase illiquid domestic exposure merely because domestic investment is politically attractive.

Year 5: institutional review

Publish an independent five-year review of petroleum law, fiscal administration, environmental performance, local content, fund management and public investment. Record what changed and why. The aim is not to defend the original framework. It is to determine whether the framework still fits the evidence.

54. Adverse scenario playbooks

Scenario A: oil price drops by 40%

Immediate actions:

- update revenue forecast using realized contract terms;
- freeze assumptions based on the prior price path;
- identify legally committed expenditure;
- assess fund-withdrawal authority;
- defer projects with low completion value if permitted;
- publish a revised cash bridge.

Analytical caution: the public-revenue decline is not equal to the gross price decline because cost recovery, royalty and profit sharing interact.

Scenario B: a major development is delayed by two years

Immediate actions:

- remove projected first oil from the budget baseline;
- recalculate supplier and labour demand;
- revise port and power infrastructure timing;
- test debt-financed projects linked to expected petroleum cash;
- update the fund trajectory.

The purpose is to avoid borrowing against an operator schedule that has not been achieved.

Scenario C: cost audit identifies disputed expenditure

Immediate actions:

- record the disputed amount separately from final recovery;
- preserve all underlying documentation;
- activate contractual expert or dispute procedures;
- avoid presenting the disputed number as public cash;

- disclose process status where law permits.

Scenario D: major offshore pollution event

Immediate actions:

- activate the incident command structure;
- protect life first;
- initiate independent environmental sampling;
- notify affected communities and neighbouring states where relevant;
- secure financial assurance;
- preserve evidence;
- publish verified incident facts at regular intervals;
- separate compensation claims from cleanup operations.

A serious event should not be used to rewrite scientific findings before evidence is collected.

Scenario E: domestic economy overheats

Indicators:

- construction costs rising materially faster than inflation;
- vacancies in public technical roles;
- rapid wage divergence;
- import congestion;
- project delays and variations;
- non-resource exporters losing competitiveness.

Response options for decision-makers include rescheduling public investment, increasing external saving, changing procurement sequencing or expanding training and immigration pathways under applicable law. The manual presents the options without selecting a political preference.

Scenario F: public confidence deteriorates

Triggers include unreconciled accounts, unexplained revisions, delayed audits, weak stakeholder representation or repeated permit breaches. The response should emphasize independent verification, publication of source records, lawful investigation and correction of known errors. Public relations should not substitute for missing evidence.

55. Cabinet and regulator checklists

Monthly petroleum dashboard

- actual production by project;

- downtime;
- cargo lifts;
- government entitlement;
- realized sale price;
- royalty billed and received;
- recoverable cost submitted;
- audit status;
- fund deposits;
- fund investment income;
- withdrawals;
- environmental incidents;
- flaring and emissions where measured;
- permit compliance actions.

Quarterly fiscal dashboard

- opening fund balance;
- petroleum cash inflows;
- investment return;
- withdrawals;
- closing balance;
- public debt;
- non-oil primary balance;
- capital-program execution;
- construction-cost index;
- maintenance funding;
- forecast revision.

Annual social dashboard

- household real consumption;
- poverty/deprivation using current data;
- employment and real wages;
- local ownership and domestic value added;
- school outcomes;
- health outcomes;
- housing and utilities access;
- regional and hinterland gaps;

- migration;
- community grievances and resolution.

Annual environmental dashboard

- project permit status;
- monitoring results;
- incidents;
- emergency drills;
- response times;
- financial assurance;
- fisheries impacts;
- biodiversity indicators;
- emissions;
- decommissioning liabilities.

56. The 24-month implementation sequence for a prospective state

Months 1-3: legal and institutional map, licence register, conflict rules, geological data inventory, environmental baseline design.

Months 4-6: independent fiscal benchmark, regulator staffing plan, emergency-response map, public disclosure policy.

Months 7-9: sovereign-revenue framework options, procurement and audit protocols, beneficial-ownership system, stakeholder consultation framework.

Months 10-12: draft legal instruments, production measurement protocol, model petroleum accounting ledger, spill-response exercise.

Months 13-15: publish model contracts or fiscal principles, establish cost-audit team, environmental review panel and public-data portal.

Months 16-18: test fund accounting with synthetic transactions, test parliamentary reporting, update marine baseline, negotiate regional emergency support.

Months 19-21: run low-price, dry-hole, delay and spill scenarios; publish assumptions and institutional gaps.

Months 22-24: independent readiness review and public correction plan. Do not certify readiness merely because every policy document exists. Test whether the people, systems and data function.

57. Evidence standard for declaring readiness

A country should not declare a workstream complete until the corresponding evidence exists.

Legal readiness requires enacted and commenced law plus defined implementing authority.

Fiscal readiness requires transaction testing, not a policy paper.

Environmental readiness requires baseline data, permits, monitoring capacity and emergency drills.

Audit readiness requires staff, access rights, data retention and completed sample tests.

Local-content readiness requires verified ownership and skills data, not registration alone.

Public-investment readiness requires procurement and maintenance capacity.

Social readiness requires current surveys and grievance mechanisms.

Fund readiness requires audited financial statements and lawful budget integration.

Part X conclusion

The central principle of the operating manual is sequencing. A new producer should build evidence and institutional capacity before each irreversible stage: licence award, appraisal, development approval, first oil, first withdrawal and rapid scale-up.

Guyana's experience is useful because several of these stages occurred in a compressed period and left a substantial documentary record. The transferable product is not a single contract or statute. It is a staged system for asking the right question before the cost of a mistake rises.

Edition 2.0 deepening: the operating manual as a sequence of irreversible gates

Analytical section 58. The five irreversible gates

A newcomer petroleum state should organize readiness around five gates.

Gate 1: licence award. Before acreage is awarded, the state should know who has authority, what data exists, which work obligations are required, how beneficial ownership is checked and which environmental baseline obligations apply.

Gate 2: commercial discovery. Before the state accepts a commercial-development path, it should independently understand reserves, development cost, fiscal outcomes, infrastructure needs and environmental exposure.

Gate 3: final investment decision and development approval. Before irreversible capital is committed, the state should have completed legal, fiscal, environmental, decommissioning and emergency-response reviews.

Gate 4: first oil or first gas. Before production begins, the state should prove that metering, cost accounting, royalty, tax, sales, sovereign-revenue custody, incident command and public reporting systems function.

Gate 5: first major budget withdrawal. Before resource revenue is materially absorbed into the domestic economy, the state should have project appraisal, procurement, maintenance and macro-fiscal controls.

Each gate should have evidence requirements and a named institution authorized to sign readiness.

Analytical section 59. A national petroleum RACI matrix

Every major petroleum process should identify who is:

- Responsible;
- Accountable;
- Consulted;
- Informed.

The matrix should cover:

- licensing;
- contract negotiation;
- geological data;
- reserves review;
- development plan;
- production licence;
- EIA;
- permit;
- metering;
- petroleum valuation;
- cost audit;
- tax;
- cargo sale;
- NRF custody;
- budget withdrawal;
- local content;
- emergency response;
- decommissioning.

The purpose is to prevent shared responsibility from becoming no responsibility.

Analytical section 60. The discovery 100-day plan

Within the first 100 days after a material commercial discovery, government should create a single readiness programme.

Days 1-30:

- preserve all geological and contractual records;
- establish conflict controls;
- commission independent resource and fiscal review;
- freeze premature revenue assumptions in the budget;
- map institutional gaps.

Days 31-60:

- create environmental and social baseline workplan;
- define development-plan review team;
- establish cost-accounting architecture;
- begin sovereign-revenue design options;
- create public communication protocol.

Days 61-100:

- publish the readiness gap register where lawful;
- approve staffing and adviser strategy;
- run fiscal stress tests;
- establish first-oil data architecture;
- begin emergency-response planning.

The discovery should therefore trigger administration before it triggers spending.

Analytical section 61. The development-approval dossier

A development approval should be supported by one consolidated government dossier containing:

- resource estimate;
- reservoir uncertainty;
- development concept;
- capital cost;
- operating cost;
- production forecast;
- fiscal model;
- cost-recovery forecast;
- environmental assessment;

- financial assurance;
- emergency response;
- local-content plan;
- decommissioning concept;
- public infrastructure requirements.

The dossier should record both operator assumptions and independent government analysis.

Material disagreements should remain visible.

Analytical section 62. The first-oil certification exercise

Six months before first oil, government should run a full transaction test.

The exercise should simulate:

- daily production;
- meter exception;
- government entitlement;
- cargo nomination;
- benchmark price;
- royalty;
- tax;
- cost recovery;
- cash settlement;
- NRF deposit;
- public disclosure.

A second exercise should simulate an incident occurring during the same period.

Readiness is proven when institutions can perform the transaction under pressure, not when policy documents have been published.

Analytical section 63. The first-withdrawal certification exercise

The first major petroleum-funded budget programme should also be tested.

The government should demonstrate:

- legal withdrawal authority;
- transfer to the Consolidated Fund;
- parliamentary appropriation;
- project appraisal;
- procurement;
- contract management;

- completion reporting;
- maintenance funding.

This makes the first withdrawal an institutional milestone rather than a simple cash movement.

Analytical section 64. A single national resource dashboard with layered access

The state should maintain one integrated data model even if public access varies.

Layer 1 can be fully public:

- production;
- revenue;
- fund balance;
- major permits;
- public contracts;
- key outcomes.

Layer 2 can be regulator-access:

- metering details;
- cost records;
- commercially sensitive data;
- detailed environmental monitoring.

Layer 3 can be restricted for security or privileged legal material.

The important control is that all layers use consistent identifiers.

Analytical section 65. The minimum permanent specialist team

A small state does not need a huge bureaucracy.

It does need permanent capability in:

- petroleum engineering;
- geology/geophysics;
- fiscal modelling;
- petroleum accounting;
- tax;
- law;
- environmental science;
- emergency response;
- data management;
- procurement and public investment.

External advisers can supplement this team.

They should not become the only holders of state knowledge.

Analytical section 66. Adviser management protocol

Every external adviser engagement should specify:

- scope;
- conflict check;
- data ownership;
- model ownership;
- source-file delivery;
- training requirement;
- handover;
- confidentiality;
- publication rights.

The state should receive editable models and working papers, not only a final PDF.

This reduces dependence on the same adviser for future decisions.

Analytical section 67. A low-capacity state should prioritize control points

When staffing is limited, not every process can receive equal attention.

The highest-value control points are:

- licence and contract;
- reserves;
- development cost;
- metering;
- petroleum valuation;
- cost audit;
- revenue custody;
- environmental permit;
- financial assurance;
- emergency response;
- budget withdrawal.

A small regulator should secure these points before expanding into lower-risk administrative detail.

Analytical section 68. Island-state operating manual

A small island producer should add six specific controls.

1. **Tourism exposure map.** Identify beaches, reefs, ports and marine recreation.
2. **Desalination and water-intake protection.** Model spill and shutdown scenarios.
3. **Hurricane concurrency.** Test a petroleum incident during storm disruption.
4. **Port redundancy.** Identify what happens if the primary port is unavailable.
5. **Regional response agreements.** Pre-negotiate customs, aviation and equipment entry.
6. **Small-labour-market strategy.** Protect regulators and utilities from specialist drain.

These controls are not unique to islands, but concentration makes the consequences larger.

Analytical section 69. The no-oil scenario

A petroleum strategy should still create value if commercial production never occurs.

The state should ask which investments remain useful under a dry-hole outcome.

Usually these include:

- geological data;
- maritime mapping;
- environmental baselines;
- beneficial-ownership systems;
- procurement controls;
- emergency coordination;
- public financial management.

This test prevents exploration optimism from creating stranded bureaucracy or debt.

Analytical section 70. The lower-price scenario

The operating manual should assume that resource prices can remain below the planning case for several years.

The state should identify:

- protected essential services;
- deferrable projects;
- recurrent commitments;
- debt service;
- fund buffers;
- exchange-rate exposure.

The purpose is to decide before the shock which commitments are flexible.

Analytical section 71. The delayed-project scenario

A two-year project delay affects more than future oil revenue.

It can also affect:

- public expectations;
- infrastructure built for the project;
- local suppliers;
- labour;
- debt;
- regulator staffing.

The contingency plan should therefore cover the wider economy.

Analytical section 72. The major-spill scenario

The spill scenario should include:

- immediate command;
- operator action;
- financial assurance;
- public communication;
- fisheries;
- health;
- cross-border notification;
- evidence preservation;
- claims;
- independent investigation.

The exercise should be timed.

Decision delay should be treated as a risk variable.

Analytical section 73. The institutional-disruption scenario

Institutions can face leadership change, litigation, cyber incidents, natural disaster or staff turnover.

The petroleum system should remain operable.

Business-continuity plans should identify:

- delegated authority;
- backup records;
- credential recovery;
- alternate office;

- emergency signatories;
- succession.

This is administrative continuity, not a political judgment.

Analytical section 74. The annual resource-governance review

Once production begins, the state should publish an annual integrated review covering:

- production;
- reserves;
- revenue;
- audits;
- NRF;
- budget use;
- local content;
- social indicators;
- environmental performance;
- incidents;
- decommissioning;
- unresolved disputes.

The review should include corrections from prior years.

Analytical section 75. Independent readiness review

Before each irreversible gate, government should commission an independent challenge review.

The reviewer should not certify the political desirability of development.

The reviewer should test whether the evidence and controls required for the decision exist.

The final output should classify:

- ready;
- ready with conditions;
- not evidenced.

Any unresolved high-impact condition should have an owner and deadline.

Analytical section 76. Expanded Part X conclusion

The operating manual should make the book usable on Monday morning, not only intellectually persuasive.

Its central rule is simple:

Do not move from one irreversible petroleum stage to the next merely because the commercial project is ready.

Move when the state can also measure, regulate, account, respond and explain.

The manual therefore converts the Guyana case and international comparators into a sequence of gates, ledgers, drills and evidence requirements.

For a new producer, that sequence is the real blueprint.

It preserves national optionality before discovery, bargaining power after discovery, revenue integrity after first oil and development discipline after the first large withdrawal.

Evidence dossiers and implementation workbooks

Supporting dossiers preserve detailed research notes, reconciliation methods and operating tools. They are evidence annex material, not additional core chapters.

THE GUYANA SEQUENCE

From exploration to production: petroleum governance, environmental stewardship and a practical comparative case for emerging producers

Ragunauth Ramsaroop | Independent research working manuscript | Version 0.1 | 27 September 2026

Status: Evidence-led opening manuscript, not the complete 140-160-page flagship edition. External expert review and full line-by-line source audit remain pending. Citations [S01]-[S27] identify dated material in SOURCE_REGISTER.md.

Abstract

Guyana's offshore petroleum expansion offers an unusually compressed record of resource discovery, contractual negotiation, institutional preparation, sovereign revenue management and rapid commercial production. Liza was announced as a commercial discovery in 2015. The Natural Resource Fund was established by legislation in January 2019 before the first offshore crude was produced in December of the same year. Four floating production, storage and offloading facilities were operating by the first half of 2026. At June 2026 the government reported a USD 4.2942 billion NRF cash balance after USD 1.02 billion of withdrawals in the first half. These observations are meaningful, but on their own they establish neither an optimal petroleum agreement nor durable welfare gains. [S01, S04, S07]

This research reconstructs events in the order they happened. It distinguishes pre-existing foundations, policy instruments established before first oil, later reforms and results still under evaluation. It examines revenue accounting, the 2016 Stabroek production sharing agreement, environmental impact assessments, oil-spill legislation, public financial management, labour and domestic suppliers, forest protection and climate finance. It then maps administrative and technical decision points for another prospective producer. The paper treats a new island state differently from a forested continental state, and a non-commercial discovery differently from an operating offshore field.

The evidence is mixed. Formal fiscal and reporting institutions preceded first oil. Government and operator data document rapid expansion and growing petroleum revenues. IMF staff in July 2026 also called for stronger oil-sector audit capacity, better procurement and public-enterprise oversight, more complete household statistics and a firmer long-run fiscal guide. The EITI Board opened a delayed 2026 validation while stating concern about the independence and stakeholder confidence of the civil-society selection process. The legal framework for pollution preparedness expanded with a 2025 act, but the June 2026 government report described regulations and commencement as work still underway. These are central facts rather than footnotes. [S07, S08, S16, S19]

A transferable case is not a list of admired policies. It is a documented sequence of institutional choices, conditions required for operation, observable results, alternative interpretations and identified points of failure.

Method and definitions

The unit of analysis is Guyana's petroleum and related national governance system from exploratory activity to 27 September 2026. The offshore Stabroek project provides the production case, but the ecological and development analysis covers Guyana as a whole. The temporal cut-off is essential: September 2026 projections about year-end output and revenue are not presented as observations.

Five evidence categories are kept apart. Legal evidence consists of texts enacted, brought into force and amended. Administrative evidence consists of official licence, permit, audit and oversight documents. Financial evidence consists of audited or dated management accounts, with explicit cash-period boundaries. Operating evidence consists of production and commissioning records. Social and ecological evidence requires outcome data rather than expenditure or project announcements. Statements from government and companies are attributed to their authors. Independent multilateral assessments and civil-society concerns are reported with their own evidentiary limitations.

Throughout, a production capacity figure means engineering capacity, not an actual monthly mean. A crude export receipt is not public revenue. Government profit-oil lifts are not identical to sales received in the same month. A fund deposit is not a parliamentary withdrawal, and the fund's cash balance is not equivalent to the public investment portfolio. Contractual value for future carbon-credit purchases is not current cash. National GDP is not a household consumption survey. These separations prevent the apparent precision of high-level figures from concealing incompatible denominators.

1. Before discovery: institutions, geology and the timing problem

Guyana's petroleum history did not begin in 2015. Onshore work dates to the early twentieth century; offshore exploration began in the middle of that century. Seismic surveys and drilling by successive companies preceded the commercial Liza discovery. ExxonMobil's retrospective 2023 report places offshore exploration attempts in the late 1950s and describes more than forty wells drilled in the basin over decades. This history matters because a country rarely controls the exact timing of an economically viable discovery. Dry wells, capital cycles, technical interpretation and petroleum prices determine much of the interval between exploration and commercial development. [S01]

Guyana also entered the modern oil period with an existing economic and ecological agenda. Its original Low Carbon Development Strategy began in 2009. A parallel Guyana-Norway results-based forest agreement began in the same year. The government subsequently reported USD 227 million earned for forest climate services during 2009-2015. A monitoring, reporting and verification system for the forest sector developed before first oil. These were pre-petroleum policy institutions and a separate financial stream, not products of the petroleum fund. [S22]

This chronology prevents two analytical mistakes. First, attributing every later institution to the petroleum discovery erases existing public administration, regulatory experience, forest monitoring and the long history of commodity production. Second, projecting Guyana's forest-carbon results onto a small island with few forests would confuse the national resource endowment with a universally available funding instrument. The transferrable administrative lesson is the practice of mapping existing technical capacity before a commercial discovery, not the replication of Guyana's exact endowment.

In an exploration-stage country, the relevant decision documents include the geological data policy, licence register, basin environmental baseline, fiscal-risk inventory, ministerial and regulatory division of responsibilities, spill-planning jurisdiction, beneficial-ownership requirements and public disclosure rules. Each is meaningful before a first commercial find. A government may procure legal and reservoir expertise through transparent short-term appointments instead of creating a large permanent petroleum bureaucracy without proven reserves. Whether to establish a state producer, auction acreage, negotiate directly or reserve blocks requires a jurisdiction-specific legal and fiscal assessment. The case study will therefore map decision rights and procedural safeguards rather than prescribe one commercial model.

Guyana is also climatically distinctive. Much of its population and capital stock sit on a low coastal plain subject to flooding, while large forested and hinterland regions have different infrastructure, service-delivery and land-governance needs. An offshore development changes the fiscal envelope without eliminating these geographic constraints. An independent comparison must measure whether new capital expenditure reduces lifecycle exposure to flooding, power insecurity and maintenance costs. Building a new asset is an input; a working, resilient asset delivering service over time is the intended outcome.

The period before discovery is thus analytically separate from 2015-2019. A useful exploration chronology should identify not only when a well discovered hydrocarbons but also when the state gathered geological information, published licence terms, defined ownership, set environmental baselines and strengthened its ability to challenge an operator's commercial and technical submissions. Without that baseline, governments and external observers struggle to separate inherited institutional capacity from later improvements.

2. The discovery-to-first-oil window, 2015-2019

The Liza discovery announced in May 2015 changed the commercial outlook for the offshore Stabroek block. On June 27, 2016 the government and the consortium entered a petroleum agreement, with a stated October 7, 2016 effective date in the official block register. Different public records refer to signature, agreement and effective dates. Those dates should appear separately in the final chronology. The project reached first oil in December 2019. The roughly four-year interval after discovery is the institutional preparation window studied here. [S01, S02, S03]

The 2016 Stabroek agreement reflects the conditions and bargaining positions of a frontier offshore development at the time it was signed. Any contractual comparison needs the exact agreement text, hydrocarbon uncertainty at signature, oil-price conditions, deepwater capital intensity, exploration obligations, relinquishment provisions and environmental commitments.

Retrospective government revenues and later geological success do not automatically establish the value a counterparty placed on acreage before commercial certainty. Conversely, early commercial uncertainty does not remove the need to document fiscal options or expose a government to unreviewed recoverable costs.

The public Stabroek agreement has a 2% royalty, up to 75% monthly cost-recovery ceiling, and an equal division of remaining profit oil. The Guyana Revenue Authority explains that when the ceiling binds, a simplified revenue calculation gives 12.5% of gross sales as government profit oil plus a 2% royalty, or 14.5% under its stated accounting approach. That is a conditional illustration, not an unchanging government entitlement. When permitted recoverable costs use less than the cap, the profit-oil pool increases. Approved future development expenditures may replenish the cost bank. Audit treatment, allowable expenditure and valuation matter as much as headline sharing percentages. [S02, S07, S15]

The historical test of the contract therefore has at least four separate components: returns relative to the information available at signing, compliance with the signed terms, effectiveness of subsequent cost verification, and the state's ability to use later experience in different acreage awards. Public debate frequently merges these into a single judgment. This manuscript keeps them separate and will not declare a country-level winner or a contract-level overall score.

A relevant reform sequence is observable. The government later announced model terms for future offshore bidding with a 10% royalty, a 65% cost-recovery ceiling and a 10% corporate tax. Profit oil remains shared equally after cost recovery under the model. These later terms are distinct from the continuing Stabroek agreement and must never be back-applied when calculating historical or forecast revenues from the producing block. The published model documents and any final signed new licence terms require an agreement-by-agreement crosswalk. [S14]

What early fund establishment demonstrates, and what it does not

The Natural Resource Fund Act received assent on January 23, 2019, almost eleven months before first oil in December 2019. The Ministry of Finance's 2024 NRF annual report confirms the date. This is a material pre-production institutional milestone: the state adopted a statutory vehicle before recurring offshore oil cash arrived. Yet formal creation alone says little about later spending efficiency, distributional outcomes or the adequacy of subsequent withdrawal rules. Those questions require separate assessment. [S04]

The 2019 establishment also did not mean the rules remained unchanged. The 2021 law replaced the 2019 act. The withdrawal provisions changed again on February 6, 2024. A new country examining this case should distinguish statutory creation, operational readiness, independent monitoring, transparent publication, integration with annual budgets and revision procedures. A law on paper is the first link in a longer accountability chain.

3. From first oil to large-scale production, 2019-September 2026

Production commenced at Liza Phase 1 in December 2019. Liza Phase 2 followed in 2022, Payara in November 2023 and Yellowtail on August 8, 2025. Yellowtail's ONE GUYANA FPSO joined the Destiny, Unity and Prosperity vessels and took combined installed capacity past 900,000 barrels per day. ExxonMobil announced a daily production milestone of 900,000 barrels in November 2025. That one-day milestone is distinct from annual average production, and installed capacity alone never establishes actual output. [S01, S11, S12]

The national petroleum production chart extends through July 31, 2026. The Ministry of Finance's 2026 mid-year report says production from the four FPSOs exceeded 900,000 barrels per day during the first half. The same report describes the fifth Uaru project as under construction with start-up targeted for the final quarter of 2026. Future development output should remain in a prospective column until a dated commissioning announcement and verified operating record are obtained. [S07, S13]

Production is not a single variable. An operating dossier needs the count and size of producing vessels, uptime, reservoir performance, flaring events, methane monitoring, water injection, transport logistics, crude grades, lifting dates, export prices, cost-bank balances and the share delivered to the state. The government recorded 162 crude lifts during January-June 2026, including 21 government profit-oil lifts from the four producing units. Yet receipts in the same period included payments for three lifts occurring in late 2025, while payments for some mid-2026 lifts fall later. An analytical table pairing production dates directly with cash receipts without that timing adjustment would misstate the relationship. [S07]

Guyana reported USD 1,778.6 million in government profit-oil sales receipts and USD 218.4 million in royalties during the first half of 2026. End-June NRF cash stood at USD 4,294.2 million after USD 1,020 million in withdrawals during the period and including reported interest. These figures come from the government's September 2026 publication. Independent reconciliation with Bank of Guyana monthly and quarterly releases and external audit statements is part of the final source gate. [S07]

The speed of commissioning is commercially significant but creates linked challenges. Regulatory personnel need sufficient specialist expertise and access to data as vessel numbers multiply. Shipping and aviation capacity grow along with occupational risk. Contractor spending can expand faster than qualified local supplier capacity. Power demand, roads and port assets face supply-chain constraints. Large monthly tax and petroleum receipts expand budget options while also increasing the consequences of weak procurement or poor project sequencing. None of these effects is captured by a barrels-per-day graph.

The final paper will therefore present a development-by-development table with first-oil dates, sanctioned costs as of the applicable date, FPSO investment-basis capacity, actual quarterly output, permitted emissions and flaring, environmental reporting and the regulator's approval and inspection milestones. An approved project not yet producing will have blank realized-output cells rather than copied operator forecasts.

4. The Natural Resource Fund as a fiscal institution

The conceptual purpose of a natural-resource fund is to separate volatile extractive cash inflows from the immediate authorization of public expenditure. In Guyana's case, the fund was legally established before first oil. The 2021 replacement statute specified a governance structure including a board, revised accountability arrangements and withdrawal rules. The 2024 amendment increased the statutory withdrawal envelope. A claim of good fiscal practice requires analysis of each version and its observed effects rather than treating the act as an immutable institution. [S04, S05]

The fund receives the government's defined petroleum revenue streams and investment income. Withdrawals move money to the consolidated fund under parliamentary budget authorization. The public account must allow a reader to reconstruct each stock-and-flow identity:

Opening fund cash + petroleum receipts + investment income + other recognized movements - authorized transfers = closing fund cash.

Each item needs an observation period, currency and treatment of accruals or late receipts. The initial paper uses the figures disclosed for June 2026 rather than extrapolating a final year-end outcome. The Ministry of Finance reported USD 4.2942 billion in cash at end-June after USD 1.02 billion in first-half transfers. Its revised 2026 estimate raised full-year petroleum deposits to USD 6.4976 billion, from the much lower January budget assumption. This updated full-year estimate remains a projection, influenced in the ministry's explanation by oil prices and reduced use of the cost-recovery ceiling. [S06, S07]

The revision illustrates two fiscal hazards. First, a large increase in projected receipts can be mistaken for an equally large increase in prudent current-year spending space. The withdrawal law and parliamentary budget framework are separate from the projection. Second, a rapid decline in historical cost recovery could create short-term windfall receipts that do not recur in the same amount after a new field enters development. The 2026 mid-year report expressly notes that costs from additional projects may increase the cost bank again. [S07]

IMF staff's July 2026 statement recognized the accumulation of substantial NRF balances and Guyana's debt position, while proposing consideration of a longer-run operational fiscal guide based on the non-oil primary balance. Staff argued that unexpected gains under sustained high oil prices should be examined against the economy's ability to absorb investment and called for stronger monitoring of public spending outcomes and public-enterprise risks. These are attributed external assessments. They do not prove one single withdrawal formula suitable for all emerging producers. [S08]

Cash management is not public investment evaluation

A fund statement answers where petroleum cash is and how money moves. A public investment system answers which projects are funded, procured, built, independently inspected, maintained and used. The two systems are linked but not interchangeable. A budget line for a hospital is not a verified improvement in service quality. A construction contract is not a functional road that survives flooding. A scholarship allocation is not a measured increase in skills or labour-force productivity.

The transfer framework should therefore include a separate public investment register. Each large project would record a published business case, procurement method, contract variations, delivery date, capital and lifecycle cost, disaster exposure, expected users and independent completion evidence. Where data are absent, the final manuscript will mark an evidence gap instead of inferring success or failure. Even with high revenues, capital-intensive programmes compete for engineers, surveyors, procurement specialists, nurses, teachers, land and imported construction inputs.

For a potential new producer, the administrative sequence to document includes the statutory revenue recipient, published deposit statements, a withdrawal authorization chain, parliamentary scrutiny, an external audit schedule and the project-level outcome ledger. The precise spending-versus-saving rule is a sovereign policy choice shaped by development needs, reserve life, debt, population, climate exposure and macroeconomic conditions. This research describes alternative designs and their trade-offs without assigning a universal answer.

5. Fiscal contract oversight and the cost bank

Production sharing is often explained as a royalty followed by a cost-oil allocation and a split of the remainder. That summary is insufficient for technical oversight. A credible government revenue forecast requires a model of recoverable cost categories, timing and approval of development expenditures, production valuation, commodity prices, crude quality adjustments, operating expenses, currency and transfer-pricing controls, decommissioning treatment and dispute procedures.

Under the 2016 Stabroek framework as summarized by the Guyana Revenue Authority, the simplified case in which the 75% cap is fully used allocates 75 of each 100 normalized revenue units to cost recovery, 12.5 to government's half of remaining profit oil and 12.5 to the operator group. A separate 2-unit royalty produces the familiar 14.5-unit combined government royalty/profit share under the authority's calculation. This is an explanatory model rather than a substitute for detailed contract accounting. In other periods the ceiling need not bind. [S02, S15]

At a hypothetical 50-unit cost share with a 100-unit revenue base, a simple 50/50 remainder gives the state 25 units of profit oil plus 2 units of royalty. At a hypothetical 20-unit share, the same pedagogical formula gives 40 plus 2. Actual payments need the contract's detailed oil-volume and valuation procedures, and this illustration excludes tax treatment, timing, carried costs and cash-versus-kind settlement. No reader should interpret 27% or 42% as an observed 2026 fiscal take or a forecast.

The 2026 mid-year report reported accelerated cost recovery as the historical cost bank was reduced and projected a substantially higher number of government lifts than expected in January. The report warns that new projects enter the cost bank and that clearing historical costs does not create a permanent government claim to half of gross sales. Such cautions belong in the main text because the choice of production denominator changes the interpretation of public revenue. [S07]

Contract administration also depends on verifiable audit capacity. The July 2026 IMF mission noted ongoing efforts to resolve cost-oil audits, including arbitration, and emphasized timely resolution, specialist audit staffing and the quality of government oversight. An administrative case study must distinguish allegations made in disputes, expenditures accepted as recoverable,

auditor adjustments proposed, adjustments finally collected and open proceedings. A proposed disallowance is not recovered public revenue. [S08]

The later model fiscal terms illustrate a second issue: the government's bargaining position and acceptable return vary across the discovery cycle. The announced model's 10% royalty and 65% maximum cost recovery, together with equal residual profit oil and a 10% corporate tax, differ from the 2016 block agreement. A comparison should show gross expected government receipts, investor breakeven and downside geological outcomes under common assumptions; it should not infer comparative welfare from two headline royalty rates. [S14]

6. Environmental law, liability and offshore operations

Petroleum regulation should begin with an accurate inventory of harms, affected populations, potential transboundary effects and the physical capacity to respond. For a coastal state the issues include coastal water and fisheries, tourism, biodiversity, emergency maritime aviation, ocean currents, offshore worker safety, greenhouse-gas emissions, shore-based waste and possible liability beyond national waters. Guyana's EPA operates a formal environmental authorization process for projects with environmental impacts. The existence of a permit is a necessary legal record, not proof that its terms have been independently monitored or that every risk is extinguished. [S21]

The sequence should distinguish environmental assessment before project approval, permit conditions during construction, real-time operating monitoring, incident reporting, post-incident enforcement, insurance coverage and financial recourse. It should also distinguish public access to raw monitoring results from operator-authored summaries. A complete research exhibit would list each operating FPSO, its applicable environmental permit and environmental impact assessment, permitted flaring limits, reporting frequencies, available inspections, observed incidents and documented remediation.

Guyana enacted the Oil Pollution Prevention, Preparedness, Response and Responsibility Act in 2025. It addresses oil-spill prevention, planning, national response and financial responsibility. The 2026 mid-year report, covering January-June, says implementing regulations were still being drafted and that commencement was expected by the end of 2026. At the 27 September cut-off this report does not establish that the entire statute was already operational. The final legal chapter must check the Official Gazette and commencement instruments before moving a provision from 'enacted' to 'in force'. [S19, S07]

Financial assurance illustrates why precise legal language matters. In May 2026 the Court of Appeal overturned a 2023 High Court order requiring an unlimited guarantee under an offshore permit. The government's description of the appellate decision says the EPA had lawful discretion over the amount of financial assurance and that the High Court had improperly substituted its own decision for the regulator's judgment. A contemporary independent report also distinguishes an operator's continuing liability from the amount of security furnished in advance. Neither point, alone, resolves the empirical question of whether funds available following a catastrophic release would meet cleanup, compensation and ecological restoration costs. An authenticated judgment and actual policy documents are required for final legal analysis. [S20]

An offshore assurance model has multiple layers. Direct operator resources, insurance, any parent-company guarantee, rapid-access emergency cash, contracted response assets, reciprocal regional assistance and state contingency planning differ in legal trigger, latency, currency, scope and enforceability. They should not be merged into a headline figure without examining policy exclusions, aggregate limits, arbitration, who controls claim settlement and conditions after a multi-jurisdiction spill. For island and continental states alike, cross-border current modelling and neighbours' ability to respond are material planning inputs.

A future final chapter will include a reproducible response drill template: discovery-to-notification time, mobilization time, dispersant approval procedures, mechanical recovery resources, sensitive-habitat priority maps, fishery-closure authority, independent incident sampling, medical surveillance and claims processing. Without disclosed observed drill performance, the template remains a prospective audit instrument rather than a finding about current operational readiness. Forest conservation and offshore pollution are separate environmental systems and must be evaluated on their own evidence.

7. Forest conservation, carbon finance and the fossil-fuel coexistence problem

Guyana's 2009 LCDS predated oil production by a decade. Its results-based forest partnership with Norway covered 2009-2015 and the 2024 LCDS update reported USD 227 million earned in that period. In 2022 the country introduced an expanded LCDS 2030 and arranged a multiyear sale of ART TREES jurisdictional forest credits to Hess. The published minimum purchase commitment is USD 750 million, with specific prices and volumes by credit vintage. The policy states that 15% of associated receipts are dedicated to eligible Indigenous village-led plans, and the remaining share finances wider LCDS priorities. A contract for future delivery is distinct from credits already issued, transferred, paid for and spent. [S22, S23]

This history permits analysis of two distinct resource-finance channels: extractive revenue deposited in the NRF and ecosystem-service payments managed through LCDS financial structures. A combined national wealth view is useful for climate and budget choices, but it would be inaccurate to describe all carbon receipts as oil revenue or add contracted future carbon sales to cash in the NRF. Oil production is a fossil-fuel supply activity; forest-credit issuance depends on independently specified carbon-accounting baselines, monitoring, additionality and leakage treatment. One does not mechanically cancel the other. Any claim of national carbon neutrality requires economy-wide emissions and accounting safeguards, not a simple subtraction of a contracted offset volume from petroleum exports.

The transferable question is whether a resource-producing state maintains independently verifiable environmental accounting for different assets. In Guyana's case, the forest monitoring and results-payment experience is relevant to land-use data quality and international partnership design. Its applicability to another jurisdiction depends on ecosystem type, forest tenure, historical deforestation, Indigenous rights, corresponding adjustments and carbon buyer eligibility. A reef-based island has different biodiversity and marine accounting issues. Carbon-market prices, certification methods and buyer eligibility are also vulnerable to change.

LCDS policy statements about village allocations, sea defenses, electricity investment and community livelihoods need outcome-level tests. A national transfer proves expenditure

authorization, but independent review would examine arrival in designated accounts, locally agreed development plans, procurement documentation, functioning assets and households reached. Consultation must identify which communities participated, which declined and how grievances were addressed. A strategy can carry the language of inclusion while still requiring village-level verification.

IMF staff in July 2026 highlighted Guyana's market-based forest conservation, disaster-resilience investments and energy transition, while also identifying needs for stronger official statistics and procurement. These assessments do not eliminate the analytical tension between increasing fossil output and the global energy transition. The final analysis will report domestic operational emissions separately from combustion of exported oil, land-use emissions and carbon-credit issuance. A policy commitment and a measured atmospheric outcome belong in different columns. [S08]

8. Local value, social outcomes and accountability

Guyana's Local Content Act was passed in December 2021. The law establishes a formal route for specified Guyanese participation and reporting by petroleum operators and suppliers. The 2026 mid-year report estimates roughly USD 466.2 million in procurement across sectors prioritized for Guyanese participation during the first half of the year, while government work on possible expansion of the law's scheduled categories was still underway. The number measures gross procurement in designated sectors. It does not independently reveal domestic value added, local beneficial ownership, wage quality, profits retained, product standards or the effect on non-petroleum employers. [S05, S07]

The final local-value chapter will disaggregate contractors, service categories and ownership. It will use an explicit distinction between a registered Guyanese supplier and domestic production content. Examples include port services, welding, catering, transport, engineering, aviation, environmental monitoring and professional services. An imported piece of equipment resold by a locally registered firm generates different domestic value from an engineered product manufactured and maintained in Guyana. The public data should be audited against supplier certification and customs records where legally available, with confidentiality and commercial rights preserved.

Rapid resource expansion also changes wages and the availability of teachers, nurses, civil servants and smaller-firm engineers. Government's 2026 mid-year report discusses technical and professional skills shortages and potential execution delays. Fiscal transfers and construction can raise activity while pushing against constraints in local land, machinery and skilled labour. An evidence-led study needs salary indexes, migration measures, training completions and occupational vacancy data by region. National job announcements or scholarships alone do not establish net employment and long-run earnings effects. [S07]

Social outcomes are especially susceptible to mistaken timing. The World Bank's country overview reports an older national poverty estimate based on 2019 data and explicitly notes the absence of recent comparable poverty data in that published summary. It would be misleading to treat that figure as the current 2026 poverty rate. An updated household survey, real regional consumption, nutrition, access to safe water, school completion and hinterland service coverage

offer a stronger basis for later conclusions. The final paper will identify missing indicators rather than invent a post-oil poverty trajectory. [S10]

Transparency has several layers: formal contract publication, beneficial ownership, production volumes, government receipts, cost-oil audits, independent regulation, public participation and spending outcomes. Guyana has published several petroleum documents and EITI national reports. Its December 2025 EITI country report covered the 2023 fiscal year. The time lag matters for a rapidly expanding industry. [S18]

The EITI Board's May 2026 decision launched a full validation starting May 15 following a delay and stated concern about a lack of stakeholder confidence in the independence of civil-society representatives' selection process. This is a documented disagreement about the quality of participation, not grounds for this manuscript to pronounce on the legitimacy of a political actor. The final report will present the EITI decision, views of the multi-stakeholder parties and subsequent verified developments without asserting the outcome of an unfinished evaluation. [S16, S17]

More frequent digital publication can improve timeliness but will not replace independent challenge. A public resource ledger becomes useful when readers can reconcile production, sale prices, contract obligations, royalty payments, NRF deposits and authorized budget transfers. That ledger should carry dataset versions, source URLs and correction notices. A complete digital publication should be intelligible to citizens and technically testable by researchers.

9. Five international comparator designs and limits

Ghana's Petroleum Revenue Management Act 2011 established a stabilization fund and a heritage fund, along with publication and parliamentary accountability provisions. This provides a comparator for separating short-run expenditure stabilization from long-run financial savings and for setting statutory reporting frequencies. Comparing statutes alone would not show which country delivered better outcomes: data on implementation, withdrawals, public debt and actual welfare need to be paired with each legal instrument. [S24]

Norway's Government Pension Fund Global receives the state's net petroleum cash flow and sits within a fiscal framework linked to an estimated 3% long-term real return. Its 2025 official white paper discusses a guideline for keeping normal-year withdrawals below that rate and maintaining buffers for adverse years. Norway has a different institutional history, population, capital market environment and mature stock of financial assets. A new producer cannot translate its numerical fiscal guideline without modelling local infrastructure deficits, expenditure absorption, debt and reserve uncertainty. [S25]

Timor-Leste established a Petroleum Fund by law in 2005 and amended that framework in 2011. Its early fund architecture offers a comparison for a small developing state facing large prospective petroleum receipts. The final paper will add verified 2026 data on petroleum-field production, fund withdrawals, budget reliance, and development outcomes before drawing conclusions about the performance of its particular formula. A statutory rule and its fiscal sustainability are separate propositions. [S26]

Suriname offers a regional pre-first-oil comparison. TotalEnergies' September 19, 2026 update reports continuing construction at the GranMorgu offshore development, with first oil planned for 2028. It is therefore a prospective comparison, not another operating case. The administrative question concerns which Guyanese preparations could be examined before Suriname's commissioning and which differ under its national oil company, contract terms, marine environment and institutional structure. [S27]

Small island states must be handled on their own terms. A prospective producer with no tropical forest sector cannot finance development through Guyana's precise jurisdictional forest-credit structure. A shallow-water or nearshore field presents a different spill-response geography from deepwater Stabroek operations. Tourism and reef exposure, narrow fiscal revenue bases, hurricane risk, desalination, land scarcity and cross-border response agreements change the feasibility of proposed institutions. The resulting handbook will be a configurable set of decision records, not a claim that one nation's contract and fund law suit every geology or society.

10. Transferable administrative sequence and adverse cases

A sequence-based framework begins with the premise that sovereign choices are made under different degrees of uncertainty. During exploration, the uncertainty is commercial: hydrocarbons may never be discovered in economic quantities. After a discovery, the challenge is negotiating rights and environmental protections before irrecoverable infrastructure is commissioned. At first oil, financial accounting and incident reporting need to work reliably. During scaling, the bottleneck often shifts toward public investment capacity, cost-oil review, labour, supplier quality, maintenance and exposure to global prices.

An early-country workbook would document the following actions and evidence without deciding the jurisdiction's political choices: establish a public chronology of licence and environmental authorities; inventory all fiscal obligations and contingent liabilities; map marine and coastal ecosystem baselines; obtain independent reservoir, legal, accounting and spill expertise; publish contract versions and define justified confidentiality exceptions; describe who verifies volume and price; and simulate an offshore emergency with available regional partners. Each item has a named evidence document and an owner, rather than a generic assertion of preparedness.

The second workbook starts when a commercial discovery is established. It records the development proposal's production range, cost envelope, decommissioning plan, environmental alternatives and labour forecast. It creates a source-and-use-of-funds bridge under different oil prices and differentiates operator spending from direct government expenditure. It identifies whether fiscal rules would operate before or after first cash receipts, which body publishes fund statements and which institution investigates alleged audit anomalies. Political decisions about contract models, ownership structures or spending ratios remain open; the workbook presents their quantifiable constraints.

A first-oil readiness audit would test whether field meters, cargo valuation, royalty invoices, cost-bank records, central bank reconciliations, incident-response roles and legislative reporting function under realistic load. A regulator should not assume every new FPSO offers the same risk profile or environmental permit conditions. A simple flowchart linking the operator's field data to

the revenue authority, finance ministry, NRF custodian and parliamentary appropriation creates a traceable basis for independent scrutiny.

Adverse case A: a long low-price interval

A sustained fall in realized crude prices compresses gross sales, royalties and profit oil. If economic activity depends on imported equipment, construction can also slow. A test should compare projected authorized spending against revised legally available fund transfers, fiscal buffers, project postponement clauses, debt service and protection of essential operations. The same price shock has different effects when the PSA cost cap binds and after capital costs decline. Never assume a fixed state share of gross revenue across all phases.

Adverse case B: newly approved development replenishes the cost bank

A field reaches high production and government receipts rise as historical development costs decline. The operator then proceeds with further sanctioned projects. New recoverable expenditures affect the profit-oil calculation. A model based on a single high-receipt period overstates durable public cash if it ignores this cycle. Preserve fields, vintages, cost categories and contractual cross-charging constraints separately.

Adverse case C: a major offshore spill

The study assumes a serious release during adverse sea conditions and asks which legal provisions are in force, who orders emergency action, when cash becomes available, which state pays upfront, what coverage exclusions apply, how claims enter foreign jurisdictions and how independent evidence is preserved. The hazard exercise tests institutions, not an asserted probability of catastrophe. The final paper requires independent review of actual emergency plans and contractually enforceable insurance terms.

Adverse case D: overheating without fiscal insolvency

Even with a rising financial fund balance, public construction demand may exceed available contractors, engineers and aggregate supply. Prices and wages rise, project delays accumulate and the non-oil economy loses competitiveness. The July 2026 IMF statement highlights absorption, procurement and monitoring concerns as questions for ongoing policy review. The stress test therefore tracks real input costs and completed units of service, not budgeted nominal spending alone. [S08]

Adverse case E: public confidence breaks despite disclosure

An extractive revenue portal posts thousands of data rows while representatives of affected communities question the independence of stakeholder selection or regulators' responses. Mechanical disclosure does not substitute for meaningful access to information, independent representation and a process for correcting data. The EITI Board's May 2026 statement makes the distinction concrete. The audit would trace how complaints are recorded, decisions documented and adverse findings disclosed. [S16]

11. Initial findings and publication boundaries

Five verifiable facts structure the emerging case. The first commercial discovery preceded first oil by roughly four years. A Natural Resource Fund statute was assented to before first production. Subsequent fund governance and withdrawal rules changed in 2021 and 2024. Four offshore FPSOs were producing by the first half of 2026, with growth in reported profit-oil receipts and substantial cash held in the NRF. Guyana also has a distinct pre-petroleum low-carbon and forest-finance history that continued alongside expanding oil production. [S01, S04, S07, S22]

The next layer concerns institutional performance rather than chronology. IMF staff documents continuing needs in fiscal anchors, public investment monitoring, procurement, petroleum audit capacity, statistics and public-enterprise oversight. EITI 2026 validation is an ongoing process with publicly recorded stakeholder concerns. Environmental policy includes expanded statutory architecture whose commencement and practical implementation require careful verification. These issues are integral to a useful international case study. [S07, S08, S16, S19]

What an emerging producer can extract from the evidence is a disciplined research method: establish the historical sequence, date each legal change, distinguish entitlement from receipt and expenditure, record ecological hazards at the licence level, publish cross-checkable financial reconciliations, separate climate assets from fossil-fuel accounting, and verify public-service outcomes independently of announcements. Those administrative questions apply across different policy choices. Specific contract terms, saving formulas and energy strategies remain jurisdiction-dependent decisions.

This is a first research manuscript, not a finished monograph. Before final publication the author must complete annual fund reconciliations, contractual and judicial text review, independent environmental and social evidence, a dated 2026 EITI outcome check, quantitative scenarios, comparative-country datasets, external technical review and page-specific reference verification. The proposed 140-160 substantive pages are a future acceptance gate rather than a page count claimed here.

Technical appendix A. Analytical definitions and worked examples

A1: Stock-flow reconciliation. Let F_0 be opening fund cash, D be period petroleum deposits, I be recognized cash investment income, O be other permitted cash flows and W be lawful disbursements. Then $F_1 = F_0 + D + I + O - W$. Disclose transaction timing, currency, investment valuation and whether interest has already been included in D . The Ministry's June 2026 fund balance was USD 4.2942 billion, after USD 1.02 billion in first-half withdrawals. An annual account cannot be assembled by combining that figure with full-year projected deposits without dates. [S07]

A2: Production-to-cash bridge. Petroleum units produced in month M differ from saleable units, cargo lifted in month M and receipts recognized by government in month M . A dated lift ledger should identify cargo, lift date, entitlement owner, grade, price determination, settlement currency, invoice date, deposit date and exchange rate. In H1 2026 the government's 21 lifts included late-recognition effects from 2025. [S07]

A3: Simplified production-sharing example. Normalize a period's gross sales revenue to 100. In the binding 75% ceiling case, assign 75 to illustrative cost recovery and split 25 profit units equally. The state receives 12.5 profit units, plus the separately calculated 2% royalty described by the GRA, giving 14.5 combined units. At a 50% illustrative actual cost share, the split of remaining 50 units yields 25 state profit units plus royalty. At 20%, the corresponding illustration yields 40 plus royalty. These simplified examples omit detailed PSA adjustments and are neither observed 2026 figures nor revenue forecasts. [S02, S15]

A4: Illustrative oil-price sensitivity. A purely hypothetical field selling 1,000,000 barrels per day for 365 days produces 365,000,000 annual barrels. A USD 10 per barrel realized price change moves gross oil sales by USD 3.65 billion before operating expenses, royalty, recoverable costs, fiscal terms, discounting or cash timing. Applying that change directly to the NRF would be incorrect. The case-study model must instead use contract-specific entitlement and fund law.

A5: Capacity versus actual production. Nameplate daily capacity multiplied by 365 describes a theoretical annual maximum before downtime and operational constraints. Actual barrels are the integral of daily measured output, adjusted for reporting boundaries. Government monthly data and independently reported operator annual results should be reconciled.

A6: Carbon versus petroleum cash. A forest-credit agreement describes volume, price, vintage, payment conditions, registry issuance and benefit-sharing obligations. Petroleum cash follows royalty and profit-oil provisions. These may be reported on the same national fiscal page only if separately attributed. Contracted Hess purchase value is not the same thing as realized forest-carbon receipts. [S23]

Technical appendix B. The newcomer decision record

For every proposed instrument, the handbook has eight fields: legal authority; public purpose and measurable output; geological or ecological dependency; evidence required before adoption; implementing institution and budget; independent reviewer; disclosure frequency; and evidence that would trigger revision or suspension. Entries will cover the petroleum licence, environment permit, benefit-sharing instrument, revenue fund, procurement contract, local supplier registry, public investment project and transboundary spill protocol.

Technical appendix C. Questions for external reviewers

- Did the paper apply the legal text in force on the relevant date?
- Did the cost-oil model follow the original executed PSA, including valuation and royalty provisions?
- Did any model count contracted credit sales as cash already received?
- Were unpublished operator assumptions mistakenly treated as regulator-confirmed data?
- Were spill-insurance limits, legal liability and access to immediately liquid compensation distinguished?

- Do Indigenous communities' documented positions appear beyond official summaries?
- Did any poverty estimate use survey data older than the observation year claimed?
- Are state receipts, deposits, parliamentary transfers and spending outputs separately traceable?
- Does the case transfer instruments without recognizing island-state ecological and administrative limits?
- Has an independent expert performed the review, or is a review merely proposed?

References

Full URLs, source types, dates, numerical facts and remaining challenges are recorded in SOURCE_REGISTER.md (S01-S27). Citations inside the final PDF must point to exact pages/sections where available. This edition cites the public original sources by source ID and deliberately does not represent external peer review as completed.

THE NEW-PRODUCER HANDBOOK

A staged institutional workbook for an emerging oil-producing country or island state

Independent companion annex to The Guyana Sequence | Ragunauth Ramsaroop | 27 September 2026

This handbook is a transferability instrument, not a claim that Guyana's history is a universal model. It offers defined questions, document templates and alternative institutional structures from published country cases. Sovereign authorities remain responsible for policy, legislation, fiscal choices, treaty obligations and public consultation. Statements of Guyana's actual history refer to source IDs in SOURCE_REGISTER.md.

1. Start with a resource and institutional map rather than a model law

The first analytical task for a prospective petroleum state is to describe the country before petroleum income changes its financial structure. Guyana's 2009 Low Carbon Development Strategy, earlier commodity economy and decades of offshore exploration predated the first major commercial discovery. The country did not begin building every relevant technical or financial capability only after oil arrived. An island without large tropical forests or comparable offshore acreage has a different starting set of options. [S01, S22]

An initial country profile records offshore geology and the uncertainty of commercial reserves, coastal population and infrastructure, current export composition, public debt, annual tax revenue, national disaster losses, fisheries and tourism dependence, existing environmental agencies, maritime boundaries, regulatory staff, procurement integrity processes, public finance reporting and access to international specialist advice. These are separate datasets. A positive resource discovery changes the geological column but does not erase pre-existing environmental or fiscal exposure. The profile should state the latest observation year for every variable. Missing observations are recorded as missing.

The profile also identifies the institutional decision-makers whose legal mandates may overlap. Typical actors include a parliament, a ministry responsible for resources, finance authorities, a revenue agency, an environmental regulator, a national audit institution, a central bank, a coast guard or maritime authority, regional government, Indigenous or customary-rights institutions, professional regulators and an independent judiciary. Actual institutions differ under civil-law, common-law, hybrid and constitutionally decentralized jurisdictions. The workbook records the current legal mandate rather than silently assigning a new one.

A useful opening deliverable is a two-page public evidence map. The first page identifies the legal source, date of enactment and commencement, administering authority and current implementing regulations. The second page identifies data held by each institution, reporting lag, legal publication restrictions, unresolved boundary disputes and any independent review process.

If existing law does not authorize the proposed activity, the gap is visible before budgets or investment agreements assume implementation.

A separate social and ecological baseline sets the denominator against which later claims will be tested. A coastline with dense tourism and reef ecosystems has a different vulnerability profile from Guyana's low coastal plain and broad forested interior. Climate adaptation, coastal fishery economics, drinking-water resilience and household service levels require direct measurement. Petroleum is a potential revenue source, not a substitute for obtaining the baseline.

2. Exploration stage: six documentary systems

The exploration stage has no guaranteed development or receipts. Guyana's experience includes decades of surveys and unsuccessful wells before Liza. A new state considering costly petroleum-specific institutions should record whether they are scaled to known commercial activity or to uncertain potential. In a country with limited human resources, time-bound specialist contracts and regional sharing agreements are among the institutional forms to compare against a permanent bureaucracy. Neither arrangement removes the need for retained institutional memory or independent conflict-of-interest screening. [S01]

The first documentary system is geological. A searchable block map links the licensing unit, coordinates, bathymetry, historical well records, independent reserve interpretation and uncertainty classification. All petroleum-in-place, contingent-resource and commercially recoverable-reserve statements use explicit definitions. Geological prospects are not parliamentary assets until commerciality is established and sales cash materializes.

The second system is fiscal. Each licence award or proposed agreement has a versioned record of fiscal instruments, stabilization clauses, obligations carried by the contractor, state obligations, allowable costs, relinquishment, abandonment and disputed-area exposure. Where parties negotiate confidentially, the public register states the legal basis and duration of nondisclosure and identifies which post-award terms enter the public domain. Tender dates and the grounds for different procurement methods are traceable.

The third system is ecological. Before drilling, the regulator receives baseline observations of water quality, seabed habitats, fisheries seasons, migration patterns, coastal protected areas, maritime traffic and prevailing currents. For transboundary risks, the record identifies neighboring states and the existing international response agreements. Geological survey and drilling permit terms should not be conflated with consent for a commercial producing field.

The fourth system is technical independence. A national authority can appoint external petroleum engineers, transaction counsel, public accountants and environmental scientists, subject to published conflict controls, confidentiality protections and knowledge-transfer requirements. The record lists each adviser, scope, deliverable, price, tenure, supervision and training obligation. Independent review is not established solely by naming an international consultancy; the published decision file needs evidence of the questions advisers examined and competing options considered.

The fifth system is public participation and rights. A dated register describes legally relevant consultations with fisher groups, coastal settlements, Indigenous and customary-rights holders, tourism representatives and other affected communities. Participation evidence records who received notice, technical materials and translation support, the matters raised and the regulator's reasoned response. National constitutional and treaty obligations determine how consultation, consent, compensation and review operate. The register should not expose personal data or confidential cultural sites.

The sixth system is contingency. A prospective exploration state maps the institutional owner of offshore emergency notification, regional deployment, oil-spill equipment, public-health coordination, aviation and maritime rescue, and access to claims assistance. At this stage a tabletop exercise may demonstrate gaps more clearly than a general emergency-policy statement. Its report identifies response time, jurisdiction and unresolved procurement, not a presumed ability to contain an unmodelled release.

These six systems can be prepared without predicting that commercial oil will be discovered. In an unsuccessful exploration outcome, their data and procedures may still inform geological archives, marine science and public procurement. The financial case for building them at a particular scale remains country-specific.

3. After discovery: the contract and approval decision file

A major commercial discovery introduces a different information problem. The state needs independent evidence about the field's technical and financial viability, while the operator controls substantial reservoir and project-cost information. The period between Guyana's 2015 Liza discovery and December 2019 first oil illustrates why agreements, environmental decisions and fiscal arrangements should be dated separately. Guyana's first NRF law was enacted during this pre-production period. [S01, S02, S04]

The development decision file begins with a technical summary listing verified discovered resources, development phases, forecast decline, required subsea facilities, FPSO or platform capacity, export route, power requirements, drilling schedule and decommissioning concept. A high production forecast must remain labelled as an operator case unless independently confirmed. Where the geology is uncertain, the file records alternative development sequences and their effect on public revenue rather than presenting one central production curve as an inevitable trajectory.

The commercial file compares the actual investment risks allocated by each contract form under the assumptions prevailing when signed. Production-sharing agreements, royalties and taxation, conventional licences and concession-tax systems, service contracts and state equity offer different combinations of upfront funding, price exposure, cost audit, abandonment liability and public oversight. The workbook does not presume any one form is appropriate for a given jurisdiction. It requires internally consistent cash flows, investor-return assumptions and explicit responsibility for failure to reach commercial production.

The fiscal model maintains separate input ledgers for price, recoverable costs, operating costs, investment timing, production, depreciation where relevant, state equity, royalties, profit

sharing, corporate taxation, bonuses and decommissioning. The output ledger distinguishes gross petroleum sales, contractor receipts, the state's profit oil, royalty payments, other taxes, NRF deposits, investment income and money later appropriated through the budget. Each ledger must reconcile on a dated basis.

Guyana's signed 2016 Stabroek agreement includes a 2% royalty, a 75% cost-recovery ceiling and equal sharing of profit oil. Its later model announced a 10% royalty, a 65% recovery ceiling and 10% corporate tax for different licensing circumstances. The older contract's cost bank and the later model should occupy separate scenarios in any comparison. A 2026 cash outcome is not enough to reconstruct the entire risk and bargaining position at the date of the first agreement. [S02, S14, S15]

The environmental approval file is linked to but institutionally distinct from the financial model. A field's commercial value does not establish that mitigation, marine response, compensation, monitoring or financial assurance are adequate. The file identifies each applicable environmental impact assessment, alternatives considered, legal right to inspect and halt operations, monitoring data ownership, baseline methods, spill modelling assumptions and required independent studies.

The public record should preserve dissenting technical views. A regulator's reasoned decision has different evidentiary status from an operator's response to comments, a government announcement or a court ruling. Where litigation is pending, a risk register distinguishes the order in force, the relief requested and legal contingencies still unresolved. Guyana's May 2026 appeal concerning environmental financial assurance demonstrates why final manuscripts must read authenticated judgments instead of relying on competing summaries. [S20]

4. The first-oil readiness rehearsal

First oil often arrives before the initial public development model has stabilized. Under a production-sharing arrangement, cost recovery, crude lifting schedules and sales settlement affect the precise timing of government cash. A national oil accounting system must therefore function during commissioning, not several years later when petroleum receipts already dominate fiscal reporting. Guyana's H1 2026 figures demonstrate the importance of this distinction: some payments received in the first half corresponded to cargoes lifted near the end of 2025. [S07]

The rehearsal opens with a fictitious cargo. The operator supplies a metered production record and government entitlement calculation. The designated fiscal authority checks the volume, crude quality adjustment, eligible costs and royalty; the sales entity checks invoicing and settlement. The central bank confirms receipt. The finance ministry reconciles the money to the NRF statement. The parliamentary record then tracks any authorized transfer from the fund to the consolidated budget. Each step carries a date, accountable official, audit trail and route for a discrepancy.

No example needs to use confidential, live commercial numbers. A deliberately fictional first exercise can contain a missing invoice, late shipment, wrong exchange rate, duplicated cost invoice and incorrect posting of investment income as oil sales. The purpose is to test whether the participating institutions identify and correct differences without an opaque manual

adjustment. Material issues found in exercises should appear in a remediation log with a revised protocol and retest date.

At the same time, a parallel environmental drill begins with a hypothetical subsea discharge in poor visibility and a second simulated warning from a neighboring maritime authority. The exercise records the emergency command structure, notification triggers, helicopter and vessel mobilization, sampling, shoreline protection priorities, public communication, legal evidence preservation and claims intake. A state with limited equipment may document formal agreements for shared regional response. The record should distinguish an agreement on paper from a tested mobilization time.

A third rehearsal concerns access to information. Citizens, parliamentarians and researchers try to trace a fictional cargo from the project's regulatory record to the public revenue ledger and verify an appropriation against the published budget. If essential data are lawfully confidential, the portal should specify why and what aggregated fields remain public. A portal with attractive graphics but irreconcilable periods or undefined terms is not a substitute for usable records.

The rehearsals create three different types of evidence: account reconciliation, operating-response capability and public accessibility. None is a measure of the ultimate success of an oil development, but each identifies specific institutional functions before first-year petroleum revenue becomes an ordinary budget assumption.

5. Designing a fund: alternatives rather than a copied withdrawal percentage

Guyana established a statutory Natural Resource Fund before first oil, replaced the first law in 2021 and changed withdrawal provisions in 2024. Ghana's 2011 revenue framework has dedicated stabilization and heritage components. Timor-Leste's Petroleum Fund dates to 2005 with a 2011 amendment. Norway has a mature global fund with a long-run real-return-linked fiscal guideline. The relevant analytical insight is the range of statutory and fiscal architectures, not a ranking of countries or a single numerical withdrawal formula. [S04, S24, S25, S26]

A country examining a resource fund first documents the problem the fund addresses. Annual petroleum cash is exposed to price and production changes. Underground reserves cannot be valued as risk-free cash. Fiscal law often needs a legal method for receiving proceeds, reporting investment performance, transferring a defined amount to the budget and exposing disbursements to ordinary procurement and audit rules. Different development needs, debt service, age structure, exchange-rate arrangements and climate liabilities alter the budget and savings trade-offs.

One analytical model separates stabilization assets from a long-term savings portfolio, with a published transfer rule for each. A second holds a single sovereign portfolio and ties normal-year spending to an estimate of sustainable returns. A third adopts a formula linked to a defined share of verified revenue or assets within constitutional budget procedures. A fourth hybrid model retains a legal spending envelope but adds a multi-year expenditure ceiling, emergency exception and a clear independent disclosure process. Each model creates different accounting, forecasting and governance demands.

The workbook requires a country-specific estimate of the usable cash buffer under a joint commodity-price and natural-disaster stress event. It identifies the unit in which spending obligations arise, the currency and liquidity of fund investments, the risk of simultaneous domestic recession, and a credible re-entry path after an emergency exception. If a rule permits extra withdrawals during a disaster, the document states who declares the exception, the amount and duration, oversight route and subsequent repayment or recalibration mechanism where applicable.

The core fund reconciliation is ordinary accounting. Opening cash, new petroleum deposits, investment return, other permitted adjustments and withdrawals together explain closing cash. Any transfer into the budget must reconcile to both the fund's statement and the government's budget execution statement. If royalties appear in oil-sale proceeds, they should not be added again under a different financial label. Cash received for prior-year liftings needs a visible timing bridge. One annual press release is not enough to replace account-level controls.

External audits establish whether statements are fairly prepared under an identified accounting standard. They do not establish whether the government selected or delivered the most valuable public projects. That second question requires public investment appraisal, procurement oversight, engineering completion evidence and independent service-delivery evaluation. Guyana's large fund cash balance in June 2026 and the IMF mission's July comments about investment monitoring illustrate the need to keep the two questions separate. [S07, S08]

A useful funding-options dossier therefore contains side-by-side legislation, liquidity simulations, administrative costs, governance responsibilities, spending plans, unknowns and public consultation records. The sovereign authority chooses the design under its constitutional and fiscal system. Publication of alternative assumptions allows later researchers to reconstruct the decision.

6. The petroleum revenue transparency ledger

An extractive revenue website serves different audiences. Parliament needs statutory compliance and account reconciliations. A central bank needs cash management and portfolio controls. Tax authorities need expenditure verification and settlement records. Local communities need understandable information about revenues, distribution programmes and environmental incidents. Researchers need data definitions and historical versions. A public ledger therefore needs more than a current total.

The minimal petroleum portion lists the project and contract identifier, fiscal instrument, observation period, quantity and unit, price basis, lift date, government entitlement, invoice period, payment date, receiving account and audit status. Project-specific records with legally protected commercial information may be aggregated, but the scope and legal basis of aggregation should be documented. An entity-level register records beneficial-ownership disclosures and changes subject to applicable privacy and security law.

The fund portion lists opening assets, new deposits by category, investment interest or income, valuation adjustment, lawful budget transfers and ending cash. Each published table includes its accounting basis and revision history. The budget portion separately identifies appropriations,

procurement contract values, actual spending, verified physical completion, maintenance allocations and beneficiary results. A reader should be able to follow cash without assuming every transfer immediately produces a completed asset.

Environmental transparency is part of the same programme but has a different standard of evidence. The public permit register identifies drilling and production authorizations, impact assessment dates, permit expiry, reportable incident definitions, measured emissions, significant enforcement decisions, available monitoring data and any material remediation obligations. The administrative record captures which regulator confirmed a fact and which claim rests on operator self-report.

Civil-society representation cannot be reduced to web access. The EITI Board's May 2026 decision recorded stakeholder concerns about civil-society selection and initiated a full validation with external participation. An implementation workbook therefore documents who selected representatives, applicable rules, whether affected groups had access to technical support, how dissenting submissions were considered and whether public records provide a traceable response. The 2026 process should be described according to its final verified status before publication. [S16, S17]

A release process protects analytical integrity. Each dataset includes version, publication date, source agency, applicable period, methodological changes and a clear correction notice. A monthly series that changes definitions halfway through the year should not appear as a continuous trend without reconciliation. Differences between production, exports, oil sales and deposits should be shown rather than obscured.

7. Local participation and the value-added question

Governments use several forms of local participation: supplier qualification, procurement preferences or reservations, mandatory reporting, technical training, joint ventures, domestic processing, public education and access to finance. The domestic economic outcome depends on the mix of actual productive inputs rather than one gross procurement figure. Guyana's 2021 Local Content Act and its 2026 procurement reporting offer a case for examining the difference between an administratively eligible supplier and independently measured domestic value added. [S05, S07]

The workbook separates business registration, beneficial ownership, share of labour paid locally, imports embedded in purchased goods, plant and equipment owned within the jurisdiction, local subcontracting, professional accreditation, technology transfer and taxes retained. A service supplier may be a meaningful domestic employer even with imported equipment. Another company may book large contract value through a local subsidiary with most technical work completed abroad. Without disaggregation, both transactions look similar in gross purchase statistics.

A registry should permit independently checkable ownership and skills claims while respecting commercial confidentiality and individual privacy. Independent verification might compare procurement statements with audited supplier accounts, customs classifications, professional training completion and aggregate labour statistics. Alternative explanations for spending

increases, including domestic construction inflation or reclassification of supplier categories, belong in the same analytical report.

Geography is equally important. Offshore supply chains often concentrate near ports and the capital, while inland and remote districts face different infrastructure needs. A national local-content percentage says little about benefits to hinterland communities. A map of procurement distribution, pay and completed infrastructure is therefore more informative than a single national total. For a small island, limited land, port throughput, environmental carrying capacity and seasonal tourism affect local supply opportunities.

The analysis must also measure possible displacement. Large operator contracts may raise wage rates and absorb engineers, drivers and electricians previously employed by clinics, schools, municipal agencies and non-oil firms. The relevant question is whether new training and productive capacity expand the overall skilled workforce faster than demand, not merely whether the petroleum industry hires more people. Labour-force and wage data require source, sample period and inflation adjustment.

8. Indigenous peoples, coastal livelihoods and public participation

An ethical and analytical case study makes participation visible as a process and an outcome. Guyana's forest-carbon policy commits part of eligible proceeds to Indigenous village-led development plans. Offshore petroleum activities primarily expose marine and coastal interests, while national spending decisions affect interior communities. Both require independently documented benefits and grievances. A small island's customary marine tenure, coastal fishery governance and tourism land rights may present different institutions and legal obligations. [S22, S23]

The country profile maps statutory land tenure, customary resource rights, relevant constitutional provisions, applicable international obligations, agreed consultation procedures and environmental justice claims. The appropriate consultation and consent process depends on those instruments and the circumstances of the project. Evidence includes dates, disclosure formats, translations, support for independent technical review and a record of the response to documented objections. A signed attendance sheet alone does not establish informed participation.

Community-finance claims require four distinct records: funds legally designated, sums transferred, locally selected projects and independently observed outcomes. A documented transfer to a village account is one fact. Community consent to a project, completion of the project, continuing operation and resulting improvements are additional facts requiring their own evidence. The full white paper should preserve contrary testimony and avoid extrapolating from a handful of cases to every participating village.

Maritime impacts require similar attention. Fishers' exposure depends on where vessels operate, exclusion zones, changes in landings, port congestion, prices, compensation procedures and baseline variability. An increase in national GDP does not prove unaffected fishers benefited. A properly specified marine impact study accounts for weather, seasonal migration, fisheries

management, other industries and changes in reporting. The state and operator's ecological claims need independent verification where data are available.

The publication protocol should provide a right of factual response for materially criticized institutions and stakeholder organizations without allowing any source to veto findings. It should protect respondents who face credible confidentiality or retaliation risks. A clear chronology of complaints, agency responses and court decisions is more useful than a general assertion of agreement.

9. Environmental assurance across the full project lifecycle

Environmental permitting and oil-spill financial assurance are related but distinct. Licensing decides whether and under what conditions exploration or production is lawful. Financial assurance addresses availability and enforceability of resources after a covered event. Emergency response identifies assets and decisions available within the first hours and days. Ecological restoration addresses losses long after the emergency ends. Confusing these categories can create false confidence or understate actual liabilities.

Guyana's 2025 oil-pollution legislation and May 2026 appeal over permit-linked financial assurances offer detailed subjects for legal verification. The 2026 mid-year report described implementing regulations as under development and commencement as a future event. At a new publication cut-off, enacted legislation, operational regulations, permit guarantees and appellate orders must each be verified separately. This handbook does not assume any quoted insurance total without the actual contractual documents. [S07, S19, S20]

A project-specific risk register records the hazard, expected pathways, sensitive receptors, probability model and its uncertainty, monitoring threshold, legal reporting trigger, escalation owner, containment strategy, finance instrument and external response partner. Where the country lacks credible dispersion modelling or ocean-current data, the register states the limitation instead of displaying unwarranted decimal precision. Regional assistance agreements require contact details, transit permissions, operational compatibility and periodic exercises.

Insurance due diligence examines covered activities, limit per occurrence and aggregate, exclusions, deductible, insurer creditworthiness, jurisdiction for claims, financial assurance beneficiary, rapid-access liquidity, environmental restoration scope, third-party claims and decommissioning. A parent-company guarantee raises separate questions about capacity, jurisdiction, enforceability and its interaction with an operator's continuing liability. Evidence of a guarantee's existence is different from evidence that public liabilities are fully secured.

Monitoring covers normal operations as well as exceptional spills. Discharge records, produced-water treatment, routine and non-routine flaring, methane measurement, drilling waste, ballast handling, vessel collisions, protected species, beach contamination, accidental chemical release and supply-base impacts need project-specific methods and reporting thresholds. Annual mean conditions should not conceal short high-intensity incidents.

A small-island adaptation module places reef health, mangroves, beaches, freshwater lenses, tourism seasonality and evacuation logistics into the response plan. Guyana's forest MRV and

coastal flood adaptation experience contain relevant monitoring methods, but the appropriate ecological baseline differs. A transferability assessment states precisely which laboratory methods, satellite products, emergency drills or public disclosure practices travel between jurisdictions and which require new expertise.

10. Public investment and durable service outcomes

Converting exhaustible revenue into durable national assets is a broader question than saving revenue in an investment fund. Infrastructure and human capital have different delivery schedules and lifecycle liabilities. Large petroleum receipts create an opportunity to finance capital projects but also expose a small economy to construction cost inflation, workforce shortages, poor maintenance planning and irreversible land-use changes. IMF staff's July 2026 mission highlighted Guyana's need for sound public investment appraisal and monitoring. [S08]

The public investment record begins with a published definition of the service problem, baseline conditions, alternatives, demand assumptions, environmental costs, construction schedule and lifecycle operation. For a hospital, the completion test includes staffing, power, maintenance, service hours and patient outcomes, not merely the ceremonial opening. For roads, the test covers load conditions, all-weather access, drainage and maintenance. For coastal barriers, the test covers projected flood risk, residual exposure and local ecological impacts.

The procurement register connects each project to tender design, qualified bidders where publicly available, evaluation criteria, contract award, change orders, schedule extensions, independent inspection and defects liability. It distinguishes price savings during tender from total cost after scope changes. A public project portfolio report should identify projects whose maintenance liabilities exceed planned operating budgets before those assets enter service.

Fiscal analysis compares infrastructure delivery capacity with projected annual petroleum cash under multiple prices and production paths. A legally available withdrawal is not evidence that domestic construction supply can absorb the resulting spending without delays or price escalation. A separate model estimates import demand, skilled labour, land requirements and potential crowding of non-oil investment. Uncertainty in domestic absorption is reported alongside uncertainty in the oil price.

Social outcomes need disaggregation. Regional living standards, access to water and electricity, education completion, maternal and child health, housing affordability, travel time and employment among women and younger workers are plausible indicators. Their applicability and target levels follow national development plans and community consultation. Surveys taken years before first oil should remain marked historical rather than be presented as current evidence.

11. Five-year observation schedule

The workbook groups observation by operational stage rather than guaranteeing a calendar date for oil production. Year 1 begins when a jurisdiction formally launches a resource-governance work programme, whether or not a commercial discovery exists. Every item below is a record requested for research and possible administrative review, not a mandatory legal prescription.

During months 0-3 the record covers petroleum licence and geological holdings, current law and commencement status, fiscal exposures, marine baseline, existing disaster protocols and the capacity of existing agencies. The finance ministry assembles a preliminary risk and revenue-uncertainty register. A ministry or regulator charged with petroleum maintains the licensing and contract inventory. Statutory audit bodies map their access to records.

Representative stakeholder institutions identify consultation requirements under applicable law.

During months 4-6 the record includes independent technical advice procured, review of existing disclosure obligations, available fiscal and contract design alternatives, cost-audit capacity, ecological knowledge gaps and an initial incident communications exercise. Where petroleum prospects are not commercially proven, the budget scenario uses zero commercial oil revenue as its central binding reality rather than projected receipts. If a commercial discovery has occurred, engineering and fiscal scenarios identify range and timing.

During months 7-12 the country dossier details chosen regulatory and contract procedures, EIA scoping, transboundary notifications where required, future fund legislation options, sample reconciliation records, independent audit mandates and a funded skills plan. Publication may be staged in line with lawful data protection, but the record explains disclosure omissions. A tabletop exercise tests resource and environmental decisions across agencies with no assumptions of operational perfection.

During months 13-24 the observed documents reflect the actual development stage: environmental permit record, drilling or commissioning inspections, contract-cost audits, any effective resource fund legislation, validated production metering where applicable and reporting systems. A jurisdiction still in exploration does not need fictitious profit-oil tables; it records progress against exploration-stage tasks instead. Any delay in production updates the fiscal scenario and budget expectations.

During years 3-5 the portfolio review examines signed versus actual development schedules, year-by-year revenue reconciliations if production has begun, actual fund deposits and transfers, fiscal stress-test assumptions, public investment delivery, domestic-value measurements, environmental incidents, administrative appeal outcomes and independent community outcome evidence. Comparable historical series require a fixed base and disclose methodological revisions.

At each stage the workbook reserves a correction section. A delayed field, adverse environmental judgment, commodity shock or public procurement failure changes facts, which changes the scenario. Reviewers should be able to reconstruct the original assumptions and the date those assumptions were revised. The workbook does not erase an earlier forecast when a later publication supersedes it.

12. Ten cross-agency exercises for the implementation team

Exercise one asks the petroleum ministry and environmental regulator to trace a fictional exploration block from acreage designation through EIA scope and final approval. Record the public legal documents and any unresolved overlap of powers.

Exercise two asks independent technical counsel to quantify how a dry-well outcome changes the proposed administrative budget. Document the difference between spending on general-purpose geological data and spending that depends on an operating hydrocarbon industry.

Exercise three uses the executed PSA or concession terms to trace a fictional field's sales, allowable cost, royalty, taxation and state receipt. The revenue authority records how disputes and payment delays affect cash timing.

Exercise four asks the central bank, finance ministry and auditor to reconcile the same fictional oil payment across accounts, including a settlement posted one month after physical lifting. A mismatch creates a correction record, not a hidden balancing item.

Exercise five asks the parliamentary budget office to distinguish legally available fund transfers from cash already collected, projected year-end receipts and operational spending commitments. Run an illustrative low-price year and document different buffer designs without prescribing a single fiscal rule.

Exercise six asks the environmental regulator and coast guard to test a simulated spill with a neighboring state affected. Record who authorizes sampling, emergency procurement, shoreline access and claims intake. Repeat the same exercise under a power outage or severe coastal storm.

Exercise seven asks the procurement authority to evaluate a large petroleum-financed infrastructure project before and after a hypothetical 25% construction-cost overrun. Include maintenance, skill shortages and the consequences of delaying other projects.

Exercise eight asks the local supplier registry and an independent economist to reconcile a high gross procurement announcement with wage payments, imported equipment, local ownership, subcontracting and retained domestic profits.

Exercise nine asks a protected independent panel to document stakeholder participation in a fictional coastal development approval. The test is whether different views reach the decision-maker and receive reasoned responses, not whether every party agrees.

Exercise ten asks a mixed technical review team to assess a hypothetical halt in offshore output coinciding with flooding and a sharp oil-price fall. Recalculate cash, essential services, contract obligations, ecological emergency funding and the credibility of public communications.

Every exercise produces a date, participating institution, source documents, observations, corrective actions, verification method and retest date. An uncompleted exercise is shown as outstanding rather than assigned a favorable qualitative label.

13. What Guyana's case does not establish

Guyana's early NRF legislation is a historical observation. It does not by itself prove that a particular savings rate, withdrawal formula or investment portfolio has maximized intergenerational welfare. Such a claim would need long-run counterfactual analysis, accounting and household outcomes. A large fund balance and rapid production growth are not interchangeable with measured school performance, flood resilience or reductions in poverty. [S04, S07, S10]

The 2016 and later model PSAs document different fiscal terms in different commercial circumstances. These documents do not establish that the same contract works for a shallow-water field, a field with state equity, a landlocked onshore producer or an island whose public balance sheet cannot bear a large contingent spill liability. The exact risk allocation and counterparty financing capacity must be reconstructed for each case. [S02, S14]

Guyana's forest programme offers a separate climate-finance case, but a country without forests of comparable scale has no identical carbon asset. A country's petroleum revenues and forest-credit receipts have different legal rights, accounting bases and environmental effects. Carbon-credit eligibility and contract value do not establish measured nationwide carbon neutrality or a permanent right to petroleum expansion. [S22, S23]

The existence of enacted pollution legislation is not synonymous with commencement, funded enforcement or tested emergency operations. An appeal about financial assurance does not settle every possible ecological-loss scenario. The research must retain authenticated legal sources and independent insurance analysis as open requirements. [S19, S20]

Finally, country comparisons have different production histories and social baselines. Ghana's revenue law, Norway's mature financial fund, Timor-Leste's sovereign petroleum fund and Suriname's pre-first-oil preparations are documentary comparators rather than a league table. A useful successor study would collect consistent country-year data, label changes in accounting methodology, document the distinct welfare measures available and subject any causal claims to a defensible identification strategy. [S24-S27]

14. Compact reusable templates

Country opening record: Name of jurisdiction; programme initiation date; discovery status; resource classification; public debt observation year; existing royalty and tax base; existing environmental and marine protections; sensitive ecosystems; customary-rights framework; coastal and inland settlements; responsible institutions; independent audit mandate; unresolved treaty and maritime questions; last household-survey observation year; central-bank data accessibility.

Agreement comparison record: Contract identifier and version; signature and effective dates; contract model; legal approval; resource and price uncertainty at the time; development and abandonment obligations; royalty, tax and profit-sharing terms; cost eligibility and audit rights; stabilization and dispute resolution; assignment and beneficial ownership; environmental and financial assurance; public disclosure status; independent counsel and conflict review.

Fund reconciliation record: Fiscal period; audit status; opening cash and investment portfolio value; petroleum deposits by source; investment income and valuation adjustments; authorized withdrawal; receiving treasury account; budget-law reference; closing cash; reconciliation difference; publication and revision dates; public audit opinion.

Environmental assurance record: Project and licence identifier; baseline surveys; EIA date; permit conditions; discharge and emissions thresholds; incident reports; independent monitoring; response assets and mobilization results; insurance contract and exclusions verified; parent or other guarantee where applicable; claim jurisdiction; decommissioning security; statutory effective dates; public grievance and judicial review procedures.

Community outcomes record: Affected stakeholder class; legal consultation rule; disclosure and meeting dates; dissenting and supporting submissions; reasoned agency response; monetary entitlement and transfer; locally approved projects; construction and operation verification; measured benefit and distribution; continuing grievances; independent reviewer.

Public investment outcome record: Project identifier; approved business case; procurement method; financing mix; cost baseline; contract award and variations; completion evidence; maintenance provision; climate or environmental exposure; service baseline; actual beneficiaries; operation performance; auditor's findings; outstanding remedial work.

Correction and disclosure record: Dataset or legal assertion corrected; original publication date; conflicting source; confidence and reason for correction; old and new values; whether previous analytical conclusions change; approving researcher; next source verification date; cross-referenced exhibits.

The final 140-160 substantive-page monograph will integrate these templates with year-by-year datasets, a verified comparison of fund rules, legal text citations, multi-scenario modelling and independent external technical review. Until then, this playbook is an openly labelled research instrument attached to the working edition.

New Producer Implementation Blueprint

A configurable exploration-to-production governance system derived from the Guyana case

How to use this blueprint

This document converts the Guyana research into a sequence of decisions for a country or island state. It does not prescribe a political choice, ownership structure, fiscal regime or spending ratio. Each government should adapt the sequence to its constitution, legal tradition, geology, marine ecology, population, debt position and institutional capacity.

Phase 0: Before a commercial discovery

A. Legal and institutional map

Create one public diagram showing:

- ministry responsible for petroleum policy;
- licensing authority;
- environmental regulator;
- revenue authority;
- central bank;
- auditor general;
- procurement authority;
- maritime safety authority;
- disaster-response command;
- local-government responsibilities;
- Indigenous or customary-rights authorities;
- parliamentary oversight bodies.

For every institution list statutory authority, decision rights, staffing, conflicts-of-interest rules and publication duties.

B. Geological information governance

Create:

- acreage map;
- licence register;
- seismic and well-data inventory;

- data-room access policy;
- relinquishment schedule;
- beneficial-ownership register for licence holders;
- retention and confidentiality schedule.

Avoid converting exploration forecasts into public-revenue forecasts.

C. Environmental baseline

Before drilling expands, record:

- currents and seasonal circulation;
- sensitive marine habitats;
- fisheries;
- mangroves, reefs and wetlands where relevant;
- coastal communities and water intakes;
- seabird and marine-mammal observations;
- baseline hydrocarbon chemistry;
- emergency ports and airfields.

Publish versioned datasets where safety and commercial restrictions permit.

D. Fiscal readiness

Build a transparent petroleum fiscal model with:

- price;
- production;
- development cost;
- operating cost;
- royalty;
- cost-recovery rule;
- profit sharing;
- tax;
- depreciation;
- decommissioning;
- financing;
- timing.

Run downside cases before upside cases.

Phase 1: Commercial discovery

Gate 1: Independent discovery review

Before public spending plans rely on oil:

- independently review the resource range;
- distinguish discovery from commerciality;
- publish uncertainty bands;
- require a development plan before treating reserves as production-ready.

Gate 2: Contract and fiscal review

Use independent petroleum fiscal expertise to model:

- investor breakeven;
- government revenue under low/base/high prices;
- delayed startup;
- cost overruns;
- lower-than-expected reservoir performance;
- decommissioning and abandonment.

Keep negotiation strategy confidential where law permits, but publish executed agreements unless a specific lawful exception applies.

Gate 3: Sovereign revenue architecture

Before first recurring petroleum receipt:

- enact the revenue-fund legal basis;
- specify deposit and withdrawal rules;
- define investment objectives;
- prohibit off-budget expenditure from the fund unless the constitution expressly provides otherwise;
- appoint accountable fund management;
- define audit frequency;
- specify public reporting;
- create conflict-of-interest rules;
- state how emergency deviations work.

The fund should integrate with the national budget, not become a parallel budget.

Phase 2: Development approval

Technical approval packet

Require the operator to submit:

- field development plan;
- reservoir depletion strategy;
- FPSO or platform capacity;
- metering design;
- gas handling;
- flaring controls;
- produced-water management;
- waste plan;
- decommissioning concept;
- local-content plan;
- emergency-response plan.

Environmental decision packet

Require:

- EIA;
- alternatives analysis;
- cumulative-impact analysis where multiple fields are planned;
- spill modelling;
- sensitive-receptor map;
- consultation record;
- financial assurance;
- monitoring plan;
- incident notification thresholds;
- restoration and compensation procedures.

Public revenue packet

Before sanction, publish a non-confidential fiscal sensitivity:

- \$40 oil;
- \$60 oil;
- \$80 oil;
- low/base/high production;

- cost-overflow case;
- delay case.

Do not publish a single-point revenue promise.

Phase 3: Twelve months before first oil

Revenue rehearsal

Run a simulated month:

1. operator reports production;
2. regulator verifies meters;
3. entitlement is calculated;
4. cargo is lifted;
5. sale price is determined;
6. royalty is invoiced;
7. state profit share is settled;
8. cash reaches the designated account;
9. central bank verifies receipt;
10. sovereign fund statement is published;
11. auditor tests the reconciliation.

Publish the exercise and corrections.

Emergency rehearsal

Run a live multi-agency spill exercise:

- detection;
- notification;
- command transfer;
- offshore containment;
- shoreline protection;
- fisheries closure;
- independent sampling;
- public communication;
- claims intake;
- cross-border notification.

Record mobilization times and corrective actions.

Public investment readiness

Create a ranked pipeline of projects, but require:

- economic and social appraisal;
- climate-risk review;
- procurement plan;
- land status;
- maintenance budget;
- delivery agency;
- measurable service output.

Do not equate the size of petroleum receipts with the size of the executable capital programme.

Phase 4: First oil through rapid scale-up

Quarterly production-to-cash statement

Publish:

- barrels produced;
- barrels lifted;
- government entitlement;
- realized sale price;
- royalty;
- profit oil;
- tax where applicable;
- sovereign fund deposit;
- investment income;
- withdrawal;
- closing balance.

Use the same definitions each quarter.

Cost-audit programme

Maintain:

- audit period;
- expenditures tested;
- amount challenged;
- amount accepted;
- amount recovered;

- disputed amount;
- arbitration or court status.

Do not announce proposed disallowances as recovered revenue.

Local-value ledger

Track:

- supplier beneficial ownership;
- domestic payroll;
- domestic procurement;
- imported component;
- training;
- certifications;
- technology transfer;
- export capability gained.

Environmental performance ledger

Track by project:

- permit;
- monitoring reports;
- flaring;
- discharges;
- spills;
- enforcement;
- financial-assurance status;
- emergency-drill date.

Phase 5: Production maturity

Fiscal transition

Move from startup assumptions toward:

- structural non-resource balance;
- long-term fund-return assumptions;
- public-asset maintenance;
- contingent liabilities;
- post-peak production scenarios.

Economic transition

Measure:

- productivity outside petroleum;
- non-oil exports;
- energy cost;
- food production;
- manufacturing;
- logistics;
- tourism where relevant;
- workforce participation;
- real household income.

Resource depletion plan

Before production declines:

- update abandonment funding;
- define post-production regulatory responsibilities;
- model reduced royalties;
- stress-test the national budget;
- determine required financial assets to sustain priority services.

A special module for small islands

Small islands face conditions that often differ from Guyana.

Marine concentration risk

A serious spill may affect a large share of fisheries, tourism coastline or desalination infrastructure. Financial assurance and regional response agreements should reflect this concentration.

Land and labour limits

A small workforce may be unable to supply both petroleum services and existing sectors. Local-content policy should prioritize capability and wage sustainability rather than gross percentage targets alone.

Fiscal scale

A small public administration may receive revenue far larger than its annual capacity to procure and manage projects. Offshore financial saving becomes more important when rapid domestic spending creates severe inflation or construction bottlenecks.

Natural-capital differences

Guyana's forest-carbon model depends on its forest endowment and monitoring history. Island comparators should assess mangroves, reefs, seagrass, fisheries and blue-carbon opportunities separately.

Disaster overlap

Hurricanes, storm surge, coastal erosion and petroleum accidents can coincide. Emergency planning should include compound events.

Twenty-four-month government action calendar after a commercial discovery

Months 0-3:

- confirm commercial-resource range;
- freeze unsupported revenue promises;
- publish licence chronology;
- appoint independent fiscal/legal/environmental advisers;
- launch baseline environmental update.

Months 4-6:

- publish fiscal-model methodology;
- draft revenue-fund framework;
- create inter-agency petroleum working group;
- publish beneficial-ownership requirements;
- establish contract and cost-data repository.

Months 7-12:

- enact fund and reporting rules;
- complete development EIA;
- define financial assurance;
- simulate first-oil revenue reconciliation;
- establish procurement and local-value baselines.

Months 13-18:

- appoint and train fund oversight bodies;
- conduct spill exercise;
- publish first public investment pipeline;
- test metering and cargo valuation controls;
- train audit teams.

Months 19-24:

- publish readiness audit;
- resolve material gaps before production;
- finalize quarterly disclosure templates;
- establish independent corrections process;
- publish downside fiscal scenarios.

Five-year scorecard after first oil

The scorecard records evidence, not political grades.

1. Revenue traceability: share of material petroleum receipts reconciled from cargo to sovereign fund.
2. Audit closure: percentage of audited costs with final disposition.
3. Fund transparency: statements published on schedule and externally audited.
4. Project delivery: planned versus completed public infrastructure, plus maintenance readiness.
5. Inflation and absorption: construction costs, wage pressure and execution delays.
6. Local value: domestic wages, ownership and higher-skill services retained.
7. Environment: permit compliance, incidents, spill drills and assurance coverage.
8. Social outcomes: current household survey, water, health, education and regional access.
9. Economic diversification: non-oil exports and productivity.
10. Intergenerational balance: financial assets, liabilities and decommissioning commitments.

Red flags

- budgeting against undiscovered or unsanctioned reserves;
- a sovereign fund with no binding publication rule;
- fund spending outside the normal budget;
- a single-point oil-price forecast;
- environmental permitting after commercial commitments become difficult to reverse;
- local-content targets without supplier ownership checks;
- gross procurement presented as domestic value added;
- carbon-credit contracts presented as immediate cash;
- production capacity presented as actual output;
- an enacted environmental statute presented as operational before commencement and regulations;
- an unresolved audit dispute presented as revenue already recovered;

- public investment measured by money spent rather than services working.

Final adaptation rule

Copy the decision discipline, not the exact Guyanese numbers.

Every country should ask which Guyana practice solved a general governance problem, which depended on Guyana's own natural endowment, and which remains under evaluation. The strongest blueprint is one that preserves this distinction.

Comparative Cases Matrix

How Guyana should be compared without creating a political ranking

This matrix supports the flagship paper. It compares institutional mechanisms, not governments.

Comparative research protocol

For each comparator, the final paper will:

1. use the statute or official fund report as the primary legal source;
2. record the date the mechanism was enacted and the date commercial production began;
3. show actual fund flows for the latest comparable year;
4. separate official objectives from observed fiscal and social outcomes;
5. include at least one independent multilateral or peer-reviewed assessment;
6. identify a failure mode or constraint as well as a documented strength;
7. avoid scores, rankings, winners or political labels.

Ghana

Ghana's Petroleum Revenue Management Act provides a useful design comparison because it separates near-term stabilization from heritage savings and creates a dedicated public accountability mechanism. The research question is not whether this system is superior to Guyana's. The useful comparison is whether multiple-purpose funds or a single fund with layered rules generate clearer decision rights, fewer off-budget pathways and stronger public comprehension.

Norway

Norway is relevant for long-horizon financial discipline, integration between petroleum cash and the national budget, professional asset management and extensive disclosure. It is a poor direct template for early state capacity, infrastructure scarcity or immediate development gaps. The flagship will use Norway to illustrate institutional maturity and long-term asset conversion, while clearly stating why a new producer should not import a numerical spending guideline without its own macroeconomic model.

Timor-Leste

Timor-Leste is a small-state comparison for high petroleum-revenue concentration, public investment needs and the tension between saving and domestic development. Its experience helps test whether a strict benchmark remains credible when infrastructure deficits are large. The research will use updated Petroleum Fund reports before drawing conclusions about current sustainability.

Suriname

Suriname is the most useful regional pre-first-oil comparison at the 2026 cut-off. It offers a live test of which preparations can be completed before GranMorgu begins production and where national oil company arrangements, fiscal law and public-finance constraints differ. Planned production is not counted as observed output.

Small-island adaptation

For a small island state, the comparative chapter will add:

- reef and fisheries dependence;
- tourism concentration;
- hurricane and storm-surge overlap;
- desalination or freshwater intake exposure;
- limited ports and response vessels;
- narrow skilled-labour pools;
- smaller domestic supplier markets;
- limited land for shore bases;
- high import shares;
- regional mutual-aid requirements.

The island blueprint therefore emphasizes emergency cooperation, financial assurance, phased local-content targets, offshore savings during construction bottlenecks and independent ecological baselines before first development approval.

PART I RESEARCH DOSSIER

Before the Discovery: Guyana's Geological, Institutional, Environmental and Fiscal Foundations

Working research dossier for Chapters 1-4 of the 520-560-page flagship publication. Evidence cut-off: 27 September 2026. Status: research-stage material. External review pending.

Purpose of Part I

Part I establishes what existed before the 2015 Liza discovery. This prevents a common analytical error: treating every later capability, law, institution or environmental policy as a consequence of oil.

The working hypothesis is narrower. Guyana entered the offshore petroleum era with several inherited legal, geological, public-finance and environmental institutions. Some were decades old. Others developed in the years immediately before commercial discovery. Their existence does not prove they were sufficient for the scale of later petroleum production. It does establish the institutional starting point against which post-2015 reforms should be measured.

Part I therefore asks four questions:

1. What economic, legal and institutional conditions existed before the commercial discovery?
2. What did Guyana already know about its petroleum geology and maritime acreage?
3. Which environmental and natural-capital institutions existed before first oil?
4. Which capacities had to be expanded after discovery because the scale and speed of petroleum development changed the demands on government?

The final chapters will distinguish inherited capacity from later adaptation.

Chapter 1 research dossier

Guyana before offshore petroleum

1.1 Economic starting point

The pre-discovery economy should be reconstructed from official national accounts, fiscal data, external-debt statistics, export structure, employment and infrastructure indicators. The chapter will not use present-day GDP as a proxy for historical administrative capacity.

The baseline dataset should include at least:

- nominal and real GDP for 2000-2015;
- export composition by gold, bauxite, rice, sugar, timber and other major categories;

- central-government revenue and expenditure;
- public debt and external debt;
- current-account position;
- inflation;
- electricity generation and reliability;
- road, port, aviation and telecommunications indicators;
- labour-force participation and skills data;
- poverty and household survey dates;
- regional population distribution;
- coastal flood exposure;
- public-sector employment in technical agencies;
- higher-education output in engineering, law, finance and environmental disciplines.

The reason for building a broad baseline is methodological. A country does not receive a petroleum windfall into an institutional vacuum. Its ability to negotiate contracts, supervise development, absorb capital investment and verify environmental performance depends on institutions and labour markets built for earlier economic conditions.

1.2 Commodity and extractive experience

Guyana's prior experience in mining, forestry and agriculture matters because extractive administration, land rights, environmental permitting and commodity-price volatility were not new concepts in 2015. The petroleum sector differed in scale, capital intensity, marine risk and contractual complexity, but several government functions had earlier analogues.

The final chapter should identify which capabilities transferred from mining and which did not. Examples include:

- licensing and concession administration;
- royalty assessment;
- environmental permits;
- occupational safety;
- customs and import controls;
- geological data management;
- community relations;
- Indigenous land issues;
- export documentation;
- international commodity-price exposure.

This section should avoid claiming that prior mining regulation automatically prepared the state for deepwater petroleum. It will instead map transferable administrative functions and identify where offshore petroleum required new expertise.

1.3 Geography and infrastructure constraints

A national map should show the concentration of population and economic activity along the coast, the location of major ports and airports, hinterland communities, mining corridors, protected areas and major drainage infrastructure. This provides context for later public-investment choices.

The final analysis should answer:

- Which infrastructure constraints predated oil?
- Which constraints later became more visible because rapid growth increased demand?
- Which petroleum-sector facilities were privately financed?
- Which supporting infrastructure involved public expenditure?
- Which investments primarily served the wider economy rather than the oil sector?

This distinction matters for transferability. A small island producer may have shorter internal transport distances but much higher exposure to port disruption, hurricanes, reef damage and tourism losses. Guyana's continental geography produces different infrastructure and environmental trade-offs.

Chapter 2 research dossier

A century of petroleum exploration

2.1 Exploration did not begin in 2015

The official Petroleum Management Programme records a long exploration history. It lists early signs of hydrocarbons, a first oil prospection licence in 1938, renewed offshore exploration activity in 1965, regulatory changes in the late 1960s, later exploration programmes and several interruptions caused by financing and boundary disputes. [S35]

This history is important because it shows the difference between petroleum potential and commercial success. A basin may attract exploration for decades before a commercially viable discovery. Governments designing institutions during exploration therefore face a difficult problem: they need sufficient capacity to protect national interests without committing large recurring public expenditures against an uncertain future industry.

The country blueprint should extract this lesson directly. Before commerciality is established, governments need scalable legal, geological, environmental and fiscal capability. They do not need to build an oversized production bureaucracy funded by hypothetical future revenue.

2.2 Public investment in geological knowledge

The final chapter should reconstruct the role of geological surveys, seismic work, public data, international technical assistance and promotional activity. The source register already records the government's summary of 1980s efforts to attract investors and synthesize petroleum data. [S35]

A stronger evidence package should add:

- original World Bank project documentation;
- Guyana Geology and Mines Commission records;
- historical licensing regulations;
- original seismic and well-data catalogues where public;
- records of government petroleum units;
- model agreements used before the modern offshore era.

The key question is not whether external assistance was good or bad. The question is what public capability remained after each programme ended. The final chapter should track data ownership, staff training, legislation and institutional continuity.

2.3 The 1986 statutory foundation

The Parliament of Guyana records the Petroleum (Exploration and Production) Act 1986. The law established a statutory framework for exploration and production and applied to petroleum in Guyana's land and maritime areas. [S38]

The full legal chapter should later determine:

- which provisions governed licensing before 2015;
- which regulations were in force at the time of the Stabroek agreement;
- what powers were held by the minister;
- how petroleum prospecting licences, production licences and agreements interacted;
- how confidentiality and data ownership worked;
- how environmental duties interacted with petroleum rights;
- which provisions have since been superseded or amended.

No 1986 provision should be quoted as current law without checking later legislation and commencement.

2.4 Maritime boundaries as a petroleum-governance issue

Offshore petroleum governance depends on uncontested or lawfully managed maritime rights. The Permanent Court of Arbitration's case record states that the Guyana-Suriname dispute involved oil concessions and a 2000 incident in which an oil rig and drill ship were ordered to leave a disputed maritime area. Guyana initiated arbitration in 2004. [S40]

This belongs in the petroleum history because boundary certainty affects exploration rights, company risk, seismic acquisition and development planning. The final chapter should use the award itself for legal propositions and the case record only for procedural history.

The transferable lesson is procedural. A prospective producer should map unresolved maritime boundaries before commercial development, identify applicable international dispute mechanisms and avoid treating acreage promotion as a substitute for clear sovereign title.

2.5 The Liza-1 discovery record

The official well register records Liza-1 as an exploration oil well in the Stabroek Block with a well date of 15 May 2015. [S36]

The final discovery chapter should separate:

- spud date;
- discovery announcement;
- well date;
- appraisal results;
- commerciality determination;
- reserve/resource estimates;
- development approval;
- final investment decision;
- first oil.

These are different milestones. Combining them into a single "discovery date" hides the development sequence.

2.6 Exploration evidence gaps

Before Part I is source-audited, the following items remain required:

1. Original licence instruments for historically significant offshore blocks.
2. A verified timeline of the 1999 Stabroek agreement and later contractual changes.
3. Original pre-2015 model agreements or fiscal templates where public.
4. Historical seismic acquisition and well chronology from regulator records.
5. Contemporary oil-price and frontier-basin investment conditions when material to negotiations.
6. A clear chronology of boundary disruptions and resolution.
7. Independent academic petroleum-geology literature for the Guyana-Suriname Basin.

Chapter 3 research dossier

Environmental and natural-capital foundations before oil

3.1 Environmental law predates first oil

Parliament records the Environmental Protection Act 1996, establishing the statutory foundation for environmental management, pollution control, impact assessment and sustainable use of natural resources. [S39]

This matters because environmental governance did not begin when oil production began. The analytical task is to determine how a general environmental framework designed before large-scale offshore petroleum was adapted to deepwater projects.

The final chapter should establish:

- EPA institutional authority;
- environmental authorization requirements;
- EIA triggers;
- public participation rules;
- permit enforcement powers;
- pollution and waste provisions;
- appeal rights;
- monitoring and reporting;
- interaction with petroleum approvals.

The existence of the 1996 act should not be treated as evidence of adequate later enforcement. The project-by-project chapter in Part V will examine permit implementation separately.

3.2 The 2009 Low Carbon Development Strategy

Guyana's original Low Carbon Development Strategy predates the 2015 commercial discovery and first oil by a decade. The official LCDS record describes the strategy as beginning in 2009 and the Guyana-Norway partnership as part of the first phase of results-based forest climate finance. [S22, S35]

This timing is central to the flagship argument. Forest conservation, climate finance and national natural-capital policy should not be described as petroleum-offset programmes invented after the oil discovery.

The final section should track:

- the original 2009 policy objectives;
- the 2009 Norway partnership;
- forest-monitoring system development;
- payment conditions;
- safeguards;
- investment channels;
- community and land-title components;
- later transition to LCDS 2030 and jurisdictional carbon markets.

The working source register records the government's 2024 LCDS update as reporting USD 227 million earned for forest climate services for 2009-2015. That figure is a government programme record and should be reconciled with Norway's published disbursement records before the final chapter. [S22]

3.3 Why natural capital belongs in a petroleum case study

A petroleum case study that measures only underground hydrocarbons and financial assets understates national wealth. Guyana's forests, biodiversity, fisheries, agricultural land, freshwater and coastal protection also produce economic and social value.

The final national wealth framework should classify at least:

- petroleum reserves;
- financial assets;
- public infrastructure;
- human capital;
- forests;
- biodiversity;
- fisheries;
- agricultural land;
- water resources;
- coastal protective systems;
- environmental liabilities.

The purpose is not to convert every ecosystem into a speculative monetary value. The purpose is to prevent petroleum revenue from being treated as the only national asset affected by development.

3.4 Coastal exposure

Guyana's coastal settlement pattern makes flood resilience and drainage material to national development. The final chapter should use official coastal-risk, drainage, sea-defence and climate-adaptation sources.

Required evidence includes:

- population distribution by region;
- coastal elevation;
- historical flood events;
- sea-defence coverage;
- drainage and irrigation systems;
- insured and uninsured exposure;
- projected climate impacts;
- adaptation expenditure and delivered outcomes.

This baseline becomes important later when petroleum-financed public investment is evaluated. A sea-defence allocation should be assessed against completed resilient infrastructure and reduced risk, not expenditure alone.

3.5 Environmental baseline for offshore petroleum

The flagship should identify which marine baseline datasets existed before large-scale offshore production and which were produced through project-specific EIAs.

The evidence register should include:

- currents;
- water quality;
- marine mammals;
- fisheries;
- seabirds;
- turtles;
- benthic habitat;
- mangroves;
- coastal protected areas;
- fishing grounds;
- shipping routes;
- sensitive shoreline receptors.

Where baseline data are operator-funded, the paper should record who collected the data, who owns it, whether raw data are public, and whether independent researchers have replicated the findings.

Chapter 4 research dossier

The state-capacity baseline before the petroleum windfall

4.1 Capacity is an institutional system

State capacity is not a single agency. For petroleum governance, it includes the ministry responsible for petroleum policy, licensing authorities, geological institutions, the environmental regulator, revenue administration, customs, central bank, finance ministry, auditor general, procurement bodies, parliament, courts, maritime authorities, emergency services and national statistics.

The final chapter should build a dated institutional map for 2014 and then compare it with 2019 and 2026.

4.2 Legal and fiscal capabilities before commercial production

The existence of the 1986 petroleum law, the 1996 environmental law and older public-finance institutions demonstrates an inherited legal base. [S38, S39]

The more difficult research question is operational depth. The flagship should estimate, where reliable records exist:

- petroleum-specialist staffing;
- environmental technical staff;
- reservoir and petroleum engineering expertise;
- transfer-pricing and international tax capability;
- cost-audit capacity;
- petroleum-law expertise;
- maritime response capability;
- procurement professionals;
- statistical staff;
- public investment appraisal skills.

This should use published staffing and budget evidence rather than retrospective assumptions.

4.3 The discovery-to-first-oil capacity race

The final publication will compare the institutional baseline with the four-year interval from the 2015 discovery to 2019 first oil.

Required evidence includes:

- technical assistance;
- new petroleum units;
- legal reforms;
- contract-administration systems;
- cost-audit arrangements;
- environmental review;
- sovereign-fund design;
- first-cargo measurement;
- central-bank custody;
- EITI implementation;
- local-content policy development.

The core question is where preparation occurred before revenue arrived and where reforms followed later.

4.4 The Natural Resource Fund as a pre-first-oil milestone

Parliament's official record lists the Natural Resources Fund Act 2019 as passed on 3 January 2019 and published on 23 January 2019. The act itself records presidential assent on 23 January 2019. [S37]

This gives the flagship a precise chronology point: the first statutory NRF framework preceded first oil later in 2019.

The analysis must still distinguish:

- legal enactment;
- commencement;
- operational account structure;
- first deposit;
- first investment return;
- first withdrawal;
- later replacement legislation;
- later withdrawal-rule amendments.

This is a central example of the paper's broader method. Early formal preparation is a documented fact. Long-term effectiveness is a separate empirical question.

4.5 Measuring institutional adaptation

The final state-capacity chapter should score no government and rank no political actor. Instead, it should present time-series measures where possible:

- petroleum regulator staffing;
- number of contracts or permits administered;
- audit completion time;
- publication lag;
- environmental inspection frequency;
- procurement processing time;
- investment project completion;
- statistical publication lag;
- unresolved legal or fiscal disputes.

The objective is to test whether institutional capacity kept pace with industry scale.

Part I exhibit programme

Exhibit I-1. Petroleum exploration chronology

Columns:

- date;
- licence/well/project;
- authority;

- operator;
- event type;
- evidence source;
- commercial result;
- legal significance.

Exhibit I-2. Pre-oil institutional map

Nodes:

- Ministry of Natural Resources;
- petroleum administration;
- GGMC/geological data functions;
- EPA;
- GRA;
- Ministry of Finance;
- Bank of Guyana;
- Audit Office;
- Parliament;
- courts;
- maritime and emergency authorities.

Each node should show statutory basis, key petroleum function and pre-2015 capability evidence.

Exhibit I-3. Environmental and natural-capital baseline

Layers:

- forests;
- protected areas;
- coastal population;
- fisheries;
- major rivers;
- oil blocks;
- shore-based petroleum support infrastructure;
- sensitive receptors.

Exhibit I-4. Before/after institutional capability matrix

Compare 2014, 2019 and 2026 for:

- petroleum law;
- revenue fund;
- local content;
- EITI;
- petroleum taxation;
- environmental permitting;
- spill legislation;
- production monitoring;
- public petroleum data;
- contract audit;
- public investment management.

No qualitative score should be assigned unless based on a transparent measurable indicator.

Exhibit I-5. What a newcomer should do before discovery

A staged checklist:

- resolve or manage maritime boundaries;
- maintain a public licence register;
- preserve state ownership of geological data;
- define environmental baseline standards;
- budget scalable technical expertise;
- model several fiscal regimes;
- design contract disclosure rules;
- establish conflict-of-interest controls;
- map emergency response partners;
- avoid debt or recurrent spending against unproven petroleum revenue.

Part I source package currently available

S22: official LCDS 2030 update and historical programme account. S28: pre-oil sovereign wealth fund Green Paper. S35: official petroleum exploration history. S36: official Liza-1 well record. S37: parliamentary NRF Act 2019 record. S38: parliamentary Petroleum Act 1986 record. S39: parliamentary Environmental Protection Act 1996 record. S40: Permanent Court of Arbitration Guyana-Suriname case record.

Part I next-source requirements

1. Original World Bank petroleum exploration promotion project records.
2. Original 1980s petroleum regulations and model agreements.
3. Guyana Geology and Mines Commission historical petroleum datasets.
4. Full PCA Guyana-Suriname award.
5. Norway's official disbursement records under the forest partnership.
6. Historical Bank of Guyana and Ministry of Finance macro series.
7. Historical EPA staffing, budget and authorization records.
8. Independent peer-reviewed Guyana-Suriname Basin geology.
9. Historical public-debt and external-sector records.
10. Coastal flood and sea-defence datasets.
11. Pre-2015 household and labour-force evidence.
12. Historical port, aviation and electricity capacity.

Part I research boundary

No statement in this dossier should be interpreted as a political rating of Guyana's governments before or after the petroleum discovery. The finished case study will document institutional sequence, measurable outputs, evidence gaps and transfer conditions. Disputed interpretations will be attributed to the relevant source.

PART II RESEARCH DOSSIER

Discovery, Contractual Architecture, Cost Recovery and Revenue Assurance

Working research dossier for Chapters 5-9 of the 520-560-page flagship publication. Evidence cut-off: 27 September 2026. Status: research-stage material. External legal and petroleum-fiscal review pending.

Research objective

Part II reconstructs the commercial and legal architecture governing the Stabroek Block without reducing the agreement to a single royalty rate or a single government-take figure.

The final analysis must distinguish:

- information available when the agreement was signed;
- geological and commercial risk at the time;
- signed fiscal terms;
- the mechanics of cost recovery;
- actual project costs submitted by the contractor;
- government audit findings;
- the status of disputed adjustments;
- later model fiscal terms developed under different geological conditions;
- later licences and agreements;
- petroleum revenues actually received.

No overall political score or contract ranking will be assigned.

Chapter 5 research dossier

Liza 2015 and the information shock

The Liza-1 well is recorded by the Petroleum Management Programme as an exploration oil well dated 15 May 2015. [S36]

The final discovery analysis should establish a chronology of information becoming available to both state and contractor:

1. pre-drill geological interpretation;
2. well spud and drilling;
3. hydrocarbon indications;

4. public discovery announcement;
5. appraisal drilling;
6. early resource estimates;
7. commerciality assessment;
8. concept selection;
9. development planning;
10. production-licence approval.

This chronology matters because a negotiating position changes as geological uncertainty declines.

The paper should not infer that information known after appraisal was available to negotiators at an earlier date. The correct counterfactual question is: what would a technically informed government and investor have known at each decision point?

Required evidence:

- original Liza discovery announcements;
- appraisal well records;
- resource-estimate revisions by date;
- oil-price environment during 2015-2016;
- deepwater development cost estimates available at the time;
- comparable frontier-basin fiscal terms;
- financing and execution risk.

Chapter 6 research dossier

The 2016 Stabroek petroleum agreement

The official Stabroek Block register identifies the petroleum agreement date as 27 June 2016 and the effective/licence date as 7 October 2016. [S41]

These dates must remain separate. The audit material for 2018-2020 also treats 7 October 2016 as the effective date for recoverability analysis. [S42]

6.1 Contract crosswalk required

The final flagship needs a clause-level table with at least:

- parties;
- contract area;
- term;
- exploration phases;

- relinquishment;
- work obligations;
- discovery notification;
- commerciality;
- development plans;
- petroleum production licences;
- royalty;
- cost recovery;
- profit oil;
- valuation;
- natural gas treatment;
- taxation;
- accounting procedure;
- audit rights;
- currency;
- imports;
- local purchasing;
- employment and training;
- assignment;
- change in control;
- abandonment;
- insurance;
- environmental duties;
- dispute settlement;
- stabilization;
- confidentiality;
- force majeure;
- effective date.

Every row should include:

- exact article/annex;
- short neutral description;
- fiscal or governance consequence;
- audit evidence;
- later interpretation or dispute;
- transferability note.

6.2 Headline fiscal mechanics

Existing official and audit sources establish the need to model the 75% cost-recovery ceiling as a ceiling, not as an assertion that actual economic costs always equal 75% of revenue.

The flagship should show at least four numerical states:

A. recoverable costs exceed the monthly ceiling; B. recoverable costs are below the ceiling; C. production exceeds sales; D. later project costs replenish the cost bank.

The simplified waterfall is pedagogical only. The final fiscal model must trace actual contract accounting.

6.3 Effective date and recoverability

The government-published 2018-2020 audit material records an exception involving insurance costs incurred before 7 October 2016 and states that, subject to specified pre-contract-cost treatment, costs before the effective date were not recoverable. [S42]

This is important because it demonstrates why the contract chapter must distinguish:

- signing date;
- effective date;
- pre-contract costs expressly recognized;
- costs incurred after effective date;
- costs later challenged by auditors.

The final manuscript should not infer final financial recovery from an audit exception. It must identify whether the contractor agreed, whether a credit was processed, whether government accepted the resolution and whether any dispute remained.

Chapter 7 research dossier

Cost recovery, accounting and the cost bank

Cost recovery is where fiscal theory becomes accounting practice.

The 2018-2020 audit published through the Petroleum Management Programme reviews cost and revenue components and includes detailed exceptions. [S42]

The final chapter should group audit issues by category rather than listing them randomly.

7.1 Proposed audit taxonomy

Allocation across blocks

Costs that benefit multiple licences or operations require a defensible allocation basis.

Timing

Costs incurred before or after the effective date, capitalized costs and costs recognized in later periods require correct treatment.

Duplicate or erroneous charges

Duplicate invoices, coding mistakes and costs assigned to the wrong block should be isolated.

General and administrative costs

Office, overhead and shared-services allocation should follow contractual accounting rules.

Emergency and environmental costs

Studies and response systems may benefit more than one project or licence and should be allocated consistently with the agreement.

Valuation

Cost oil and profit oil depend on volume and price rules. Valuation methods must be audited separately from cost classification.

Affiliate transactions

Related-party charges require particular attention to arm's-length principles and contract rules.

Decommissioning

Future abandonment and restoration obligations require explicit treatment.

7.2 The cost bank

A transparent cost-bank model should include:

Opening unrecovered costs + new recoverable costs + permitted adjustments

- cost oil recovered

= closing unrecovered costs.

The model needs project and category tags.

A single block-wide number can hide the effect of new project approvals. If development spending rises while older costs fall, government profit-oil entitlement can change materially without any amendment to the contract.

7.3 Audit status categories

Every audit item in the paper should carry one of these labels:

- identified;
- contractor response received;
- government response recorded;
- credit accepted;
- credit posted;
- dispute unresolved;
- arbitration;
- final settlement;
- no public evidence of disposition.

This prevents a proposed adjustment from being reported as money already recovered.

Chapter 8 research dossier

Later model terms and changing information

Later fiscal terms were developed after commercial success had reduced frontier uncertainty.

The flagship should compare agreement generations without implying that later terms were available under identical conditions in 2016.

Comparison fields should include:

- royalty;
- cost-recovery ceiling;
- profit sharing;
- corporate tax;
- signing bonus;
- acreage size;
- relinquishment;
- ring-fencing;
- local-content obligations;
- environmental obligations;
- decommissioning;
- audit rights;
- beneficial ownership;
- dispute settlement.

The core research principle is temporal fairness. Contract terms negotiated after multiple commercial discoveries should not be used as if they were contemporaneous market evidence for an earlier frontier negotiation.

At the same time, the historical agreement should be assessed against comparable agreements and available expertise at the time, not insulated from scrutiny by uncertainty.

Both propositions can be true.

Chapter 9 research dossier

Contract governance for a newcomer

A transferable contract-governance system should focus on process quality rather than copying Guyana's percentages.

Before negotiation

- establish a multidisciplinary negotiating team;
- preserve state control over geological data;
- obtain independent reservoir and petroleum-economics advice;
- model investor returns under several price and discovery outcomes;
- map fiscal interactions with tax law;
- identify environmental and decommissioning liabilities;
- publish conflict-of-interest rules;
- define confidentiality boundaries.

During negotiation

Maintain a decision log showing:

- clause proposed;
- government's objective;
- contractor position;
- independent technical assessment;
- fiscal effect;
- legal effect;
- environmental effect;
- final negotiated result;
- approval authority.

Before signature

Complete:

- fiscal model review;
- legal consistency review;
- environmental-law review;
- tax review;
- audit-rights review;
- beneficial-ownership check;
- dispute-resolution analysis;
- decommissioning-liability review;
- parliamentary/public disclosure assessment.

After signature

Create a contract administration calendar:

- work commitments;
- relinquishment dates;
- development submissions;
- production licences;
- annual budgets;
- cost statements;
- audit windows;
- royalty due dates;
- tax filings;
- environmental reporting;
- training obligations;
- decommissioning provisions.

Initial documentary findings from government-published audit material

The 2018-2020 audit provides useful evidence about the operation of the agreement in practice.

It includes exceptions relating to:

- costs coded to the wrong block;
- allocation of office-related construction;
- allocation of emergency-response study costs;
- insurance costs before the effective date;
- duplicate charges.

The paper should use these examples to demonstrate the role of audit rights rather than to generalize from individual exceptions.

The audit also discusses government profit-oil validation and cost-oil valuation. This gives the flagship a basis for explaining how volume, valuation and cost accounting interact.

A key publication control follows:

Audit findings are evidence of an audit process. They are not, by themselves, evidence that every exception was finally resolved or collected.

Contract chapter exhibits

Exhibit II-1. Agreement chronology

- original block agreement;
- 2015 discovery;
- June 2016 agreement execution;
- October 2016 effective date;
- production licence;
- later amendments if any;
- later model agreement;
- later block agreements.

Exhibit II-2. Fiscal waterfall

Show:

- gross petroleum available for allocation;
- royalty;
- cost recovery;
- residual profit oil;
- government share;
- contractor share;
- tax treatment;
- cash timing.

Exhibit II-3. Cost-bank bridge

Annual or quarterly:

- opening unrecovered costs;

- additions;
- disallowed/credited items;
- recoveries;
- closing balance.

Exhibit II-4. Audit exception register

Fields:

- audit period;
- exception;
- amount;
- contract basis;
- contractor response;
- government status;
- public resolution evidence.

Exhibit II-5. Agreement-generation crosswalk

2016 Stabroek versus later model terms, with a separate column describing the information environment at each date.

Required quantitative tests

1. Oil price at USD 40, 60, 80 and 100 per barrel.
2. Cost-recovery use at 25%, 50%, 65% and 75%, where contractually relevant.
3. New-project cost-bank additions.
4. Production downtime.
5. Sales timing versus production timing.
6. Government lift timing versus cash receipt.
7. Valuation difference under third-party and affiliate sale cases where permitted by evidence.
8. Disallowed-cost sensitivity.
9. Decommissioning liability scenarios.
10. Discounted cash flow under several project-cost assumptions.

All model outputs must be labelled illustrative unless calibrated to published actual data.

Source package currently available

S02: official Stabroek agreement publication page. S14: later model fiscal terms. S15: Guyana Revenue Authority royalty explanation. S29: IMF 2023 Article IV analysis of fiscal architecture. S41: Stabroek Block official register. S42: 2018-2020 cost-recovery audit publication. S43: regulatory document index including cost audits and licences.

Remaining Part II evidence gates

1. Machine-extract and independently review the complete executed 2016 agreement.
2. Build the full article-by-article crosswalk.
3. Obtain and review the 1999-2017 final audit.
4. Determine final public status of 2018-2020 exceptions.
5. Determine status of 2021-2022 and later audits.
6. Reconcile cost-oil volumes to monthly production reports.
7. Reconcile profit-oil entitlement to government lift records.
8. Verify later signed PSAs, not only model documents.
9. Obtain independent petroleum-fiscal expert review.
10. Obtain independent petroleum-law review.

Publication boundary

This dossier does not determine whether the 2016 agreement was "good" or "bad" as an overall political judgment.

The flagship will provide the contractual facts, fiscal mechanics, information available at relevant dates, actual audit evidence, later reforms, international comparison mechanisms and downside cases. Readers can assess policy choices from the documented record.

PART III RESEARCH DOSSIER

The Pre-First-Oil Window, First Revenue and Production Scale-Up

Working research dossier for Chapters 10-14 of the 520-560-page flagship publication. Evidence cut-off: 27 September 2026. Status: research stage. Monthly production and cargo-level reconciliation remain to be completed.

Research objective

Part III reconstructs the interval from commercial discovery through large-scale production.

The analytical sequence is:

commercial discovery → appraisal → development approval → environmental authorization → production licence → project construction → first oil → production ramp → cargo lifting → government entitlement → petroleum cash receipt.

Each stage has a different date and a different source.

The final publication will avoid a common error: using a single "first oil" date as if it represented the full development and revenue cycle.

Chapter 10 research dossier

Discovery to development decision

Liza-1 is recorded in the official well register with a well date of 15 May 2015. [S36]

The first development followed a sequence of:

- appraisal;
- field-development planning;
- EIA;
- environmental authorization;
- production licence;
- engineering/procurement;
- FPSO conversion/construction;
- subsea installation;
- drilling;
- commissioning.

The official Liza Phase 1 permit record states that the environmental application was submitted on 5 July 2016 and the environmental permit was issued on 1 June 2017. [S48]

The official petroleum-management site publishes the Liza Phase 1 production licence and Phase 2 addendum. [S86]

The final chronology must identify the exact production-licence issue date from the executed licence, rather than relying on later summaries.

Development-decision evidence

For each project the final table should include:

- commercial discovery/appraisal basis;
- field development plan;
- environmental application;
- EIA approval;
- petroleum production licence;
- final investment decision;
- sanctioned capital;
- FPSO name;
- design capacity;
- estimated wells;
- first oil;
- actual ramp-up;
- subsequent optimization.

Where company and government dates differ because one refers to FID and another to a production licence, both should be preserved.

Chapter 11 research dossier

The pre-first-oil governance window

The four-year interval between the 2015 commercial discovery and December 2019 first production gave government a finite preparation period.

A central documented milestone is the first Natural Resource Fund law, passed in January 2019 before recurring petroleum production. [S37]

The pre-oil Green Paper also documents government work on sovereign-wealth-fund design beginning before first oil, including external technical assistance and inter-agency work. [S28]

This chapter should build an institution-by-institution chronology:

Petroleum administration

- licensing;
- development-plan review;
- production measurement;
- operator reporting;
- contract administration.

Fiscal

- royalty assessment;
- cost audit;
- profit-oil entitlement;
- petroleum taxation;
- sales/marketing.

Sovereign revenue

- NRF legislation;
- central bank operational agreement;
- bank accounts;
- investment rules;
- reporting.

Environmental

- project EIAs;
- permits;
- spill plans;
- regulator staffing;
- monitoring.

Transparency

- contract publication;
- EITI;
- public petroleum data;
- parliamentary reporting.

The chapter should show which systems were operational before December 2019 and which were strengthened later.

Chapter 12 research dossier

First oil and the first revenue cycle

Production from Liza Phase 1 began in December 2019. Official and operator records consistently place the first project startup in that month. [S81, S83]

First production is not first government cash.

A cargo-based accounting chain should include:

1. production measured on the FPSO;
2. crude stored;
3. cargo nominated;
4. entitlement assigned;
5. tanker lifted;
6. cargo priced;
7. invoice settled;
8. royalty/profit-oil receipt recognized;
9. money deposited in the NRF.

The 2020 audited NRF statements show the practical timing difference. The fund recognized four 2020 profit-oil lifts while the fourth was still a receivable at year end and settled in 2021. [S45]

This makes the first-year case useful for newcomer countries.

A petroleum ministry, revenue authority, central bank and finance ministry must agree on:

- the production period;
- lift date;
- price;
- entitlement;
- invoice date;
- settlement date;
- accounting year.

Without this bridge, public reports can contain different numbers that are each correct under different accounting bases.

Chapter 13 research dossier

FPSO-by-FPSO production expansion

The production sequence verified so far is:

Liza Destiny

First production: December 2019. [S81, S83]

Liza Unity

First production: 11 February 2022. [S82]

Prosperity

First oil at Payara: November 2023. The government release dated 14 November 2023 states first oil had occurred and identifies Destiny and Unity as the earlier projects. [S83]

ONE GUYANA

Yellowtail first production: 8 August 2025. ExxonMobil stated that the fourth development increased combined installed capacity above 900,000 barrels per day. [S84]

In November 2025 the operator reported a point-in-time daily production milestone of 900,000 barrels. [S85]

The regulator's oil-production dataset covers actual production from 20 December 2019 through 31 July 2026. [S81]

The final production chapter should use the regulator's monthly data to calculate:

- monthly barrels;
- daily average;
- rolling 12-month production;
- FPSO share;
- production ramp;
- downtime;
- year-over-year growth.

Installed capacity and reported actual production must remain separate.

Project table to build

The final publication will update status only when a dated primary source establishes a startup.

Chapter 14 research dossier

Production data quality

Production data should be stored at the finest reliable time resolution available.

The core dataset needs:

- date;
- field/project;
- FPSO;
- oil produced;
- water produced;
- gas produced;
- gas reinjected;
- gas flared;
- downtime;
- oil lifted.

Five measures that must not be confused

Nameplate/design capacity

Engineering production capability under specified conditions.

Daily production

Observed output on a particular day.

Monthly daily average

Monthly barrels divided by days.

Annual average

Annual barrels divided by 365/366.

Cargo lift

A shipment event that draws from stored produced crude and need not match the same month's production.

A headline such as "900,000 barrels per day" must therefore identify whether it is:

- installed capacity;
- target;
- one-day milestone;
- period average.

Production and revenue bridge

The flagship should build a reconciled table with three linked but distinct layers.

Layer A. Physical production

- monthly barrels by FPSO;
- total field production;
- downtime.

Layer B. Commercial lifting

- cargo count;
- barrels lifted;
- entitlement owner;
- lift date;
- price.

Layer C. Public cash

- government profit-oil invoice;
- royalty;
- settlement date;
- NRF deposit.

The bridge will explain timing differences instead of forcing the three layers into the same calendar month.

Production-scale regulatory pressure

Production growth also changes administrative workload.

Each additional project creates:

- production reporting;
- cost statements;
- environmental monitoring;
- permit review;
- spill-response exposure;
- decommissioning obligations;
- local-content plans;
- cargo sales;
- tax filings.

The flagship should compare regulatory staffing and audit resources against:

- number of FPSOs;
- number of active wells;
- production volume;
- annual contractor spend.

This is a more meaningful capacity test than the number of laws enacted.

New-producer blueprint extracted from Part III

Before first oil, a newcomer should test:

1. production-meter verification;
2. crude-quality measurement;
3. cargo nomination;
4. sales/pricing protocol;
5. royalty invoice;
6. cost-accounting submission;
7. government entitlement calculation;
8. bank settlement;
9. sovereign-fund deposit;
10. public notification;
11. environmental incident reporting;
12. regulator emergency command.

The country should run a simulated first cargo before the actual first cargo.

Required Part III exhibits

1. Liza discovery-to-first-oil critical path.
2. Institution-readiness timeline 2015-2019.
3. FPSO commissioning timeline.
4. Monthly realized production by project.
5. Installed capacity versus actual output.
6. Cargo lift versus production bridge.
7. Profit-oil lift versus NRF deposit.
8. Revenue-recognition timing example from 2020.
9. Regulatory-workload growth.
10. First-oil readiness checklist.

Remaining Part III evidence gates

1. Extract exact Liza Phase 1 production-licence date and conditions.
2. Extract Liza Phase 2 addendum.
3. Extract Payara production licence.
4. Extract Yellowtail production licence.
5. Complete monthly production series through July 2026.
6. Build cargo-lift series.
7. Link government lifts to cash deposits.
8. Record project FIDs and sanctioned costs by date.
9. Compile downtime and production optimization.
10. Add independent petroleum-engineering review.

Publication boundary

Part III will not describe installed capacity as actual annual output.

It will not treat operator forecasts as achieved production.

It will not describe first oil as the date government first received cash unless the cash record proves that timing.

PART IV RESEARCH DOSSIER

The Natural Resource Fund, Fiscal Absorption, Public Investment and National Wealth

Working research dossier for Chapters 15-20 of the 520-560-page flagship publication. Evidence cut-off: 27 September 2026. Status: research stage. Annual audited account extraction and public-investment outcome review remain in progress.

Research objective

Part IV asks what happens after petroleum revenue becomes public financial wealth.

The central accounting sequence is:

petroleum entitlement → sale and settlement → NRF deposit → investment return → statutory withdrawal calculation → parliamentary appropriation → Consolidated Fund transfer → procurement → completed public asset or service → maintenance → measurable outcome.

Each stage has a different evidence source.

A country can have a well-reconciled sovereign fund and weak project delivery. It can also have urgent development needs and choose a larger legal withdrawal envelope while maintaining transparent accounting. The final paper will document choices and effects without assigning an overall political score.

Chapter 15 research dossier

NRF law through time

The first Natural Resource Fund law was enacted in January 2019 before first oil later that year. [S37]

The framework was subsequently replaced in 2021 and amended again in 2024. [S04, S05]

The flagship should treat these as separate legal generations.

Legal crosswalk fields

For each legal version:

- date passed;
- date assented;
- commencement;
- fund objective;

- petroleum revenue definition;
- custodian;
- board/governance bodies;
- investment mandate;
- reporting;
- audit;
- withdrawal formula;
- parliamentary authorization;
- emergency provisions;
- prohibited uses;
- public notification.

A later law should not be retroactively applied to earlier withdrawals.

Operational agreement

The Petroleum Management Programme publishes the December 2019 operational agreement between the Minister of Finance and the Bank of Guyana for management of the NRF. [S04/S43 context]

The final legal analysis should connect statutory responsibility to:

- bank custody;
- investment execution;
- records;
- reporting;
- expenses;
- internal controls.

Chapter 16 research dossier

Reconstructing fund cash flows

The NRF_RECONCILIATION_WORKBOOK.md is the accounting companion to this chapter.

The core distinction is between:

- cash receipts;
- revenue recognized;
- receivables;
- investment income;

- market value;
- withdrawals;
- closing net assets.

The 2020 audited statement demonstrates the distinction clearly. Four profit-oil lifts were recognized during 2020, while the fourth was still a receivable at year end. [S45]

This is why a government petroleum-revenue chart must state whether it uses:

- production;
- cargo lifts;
- invoiced entitlement;
- recognized accounting revenue;
- cash received.

Annual reconciliation standard

For each year: Opening audited balance + petroleum revenue recognized + investment income + other accounting movements

- withdrawals
- expenses

= closing audited balance.

A separate cash bridge will reconcile: Opening cash + cash deposits + cash investment receipts

- cash transfers
- cash expenses

= closing cash.

Restated comparatives from later audited statements take precedence over earlier presentation where the auditor explicitly restates the figures.

Chapter 17 research dossier

Saving, spending and intergenerational equity

Petroleum is a depleting asset.

When crude is extracted, the country exchanges an underground asset for:

- contractor/private income;
- public petroleum revenue;
- taxes;

- domestic activity.

The public-policy question is how public petroleum revenue is converted into durable wealth.

Potential destinations include:

- external financial assets;
- public infrastructure;
- human capital;
- debt reduction;
- social transfers;
- climate resilience;
- productive institutions.

Each has a different return profile and risk.

Intergenerational framework

The final paper should model:

- reserve life;
- production path;
- oil-price uncertainty;
- population growth;
- existing infrastructure gap;
- debt;
- public-investment return;
- financial-asset return;
- maintenance liabilities;
- disaster exposure.

No universal savings percentage follows from the existence of oil.

International comparison

Norway links normal-year fiscal use to long-run expected real returns from a mature external fund. [S74]

Ghana separates stabilization and heritage objectives. [S75]

Timor-Leste uses a petroleum fund with explicit liquidity management and substantial budget relevance. [S77, S78]

Guyana's architecture has its own withdrawal rules and development context.

The comparison should focus on fiscal mechanisms and conditions, not a ranked verdict.

Chapter 18 research dossier

The non-oil primary balance and absorption

An economy can face spending constraints before it faces a financing constraint.

The IMF July 2026 mission statement discusses absorption capacity, investment monitoring and a possible longer-run fiscal guide linked to the non-oil primary balance. [S64]

This is IMF staff analysis.

The flagship should explain the non-oil primary balance in plain terms:

Non-oil revenue

- non-interest public expenditure

= non-oil primary balance.

A larger deficit indicates more government spending financed by petroleum or borrowing rather than recurring non-oil revenue.

Absorption constraints

Rapid spending may encounter:

- limited contractors;
- limited engineers;
- imported materials;
- port congestion;
- land constraints;
- public-sector vacancies;
- procurement delays;
- inflation.

The correct response is an empirical policy question.

The research should measure:

- tender participation;
- contract variation;
- completion delays;
- construction inflation;
- wage pressure;
- import growth;
- project commissioning.

Chapter 19 research dossier

Public investment management

A withdrawal from the NRF does not identify which project ultimately receives the money.

Once transferred to the Consolidated Fund, public spending follows budget and procurement systems.

The flagship should therefore create a separate public-investment evidence register.

Project lifecycle

Need → appraisal → budget → procurement → contract → construction → completion → commissioning → operation → maintenance → service outcome.

A project should not be marked complete at contract award.

Outcome categories

For a road:

- travel time;
- vehicle operating cost;
- safety;
- flood resilience;
- maintenance condition.

For a hospital:

- staffed beds;
- waiting time;
- service availability;
- clinical outcomes.

For electricity:

- dependable capacity;
- generation;
- outages;
- tariff;
- losses.

For water:

- access;

- pressure;
- quality;
- continuity.

For schools:

- attendance;
- completion;
- learning;
- teacher availability.

Lifecycle liabilities

Every capital asset creates future costs:

- maintenance;
- staffing;
- utilities;
- insurance;
- replacement.

The national-balance-sheet chapter should record those liabilities.

Chapter 20 research dossier

National balance-sheet accounting

GDP is an annual flow.

A national balance sheet asks what assets and liabilities remain.

Assets

Financial

- NRF investments;
- reserves;
- other public financial assets.

Produced capital

- roads;
- bridges;

- ports;
- power systems;
- hospitals;
- schools;
- water systems.

Human capital

- education;
- health;
- technical capability.

Natural capital

- petroleum remaining;
- forests;
- biodiversity;
- fisheries;
- water;
- agricultural land;
- coastal protection.

Liabilities

- public debt;
- maintenance obligations;
- pensions;
- contractual guarantees;
- decommissioning exposure where borne publicly;
- environmental remediation exposure;
- disaster liabilities.

Petroleum depletion

Petroleum production reduces the remaining physical resource.

A comprehensive wealth measure should avoid describing gross petroleum revenue as a pure addition to wealth without recognizing depletion.

The country becomes wealthier over time when the converted assets and capabilities deliver durable value exceeding associated liabilities and resource depletion, subject to distribution and environmental outcomes.

The final research should avoid a single aggregate monetary valuation where evidence is too uncertain. Physical indicators and separate asset classes are acceptable.

NRF dashboard design

The public dashboard should contain at least:

Petroleum receipts

- royalty;
- profit oil;
- taxes where applicable;
- cash deposit date.

Fund performance

- opening balance;
- deposits;
- realized return;
- unrealized return;
- expenses;
- withdrawals;
- closing value.

Legal withdrawal

- formula;
- calculated ceiling;
- amount appropriated;
- amount transferred.

Public expenditure

This should link to a separate budget/project dataset rather than imply that fund money remains traceable to one project after consolidation unless the budget system supports that mapping.

Adverse fiscal scenarios

Price shock

A sustained price decline reduces petroleum receipts.

Test:

- legal withdrawal space;
- essential expenditure;
- project postponement;
- debt service;
- fund liquidity.

Production interruption

A technical outage reduces cargo availability.

Test:

- cash timing;
- budget assumptions;
- fund buffer;
- emergency expenditure.

Market loss

A sovereign fund invested in financial markets can experience negative returns.

Test:

- liquidity;
- withdrawal rule;
- forced-sale risk;
- rebalancing.

Construction inflation

High public investment raises project costs.

Test:

- nominal budget;
- physical output;

- contractor capacity;
- maintenance affordability.

Natural disaster

Test simultaneous:

- petroleum revenue disruption;
- emergency spending;
- infrastructure repair;
- fund withdrawal.

New-producer blueprint extracted from Part IV

Before first revenue, a newcomer should:

1. enact a lawful petroleum-revenue custody structure;
2. define petroleum revenue precisely;
3. define cash and accrual reporting;
4. publish deposit notices;
5. establish independent annual audit;
6. define withdrawal authority;
7. integrate withdrawals with the ordinary budget;
8. establish public-investment appraisal;
9. stress-test low-price and disaster cases;
10. publish a national-balance-sheet framework.

The numerical withdrawal rule remains a national political and fiscal choice.

Required Part IV exhibits

1. NRF legal-version timeline.
2. Annual audited stock-flow table.
3. Cash versus accrual reconciliation.
4. Withdrawal-rule crosswalk.
5. Petroleum deposits versus withdrawals.
6. Market-value versus cash balance.
7. Non-oil primary balance.
8. Absorption-capacity indicators.

9. Public investment lifecycle.
10. National balance sheet.
11. Oil-price/fund stress tests.
12. Disaster plus revenue-shock scenario.

Remaining Part IV evidence gates

1. Extract audited NRF accounts 2022-2025.
2. Verify legal commencement/version dates.
3. Reconcile annual receivables.
4. Reconcile all withdrawals to budget notifications.
5. Build non-oil primary-balance series.
6. Build project-level public investment register.
7. Verify lifecycle operating costs.
8. Build national wealth baseline.
9. Complete fund-stress model review.
10. Obtain independent accounting/public-finance review.

Publication boundary

Part IV will not equate a growing fund balance with complete development success.

It will not equate a large withdrawal with poor policy or a small withdrawal with good policy.

It will document the fiscal rule, conditions, trade-offs, accounting evidence and observed investment outcomes.

PART V RESEARCH DOSSIER

Environmental Governance, Offshore Risk, Financial Responsibility and Climate Coexistence

Working research dossier for Chapters 21-27 of the 520-560-page flagship publication. Evidence cut-off: 27 September 2026. Status: research stage. External environmental-law, marine-science and spill-response review pending.

Research objective

Part V tests the environmental governance system as a sequence rather than a label.

The sequence is:

baseline science → environmental application → environmental impact assessment → public participation → permit → project design controls → operating monitoring → incident reporting → regulator inspection → enforcement → remediation → financial responsibility → long-term ecological outcome.

The existence of one link does not establish performance of the full chain.

The final chapter will therefore distinguish legal requirements, permit conditions, operator implementation, regulator action, observed environmental outcomes and unresolved evidence.

Chapter 21 research dossier

The environmental authorization system

Guyana's environmental framework predates offshore oil production. The Environmental Protection Act 1996 established the general statutory basis for environmental management. [S39]

The Liza Phase 1 record gives a dated example of the environmental authorization chain. The official permit page records an application submitted on 5 July 2016, an approved EIA and management plan dated 1 June 2017, and an environmental permit issued on the same date. The management plan referenced a revised Oil Spill Response Plan and Wildlife Response Plan. The permit was modified in May 2021. [S48]

The final chapter should map statutory roles across:

- Environmental Protection Agency;
- Ministry responsible for petroleum;
- maritime authorities;

- Civil Defence Commission;
- coast guard and emergency services;
- public-health agencies;
- fisheries authorities;
- local government;
- neighbouring-state notification channels.

The chapter should also separate:

- environmental authorization;
- petroleum production licence;
- occupational safety;
- maritime navigation;
- oil-spill liability;
- emergency command.

These legal functions may interact but are not interchangeable.

Public participation

The Yellowtail public notice records an EPA determination that an EIA was required and invited written public submissions during the statutory review period. [S52]

A complete participation assessment should measure more than whether a notice appeared.

Required fields:

- notice date;
- consultation locations;
- language and accessibility;
- stakeholder groups;
- questions raised;
- written response;
- changes made to project design or permit;
- unresolved objections;
- appeal route.

The flagship should identify whether consultation occurred before material project decisions became irreversible.

Chapter 22 research dossier

Project-by-project environmental evidence

Environmental controls should be assessed by development, not generalized across the Stabroek Block.

The evidence register currently confirms:

- Liza Phase 1 original environmental permit and later modification;
- an official index containing Liza Phase 2 amended permit/EIA material;
- an official index containing Payara environmental permit/EIA material;
- Yellowtail EIA application/public notice;
- Uaru environmental permit;
- Whiptail environmental permit publication. [S48-S52]

The final project matrix should include:

- application date;
- EIA requirement decision;
- terms of reference;
- consultant;
- public consultation;
- final EIA;
- permit issue date;
- permit expiry;
- amendment date;
- production licence date;
- flaring condition;
- produced-water condition;
- waste condition;
- oil-spill plan;
- wildlife response;
- monitoring plan;
- independent audit requirement;
- financial assurance;
- inspection record;
- incident record.

Liza Phase 1 as an operating example

The 2021 EPA notice says the agency modified Liza Phase 1's permit as the project approached the EIA's estimated flaring level. The modified permit added emissions reporting requirements, flaring time constraints and a US\$30/tCO₂e payment beyond specified timelines. [S55]

This provides a testable regulatory chain:

- EIA estimate;
- observed operating issue;
- regulator action;
- modified permit;
- financial consequence;
- later performance.

The final paper should reconstruct actual monthly flaring and regulator payments before drawing conclusions about effectiveness.

Uaru as a later permit-generation example

The Uaru permit is effective from 27 April 2023 through 26 April 2028 and contains explicit EPA powers relating to sampling, analysis, annual review and permit modification. [S50]

The Uaru production licence separately requires environmental-law compliance and response capability. [S56]

A later-generation permit should not automatically be assumed stronger in practice. The flagship should compare measurable conditions and monitoring requirements across permit generations.

Chapter 23 research dossier

Oil-spill prevention, preparedness and response

Offshore spill governance has at least five layers:

1. prevention through engineering and operating controls;
2. operator emergency response;
3. national incident command;
4. regional/transboundary cooperation;
5. financial resources for cleanup and compensation.

A robust case study should test all five.

Response resources

The Uaru production licence requires expert personnel and equipment to prevent or respond to petroleum releases and includes capping-stack and dispersant provisions. [S56]

The final review must verify implementation:

- location of response equipment;
- ownership;
- service contracts;
- maintenance records;
- mobilization time;
- stock quantities;
- expiry and replenishment;
- trained personnel;
- drill history;
- regulator observation.

A written obligation is not the same as demonstrated readiness.

Worst-case discharge modelling

EIA spill models should be recorded with:

- release location;
- release duration;
- release rate;
- oil properties;
- season;
- wind;
- current data;
- weathering assumptions;
- response assumptions;
- shoreline probabilities;
- transboundary paths.

The final report should not repeat a model's output as a guaranteed prediction. It should compare assumptions against measured regional oceanographic data and explain uncertainty.

National command

The 2025 oil-pollution law adds a dedicated statutory layer to Guyana's oil-spill framework. Parliament records Act 6 of 2025 as passed and published in May 2025. [S53, S54]

The research register does not yet treat the full regime as operational. Before the final chapter, the project must verify commencement, regulations, appointment of the competent authority, forms, penalties, response-plan requirements and financial-responsibility rules.

Chapter 24 research dossier

Liability, insurance and financial assurance

Four concepts are commonly collapsed into one number:

- legal liability;
- insurance coverage;
- parent or affiliate financial assurance;
- cash available immediately after an incident.

They are not the same.

The final chapter should ask:

- Who is legally liable?
- For which categories of damage?
- Is liability capped?
- Which losses are insured?
- What exclusions apply?
- What is the aggregate limit?
- Is a parent guarantee callable directly?
- Who determines the amount due?
- Which jurisdiction governs disputes?
- How quickly is cash available?
- What public funds might be required before recovery?

Court decisions

A 2026 appellate dispute concerning financial assurance is material to the chapter, but the current source package includes only government characterization and secondary reporting. The final legal interpretation must use an authenticated judgment.

Until then, the manuscript should not state an independent conclusion about the legal adequacy of the financial-assurance regime.

Financial-stress framework

The flagship should present illustrative layers:

- operator working capital;
- insurance;
- parent support;
- contractual response services;
- public emergency expenditure;
- compensation;
- restoration;
- long-term monitoring.

Scenario values should be illustrative unless tied to public instruments.

Chapter 25 research dossier

Coastal, fisheries and biodiversity exposure

The ecological consequences of an offshore incident depend on where oil travels and what resources lie in its path.

The final environmental atlas should identify:

- artisanal and commercial fishing grounds;
- nursery and spawning areas;
- mangroves;
- coastal wetlands;
- marine protected areas;
- seabird areas;
- marine mammal habitat;
- turtle migration;
- ports;
- water intakes;
- tourism assets;
- neighbouring coastlines.

Fisheries

A strong fisheries analysis should separate:

- reported catch;
- landed value;
- employment;
- subsistence use;
- export markets;
- seasonality;
- vessel size;
- spatial distribution.

Compensation design requires evidence linking an incident to measurable loss. Baseline fishing effort and catch data therefore have value before an incident occurs.

Biodiversity

Guyana's biodiversity analysis should link offshore activity to national conservation policy without assuming either compatibility or incompatibility.

The paper should assess:

- project footprint;
- vessel traffic;
- underwater noise;
- light;
- discharge;
- spills;
- cumulative project effects;
- climate-related ecological pressures.

Chapter 26 research dossier

LCDS, forest finance and petroleum

Guyana's original Low Carbon Development Strategy began before commercial offshore discovery. This prevents the research from describing the strategy as a response created to offset petroleum production. [S22]

The later LCDS 2030 framework and jurisdictional forest-credit activity should be treated as a separate environmental-finance system.

A complete accounting bridge should show:

- forest-credit vintage;
- issuance;
- sale agreement;
- minimum contracted value;
- cash received;
- registry transfer;
- community allocation;
- national allocation;
- project spending;
- independently measured outcomes.

The working research already distinguishes a contracted purchase commitment from realized cash.

Petroleum and forest accounting boundaries

The final paper should separately disclose:

- upstream petroleum operational emissions;
- electricity and shorebase emissions;
- flaring;
- methane where measured;
- land-use emissions;
- forest-carbon credits;
- downstream combustion of exported petroleum.

No simple subtraction should be used to claim national carbon neutrality unless the underlying accounting standard supports the boundary and all material emissions are included.

Chapter 27 research dossier

Energy transition and emissions accounting

The environmental case includes more than oil-spill risk.

Offshore production affects:

- direct fuel combustion;
- flaring;
- methane;

- power use;
- shipping;
- aviation;
- material and construction emissions.

National energy policy separately affects:

- electricity generation;
- transmission;
- household energy cost;
- reliability;
- renewable integration;
- industrial demand.

The final paper should state which emissions are inside Guyana's national inventory and which occur after petroleum export.

Measurement hierarchy

For emissions:

1. direct measured fuel and flare data;
2. project-specific engineering calculations;
3. regulator-reviewed reports;
4. standard emission factors;
5. broad estimates.

Every chart should identify its measurement method.

Cumulative environmental governance

Six or more offshore developments cannot be assessed only as isolated projects.

The cumulative chapter should examine:

- vessel traffic;
- noise;
- air emissions;
- shorebase expansion;
- waste;
- cumulative spill probability;
- emergency-response concurrency;
- regulator workload;

- overlapping fishery effects;
- cumulative greenhouse-gas emissions.

A permit-by-permit system needs a parallel basin-wide view.

New-producer blueprint extracted from Part V

A newcomer country should establish the following before first production:

1. legally defined environmental authority;
2. marine baseline data owned or accessible by the state;
3. transparent EIA terms of reference;
4. documented public consultation;
5. project-specific permit conditions;
6. independent monitoring access;
7. incident-reporting deadlines;
8. spill command structure;
9. capping and containment capability;
10. regional response agreements;
11. enforceable financial assurance;
12. claims and compensation process;
13. restoration standards;
14. published annual compliance reporting;
15. rule for permit amendment when operating evidence changes.

For an island state, additional emphasis belongs on:

- reef systems;
- tourism dependence;
- hurricane disruption;
- single-port risk;
- limited storage;
- cross-border currents;
- imported response equipment.

Required Part V exhibits

1. Environmental-law chronology.
2. Project-by-project permit matrix.
3. EIA/public-participation timeline.

4. Flaring-control chronology.
5. Produced-water standards and volumes.
6. Spill command structure.
7. Response-equipment locations and mobilization time.
8. Financial-assurance layers.
9. Marine sensitivity atlas.
10. Oil-spill scenario pathways.
11. Environmental incident register.
12. Forest-credit and petroleum-accounting boundary.
13. Cumulative offshore-activity dashboard.

Remaining Part V evidence gates

- extract full Liza Phase 2 permit;
- extract full Payara permit;
- obtain and extract Yellowtail final permit;
- extract full Whiptail permit;
- collect environmental audit reports;
- collect inspections and regulator compliance notices;
- build observed flaring time series;
- build produced-water time series;
- verify Oil Pollution Act commencement;
- obtain authenticated 2026 appellate judgment;
- inspect public insurance/guarantee instruments;
- obtain marine and fisheries independent research;
- compare spill modelling with regional oceanography;
- obtain evidence of drills and response readiness;
- seek external environmental-law and marine-science review.

Publication boundary

Part V will not call Guyana's environmental system a success or failure as an overall political conclusion.

It will document the legal architecture, permit evolution, observed operating data, regulator actions, unresolved liabilities, environmental outcomes, stakeholder concerns and the conditions another country would need to adapt the mechanisms responsibly.

PART VI RESEARCH DOSSIER

Local Content, Domestic Value, Workforce, Household Welfare and Indigenous/Hinterland Outcomes

Working research dossier for Chapters 28-33 of the 520-560-page flagship publication. Evidence cut-off: 27 September 2026. Status: research stage. Independent labour, household-economics and Indigenous-rights review pending.

Research objective

Petroleum development affects national welfare through several channels:

- government revenue;
- direct employment;
- supplier contracts;
- wages;
- training;
- public investment;
- price changes;
- land and housing markets;
- infrastructure;
- community programmes;
- environmental effects.

These channels should not be collapsed into one local-content expenditure number.

The final flagship will distinguish:

- procurement booked to a Guyanese supplier;
- local beneficial ownership;
- imported content;
- domestic labour;
- domestic wages;
- domestic profit retained;
- domestic manufacturing or technical value added;
- training;
- permanent capability after a contract ends.

Chapter 28 research dossier

Local content law and implementation

Parliament records the Local Content Act 2021 as Act 18 of 2021, passed on 29 December 2021 and published on 4 January 2022. [S57]

The official Local Content Register provides a system for supplier certification and employment registration. [S58]

The final legal chapter should extract:

- statutory definition of Guyanese national;
- definition of Guyanese company;
- ownership and management requirements;
- scheduled services and targets;
- local-content plans;
- reporting frequency;
- supplier registration;
- employment requirements;
- capacity-development requirements;
- audit and inspection powers;
- offences and penalties;
- ministerial/regulatory powers.

Every statutory target should be recorded with the exact schedule and date in force.

Reporting evolution

Government mid-year reporting documents changes in implementation and reporting scope. The 2025 report states that half-year reporting guidelines were expanded to capture procurement activity in ten additional sectors and subsectors already identified in the legislation, with local-capacity assessment informing future targets. [S60]

The final paper should explain when reporting scope changes, because a rising reported procurement figure may partly reflect wider reporting coverage rather than only underlying economic growth.

Chapter 29 research dossier

Domestic value added versus local procurement

A supplier invoice does not reveal how much value stays in the country.

Consider four stylized cases:

Case A: imported item, local reseller

A Guyanese supplier imports a complete product and resells it. Domestic value may consist mainly of local wages, warehousing, financing and margin.

Case B: imported inputs, local fabrication

Raw material is imported but cutting, welding, engineering, testing and maintenance occur locally. Domestic value is higher.

Case C: local service

A locally owned engineering, logistics or professional-services firm supplies labour and expertise primarily from Guyana.

Case D: foreign-controlled local operation

A locally incorporated company employs Guyanese workers but profit, intellectual property and management services are largely external.

All four may contribute to the economy, but they should not be reported as economically identical.

Domestic value-added formula

The final research model should estimate:

Gross local-content procurement

- imported goods embedded in those purchases
- imported subcontracted services
- foreign management/IP charges attributable to delivery

= estimated domestic gross value added.

Then split domestic value into:

- wages;
- local operating surplus;

- taxes;
- domestic intermediate inputs.

This requires customs, tax, ownership and firm-level evidence.

Ownership verification

The local-content register should be sampled against:

- beneficial ownership;
- shareholder records;
- management control;
- bank/payment destination;
- payroll;
- customs imports;
- related-party contracts.

Certification status should remain separate from independent economic measurement.

Chapter 30 research dossier

Workforce and skills

Government petroleum reporting provides employment counts, but a durable skills analysis needs occupational detail.

The final workforce dataset should include:

- occupation;
- nationality;
- gender;
- age;
- region;
- permanent/temporary;
- offshore/onshore;
- employer;
- wage band;
- certification;
- training;
- promotion;
- retention;

- safety outcomes.

Skills pipeline

Map: secondary school → technical/vocational training → apprenticeship → entry role → technical certification → supervisor → professional/management role.

The oil sector may increase returns to technical training while also drawing engineers and tradespeople away from construction, mining, utilities and public agencies.

A national assessment should therefore measure both petroleum-sector employment and vacancies/wage pressure elsewhere.

Training outcomes

Training expenditure is an input.

Outcome indicators include:

- certificates completed;
- internationally recognized qualifications;
- post-training employment;
- wage progression;
- retention;
- local succession into positions previously expatriate;
- safety performance.

Chapter 31 research dossier

Household welfare and distribution

The Bureau of Statistics states that the Household Budget and Living Condition Survey fieldwork ran from November 2018 through November 2019 across all ten administrative regions. [\[S61\]](#)

This creates a major evidence boundary.

A pre-first-oil or first-oil-era household survey should not be reported as a 2026 poverty or living-standard measurement.

The final social chapter should build a timeline of available household evidence and label the latest observation year for every indicator.

Welfare dashboard

Required measures:

- real household consumption;
- income;
- food expenditure share;
- housing cost;
- electricity access and reliability;
- water and sanitation;
- internet;
- transport;
- school attendance/completion;
- health access;
- employment;
- regional inequality;
- gender;
- hinterland/coastal differences.

Macroeconomic GDP growth should appear separately.

Price effects

Rapid investment and public expenditure may affect:

- rent;
- land;
- construction materials;
- food;
- transport;
- skilled labour;
- services.

A household can face rising nominal income and rising costs at the same time.

The final analysis should therefore use real purchasing power rather than nominal income alone.

Chapter 32 research dossier

Amerindian and hinterland development

Indigenous and hinterland outcomes arise through more than petroleum.

Relevant systems include:

- land titling;

- public infrastructure;
- health and education;
- telecommunications;
- renewable/off-grid energy;
- carbon-credit benefit sharing;
- public employment;
- resource governance.

The LCDS programme states that 15% of carbon-credit revenues are made available for village-led plans and describes a village consultation/planning process. [S62, S63]

The final paper should distinguish:

- government programme design;
- village decision;
- funds allocated;
- funds received;
- project procured;
- project completed;
- project operating;
- measured community outcome.

Consent and participation

The research should seek evidence from:

- National Toshaos' Council;
- individual village councils;
- Indigenous organizations;
- legal cases;
- independent rights specialists;
- UN/other safeguard records.

Government programme descriptions should not substitute for affected-community evidence.

The chapter should also record disagreement where consultation, land rights or benefit-sharing processes are contested.

Chapter 33 research dossier

Social licence, grievance and participation

A large national development programme needs mechanisms for affected people to challenge decisions.

The final institutional map should include:

- EPA public-comment process;
- petroleum/ministry grievance mechanisms;
- judicial review;
- EITI multi-stakeholder engagement;
- village governance;
- procurement complaints;
- labour complaints;
- local-content supplier complaints.

For each mechanism:

- who has standing;
- submission process;
- deadline;
- decision maker;
- publication rule;
- appeal;
- observed case duration.

Evidence versus sentiment

Public opinion surveys, if methodologically sound, may measure attitudes.

They should not substitute for:

- statutory rights;
- environmental results;
- actual household outcomes;
- audited financial flows.

The flagship should report them as a separate evidence category.

Longitudinal local-content evidence

Government reports should be transformed into a consistent time series with fields for:

- reporting companies;
- sectors covered;
- gross procurement;
- training expenditure;
- local hires;
- total Guyanese employment;
- methodology changes.

For example, the 2024 Mid-Year Report described reporting from 40 companies, H1 procurement of US\$314 million, more than 454 new hires during the review period and 4,467 Guyanese employed by those reporting companies, while the government estimated total Guyanese oil-industry employment at about 6,500. [S59]

These figures are government-reported and should be retained with their stated population and reporting scope.

They should not be merged with later periods unless definitions are stable.

Local-value audit sample

The flagship should propose a stratified sample of suppliers across:

- engineering;
- fabrication;
- logistics;
- marine;
- aviation;
- catering;
- accommodation;
- transportation;
- legal/accounting;
- environmental services;
- manpower;
- equipment rental.

For each sampled contract:

1. contract value;
2. imported goods;

3. imported services;
4. Guyanese payroll;
5. domestic purchases;
6. taxes;
7. retained local profit;
8. training;
9. capital equipment left in country;
10. capability gained.

This produces a more meaningful measure of domestic value than invoice totals alone.

Distributional analysis

Petroleum benefits and costs should be mapped by:

- region;
- urban/rural;
- coastal/hinterland;
- income group;
- gender;
- age;
- occupation.

The final paper should avoid an unsupported statement that national gains were evenly distributed.

Likewise, absence of current household data should not be used to infer poor outcomes.

Where evidence is unavailable, state the gap.

New-producer blueprint extracted from Part VI

A prospective producer should establish before production:

1. legal definitions of local company and local worker;
2. transparent beneficial-ownership tests;
3. realistic sector-specific targets based on capacity;
4. supplier-development programmes;
5. tender visibility;
6. payment-timing rules;
7. workforce data standards;

8. training outcome measures;
9. independent domestic-value estimation;
10. updated household baseline;
11. regional distribution indicators;
12. grievance mechanisms.

For a small island state, targets need to reflect:

- labour-force size;
- tourism-sector competition for workers;
- limited industrial land;
- port capacity;
- import dependence;
- education pipeline.

Required Part VI exhibits

1. Local Content Act legal architecture.
2. Scheduled target matrix by service category.
3. Procurement and employment time series.
4. Domestic-value-added waterfall.
5. Supplier ownership map.
6. Occupational skills pipeline.
7. Wage and vacancy comparison across sectors.
8. Household-data timeline.
9. Regional welfare dashboard.
10. Indigenous/hinterland funding flow.
11. Village-plan outcome sample.
12. Grievance and remedy map.

Remaining Part VI evidence gates

- extract full Local Content Act schedules;
- compile reporting-company counts by year;
- reconcile official procurement data from 2022-2026;
- obtain supplier-register aggregate data;
- test beneficial ownership;
- obtain customs and labour evidence;

- build skills/training time series;
- identify latest representative household data;
- avoid updating old poverty data through inference;
- collect Indigenous/community sources;
- sample village-plan outcomes;
- collect grievance/appeal cases;
- external labour-economics review;
- external Indigenous-rights review.

Publication boundary

The paper will not equate gross local procurement with domestic value added.

It will not equate petroleum GDP with household welfare.

It will not equate a programme allocation with a verified community outcome.

Each will be measured on its own evidence.

PART VII RESEARCH DOSSIER

Economic Transformation, Public Investment, Energy, Diversification and Competitiveness

Working research dossier for Chapters 34-38 of the 520-560-page flagship publication. Evidence cut-off: 27 September 2026. Status: research stage. Macroeconomic and public-investment datasets require further reconciliation.

Research objective

Part VII asks whether petroleum income is being converted into durable productive capacity.

The analytical frame is broader than GDP.

The chapter separates:

- oil output;
- non-oil output;
- public investment;
- private investment;
- household welfare;
- infrastructure service quality;
- financial savings;
- human capital;
- natural capital;
- competitiveness.

No single indicator answers the development question.

Chapter 34 research dossier

Growth beyond headline GDP

IMF staff reported in July 2026 that real GDP grew by more than 19% in 2025 after average growth near 40% during 2023-2024. Staff also described broad non-oil expansion and similar high petroleum production during the first half of 2026. [S64]

These IMF observations provide a macroeconomic snapshot, but the final chapter should use official national accounts and revised historical series for quantitative decomposition.

GDP decomposition

Required time series:

- total real GDP;
- oil GDP;
- non-oil GDP;
- construction;
- agriculture;
- mining;
- manufacturing;
- transport;
- services;
- government;
- per-capita measures.

The dataset should preserve base-year changes and statistical revisions.

Why oil GDP requires separate treatment

Petroleum production can increase national output rapidly while:

- employing a small share of the labour force directly;
- importing substantial specialized equipment and services;
- generating profits accruing to foreign investors;
- generating government revenue on a different timetable.

This does not make oil GDP irrelevant. It means GDP must be linked to national income, fiscal receipts, domestic value added and household outcomes.

Non-oil growth

The final chapter should test:

- whether non-oil growth is concentrated in construction;
- whether tradable sectors expand;
- whether private productivity rises;
- whether public investment reduces logistics and power constraints;
- whether sector growth persists when oil-linked construction moderates.

Labour market

The IMF July 2026 statement reports unemployment at 6.2% at end-2025 and notes labour shortages as a policy issue. [S64]

The flagship should pair unemployment with:

- participation;
- vacancies;
- wage growth;
- migration;
- occupation;
- public/private sector;
- underemployment.

Chapter 35 research dossier

Infrastructure transformation

Petroleum revenue increases fiscal capacity, but infrastructure conversion has several stages:

appropriation → procurement → contract → construction → commissioning → operation → maintenance → service outcome.

The final project register should never mark an allocation as a completed asset.

Infrastructure classes

Transport

- national roads;
- bridges;
- community roads;
- hinterland roads;
- ports;
- airports.

Water and drainage

- potable water;
- drainage;
- canals;

- pumps;
- sea defence.

Social

- hospitals;
- schools;
- housing;
- digital infrastructure.

Productive

- power generation;
- transmission;
- industrial sites;
- logistics facilities.

Project register fields

For every major project:

- approved objective;
- implementing agency;
- original budget;
- revised budget;
- financing source;
- procurement method;
- contractor;
- award date;
- original completion date;
- actual completion;
- variations;
- maintenance plan;
- users;
- service performance.

This allows the paper to distinguish financial execution from delivery.

Lifecycle cost

Large petroleum-financed capital programmes create future recurrent costs.

The national balance sheet should estimate:

- maintenance;
- staffing;
- energy;
- insurance;
- replacement;
- debt service where applicable.

A project that requires high future subsidies has a different wealth profile from a financial asset earning external returns.

Chapter 36 research dossier

Energy security and gas-to-energy choices

The 2026 Budget Speech describes a 300 MW first phase of the Gas-to-Energy project and a broader programme of generation and grid investment. [S65]

The Guyana Energy Agency describes natural gas as part of a transition in the integrated grid alongside solar, wind, hydro and other resources. [S67]

These sources state government/agency plans. The flagship should separately measure actual commissioning, delivered megawatt-hours, reliability, cost and emissions.

Power-system metrics

Required:

- installed MW;
- dependable MW;
- peak demand;
- annual generation;
- system losses;
- outage frequency;
- outage duration;
- fuel consumption;
- fuel cost;
- average tariff;
- subsidy;
- renewable share;
- emissions intensity.

Gas-to-energy evaluation

The final analysis should identify:

- offshore pipeline capital cost;
- onshore NGL/power capital cost;
- financing;
- gas supply arrangements;
- operating cost;
- plant efficiency;
- grid upgrades;
- outage assumptions;
- tariff policy;
- LPG output;
- emissions relative to displaced HFO/diesel;
- methane leakage;
- project delay risk.

A government statement that a project will reduce prices is a policy claim until observed tariffs and full system costs are available.

Demand-side management

The GEA July 2026 update states the new action plan targets 516 GWh of energy savings by 2030 against projected generation of 2,508 GWh. [\[S66\]](#)

The flagship should track:

- implemented measures;
- verified savings;
- rebound effects;
- programme cost;
- household/business participation.

Chapter 37 research dossier

Private-sector development and diversification

Diversification should mean more than an increase in construction linked to petroleum.

The final chapter should test:

- export diversification;
- productivity;
- firm formation;
- domestic manufacturing;
- agriculture and food exports;
- services exports;
- mining value added;
- tourism;
- technology;
- regional trade.

Diversification metrics

Possible metrics:

- non-oil export share;
- Herfindahl export concentration;
- non-oil productivity;
- manufacturing value added;
- food import dependence;
- firm survival;
- formal employment;
- private investment outside petroleum support;
- research and technical capacity.

The paper should avoid a single composite political score.

Linkages

Petroleum can support non-oil sectors through:

- cheaper energy;
- logistics investment;
- skilled labour;
- demand;
- finance;
- public investment.

It can also create pressures through:

- higher wages;
- higher rents;
- construction demand;
- exchange-rate pressure;
- imported services;
- concentration of finance.

The net effect requires data by sector.

Chapter 38 research dossier

Inflation, competitiveness and Dutch-disease channels

IMF staff in July 2026 stated that available indicators did not show clear signs of overheating or resource-driven competitiveness pressure at that point, while noting that strong wage growth and wage-based real exchange rate indicators required monitoring. The same statement identified stronger foreign-exchange demand associated with import-intensive private investment. [S64]

These are dated IMF staff assessments, not permanent findings.

The flagship should construct its own indicator dashboard.

Indicator set

- CPI overall;
- food inflation;
- housing/rent;
- land prices;
- construction materials;
- nominal wages;
- real wages;
- unit labour cost;
- real effective exchange rate;
- imports;
- non-oil exports;
- vacancy rates;
- construction prices;
- public procurement prices.

Dutch-disease channels

The chapter should explain three mechanisms separately:

1. Spending effect: resource income increases domestic demand for non-tradables.
2. Resource movement: labour/capital shift toward petroleum-linked activity.
3. Exchange-rate/price effect: domestic costs rise relative to trading partners.

Guyana's stabilized exchange-rate framework changes how adjustment appears in observed indicators.

Policy response framework

The paper should describe, without prescribing a political choice:

- saving more petroleum revenue;
- pacing public investment;
- raising productive capacity;
- importing scarce inputs;
- labour training;
- migration policy;
- competition policy;
- productivity investment;
- exchange-rate/monetary options.

Each has trade-offs.

National wealth conversion framework

Petroleum extraction reduces an underground asset.

Long-run wealth conversion can be examined through changes in:

- sovereign financial assets;
- public infrastructure;
- human capital;
- private productive capital;
- environmental/natural capital;
- public debt;
- contingent liabilities.

A national wealth table should avoid double counting.

For example:

- petroleum revenue deposited into the NRF increases financial assets;
- a withdrawal reduces financial assets;
- a completed road increases produced capital;
- maintenance liabilities affect future public spending;
- environmental damage reduces natural capital.

The quality of wealth conversion depends on asset performance, not spending volume alone.

IMF 2026 evidence relevant to Part VII

The July 2026 mission statement provides several dated observations and recommendations:

- rapid oil and non-oil growth;
- strong construction;
- large public investment;
- increased FX demand with heavy import content;
- high petroleum inflows and NRF accumulation;
- low assessed risk of debt distress;
- need to align spending with absorptive capacity under sustained high oil prices;
- recommendation to consider a non-oil primary-balance guide;
- need for stronger spending-outcome monitoring;
- need to strengthen statistics. [S64]

The flagship should identify each as an IMF staff statement, not an independent fact where the claim is interpretive.

Required Part VII exhibits

1. Oil versus non-oil GDP.
2. Sector contribution to non-oil growth.
3. Public investment by sector.
4. Project completion versus appropriation.
5. Capital stock and maintenance liability.
6. Electricity capacity, generation and reliability.
7. Fuel mix and emissions intensity.
8. Gas-to-energy cost/reliability bridge.
9. Export concentration.
10. Wage and price dashboard.
11. Real exchange-rate indicators.
12. National wealth conversion table.

New-producer blueprint extracted from Part VII

Before spending large resource revenue, a new producer should build:

- public-investment appraisal;
- procurement transparency;
- cost and schedule control;
- lifecycle maintenance budgeting;
- project commissioning verification;
- national accounts capable of separating resource and non-resource sectors;
- household price monitoring;
- labour-market data;
- infrastructure service indicators;
- national balance-sheet reporting.

A small island state should add:

- hurricane resilience;
- port redundancy;
- imported construction inflation;
- tourism competitiveness;
- desalination/power exposure;
- limited contractor capacity.

Remaining Part VII evidence gates

1. Extract official 2026 national accounts.
2. Reconcile Ministry/IMF growth estimates.
3. Build 2010-2026 oil/non-oil series.
4. Build public-investment project register.
5. Verify completion and operating status for major assets.
6. Build electricity performance time series.
7. Complete gas-to-energy project cost and schedule record.
8. Calculate export concentration.
9. Build wage/price/real exchange-rate dashboard.
10. Build national wealth accounts.
11. Obtain independent macroeconomic review.
12. Obtain infrastructure/project-management review.

Publication boundary

Part VII will not infer household prosperity from GDP growth.

It will not infer infrastructure performance from budget size.

It will not infer economic diversification from non-oil construction alone.

Each proposition requires its own evidence.

PART VIII RESEARCH DOSSIER

Transparency, Cost Audits, Procurement, Parliamentary/Judicial Oversight and Public Data

Working research dossier for Chapters 39-43 of the 520-560-page flagship publication. Evidence cut-off: 27 September 2026. Status: research stage. Legal and public-finance review pending.

Research objective

A resource-governance system is not demonstrated by the number of reports published.

The flagship will test five linked questions:

1. Are material contracts, licences, production data and public receipts disclosed?
2. Are figures timely enough to support scrutiny?
3. Is independent challenge institutionally possible?
4. Are audit findings resolved and traceable?
5. Are public-finance and social outcomes visible after money leaves the petroleum account?

Transparency, oversight and enforcement will therefore appear as separate categories.

Chapter 39 research dossier

EITI and extractive disclosure

Guyana joined EITI in 2017. The EITI International Secretariat records an earlier 2022 Validation under the 2019 Standard and a new full Validation beginning 15 May 2026. [S70]

The EITI Board's 12 May 2026 decision records concern about continuing lack of full stakeholder confidence regarding the independence of the selection of civil-society representatives to the multi-stakeholder group. The Board chose to proceed with a full Validation and an external expert. [S69]

This is a formally attributed EITI concern.

The final flagship should not convert it into a broad judgment about political legitimacy.

Validation timeline to build

- EITI accession;
- first reports;

- 2022 Validation;
- corrective actions;
- report publication lags;
- 2025 multi-stakeholder representation dispute;
- deferred Validation;
- May 2026 Board decision;
- 2026 mission;
- external expert draft;
- stakeholder comments;
- final Board decision when available.

The current paper must state clearly if no final 2026 decision exists at the publication cut-off.

Disclosure quality

EITI performance should be assessed by category:

- licences;
- contracts;
- production;
- beneficial ownership;
- company payments;
- government receipts;
- state participation;
- subnational transfers;
- social/environmental payments;
- local content;
- commodity sales.

For each:

- data owner;
- observation year;
- release date;
- reconciliation method;
- source file;
- machine readability;
- unresolved discrepancy.

Timeliness

In a rapidly growing petroleum industry, a report covering fiscal year N but published in N+2 has different decision value from quarterly public data.

The paper should show disclosure lag explicitly.

Chapter 40 research dossier

Cost audits and revenue assurance

Cost-recovery audit is one of the clearest examples of why publication is not the same as final resolution.

The government-published IHS Markit final audit for 1999-2017 says the contractor submitted cumulative costs of about USD1.678 billion for review and that the audit team identified grounds to dispute about USD214.4 million plus overhead adjustments. [S71]

A later Ministry update says the government dispute was moved toward the agreement's sole-expert procedure and describes multiple audit workstreams. [S72]

These are distinct stages.

The final audit tracker should distinguish:

- amount submitted;
- amount audited;
- audit exception;
- government claim/dispute;
- contractor response;
- negotiation;
- sole expert;
- arbitration;
- final accepted adjustment;
- cost-bank credit;
- cash/revenue effect.

No audit exception will be reported as recovered revenue before the final accounting event is evidenced.

Revenue assurance institutions

The chapter should map:

- Ministry of Natural Resources;

- Guyana Revenue Authority;
- Ministry of Finance;
- Bank of Guyana;
- external petroleum auditors;
- Attorney General/legal teams;
- independent experts.

For each:

- mandate;
- staffing;
- data access;
- audit window;
- dispute authority;
- publication obligation.

Audit-cycle metric

Measure: audit period end → audit engagement → fieldwork → draft report → contractor response → final report → dispute → resolution.

A system with strong audit rights but prolonged unresolved findings has a different performance profile from a system with rapid final adjustments.

Chapter 41 research dossier

Procurement and public-spending oversight

Petroleum revenue becomes ordinary public-finance risk after transfer into the budget.

The public investment chapter must therefore connect:

- budget appropriation;
- procurement;
- contract award;
- variation;
- physical completion;
- payment;
- audit;
- service outcome.

The IMF 2026 mission statement encouraged continued strengthening of procurement practices and consistent compliance as public spending expands. [S64]

That is an IMF staff recommendation, not an author judgment.

Supreme audit institution

The Audit Office of Guyana identifies itself as the state's supreme audit institution and reports that the 2024 public accounts and performance report were presented to the National Assembly in September 2025. [S73]

The final review should extract findings relevant to major petroleum-financed expenditure without generalizing an individual audit issue across the public sector.

Project-audit register

Fields:

- ministry/agency;
- programme;
- funding source;
- contract;
- auditor observation;
- amount;
- management response;
- Public Accounts Committee response;
- corrective action;
- current status.

The final paper should differentiate:

- procedural noncompliance;
- unsupported expenditure;
- overpayment;
- delay;
- value-for-money concern;
- fraud allegation;
- proven fraud.

Chapter 42 research dossier

Parliament, judiciary and institutional checks

Formal oversight is distributed across:

- parliamentary appropriation;
- statutory reporting;
- parliamentary committees;
- independent audit;
- courts;
- administrative regulators;
- constitutional offices.

The flagship should map powers, not assign an overall political score.

Parliamentary petroleum-revenue trail

The final evidence chain should include:

- NRF receipt notice;
- annual report;
- maximum withdrawal calculation;
- budget estimate;
- Appropriation Act;
- transfer to Consolidated Fund;
- expenditure;
- Auditor General review.

Judicial review

Environmental permits and financial-assurance disputes have entered the courts.

The final legal chapter should use authenticated judgments.

For every case:

- parties;
- court;
- filing date;
- legal question;
- first-instance ruling;

- appeal;
- final order at cut-off;
- later proceedings;
- practical regulatory effect.

Government, company and civil-society descriptions of a judgment should be treated as positions unless they reproduce the authenticated text accurately.

Chapter 43 research dossier

Statistics and public data

A fast-changing petroleum economy places unusual pressure on statistical systems.

The IMF 2025 staff report described modernization work and a planned new household survey; the July 2026 mission statement again highlighted national accounts, price, external-sector, labour and household data as priority areas. [S64, S68]

The paper should distinguish:

- absence of a statistic;
- delay in publication;
- outdated base year;
- sampling limitation;
- incompatible definitions;
- actual adverse outcome.

Data-quality inventory

Required datasets:

Petroleum

- well;
- field;
- FPSO;
- production;
- flaring;
- water;
- lifting;
- price;
- government entitlement.

Fiscal

- royalties;
- profit oil;
- taxes;
- NRF;
- public debt;
- procurement;
- capital spending.

Economic

- national accounts;
- CPI;
- trade;
- balance of payments;
- wages;
- employment;
- firms.

Social

- household income;
- consumption;
- poverty;
- education;
- health;
- housing;
- regional indicators.

Environmental

- emissions;
- water quality;
- fisheries;
- biodiversity;
- incidents;
- inspections.

Reproducibility standard

Every quantitative exhibit should record:

- source;
- file version;
- download date;
- unit;
- transformation;
- missing values;
- code/formula;
- revision date.

Accountability architecture

The flagship should present accountability as six layers.

Layer 1. Rule

Law, contract, permit or regulation defines an obligation.

Layer 2. Measurement

A competent body measures performance.

Layer 3. Disclosure

The evidence is published in usable form.

Layer 4. Independent challenge

Auditor, court, parliament, civil society, affected community or expert can contest the record.

Layer 5. Correction

A substantiated problem leads to adjustment, sanction, remediation or rule change.

Layer 6. Outcome

The correction affects revenue, environmental performance, service delivery or public confidence.

An institution that performs strongly at one layer might perform differently at another.

Accountability case studies required

1. 1999-2017 cost recovery audit.
2. 2018-2020 cost recovery audit.
3. NRF annual audit and withdrawal reporting.
4. Liza Phase 1 permit modification after flaring.
5. Environmental financial assurance litigation.
6. EITI 2026 Validation.
7. Local-content supplier certification.
8. Major public-infrastructure procurement and completion.

Each case should follow the six-layer model.

New-producer blueprint extracted from Part VIII

A new producer should establish:

- contract publication policy;
- cost-audit calendar;
- beneficial-ownership register;
- production data standard;
- petroleum receipt publication;
- independent sovereign-fund audit;
- supreme-audit access;
- parliamentary reporting;
- regulator enforcement publication;
- civil-society/community participation;
- court access;
- corrections register.

For a small state with limited specialists, independent external expertise should transfer knowledge to permanent national institutions rather than remain a perpetual substitute for them.

Required Part VIII exhibits

1. Petroleum accountability system map.
2. Contract-to-cash audit trail.

3. Cost-audit timeline.
4. EITI chronology.
5. Disclosure lag dashboard.
6. Beneficial-ownership evidence flow.
7. Parliamentary petroleum-revenue flow.
8. Procurement oversight flow.
9. Environmental judicial-review chronology.
10. Statistical-data quality matrix.
11. Corrections register.

Remaining Part VIII evidence gates

- obtain final 2026 EITI decision if published by cut-off;
- extract all Guyana EITI report dates and covered fiscal years;
- build cost-audit resolution table;
- verify 2021-2025 audit workstreams;
- extract public-account findings tied to major flagship examples;
- obtain authenticated court judgments;
- map parliamentary receipt notifications;
- compile beneficial-ownership access;
- audit data publication lag;
- verify public procurement portal capability;
- obtain external public-finance and legal review.

Publication boundary

Part VIII will not treat disclosure as proof of performance.

It will not treat an audit finding as proof of recovered revenue.

It will not treat a contested EITI process as an overall judgment on a government or opposition.

It will document institutions, dates, evidence, disagreements and observable corrective actions.

PART IX RESEARCH DOSSIER

Comparative Petroleum Governance and Transferability

Working research dossier for Chapters 44-49 of the 520-560-page flagship publication. Evidence cut-off: 27 September 2026. Status: mechanism comparison in progress. No country ranking is intended.

Research objective

The purpose of comparison is not to identify a single "best" petroleum state.

The purpose is to isolate institutional mechanisms, identify the conditions under which they operate, and test whether any mechanism is relevant to an emerging producer.

The comparison uses six questions:

1. What resource and fiscal problem was the mechanism designed to solve?
2. What legal instrument governs it?
3. Which institution implements it?
4. What data are published?
5. What trade-off does it create?
6. Which national conditions limit transfer?

Chapter 44 research dossier

Norway

Norway provides a long-horizon comparison for converting petroleum cash into financial assets and integrating fund transfers into the national budget.

The Ministry of Finance's 2026 Government Pension Fund white paper describes the Government Pension Fund Global within a fiscal framework linked to an expected long-run real return of 3%. It says normal-year withdrawals should be below that level to preserve room for adverse periods. [S74]

The comparison should not reduce Norway to a 3% rule.

Relevant institutional components include:

- state petroleum taxation and ownership;
- transfer of petroleum cash to the fund;
- external portfolio investment;

- budget integration;
- parliamentary framework;
- central-bank asset management;
- transparent benchmark;
- long horizon;
- broad political continuity around the framework.

Transferability limits

Norway entered its modern fund era with:

- mature public institutions;
- high human capital;
- a developed tax system;
- a diversified economy;
- different infrastructure deficits;
- a larger historical production base.

A developing producer with major infrastructure and social gaps faces a different current-versus-future consumption trade-off.

The transferable mechanism is disciplined integration of petroleum revenue, investment and the budget.

The exact numerical spending rule is not automatically transferable.

Chapter 45 research dossier

Ghana

Ghana provides a closer institutional comparison as a developing petroleum producer with a legislated revenue-management system.

The Bank of Ghana describes two petroleum funds with distinct functions:

- Ghana Stabilisation Fund;
- Ghana Heritage Fund. [S75]

The Bank publishes financial statements and semi-annual reports. [S75, S76]

This creates a useful comparison with Guyana's single NRF architecture.

Mechanism comparison

Questions:

- Are stabilization and intergenerational objectives separated?
- How are petroleum receipts allocated before budget spending?
- What information goes to parliament?
- How often are reports published?
- How are the funds audited?
- How are transfers triggered?

The flagship should compare statutory design with observed implementation.

Transferability

A new state might choose:

- one integrated fund;
- separate stabilization and savings funds;
- a fund plus explicit fiscal rule.

The relevant design choice depends on:

- revenue volatility;
- reserve life;
- debt;
- budget size;
- financial market capacity;
- development gaps.

Chapter 46 research dossier

Timor-Leste

Timor-Leste is relevant because it is a small developing state with a petroleum fund that became central to national fiscal financing.

The Central Bank publishes annual and frequent performance reports. Its current management framework separates a Liquidity Portfolio, intended to meet expected withdrawals, from a Growth Portfolio designed for longer-term investment. [S77, S78]

This is analytically important for Guyana and island states.

A resource fund with large expected withdrawals needs a liquidity strategy separate from its long-run return objective.

Core comparison questions

- petroleum receipts versus investment return;
- annual budget withdrawals;
- fund balance;
- depletion of producing fields;
- estimated sustainable income framework;
- asset allocation;
- liquidity needs;
- domestic fiscal dependence.

The case is useful precisely because a large fund balance does not remove dependence risk.

The flagship should use current fund reporting and fiscal data before drawing conclusions.

Chapter 47 research dossier

Suriname

Suriname provides a nearby pre-first-oil comparison.

TotalEnergies stated on 19 September 2026 that the GranMorgu project has been progressing since final investment decision in October 2024 and that first oil remains targeted for 2028. [S79]

This positions Suriname in a different stage of the petroleum lifecycle from Guyana.

Guyana's most relevant lessons for Suriname concern the discovery-to-first-oil period:

- regulator capacity;
- revenue-fund readiness;
- environmental baseline;
- local-content expectations;
- contract administration;
- spill preparedness;
- public-data systems;
- public-investment planning.

Suriname's own institutions, state oil company, fiscal structure and political choices require separate analysis.

No Guyanese policy should be presumed suitable solely because the states share a basin.

Chapter 48 research dossier

Additional producer mechanisms

The flagship should add targeted mechanism cases rather than superficial country profiles.

Potential mechanisms include:

Trinidad and Tobago

The Heritage and Stabilisation Fund provides a mature Caribbean comparison for a hydrocarbon economy. The Ministry of Finance maintains annual and quarterly reporting. [S80]

Relevant issues:

- stabilization;
- heritage objective;
- mature production;
- economic diversification;
- small-state exposure.

Senegal

Potential relevance:

- new offshore production;
- domestic gas;
- local content;
- first-revenue institutional timing.

Namibia

Potential relevance:

- pre-production discovery;
- regulatory preparation;
- state participation;
- environmental baseline.

Mozambique

Potential relevance:

- large-resource expectations;
- project delay;
- security and institutional risk;

- avoiding expenditure against unrealized revenue.

Botswana

Not a petroleum case, but relevant to mineral wealth and public investment.

The final comparator set should be selected by specific analytical mechanism.

Chapter 49 research dossier

Island and small-state adaptation

A small island petroleum producer faces distinct constraints.

Marine concentration

A spill might affect:

- fisheries;
- reefs;
- beaches;
- tourism;
- desalination;
- ports;
- coastal infrastructure.

The same shoreline can support multiple core sectors simultaneously.

Infrastructure concentration

One major port, airport or power station may represent a large share of national capacity.

An emergency affecting one asset can therefore have systemic consequences.

Labour scale

A petroleum development can sharply alter wages in:

- construction;
- marine services;
- engineering;
- public administration;
- tourism.

Targets for local content should reflect the actual workforce and education pipeline.

Fiscal concentration

A small tax base can become heavily dependent on petroleum rapidly.

The blueprint should therefore stress:

- conservative revenue assumptions;
- liquidity reserves;
- multi-year budget stress tests;
- recurrent-cost discipline;
- catastrophe reserves.

Hurricane and climate exposure

An island response plan should test a spill coinciding with:

- tropical cyclone;
- port closure;
- airport disruption;
- telecommunications failure;
- evacuation.

Blue economy and natural capital

A forest-rich continental model cannot be copied literally.

An island should map:

- reefs;
- seagrass;
- mangroves;
- fisheries;
- tourism;
- blue carbon where scientifically credible;
- coastal protection.

These resources require their own evidence and rights systems.

Comparative mechanism matrix

This table compares institutional questions, not country performance rankings.

Transferability decision tree

Before adopting a mechanism from another country ask:

Resource condition

- commercial discovery?
- reserve range?
- production horizon?
- price exposure?
- development cost?

Fiscal condition

- debt?
- revenue base?
- recurrent expenditure?
- infrastructure deficit?
- disaster exposure?

Institutional condition

- central-bank capacity?
- audit capacity?
- environmental regulator?
- procurement?
- statistics?

Social condition

- population?
- regional inequality?
- Indigenous/customary rights?
- labour force?
- household baseline?

Environmental condition

- deepwater/nearshore?

- currents?
- fisheries?
- reefs?
- forests?
- hurricanes?
- transboundary risk?

Legal condition

- constitutional revenue powers?
- contract system?
- judicial review?
- maritime boundaries?
- treaty obligations?

Only after this screen should a foreign mechanism be adapted.

Comparative evidence standards

For each country:

- current legal text;
- current fund report;
- current fiscal data;
- historical context;
- independent assessment;
- observed problem;
- reform history.

Do not compare:

- one country's audited actual to another country's forecast;
- a mature fund balance to a new fund's annual inflow;
- gross petroleum revenue to government petroleum revenue;
- a policy target to observed performance.

Required Part IX exhibits

1. Petroleum lifecycle stage by comparator.
2. Fund objective and legal-design matrix.
3. Deposit/withdrawal rule matrix.

4. Reporting and audit-frequency matrix.
5. Public-investment interface.
6. Environmental governance mechanism comparison.
7. Local-content mechanism comparison.
8. Island-state risk matrix.
9. Transferability decision tree.
10. "Do not copy without adaptation" table.

New-producer conclusion from comparison

The principal transferable lesson is process discipline:

- date decisions;
- define fund objectives;
- separate cash and forecasts;
- audit petroleum costs;
- publish receipts;
- budget transparently;
- monitor environmental obligations;
- update household evidence;
- preserve institutional independence;
- revise rules transparently.

The numeric formula and institutional form remain national choices.

Remaining Part IX evidence gates

- extract Norway 2026 fund details;
- extract Ghana 2025/2026 fund reports;
- extract Timor-Leste 2025 fund report and 2026 fiscal data;
- add Suriname legal/fiscal architecture;
- extract Trinidad and Tobago HSF law and current reporting;
- select two additional new-producer cases;
- standardize all observations to common dates;
- independent comparative public-finance review.

Publication boundary

Part IX will not rank governments or declare a winning model.

It will show how different mechanisms work, what evidence exists, and what conditions control transfer.

PART X

New Producer Operating Manual

A staged implementation companion to the Guyana petroleum case study. Designed for emerging offshore producers, including island states. Evidence cut-off: 27 September 2026.

This manual translates research questions into administrative actions.

It does not prescribe a political choice, fiscal regime, spending ratio, state ownership model or energy policy.

The receiving country must adapt every action to its constitution, geology, marine ecology, debt, labour market, Indigenous/customary rights, climate exposure and international obligations.

Stage 0

Before a commercial discovery

Objective

Build enough public capacity to govern exploration without creating permanent spending commitments against hypothetical oil revenue.

0.1 Legal map

Publish a one-page legal map identifying:

- ownership of petroleum;
- licensing authority;
- environmental authority;
- maritime authority;
- tax authority;
- customs;
- central bank;
- national audit institution;
- parliament;
- courts;
- disaster-response command.

For every institution record:

- statute;

- date in force;
- responsible office;
- power;
- reporting duty;
- appeal route.

0.2 Acreage and geological-data register

Maintain:

- block coordinates;
- licences;
- expiry;
- work commitments;
- relinquishment;
- operator;
- beneficial ownership;
- well records;
- seismic ownership;
- confidentiality expiry.

The state should preserve institutional access to geological data after individual consultants or ministers leave office.

0.3 Environmental baseline

Before intensive drilling, collect:

- currents;
- bathymetry;
- water quality;
- fisheries;
- sensitive habitats;
- protected areas;
- shoreline receptors;
- tourism assets;
- marine traffic.

For an island:

- reefs;

- beaches;
- desalination intakes;
- cruise terminals;
- hurricane exposure.

Record:

- collector;
- method;
- date;
- raw-data ownership;
- public access;
- uncertainty.

0.4 Fiscal options

Model at least:

- royalty/tax licence;
- production sharing;
- state equity where legally feasible;
- hybrid options.

For each:

- dry-hole case;
- marginal discovery;
- large discovery;
- low price;
- high cost;
- development delay;
- abandonment.

Do not choose a fiscal system from headline government-take percentages alone.

0.5 Expert capacity

Create a roster for:

- petroleum law;
- reservoir engineering;
- petroleum economics;
- tax;

- audit;
- marine science;
- oil-spill response;
- insurance;
- public finance.

Every external contract should include:

- conflict check;
- scope;
- deliverable;
- price;
- knowledge transfer;
- data ownership;
- handover.

0.6 Maritime boundaries

Identify:

- agreed boundaries;
- unresolved claims;
- arbitration/treaty options;
- disputed blocks;
- notification obligations.

Do not market disputed acreage as if title risk does not exist.

Stage 0 gate

The country passes Stage 0 when it has:

- legal map;
- licence register;
- data policy;
- environmental baseline plan;
- fiscal-option model;
- independent adviser panel;
- conflict rules;
- boundary assessment;
- exploration emergency plan.

Commercial discovery is not required.

Stage 1

After a commercial discovery

Objective

Reduce information asymmetry before committing the country to a development framework.

1.1 Independent resource review

Obtain:

- discovery data;
- appraisal results;
- fluid characteristics;
- reservoir uncertainty;
- development concepts;
- recoverable-resource range.

Separate:

- petroleum in place;
- contingent resources;
- reserves;
- commercial development case.

1.2 Negotiation decision file

For each material contract clause record:

- state objective;
- operator proposal;
- independent adviser view;
- fiscal effect;
- legal effect;
- environmental effect;
- final text;
- approving authority.

Preserve draft/version history under lawful confidentiality.

1.3 Fiscal model

Model:

- production;
- price;
- capital cost;
- operating cost;
- royalty;
- cost recovery;
- profit share;
- tax;
- state equity;
- bonuses;
- decommissioning.

Stress:

- low price;
- delay;
- cost overrun;
- lower recovery;
- outage;
- future project cost additions.

1.4 Environmental decision file

Require:

- terms of reference;
- alternatives;
- baseline;
- cumulative effects;
- spill scenarios;
- climate effects;
- public consultation;
- mitigation;
- monitoring;
- financial assurance.

A development decision should show how material public comments were treated.

1.5 Decommissioning

Before production:

- estimate abandonment cost;
- define security mechanism;
- set contribution timing;
- establish residual liability;
- define asset removal and site restoration;
- audit the fund/security.

Stage 1 gate

Before final development approval:

- independent resource review complete;
- contract legally reviewed;
- fiscal model independently challenged;
- EIA complete;
- spill response tested;
- decommissioning plan funded;
- financial assurance documented;
- disclosure decision recorded.

Stage 2

First-oil readiness

Objective

Prove the revenue, environmental and emergency systems before commercial cash arrives.

2.1 Meter and production chain

Test: well → FPSO processing → storage → cargo measurement → lifting.

Independently verify calibration.

2.2 Sales and entitlement chain

Test: cargo → price → royalty → cost recovery → profit share → invoice → settlement.

Use a fictional cargo rehearsal.

2.3 Sovereign revenue chain

Test: receipt → sovereign account → public notice → accounting → audit.

If a sovereign fund exists, verify:

- legal account;
- custodian;
- signatories;
- investment rules;
- publication;
- audit.

2.4 Emergency drill

Test a serious hypothetical offshore event.

Measure:

- detection;
- notification;
- command activation;
- equipment mobilization;
- regional assistance;
- public communication;
- environmental sampling;
- claims intake.

Record actual exercise times.

2.5 Public transparency rehearsal

Ask an independent user to trace:

- licence;
- permit;
- production;
- sale;
- government receipt;
- fund deposit.

Missing definitions become a remediation item.

Stage 2 gate

First-oil readiness is not complete until:

- meter verification;
- fiscal rehearsal;
- bank reconciliation;
- environmental monitoring;
- emergency exercise;
- public data test;
- audit calendar

have each been documented.

Stage 3

First year of production

Objective

Demonstrate control under real transactions.

Month 1

Publish:

- actual production;
- first cargo;
- applicable fiscal terms;
- receipt method.

Verify:

- metering;
- invoice;
- bank receipt;
- accounting entry.

Quarter 1

Publish:

- production;
- cargo lifts;
- government receipts;
- environmental operating data;
- local employment/procurement with definitions.

Conduct:

- sample cost audit;
- permit inspection;
- spill-response review.

Quarter 2

Reconcile:

- physical barrels;
- sale barrels;
- royalty;
- profit oil;
- tax;
- fund deposits.

Review:

- contractor cost submission;
- local supplier data;
- grievances.

Quarter 3

Stress test:

- lower price;
- outage;
- spill;
- project delay.

Update:

- budget assumptions;
- liquidity;
- emergency capacity.

Quarter 4

Publish:

- audited fund accounts where timing permits;
- annual petroleum account;
- environmental annual report;
- local-content report;
- audit status;
- unresolved items.

No unresolved audit item should disappear from the public record merely because the year closes.

Stage 4

Years 2 to 5

Objective

Keep institutions growing at least as fast as the industry.

Annual institutional capacity review

Track:

- petroleum-regulator staff;
- engineers;
- auditors;
- environmental specialists;
- tax specialists;
- unresolved audits;
- permit workload;
- production volume.

Sovereign fund

Annually reconcile:

- opening balance;
- petroleum deposits;
- investment income;

- withdrawals;
- expenses;
- closing balance.

Separate market value and cash.

Public investment

For every major project:

- appraisal;
- contract;
- cost;
- completion;
- maintenance;
- service outcome.

Local value

Move from gross procurement to:

- domestic wages;
- domestic inputs;
- retained profit;
- local ownership;
- capability.

Social baseline

Update:

- household income/consumption;
- prices;
- housing;
- employment;
- education;
- health;
- regional indicators.

Do not rely indefinitely on a pre-oil survey.

Environmental system

Publish by project:

- emissions;
- flaring;
- discharge;
- incidents;
- inspections;
- permit changes;
- enforcement;
- spill drills.

Stage 4 gate

At year five conduct an independent public review of:

- legal framework;
- fiscal performance;
- fund accounting;
- public investment;
- environmental outcomes;
- local value;
- household evidence;
- institutional capacity.

The review should propose alternatives rather than assign a political score.

Crisis playbooks

A. Oil price falls sharply

Immediate:

- update realized price;
- update petroleum receipts;
- preserve cash/accounting distinction;
- recalculate lawful transfer ceiling;
- review budget exposure.

Within one quarter:

- prioritize essential recurrent spending;
- review project sequencing;
- update debt plan;
- stress fund liquidity.

Do not respond to a temporary price move using an unverified assumption of permanent decline.

B. Production outage

Immediate:

- establish technical cause;
- update production/cargo schedule;
- estimate cash timing.

Review:

- budget;
- fund deposits;
- contract obligations;
- environmental implications;
- repair logistics.

Separate capacity from actual production during the outage.

C. Major offshore spill

Immediate:

- protect life;
- activate incident command;
- stop source where feasible;
- notify affected jurisdictions;
- preserve evidence;
- initiate independent sampling.

Financial:

- activate operator liability;
- insurance;
- guarantee;
- emergency public funding if lawfully required.

Public:

- publish verified facts;

- sampling methods;
- response zones;
- claims process.

Do not publish speculative volume as established fact before measurement.

D. Cost-audit dispute

Maintain:

- amount claimed;
- auditor finding;
- contractor response;
- legal basis;
- procedure;
- final result.

Do not budget a proposed adjustment as recovered money.

E. Construction overheating

Measure:

- bid counts;
- project delays;
- construction prices;
- wages;
- imports;
- contractor concentration.

Options for public decision include:

- sequencing;
- imported capacity;
- training;
- procurement reform;
- slower rollout.

F. Natural disaster plus oil-revenue shock

Stress:

- emergency spending;

- lost tax revenue;
- infrastructure damage;
- petroleum downtime;
- fund withdrawals;
- insurance.

This combined case is especially relevant to island producers.

Island-state adaptation module

Marine economy

Baseline:

- tourism share;
- fishery share;
- reefs;
- seagrass;
- mangroves;
- beaches;
- desalination.

A marine incident can affect several sectors at once.

Single-point infrastructure

Map:

- one primary airport;
- one container port;
- one fuel terminal;
- one power station.

Create contingency for concurrent disruption.

Labour

A small labour force should not adopt unattainable local-content targets.

Prioritize:

- safety;
- technical roles;

- transferable certification;
- supervisory succession;
- supplier capability.

Revenue concentration

A small state should model petroleum receipts as a share of:

- total revenue;
- GDP;
- exports;
- foreign exchange.

The higher the concentration, the more important a low-price and production-interruption buffer becomes.

Publication and data blueprint

A newcomer should maintain one public evidence portal with linked datasets.

Legal register

- act;
- regulation;
- licence;
- agreement;
- permit;
- court decision.

Operational register

- wells;
- production;
- flaring;
- cargoes;
- incidents.

Fiscal register

- royalty;

- profit oil;
- tax;
- audit;
- sovereign fund.

Environmental register

- EIA;
- permit;
- monitoring;
- inspection;
- spill drill;
- enforcement.

Social/local register

- employment;
- supplier data;
- training;
- household indicators;
- community programmes.

Each dataset needs:

- observation date;
- publication date;
- unit;
- source;
- revision history.

24-month pre-first-oil action calendar

Months 24-18

- legal gap review;
- institutional map;
- regulator staffing;
- fiscal model;
- environmental baseline;

- maritime response agreements.

Months 18-12

- sovereign revenue framework;
- central-bank procedures;
- cost-audit procurement;
- local-content capacity study;
- household baseline;
- emergency equipment contracts.

Months 12-6

- first-cargo rehearsal;
- spill drill;
- public portal;
- production-meter assurance;
- petroleum-sales procedure;
- fund accounting test.

Months 6-3

- repeat reconciliation exercise;
- close audit gaps;
- verify insurance/guarantees;
- parliamentary reporting rehearsal;
- public communication protocol.

Months 3-0

- final readiness audit;
- regulator staffing check;
- first-cargo roles signed;
- bank accounts tested;
- public reporting calendar issued.

Five-year dashboard

Track annually:

- actual petroleum production;
- realized price;
- public petroleum receipts;
- fund balance;
- withdrawals;
- public debt;
- major project completion;
- real non-oil GDP;
- inflation;
- real wages;
- local value added;
- household indicators;
- environmental incidents;
- audit resolution time.

The dashboard should retain missing values rather than replace them with assumptions.

Final decision principles extracted from the Guyana study

1. Prepare legal and financial architecture before recurring revenue.
2. Keep legal versions dated.
3. Keep capacity, production, cargoes and cash separate.
4. Audit recoverable costs.
5. Keep sovereign-fund accounting separate from public-investment outcomes.
6. Keep petroleum emissions and forest/carbon accounting separate.
7. Verify environmental readiness, not permit existence alone.
8. Measure domestic value, not procurement labels alone.
9. Update household evidence after the economic transition.
10. Preserve stakeholder challenge and corrections.
11. Adapt foreign mechanisms to national conditions.
12. Stress-test adverse cases before relying on high-price projections.

Acceptance checklist for a country using this manual

A country should not describe itself as ready solely because every box is ticked.

The evidence behind each box should be auditable.

Minimum evidence package:

- laws;
- contracts;
- permits;
- audited accounts;
- production data;
- response drill;
- source register;
- stakeholder record;
- corrections log.

The objective is an institutional system capable of learning as evidence changes.

NATURAL RESOURCE FUND RECONCILIATION WORKBOOK

Companion technical annex for Part IV of the flagship publication. Working status: partially populated from audited and official period reports. Evidence cut-off: 27 September 2026.

Why this workbook exists

Natural Resource Fund figures appear in several valid but different forms:

- petroleum revenue recognized under financial reporting;
- cash received into the fund;
- revenue receivable at period end;
- investment income;
- government withdrawals;
- transfers to the Consolidated Fund;
- cash and cash equivalents;
- market value;
- closing net assets;
- annual forecasts.

The flagship must never place these values in one time series without reconciling their accounting basis.

Control equation

For a cash-based period bridge:

Opening cash + cash petroleum receipts + cash investment receipts + other cash inflows

- transfers/withdrawals
- cash expenses

= closing cash.

For audited accrual statements, add or subtract receivables, payables, unrealized gains/losses and any classification changes required by the financial statements.

Verified early-year audit observations

2020

Source: Bank of Guyana audited NRF financial statements. [S45]

Audited statement of income, G\$'000:

- Royalties: 2,694,223
- Profit oil: 38,651,798
- Interest on deposits: 1,581
- Net investment gain: 993
- Total income: 41,348,595

Audited cash-flow statement, G\$'000:

- Net cash flow from operating activities: 31,060,827
- Government withdrawal: 0
- Closing cash and cash equivalents: 31,060,827

The notes identify revenue receivables of G\$10,287,768, representing the fourth 2020 profit-oil lift, which was due in January 2021.

The notes report three royalty cash deposits during 2020 totaling USD 12,921,933.23.

The audited profit-oil note recognizes four 2020 lifts totaling USD 185,380,324.44, while the fourth lift was a year-end receivable rather than 2020 cash.

This produces the first key reconciliation rule:

2020 recognized petroleum revenue is not identical to 2020 cash petroleum deposits.

2021

Source: Bank of Guyana audited NRF financial statements. [S46]

Audited statement of changes in equity, G\$'000:

- Opening balance at 31 December 2020: 41,348,595
- 2021 royalties: 10,841,067
- 2021 profit oil: 74,478,746
- 2021 income: 25,902
- Deposits to Consolidated Fund: 0
- Closing balance at 31 December 2021: 126,694,310

Audited cash-flow statement, G\$'000:

- Net cash flow from operating activities: 10,313,670
- Royalties: 10,841,067
- Profit oil: 74,478,746
- Government withdrawal: 0
- Closing cash and cash equivalents: 126,694,310

The 2021 statement carries restated 2020 comparatives. The final dataset should use the latest audited comparative where later statements explicitly restate an earlier year.

Withdrawal history verified from Bank of Guyana quarterly reporting

Source: Bank of Guyana June 2025 NRF Quarterly Report. [S47]

The 2022-2025 transfer figures above describe cash withdrawals. They do not show the final use, project completion or social outcome of the transferred funds.

June 2025 fund snapshot

Source: Bank of Guyana June 2025 Quarterly Report. [S47]

For Q2 2025 the report states:

- opening market value at end-March: USD 3,339.38 million;
- end-June market value: USD 3,192.15 million;
- Q2 petroleum inflows: USD 617.95 million;
- Q2 outflows: USD 800 million;
- Q2 net return: USD 34.82 million.

The financial summary reports in G\$'000:

- Q1 opening balance: 676,725,568
- Q1 royalties: 17,549,019
- Q1 profit oil: 78,273,383
- Q1 outflow: 83,400,000
- Q1 investment result: 7,112,071
- Q1 closing balance: 696,260,041
- Q2 royalties: 17,810,764
- Q2 profit oil: 111,032,905
- Q2 outflow: 166,800,000
- Q2 investment result: 7,260,398
- Q2 closing balance: 665,564,108

This reporting should be kept separate from Table 6's transaction-by-transaction cash-receipt chronology until receivable timing is reconciled.

Cash-receipt chronology control

Bank of Guyana Table 6 supplies a dated petroleum cash-receipt history from inception through 30 June 2025.

The final reproducible dataset should store every transaction with:

- deposit date;

- petroleum source;
- FPSO/project where identified;
- receipt type: profit oil or royalty;
- USD amount;
- cargo/lift identifier where public;
- production period;
- sale/lift date;
- invoice date;
- receivable-recognition period;
- fund-deposit period.

The paper should then generate two separate annual charts:

1. **Cash deposited by calendar year.**
2. **Petroleum revenue recognized by accounting year.**

Differences between the charts are timing differences unless other accounting adjustments apply.

2026 H1 government reporting bridge

Existing source S07 records the September 2026 Ministry of Finance mid-year report as stating for the first half of 2026:

- government profit-oil cash receipts: USD 1,778.6 million;
- royalties: USD 218.4 million;
- withdrawals: USD 1,020 million;
- end-June NRF cash: USD 4,294.2 million.

These figures are official mid-year reporting, not yet a final audited 2026 annual account.

The final publication should reconcile them against:

- Bank of Guyana Q1 and Q2 2026 reports;
- monthly NRF reports;
- later audited 2026 statements when available;
- any period-end receivables.

Annual master table

The table below deliberately leaves fields blank until the relevant audited statement is extracted and reconciled.

Withdrawal-rule legal crosswalk to build

The final legal table will compare:

- 2019 Act;
- 2021 Act;
- 2024 amendment;
- annual maximum withdrawal calculations;
- parliamentary approval;
- notification requirements;
- actual transfer dates.

The table will not state a preferred political spending ratio.

Public-investment separation

The NRF workbook ends when money reaches the Consolidated Fund.

A separate public-investment dataset must follow funds into:

- appropriation;
- procurement;
- contract award;
- physical completion;
- commissioning;
- maintenance;
- service delivery;
- beneficiary outcomes.

This separation is required to avoid treating a lawful withdrawal as proof of a completed national-development result.

Audit controls

Before flagship publication:

1. Extract audited 2022-2025 statements.
2. Normalize currencies and accounting units.
3. Reconcile every annual opening balance to prior closing balance.
4. Reconcile receivables across year boundaries.
5. Reconcile Bank of Guyana cash-deposit dates to Ministry petroleum-lift records.
6. Reconcile withdrawals to parliamentary notifications and Consolidated Fund records.

7. Record investment income separately.
8. Preserve all restatements.
9. Explain differences between market value, cash, net assets and accumulated capital.
10. Have an independent accountant review the final annual bridge.

Publication rule

No NRF figure enters the final executive summary unless its accounting basis, date and currency are explicit.

ENVIRONMENTAL PERMIT AND RISK REGISTER

Companion evidence register for Part V of the flagship publication. Evidence cut-off: 27 September 2026. Status: partially populated. Permit existence, permit conditions and observed compliance are recorded separately.

Control rule

Each project is tracked through six different evidence classes:

1. application and public notice;
2. environmental impact assessment and management plans;
3. environmental permit;
4. production licence environmental obligations;
5. monitoring, incidents and regulator enforcement;
6. spill response, insurance and financial assurance.

A project is not described as environmentally compliant merely because a permit exists.

Project register

Liza Phase 1 permit chronology

The official permit record states:

- application submitted 5 July 2016;
- approved EIA dated 1 June 2017;
- Environmental and Socioeconomic Management Plan dated 1 June 2017;
- revised Oil Spill Response Plan and Wildlife Response Plan dated 1 June 2017;
- environmental permit issued 1 June 2017;
- permit modified 12 May 2021. [S48]

This sequence gives the flagship a dated example of environmental assessment occurring after the 2015 commercial discovery but before production began.

The 2021 EPA notice says the permit modification responded to resumed flaring and projected exceedance of the EIA's estimated flaring volume. The agency said the modification introduced emissions reporting, technical flaring requirements, time limits and payment of US\$30 per tonne of CO₂e for flaring beyond the specified timelines. [S55]

The publication must still test observed performance. The next dataset needs:

- monthly flaring volume;
- duration and cause;

- permitted versus exceptional flaring;
- tonnes CO2e;
- payments assessed;
- payments received;
- equipment failures;
- corrective action;
- inspection findings.

Uaru permit controls

The official Uaru environmental permit states an effective period from 27 April 2023 through 26 April 2028. The permit text gives the EPA authority to order sampling and analysis of project discharges, effluent or waste by an approved independent laboratory at the permit holder's expense. It also reserves the authority to suspend, modify or cancel the permit under specified conditions and links annual review to the previous year's required audit. [S50]

Those provisions establish regulatory powers. They do not establish the frequency with which powers have been exercised or the findings of any audit.

The Uaru production licence separately requires compliance with environmental law, adequate expert personnel and equipment for spill response, dispersant arrangements and access to capping-stack capability. [S56]

The final flagship should map every response obligation to evidence of:

- equipment physically in-country;
- subscription contracts;
- mobilization time;
- drills;
- regulator observation;
- stock inspection;
- personnel competence;
- transboundary notification;
- replacement after use or expiry.

Oil Pollution Act 2025 status control

Parliament records the Oil Pollution Prevention, Preparedness, Response and Responsibility Act 2025 as Act 6 of 2025, passed 16 May 2025 and published 30 May 2025. The Official Gazette archive records an extraordinary gazette dated 28 May 2025. [S53, S54]

The register does not yet mark the full regime as commenced.

Before publication, obtain:

- commencement order;
- implementing regulations;
- designation of the competent national authority;
- prescribed financial-responsibility requirements;
- response-plan regulations;
- reporting forms;
- enforcement procedures.

If no commencement instrument is verified by the evidence cut-off, the final paper will state only that the law was enacted/published and separately describe government statements about planned commencement.

Financial assurance register

Financial assurance must be decomposed into separate legal instruments:

- operator primary liability;
- commercial insurance;
- parent/company guarantee;
- affiliate guarantee;
- capping and containment service subscriptions;
- pollution-response contracts;
- government contingency resources;
- compensation and claims procedure.

Required fields:

- instrument name;
- parties;
- effective date;
- maximum amount if any;
- scope of covered loss;
- exclusions;
- attachment point;
- currency;
- claims jurisdiction;
- proof of financial capacity;
- termination rights;
- regulator access;
- public disclosure status.

A headline "coverage" number will not be presented unless these terms are independently verified.

Environmental incident taxonomy

Every event should be coded as:

- flaring exceedance;
- produced-water issue;
- hydraulic-fluid spill;
- crude-oil spill;
- chemical release;
- vessel incident;
- well-control event;
- air-emission exceedance;
- waste-management incident;
- wildlife impact;
- permit/reporting noncompliance;
- near miss.

Fields:

- date;
- project/FPSO;
- material;
- volume;
- regulator notification;
- immediate response;
- enforcement;
- monetary sanction;
- corrective action;
- independent sampling;
- closure evidence.

Marine ecological evidence

The final paper requires primary or independently reviewed data on:

- fisheries;
- marine mammals;

- turtles;
- seabirds;
- benthic habitat;
- coastal mangroves;
- water quality;
- currents;
- sediment;
- shoreline sensitivity;
- protected areas;
- spawning and nursery grounds.

For each dataset record:

- collector;
- funder;
- method;
- observation dates;
- raw-data availability;
- geographic coverage;
- peer review;
- limitations.

Spill response stress tests

The flagship will test institutional preparedness under at least four scenarios:

Scenario A: controlled surface release

Test local containment, boom availability, skimming, temporary storage, shoreline protection and waste disposal.

Scenario B: subsea loss of well control

Test capping stack mobilization, intervention capability, relief-well assumptions, dispersant logistics, aerial surveillance and sustained response.

Scenario C: adverse weather plus spill

Test port closure, aviation limits, responder safety, communications redundancy and delayed external support.

Scenario D: transboundary movement

Test notification, evidence-sharing, neighbouring-state coordination, compensation, fisheries closures and jurisdiction.

These are preparedness tests, not predictions of event probability.

Evidence hierarchy for environmental claims

Highest weight:

- signed permits;
- enacted/commenced law;
- regulator inspections;
- laboratory results;
- audited monitoring reports;
- authenticated judgments;
- independently verified incident records.

Supporting evidence:

- EIA models;
- operator monitoring;
- government statements;
- academic studies;
- civil-society submissions;
- media investigations.

Where sources conflict, the final chapter will state the disagreement and identify what evidence would resolve it.

Remaining environmental evidence gates

1. Extract Liza Phase 2 amended permit.
2. Extract Payara permit and amendments.
3. Obtain final Yellowtail permit.
4. Extract Whiptail permit conditions.
5. Build all-project EIA metadata table.
6. Compile flaring by project and month.
7. Compile produced-water volumes and standards.
8. Compile regulator inspection and audit records.
9. Verify Oil Pollution Act commencement and regulations.

10. Obtain authenticated 2026 Court of Appeal judgment concerning financial assurance.
11. Obtain actual insurance/guarantee documents where public.
12. Compile spill equipment and drill evidence.
13. Add independent fisheries and biodiversity analysis.
14. Record stakeholder and community submissions.
15. Have an external environmental-law specialist review the final chapter.

Publication boundary

The environmental chapter will neither assume environmental success from permit issuance nor infer regulatory failure from the existence of incidents. It will test the full chain from assessment through permit, monitoring, enforcement, remediation and financial responsibility.

MODEL METHODS AND LIMITATIONS

Working technical annex for the flagship Guyana petroleum case study. Evidence cut-off: 27 September 2026.

Purpose

The repository contains reproducible arithmetic for contract sensitivity, production shocks and sovereign-fund stress testing.

These models are decision aids. They are not forecasts of Guyana's petroleum revenue, operator returns, future production, future oil prices or future Natural Resource Fund balances.

Contract waterfall

The simplified model uses five principal inputs:

- illustrative barrels;
- illustrative realized price;
- royalty fraction;
- cost-recovery ceiling;
- requested recoverable-cost fraction.

The model then calculates:

Gross sales = barrels x price.

Actual cost recovered = gross sales x the smaller of requested cost fraction and the cost-recovery ceiling.

Profit pool = gross sales - actual cost recovered.

Government profit oil = profit pool x illustrative government profit share.

Royalty = gross sales x royalty fraction.

Government total in the teaching model = government profit oil + royalty.

The current teaching implementation uses the published 2% royalty, 75% cost-recovery ceiling and equal residual profit split associated with the 2016 Stabroek agreement's public fiscal explanation. Contract-specific accounting, valuation, taxation, timing, recoverability, decommissioning and dispute provisions require clause-level review before any result is described as an actual entitlement.

Why a percentage of gross sales is not a permanent government take

When recoverable costs use the full ceiling, the residual profit pool is smaller.

When actual recoverable costs fall below the ceiling, the residual profit pool is larger.

Later investment can add new recoverable costs.

For this reason, the 14.5% simplified binding-cap illustration should not be applied as a constant percentage across every field, month or development stage.

Contract sensitivity dataset

ILLUSTRATIVE_SCENARIOS.csv contains 16 synthetic combinations:

- oil price: USD 40, 60, 80 and 100 per barrel;
- requested recoverable cost fraction: 20%, 50%, 65% and 75%;
- illustrative barrels: 10 million in each scenario.

The purpose is to show arithmetic sensitivity while holding volume constant.

No scenario is labelled as a Guyana forecast.

Combined production shock dataset

ILLUSTRATIVE_PRODUCTION_SHOCKS.csv contains 18 synthetic combinations:

- production volume factor: 70%, 85% and 100% of an illustrative 10-million-barrel period;
- price: USD 40, 60 and 80;
- requested recoverable cost fraction: 50% and 75%.

This demonstrates that a price shock, production interruption and high cost recovery can occur together.

The cases are arithmetic stress tests. They do not estimate the probability of any operational outage.

Sovereign-fund stress model

The fund model uses a simplified annual identity:

Closing balance = opening balance + deposits + illustrative investment return - withdrawals.

The return assumption is applied to the opening balance only.

Actual sovereign-fund performance depends on intra-year cash timing, portfolio weights, market values, realized/unrealized gains, expenses and accounting standards.

The synthetic fund cases are:

1. illustrative baseline;
2. lower deposits following a low-price condition;
3. higher spending;

4. market loss followed by recovery.

Each case begins with an illustrative USD 4 billion opening balance.

The figures deliberately do not reproduce the legal Guyana NRF withdrawal formula or predict actual future petroleum deposits.

Why the fund model is separate from the NRF reconciliation

NRF_RECONCILIATION_WORKBOOK.md reconstructs observed historical public accounts.

ILLUSTRATIVE_FUND_STRESS.csv models hypothetical future paths.

Observed and hypothetical values must never appear in one chart without explicit labeling.

Model review gates

Before the fiscal models enter the final flagship:

1. Extract the executed PSA's relevant fiscal and accounting clauses.
2. Review the accounting annex and valuation provisions.
3. Reconcile simplified formulas with actual petroleum entitlements.
4. Validate cost-bank treatment.
5. Separate royalty and profit-oil timing.
6. Review tax treatment.
7. Review decommissioning treatment.
8. Test unit consistency.
9. Test extreme cases and zero cases.
10. Obtain independent petroleum-fiscal review.

Before the fund model enters the final publication:

1. Reconcile audited annual NRF statements.
2. Verify statutory withdrawal rules by legal version.
3. Use transparent deposit assumptions.
4. Use transparent return assumptions.
5. Model liquidity separately from long-horizon return.
6. Include adverse market-return periods.
7. Include oil-price and production shocks.
8. Identify any public-debt interaction.
9. Identify disaster-related fiscal exposure.
10. Obtain independent public-finance review.

Reproducibility

The source code is `illustrative_model.py`.

The generated datasets are:

- `ILLUSTRATIVE_SCENARIOS.csv`;
- `ILLUSTRATIVE_PRODUCTION_SHOCKS.csv`;
- `ILLUSTRATIVE_FUND_STRESS.csv`.

The final publication should record the Git commit associated with every model output and include a checksum manifest.

Interpretation rule

A model result establishes only what follows from its stated assumptions.

It does not establish:

- future oil prices;
- future production;
- actual recoverable costs;
- future government receipts;
- optimal fiscal policy;
- expected spill loss;
- household welfare.

Any final decision statement must return to observed evidence, legal authority and disclosed assumptions.

LEGAL AND REGULATORY EVIDENCE LEDGER

Pinpoint control for high-risk petroleum, environmental and appellate propositions

Evidence cut-off: 27 September 2026

This ledger is designed for legal and technical audit. It separates the controlling instrument, the proposition established, the pinpoint reference, and what the source does not establish.

Legal-language rules for Edition 1.2

1. Use "published in the Official Gazette" when the evidence establishes gazetting.
2. Use "commenced" only where a commencement provision or order establishes operation.
3. Use "the government states", "EPA states", or "contemporaneous reporting states" where authenticated judicial reasons are unavailable.
4. Never equate unlimited liability with an unlimited financial-security instrument unless the controlling judgment or instrument says so.
5. Never describe a negotiated or announced guarantee amount as equivalent to reviewed, enforceable and liquid financial protection without the executed instrument.
6. Separate permit wording, permit compliance, insurer strength, guarantee amount and loss-event recoverability.
7. Record appeal status by date because the legal position may change after the evidence cut-off.
8. Preserve paragraph and section pinpoints for every material legal proposition.

Residual legal evidence requests

- Authenticated full reasons from the Guyana Court of Appeal for the 7 May 2026 decision.
- Current CCJ docket or filed notice/appeal materials following the reported July 2026 leave.
- Executed parent/affiliate guarantee and indemnity instrument, including amount, term, governing law, notice, renewal/cancellation, insolvency and draw provisions.
- Current evidence of project-specific insurance limits and policy scope.
- Authoritative commencement order or commencement provision for Act 6 of 2025 and any implementing regulations in force by 27 September 2026.

These are residual evidence requests, not assumptions that the documents do not exist.

CORRECTIONS REGISTER

Guyana's Petroleum Transformation

This register records material corrections made after internal audit. It exists so later readers can distinguish substantive correction from silent replacement.

Correction policy

A future correction is material if it changes:

- a legal status;
- a material number;
- a source attribution;
- a chronology;
- a fiscal calculation;
- an environmental or social outcome characterization;
- an audit conclusion;
- a publication-status claim.

Material corrections should receive a new register entry, date, edition reference and disposition. Silent alteration of a material claim is not permitted.

Technical annexes

TECHNICAL ANNEX A

Exploration-to-production chronology, legal version control and institutional crosswalk

Purpose: provide a reproducible chronology for analysts, regulators and governments using Guyana as a case study. Source IDs refer to SOURCE_REGISTER.md. Dates in this annex distinguish discovery, signature, effective date, assent, publication, commencement, first production and cash receipt.

A1. Why chronology is a control, not a narrative convenience

Resource-governance analysis becomes unreliable when events from different legal periods are merged. A contract signed in one fiscal environment should not be evaluated as though later model terms applied to it. A statute that was passed after first oil should not be described as a pre-production safeguard. A carbon-finance programme that began years before petroleum production should not be portrayed as a response to the oil boom. Guyana's case therefore needs a strict version-controlled chronology.

The chronology has six date types. The first is geological, such as an exploration well or commercial discovery. The second is contractual, such as signature or effective date. The third is statutory, which may include passage, assent, publication and commencement. The fourth is administrative, such as a production licence or environmental permit. The fifth is operational, such as first oil or a production milestone. The sixth is financial, such as receipt, deposit, investment return or withdrawal.

A robust database should store these as separate fields. One free-text "date" field is insufficient. The Stabroek record illustrates the problem: the petroleum agreement was signed in June 2016, while the official block register records a later effective date. [S02, S03, S41] The NRF also requires several dates because the 2019 law was later replaced in 2021 and withdrawal provisions changed again in 2024. [S04, S05, S37]

A2. Pre-discovery institutional baseline

Petroleum law and historical exploration

Guyana's petroleum exploration history reaches well before Liza. The official history identifies licence activity from the twentieth century, including onshore and offshore exploration, investment-promotion efforts and periods of interruption. [S35] The 1986 Petroleum (Exploration and Production) Act provided an important legal foundation for exploration and production rights before the modern offshore boom. [S38]

The analytical implication is that the modern petroleum industry did not emerge into a legal vacuum. Any assessment of post-2015 capacity should identify which powers, offices and

procedures already existed and which were created or materially strengthened later. This avoids attributing all institutional development to the discovery itself.

Environmental law

The Environmental Protection Act 1996 established the Environmental Protection Agency and an environmental-impact framework before the petroleum discovery. [S39] The pre-discovery baseline therefore includes statutory environmental authority, even though offshore petroleum later required project-specific permits, impact assessments and technical capability far beyond the general statute.

A new producer should run the same baseline test. Before drafting petroleum-specific environmental provisions, it should identify existing environmental, maritime, fisheries, disaster-response, planning, labour, tax and customs law. Overlapping authority should be resolved before a commercial discovery produces time pressure.

Low-carbon development

Guyana's first Low Carbon Development Strategy began in 2009, and the Guyana-Norway results-based forest partnership operated during the pre-oil era. [S22] This matters for historical causality. Forest monitoring, climate-finance relationships and aspects of national development strategy preceded first oil. Petroleum revenue and forest-carbon finance later coexisted, but one should not be described as the origin of the other.

A3. Discovery and contract period, 2015-2016

The Liza-1 well record identifies the May 2015 exploration event in the Stabroek Block. [S36] Operator historical reporting also identifies the commercial discovery and subsequent development sequence. [S01]

The 2016 Stabroek petroleum agreement belongs to the discovery-to-development phase. Its terms should be evaluated against the executed agreement and the information available at the time. [S02] Later discoveries and production success can inform retrospective analysis, but they do not alter the uncertainty that existed at signature.

The legal record should retain at least these fields:

- block;
- parties;
- agreement signature date;
- effective date;
- licence date;
- work obligations;
- relinquishment terms;
- fiscal provisions;
- cost-recovery provisions;

- valuation provisions;
- tax provisions;
- audit rights;
- decommissioning duties;
- transfer and assignment provisions;
- environmental obligations;
- dispute-resolution mechanisms;
- amendment history.

This crosswalk prevents analysts from quoting one fiscal term without showing the wider contractual system.

A4. Discovery-to-first-oil institution building, 2016-2019

A major part of the Guyana case is the use of the period before recurring petroleum cash to design revenue-management institutions. The pre-oil sovereign wealth fund Green Paper documents technical work beginning in the years after discovery, including inter-agency preparation and external technical support. [S28]

The NRF Act 2019 was enacted before first oil. [S37] This is a concrete timing fact. The more demanding question is whether the entire operational architecture, including banking, reporting, investment, withdrawal authorization and audit, was ready for the first material receipts. That question should be answered through financial statements and transaction records rather than the date of the law alone. [S44-S47]

The period also included project-level environmental approvals for the first developments. Liza Phase 1 permit records document the environmental authorization framework and later modifications. [S48] This gives the chronology a second institutional track alongside finance.

A5. First oil and early production, 2019-2021

Liza Phase 1 began production in December 2019. [S01, S83] First oil is operationally distinct from the first cash deposit, because crude must be produced, lifted, sold, invoiced and settled before cash reaches the sovereign account.

The early production period should therefore record:

- first production date;
- first government entitlement;
- first government cargo;
- invoice date;
- settlement date;
- first NRF deposit;

- first audited annual account;
- first parliamentary withdrawal;
- first cost audit covering producing years.

The IMF's 2023 report states that first petroleum receipts entered the fund in 2020 and the first budget transfer occurred in 2022. [S29] These events demonstrate why "first oil" and "first fiscal use" should never be treated as the same date.

In 2021, the NRF framework was replaced and the Local Content Act was enacted. [S05, S57] Those are post-first-oil reforms. A chronology-based case study should describe them as institutional adaptation during early production rather than safeguards that existed before the first barrel.

A6. Rapid scale-up, 2022-2026

Liza Phase 2 began production in February 2022. [S82] Payara followed in November 2023. [S83] Yellowtail began in August 2025. [S84] By 2026, four FPSOs were operating and the government's production dashboard provided a monthly production series through July. [S81]

This sequence means regulatory and fiscal workload expanded quickly. Every new development adds permit conditions, production metering, cost-recovery claims, decommissioning obligations, emergency-response requirements and public revenue flows. The administrative burden scales faster than a simple count of FPSOs suggests.

The 2024 NRF withdrawal amendment belongs to this mature-early-production phase, not the pre-first-oil design. [S04] Likewise, the 2025 oil-pollution statute belongs to a period when multiple projects were already operating or under development. [S53-S54]

A7. Legal version-control table

For every legal instrument, the final publication should carry a version table with:

- title;
- bill or act number;
- parliamentary passage date;
- assent date;
- publication date;
- commencement date;
- amendment dates;
- repealed provisions;
- responsible minister or agency;
- regulations issued;
- current status at the evidence cut-off;
- authoritative source URL.

The following Guyana instruments are mandatory in the final crosswalk:

1. Petroleum (Exploration and Production) Act 1986. [S38]
2. Environmental Protection Act 1996. [S39]
3. Natural Resource Fund Act 2019. [S37]
4. Natural Resource Fund Act 2021 and 2024 amendment record. [S04-S05]
5. Local Content Act 2021. [S57]
6. Petroleum-related model fiscal or licensing terms issued after the original Stabroek agreement. [S14]
7. Oil Pollution Prevention, Preparedness, Response and Responsibility Act 2025. [S53-S54]

Where a statute has not fully commenced, the final paper should identify which provisions are operative. Enactment should never be used as shorthand for implementation.

A8. Permit and licence chronology

Petroleum projects should be tracked through separate administrative records:

- discovery well;
- appraisal;
- development application;
- EIA scoping;
- public notice;
- environmental permit;
- production licence;
- construction milestones;
- first oil;
- permit amendment;
- environmental audit;
- incident and enforcement history.

Liza Phase 1, Liza Phase 2, Payara, Yellowtail, Uaru and Whiptail should each have an individual row. [S48-S56, S86] Projects not yet producing at the evidence cut-off should retain blank actual-production fields.

This approach prevents a sanctioned project from being treated as an operating project. It also prevents environmental conditions from being generalized across different developments.

A9. Judicial and dispute chronology

A resource-governance chronology should include litigation and contractual dispute processes, but the database should not equate an allegation with a final decision.

For court matters, record:

- court;
- case number;
- filing date;
- issue;
- interim order;
- trial judgment;
- appeal;
- appellate judgment;
- further appeal;
- final status;
- operative remedy.

For contract audits, record:

- audit period;
- auditor;
- amount reviewed;
- proposed adjustment;
- operator response;
- negotiation status;
- sole-expert or arbitration step;
- final determination;
- cash consequence.

Guyana's cost-audit records and the financial-assurance litigation illustrate both types. [S20, S42-S43, S71-S72] The final publication should use authenticated legal texts before stating the legal effect of the appellate decision.

A10. Source hierarchy for dates

When dates conflict, use this hierarchy:

1. enacted statute or official gazette for legal events;
2. executed agreement for contract signature and obligations;
3. regulator or ministry licence register for administrative events;
4. audited financial statements for accounting events;
5. regulator production series for actual production;
6. operator announcement for project start where regulator data are unavailable;
7. media reporting only when the primary record is unavailable and the limitation is stated.

The hierarchy is not absolute. A later audited restatement may supersede an earlier official figure. Version history should therefore be preserved.

A11. New-producer chronology template

A prospective producer should build the chronology before commercial discovery. The minimum columns are:

This template turns chronology into an audit tool.

A12. Decision-window analysis

The Guyana case contains four decision windows.

Window 1: before discovery. The priority is legal clarity, environmental baseline science, data ownership and compact technical capability.

Window 2: discovery to final investment decision. The priority is independent commercial analysis, contract interpretation, environmental alternatives and fiscal design.

Window 3: final investment decision to first oil. The priority is revenue custody, metering, audit capacity, spill preparedness, public communication and budget rules.

Window 4: scale-up. The priority shifts toward cost verification, public investment quality, local value, macroeconomic stability, environmental compliance and statistical capacity.

The windows are more transferable than the exact Guyanese calendar. A country with a longer development period has more time to prepare. A country with a single nearshore discovery may need a different environmental sequence.

A13. Chronology failure modes

Common errors include:

- applying later fiscal terms to an earlier agreement;
- describing a law as operational before commencement;
- using an operator's target as an actual production date;
- reporting a fund deposit as budget spending;
- attributing a pre-existing climate programme to petroleum;
- using a public announcement as evidence of completed infrastructure;
- treating a disputed audit adjustment as recovered revenue;
- combining a court order and an appellate reversal without version control.

The final publication's timeline should visibly flag each risk.

A14. Chronology as a policy-transfer screen

Before transferring any instrument from Guyana, another country should ask:

1. At what stage did Guyana adopt it?
2. What problem existed at that time?
3. Which institutions already existed?
4. Was the instrument later amended?
5. What evidence shows implementation?
6. What remained contested or incomplete?
7. Does the receiving country face the same timing problem?

A rule adopted after first oil cannot be treated as proof of what must exist before first oil without explaining the difference. The purpose of the case study is to make timing explicit.

A15. Annex conclusion

Guyana's value as a case study lies partly in the density of dated transitions. Exploration, discovery, contract negotiation, pre-oil fund preparation, first production, revised legislation, local-content rules, project expansion, environmental law and ongoing accountability reforms occurred within a relatively short period. That makes chronological discipline essential.

For a newcomer, the principal operational lesson is to build a version-controlled public chronology before revenue arrives. The chronology becomes the backbone for contract audit, environmental oversight, sovereign-fund accounting and later institutional review.

TECHNICAL ANNEX B

Petroleum fiscal mechanics, cost recovery, revenue assurance and lift-to-cash accounting

Purpose: provide a transparent analytical and audit framework for understanding petroleum fiscal flows without converting a simplified example into a claim about observed government take.

Source IDs refer to SOURCE_REGISTER.md.

B1. Why petroleum revenue must be reconstructed

Public petroleum revenue is produced by a chain of physical, contractual and financial events. A barrel must be produced, measured, allocated, lifted, valued, sold and settled. Contractual royalty and profit-oil rules must be applied. Recoverable costs must be verified. Cash must then be deposited in the legally designated account.

The chain is easy to compress into a single percentage, but doing so can create serious analytical errors. Gross field revenue is not government revenue. Government entitlement is not the same as government cash. Profit oil is not the same as royalty. Recoverable cost is not the same as cash paid in the current month. An audit adjustment is not the same as money recovered.

Guyana's Stabroek agreement and public administrative explanations provide a useful case for building the full bridge. [S02, S15]

B2. Core definitions

Gross production is the measured volume produced during the period before entitlement allocation.

Sales volume is the volume commercially sold. It may differ from production because of inventory and lifting timing.

Government lift is a cargo or share of cargo representing government profit-oil entitlement or another contractual entitlement.

Royalty is the contractual payment calculated under the agreement's royalty provisions.

Recoverable cost is expenditure eligible for recovery under the contract, subject to classification, accounting, audit and any recovery ceiling.

Cost bank is the stock of eligible unrecovered cost carried forward under the contractual rules.

Profit oil is the production or revenue remaining after the applicable cost-recovery allocation, divided under the agreement's sharing formula.

Government take is a broad economic concept that may include royalties, profit oil, taxes, bonuses, state participation and other fiscal instruments. It should not be reported without defining numerator, denominator and time horizon.

B3. Simplified Stabroek illustration

Public explanations of the 2016 Stabroek system describe a 2% royalty, an up-to-75% cost-recovery ceiling and an equal division of residual profit oil. [S02, S15]

A simplified period with normalized gross revenue of 100 and the full 75 units used for eligible cost recovery leaves 25 units of profit oil. An equal split gives the state 12.5 profit units. Adding a 2-unit royalty under the stated convention produces 14.5 combined units in the simplified example.

This number is not a permanent government percentage. If only 50 units are needed for cost recovery, 50 units remain for profit oil and the state's half is 25 units, plus royalty under the applicable convention. If only 20 units are needed, the residual pool is larger again.

The calculation demonstrates why the cost bank matters. It does not replace the executed agreement's accounting provisions.

B4. Revenue waterfall model

A reproducible fiscal model should use the following sequence:

1. measured production;
2. adjustments required by the agreement;
3. valuation price;
4. gross contract revenue;
5. royalty;
6. eligible recoverable cost submitted;
7. cost-recovery ceiling;
8. actual cost recovery allowed;
9. unrecovered cost carried forward;
10. residual profit oil;
11. government profit share;
12. contractor profit share;
13. applicable tax treatment;
14. other payments;
15. government entitlement by volume and value;
16. settlement date;
17. sovereign-fund deposit date.

Each row should retain the formula and source.

B5. Price valuation

Petroleum valuation should identify:

- crude grade;
- benchmark;
- quality differential;
- loading date;
- pricing window;
- transportation adjustment;
- related-party transaction policy;
- foreign-exchange convention;
- dispute process.

A state may receive physical crude and sell it separately or receive cash under other arrangements. The revenue model should not assume the operator's average realized price equals the government's sale price.

For audit, every government cargo should have a unique transaction ID connecting entitlement, bill of lading, buyer, price formula, invoice, settlement and deposit.

B6. Cost classification

Cost-recovery audit should classify at least:

- exploration;
- appraisal;
- development drilling;
- subsea equipment;
- FPSO or production facility;
- logistics;
- shore base;
- operating expense;
- staffing;
- insurance;
- financing where permitted;
- related-party services;
- overhead;
- decommissioning;

- environmental monitoring;
- emergency response;
- taxes or duties where contractually relevant.

The audit should record whether the agreement permits, limits or excludes each category.

B7. Related-party transactions

Offshore petroleum projects frequently purchase services from related companies or global affiliates. Revenue assurance therefore requires transfer-pricing and arm's-length review.

The audit file should record:

- supplier;
- beneficial owner;
- relationship to contractor;
- service;
- procurement method;
- invoice;
- comparable market evidence;
- allocation method;
- markup;
- currency;
- contract clause;
- audit conclusion.

A cost may be genuinely incurred yet still be challenged if the contract requires reasonableness, competitive procurement or other standards.

B8. Cost ceiling versus cost bank

The recovery ceiling limits how much eligible cost can be recovered from a period's production or revenue. It does not necessarily eliminate unrecovered cost. Unrecovered eligible cost may carry forward.

A new development can therefore change future government profit oil even after earlier cost recovery declines. If additional approved capital expenditure enters the cost bank, the amount recovered in future periods may rise again.

This explains why a high-revenue period should not be extrapolated mechanically. The 2026 mid-year reporting discussed stronger receipts alongside cost-bank dynamics and future project expenditure. [S07]

B9. Audit status taxonomy

Every cost-audit item should carry one status:

1. submitted;
2. under audit;
3. auditor query;
4. proposed adjustment;
5. government position;
6. operator response;
7. negotiated resolution;
8. sole expert;
9. arbitration;
10. final determination;
11. cash effect recorded.

Guyana's published audit material demonstrates the need for this taxonomy. [S42-S43, S71-S72]
A proposed adjustment is not a cash receipt.

B10. Historical audit periods

The final paper should maintain separate audit histories for:

- pre-production or early exploration cost;
- 1999-2017 period;
- 2018-2020 period;
- later operating years as public records become available.

Each audit should identify the auditor, scope, amount submitted, amount queried, government response and final outcome where verified.

An aggregate statement such as "the audit found X" is insufficient if X remained disputed. The final manuscript should use words such as proposed, disputed, agreed, determined and recovered precisely.

B11. Production-to-lift reconciliation

The physical reconciliation begins with actual project output. The government production dashboard provides a monthly series through July 2026. [S81]

For each month: Opening inventory + production

- operational use or permitted adjustment
- cargoes lifted

= closing inventory

Each cargo should then be tagged by entitlement owner. Government cargoes should be connected to profit-oil calculations for the relevant period.

The reconciliation should recognize timing differences. A cargo lifted in December may settle in January. A government report for the first half of a year may therefore include receipts from prior-year lifts or exclude cash from recent lifts not yet settled.

B12. Lift-to-cash reconciliation

For each government cargo, record:

- cargo ID;
- project/FPSO;
- loading start;
- loading completion;
- barrels;
- crude grade;
- buyer;
- pricing basis;
- invoice value;
- settlement date;
- bank value date;
- sovereign-fund deposit;
- exchange-rate treatment;
- public disclosure reference.

The sum of settled government sales, royalties and investment income should reconcile to sovereign-fund inflows after accounting for any timing differences or permitted adjustments.

B13. Royalty accounting

Royalty should be tracked separately from profit oil. The accounting system should identify:

- contractual rate;
- valuation base;
- volume base;
- calculation period;
- payer;
- invoice date;
- due date;

- cash date;
- audit status.

Combining royalty and profit oil into one reported petroleum-revenue number may be useful for a headline, but the underlying ledger must preserve both streams.

B14. Taxes and contractual tax treatment

The final legal crosswalk should describe how income tax is treated under the executed Stabroek agreement and how later model terms differ. [S14]

A new producer should model taxes as a separate fiscal instrument rather than hiding them inside profit sharing. It should also identify whether the contract contains stabilization or special tax provisions and how those interact with future legislation.

B15. Ring-fencing

Ring-fencing determines whether costs from one field or project can be recovered against revenue from another. The detailed contract text should be reviewed before any quantitative claim is made.

The importance is practical. Without effective ring-fencing, continuing investment in later projects can delay the state's profit share from already producing projects. With strong ring-fencing, each development's economics are more isolated. The model should not assume either result without contractual evidence.

B16. Decommissioning

Decommissioning liabilities occur late but should be modeled early. The fiscal system needs to define:

- who pays;
- when reserves are funded;
- whether contributions are recoverable;
- ownership of the reserve;
- investment of the reserve;
- treatment after transfer of an interest;
- security after operator exit;
- residual state liability.

A newcomer should not treat decommissioning as a distant technical issue. If security is weak, the liability can migrate to the state after production declines.

B17. Government petroleum-sale governance

If the government sells physical crude, sale governance should include:

- transparent buyer-selection process;
- prequalification;
- sanctions and anti-money-laundering checks;
- conflict screening;
- benchmark policy;
- bid evaluation;
- approval authority;
- invoice and collection;
- public disclosure after settlement.

The state should preserve the ability to compare realized sale prices with relevant market benchmarks.

B18. Petroleum revenue forecast structure

A forecast should have at least three cases:

- lower price and lower production;
- central planning case;
- higher price or stronger production.

Each case should state:

- oil price;
- production by project;
- downtime;
- cost recovery;
- royalty;
- profit split;
- tax treatment;
- number of government lifts;
- sale timing;
- exchange rate;
- investment income;
- fund withdrawal assumptions.

A forecast is not an asset. Budgets should distinguish expected revenue from cash already held.

B19. Forecast error analysis

At the end of each year, compare forecast and actual:

- production;
- price;
- cost recovery;
- government lifts;
- royalty;
- profit oil;
- fund interest;
- withdrawals;
- closing fund balance.

The analysis should explain the drivers of error rather than merely update the next forecast.

Guyana's 2026 January budget and September mid-year revision provide an example of why versioned forecasts matter. [S06-S07]

B20. Fiscal sensitivity grid

The final publication should show a transparent grid with at least:

- oil at 40, 60 and 80 USD per barrel;
- cost recovery at 25%, 50% and 75%;
- production at 80%, 90% and 100% of the operating assumption;
- project delay;
- higher operating cost;
- lower realized price differential.

The output should separate gross revenue, royalty, profit oil, tax where relevant, fund inflow and budget withdrawal.

The model should avoid false precision. If contractual details are unavailable, results should be labelled illustrative.

B21. Audit staffing model

A new producer should estimate the audit workload in hours, not job titles. The workload grows with:

- number of contractors;
- number of projects;
- annual invoices;

- related-party transactions;
- currencies;
- development stages;
- audit periods;
- disputes;
- decommissioning arrangements.

The core government team should retain responsibility for the audit position even when external specialists perform detailed testing.

B22. Data architecture

A petroleum revenue database should connect five identifiers:

1. licence;
2. project;
3. cost item;
4. cargo;
5. government receipt.

A sixth identifier links a sovereign-fund deposit to the receipt. A seventh links a budget withdrawal to parliamentary authority.

With these identifiers, an auditor can move from a public fund statement backward to individual cargoes and contract calculations.

B23. Public disclosure without losing audit integrity

Transparency should not compromise legitimate confidential information, but confidentiality should not become a blanket barrier.

A disclosure policy should distinguish:

- executed contract terms;
- commercially sensitive bid information;
- personal information;
- security-sensitive emergency details;
- aggregate cost data;
- final audit findings;
- unresolved disputes;
- government sale prices after settlement.

The default should be a documented reason for withholding, not an undocumented assumption of secrecy.

B24. Revenue-assurance red flags

Red flags include:

- persistent unexplained difference between production and cargo volumes;
- government sale price materially below benchmark without explanation;
- repeated related-party cost growth;
- major cost categories lacking source documents;
- audit periods repeatedly extended;
- disputed amounts reported as recovered;
- fund deposits not traceable to underlying receipts;
- incompatible figures across ministry and central-bank reports;
- material late payment without disclosed interest or enforcement;
- decommissioning security below estimated liability without explanation.

A red flag triggers examination. It is not evidence of misconduct by itself.

B25. Reconciliation certificate

For each fiscal year, a publication-quality reconciliation should certify whether the following agree within documented timing differences:

- regulator production;
- operator entitlement statement;
- government cargo register;
- royalty register;
- sale invoices;
- bank receipts;
- sovereign-fund deposits;
- audited financial statements.

Any unreconciled amount should be listed individually with status.

B26. New-producer first-revenue test

Before first oil, a new government should run a synthetic transaction:

- assume one cargo;
- calculate royalty;

- calculate hypothetical cost recovery;
- calculate government profit share;
- create invoice;
- record settlement;
- deposit into a test sovereign account;
- prepare public notice;
- reconcile to a test ledger;
- prepare mock audit evidence.

If the agencies cannot complete the test, legal readiness has not yet become operational readiness.

B27. Distinguishing fiscal performance from political judgment

Fiscal analysis should describe outcomes such as revenue share, volatility, audit status, savings, withdrawals and debt. It should not convert these into a political score or endorsement.

Where commentators disagree about the quality of an agreement, the final paper should present the assumptions behind competing interpretations: information at signature, geological risk, required return, later discoveries, project cost, oil price, contractual protections and alternative fiscal regimes.

B28. Annex conclusion

The petroleum fiscal system is a chain of auditable events. The most transferable lesson is not a single percentage. It is the ability to reconstruct the chain from physical production to public cash.

A new producer should design the data model, cost-audit function and sovereign-account reconciliation before first oil. Guyana's published agreements, production records, audit material and fund reporting provide a practical case for testing how those pieces interact under rapid scale-up.

TECHNICAL ANNEX C

Natural Resource Fund reconciliation, fiscal absorption and public-investment conversion

Purpose: provide a transparent framework for moving from petroleum receipts to sovereign financial assets and then to measurable public outcomes. Source IDs refer to SOURCE_REGISTER.md.

C1. The fund has two separate jobs

A sovereign resource fund can help separate volatile petroleum cash from the annual spending decision, and it can hold financial assets for future use. Those are related but distinct functions.

Guyana's NRF was established by statute before first oil, replaced by a later framework and subsequently amended. [S04, S05, S37] The chronology is important because the design evolved as production and public revenue expanded.

The analysis should not ask only whether a fund exists. It should ask:

- which receipts enter;
- when they enter;
- who holds the cash;
- how it is invested;
- how withdrawals are calculated;
- who authorizes transfers;
- how parliament appropriates spending;
- how accounts are audited;
- how the public can reconcile transactions.

C2. Stock-flow identity

The core accounting equation is:

Closing financial assets = Opening financial assets + petroleum receipts + investment income + other permitted inflows - lawful withdrawals - permitted expenses +/- valuation movements.

Every annual and quarterly report should disclose each component separately.

A major analytical error occurs when a projected petroleum receipt is added to an observed fund balance. Forecasts belong outside the stock-flow statement until cash is received.

C3. Cash versus market value

If the fund invests in marketable assets, two values matter:

- cash movement during the period;
- end-period market value.

An unrealized market gain is not a petroleum receipt. Interest and dividend income are not royalties. A security sale can change cash without changing total financial wealth.

The reporting framework should therefore include:

- cash balance;
- securities at market value;
- accrued income;
- receivables;
- realized gain/loss;
- unrealized gain/loss;
- currency effect;
- management expense.

The Bank of Guyana's audited statements and quarterly reports provide the primary record for Guyana's fund. [S44-S47]

C4. Annual reconciliation protocol

For every fiscal year, build a table with:

1. opening audited balance;
2. profit-oil receipts;
3. royalties;
4. other petroleum receipts;
5. interest;
6. investment gains/losses;
7. withdrawals to the Consolidated Fund;
8. expenses;
9. closing audited balance.

Then reconcile:

- Ministry of Finance petroleum-revenue reporting;
- Bank of Guyana fund reporting;
- parliamentary withdrawal notices;
- audited annual statements.

Differences should be explained by timing, currency, accrual treatment or later restatement.

C5. 2020 and first receipts

The 2020 audited statement is the first full financial record after first oil and should be treated as the opening historical anchor. [S45] The final paper should identify the first royalty and profit-oil receipts, accounting currency, receivables, cash and any restatement later reported in the 2021 audited statement. [S46]

This historical reconstruction should preserve the audited presentation rather than retrofitting later reporting formats onto earlier years.

C6. 2021 statutory transition

The 2021 audited account records the replacement of the original 2019 framework by the 2021 statute. [S46] The publication should show:

- legal authority in force at each date;
- who governed the fund;
- investment mandate;
- withdrawal rule;
- reporting requirement;
- role of the central bank;
- oversight arrangements.

A legal change should not be represented as a change in cash until an actual transaction occurs.

C7. First budget transfer

The IMF's 2023 analysis records the first budget transfer in 2022. [S29] This creates an important case-study line between accumulation and use.

The final paper should trace the first transfer through:

- statutory authorization;
- parliamentary budget;
- central-bank transfer;
- consolidated-fund receipt;
- appropriation;
- spending;
- project or service outcome.

The first four establish legal and financial use. The later stages establish whether the transfer converted into a public asset or service.

C8. 2024 withdrawal reform

The 2024 amendment changed the withdrawal framework. [S04] A useful legal crosswalk should show the earlier rule beside the amended rule and simulate both using the same petroleum-revenue history.

The comparison should not decide which political choice is preferable. It should show:

- withdrawal capacity under each formula;
- remaining fund balance;
- short-run budget space;
- sensitivity to lower prices;
- implications for future financial assets.

C9. 2026 observed and projected values

The September 2026 mid-year report provides observed data through June and revised full-year projections. [S07] The final edition should present them in separate columns.

Observed first-half data include government petroleum receipts and fund withdrawals. Projected full-year values remain conditional on prices, production, cost recovery and settlement timing.

The publication should clearly label:

- actual to June;
- revised estimate for the year;
- earlier January estimate;
- variance;
- reason for variance.

C10. Intergenerational framing

Petroleum is a depleting natural asset. Extracting it converts underground wealth into cash. If all cash is consumed, the national balance sheet loses the resource without gaining an offsetting financial, physical or human asset.

A national-wealth framework therefore asks what replaces the depleted resource:

- foreign financial assets;
- resilient infrastructure;
- human capital;
- competitive firms;
- stronger institutions;
- environmental assets;

- disaster resilience.

This is broader than the fund itself.

C11. Public investment versus financial saving

A developing producer often faces legitimate infrastructure shortages. Financial saving and domestic investment are therefore competing uses of scarce petroleum wealth, not moral opposites.

The relevant variables include:

- return on external financial assets;
- expected economic return on domestic projects;
- execution capacity;
- maintenance;
- corruption and procurement risk;
- inflationary pressure;
- debt;
- disaster exposure;
- distributional effects.

A domestic project with a high nominal budget does not automatically generate a high social return.

C12. Absorptive capacity

Absorptive capacity measures how much additional spending the economy can convert into real output before prices, delays or quality problems dominate.

Indicators include:

- construction-cost inflation;
- vacancy rates for engineers and skilled trades;
- contractor backlog;
- port congestion;
- imported material delays;
- contract variation rates;
- project completion times;
- public-agency staffing;
- maintenance backlog.

IMF staff highlighted absorption and public-investment monitoring in 2026. [S64]

C13. Public-investment gate

Before a petroleum-funded project enters the capital budget, require:

1. problem definition;
2. options analysis;
3. demand forecast;
4. climate/disaster screen;
5. land and social assessment;
6. capital cost;
7. operating and maintenance cost;
8. procurement plan;
9. delivery schedule;
10. responsible operating agency;
11. outcome measure;
12. independent review.

The gate reduces the risk that large cash availability replaces project appraisal.

C14. Project register

The final flagship should include a model register with:

- project ID;
- sector;
- region;
- sponsoring ministry;
- objective;
- approved cost;
- revised cost;
- funding source;
- procurement route;
- contract date;
- contractor;
- beneficial owner;
- planned completion;
- actual completion;
- commissioning;
- maintenance budget;

- service KPI;
- climate risk;
- audit status.

A project moves from "spent" to "delivered" only after commissioning and service evidence.

C15. Maintenance as part of wealth preservation

Infrastructure without maintenance converts petroleum wealth into a depreciating asset with a short useful life.

Each major asset should have:

- design life;
- annual maintenance requirement;
- replacement cycle;
- responsible agency;
- budget line;
- condition indicator.

The publication should distinguish new construction from lifecycle funding.

C16. Debt and the fund

A country can hold a large sovereign fund while also borrowing. The national balance sheet should therefore report:

- gross public debt;
- net debt after liquid financial assets;
- debt currency;
- maturity;
- interest cost;
- contingent liabilities;
- guarantees;
- public-enterprise debt.

Borrowing may be rational for infrastructure, but the comparison should make financing cost and asset return explicit.

C17. Non-oil primary balance

The non-oil primary balance is one way to measure the fiscal position excluding petroleum revenue and interest expense. IMF staff has discussed its relevance for Guyana. [S64]

The indicator can show whether the non-oil fiscal system is becoming more dependent on petroleum transfers. It should not be used as a political score. It is a diagnostic measure requiring careful treatment of one-off capital spending and economic cycles.

C18. Fiscal sustainability scenarios

The final model should test:

- oil price at lower, central and higher paths;
- production disruption;
- project delay;
- cost-bank rebuild;
- lower financial returns;
- faster withdrawal;
- higher public investment cost;
- debt-financed infrastructure;
- decommissioning liabilities.

The output should show fund assets, withdrawals, debt and non-oil fiscal balance over at least 10-20 years.

C19. Spending floor and ceiling tests

A government may wish to protect essential services during a petroleum downturn while preventing expenditure from rising too quickly during a boom.

The analytical model can therefore calculate:

- minimum recurring service cost;
- legally committed transfers;
- capital-project commitments;
- prudent reserve;
- maximum executable capital programme.

These are analytical boundaries, not prescriptions for policy.

C20. Revenue shock protocol

When projected petroleum revenue falls materially:

1. update the production and price forecast;
2. update contract cost recovery;
3. update fund inflow;
4. test withdrawal rule;
5. identify legally committed spending;
6. rank capital projects by completion value and urgency;
7. publish revised assumptions;
8. avoid hiding the shortfall through optimistic future production.

C21. Windfall protocol

When revenue materially exceeds forecast:

1. reconcile the source;
2. determine whether the increase is temporary;
3. test capacity constraints;
4. quantify inflation risk;
5. evaluate debt reduction;
6. evaluate additional external saving;
7. evaluate a limited number of high-readiness investments;
8. publish the decision logic.

An unexpected receipt should not automatically become a new recurrent commitment.

C22. Public-enterprise exposure

Resource states often expand state-owned enterprises or create new entities for energy, transport and infrastructure.

The balance sheet should include:

- direct debt;
- government guarantees;
- purchase agreements;
- availability payments;
- minimum-revenue guarantees;
- fuel or power subsidies;
- pension liabilities;
- off-balance-sheet commitments.

The fund cannot be assessed in isolation from these exposures.

C23. Climate and disaster exposure

Guyana's coastal geography makes resilience spending relevant to national wealth. The same principle applies differently to islands exposed to hurricanes, storm surge and sea-level rise.

A project evaluation should identify:

- hazard;
- design standard;
- expected lifetime;
- residual risk;
- failure consequence;
- insurance;
- maintenance;
- recovery cost.

A resilient asset may justify higher initial cost if lifecycle risk is lower.

C24. Human capital

Petroleum wealth can be converted into human capital through education, health and skills, but expenditure alone is not the measure.

Use outcome indicators:

- learning;
- completion;
- technical certification;
- health access;
- mortality;
- specialist staffing;
- labour productivity.

A scholarship count is an input. Completion and labour-market performance are closer to outcomes.

C25. Regional distribution

A national fund may finance projects across regions with different service gaps.

The public-investment register should record region and beneficiary population. Distribution analysis should use service need and outcomes rather than crude per-capita spending alone.

For hinterland and Indigenous communities, community-level evidence and rights procedures require separate analysis.

C26. Fund governance controls

The fund operating model should define:

- board duties;
- investment mandate;
- risk limits;
- custodian;
- external manager selection;
- conflict policy;
- audit;
- performance benchmark;
- public report;
- parliamentary notification.

The institutional roles should not overlap ambiguously.

C27. Investment policy

A fund designed mainly for fiscal stabilization needs higher liquidity than a long-horizon intergenerational fund.

The investment policy should therefore derive from:

- expected withdrawals;
- volatility of petroleum receipts;
- currency needs;
- horizon;
- risk tolerance;
- domestic financial system.

A small new producer should avoid using the sovereign fund as an unstructured domestic development bank unless the law and governance are designed for that role.

C28. Performance attribution

Fund performance should separate:

- benchmark return;
- active management;
- currency;
- fees;
- cash drag;

- asset allocation.

Investment performance should not be mixed with petroleum inflow growth.

C29. Transparency dashboard

A quarterly dashboard should publish:

- opening assets;
- petroleum deposits;
- investment income;
- withdrawal;
- ending assets;
- asset allocation;
- benchmark performance;
- fees;
- currency exposure;
- pending withdrawal authorization.

A companion budget dashboard should show how transferred cash is appropriated and spent.

C30. Outcome verification

For every major petroleum-funded programme, use three measures:

- financial execution;
- physical delivery;
- service outcome.

Example: A school project can be 100% financially spent, 90% physically complete and 0% operational if staffing and utilities are absent.

This three-part test should appear across infrastructure, health, education, housing and energy.

C31. New-producer fund readiness test

Before first cash:

- enact fund law;
- appoint lawful custodian;
- adopt investment policy;
- open accounts;
- run synthetic deposit;
- run synthetic withdrawal;

- produce mock quarterly report;
- commission mock audit;
- test parliamentary notice;
- publish the procedures.

The test should fail if agencies rely on manual assumptions that cannot be independently reproduced.

C32. Ten-year review

A sovereign fund should be reviewed periodically against:

- reserve outlook;
- revenue volatility;
- debt;
- demographics;
- infrastructure stock;
- climate risks;
- public investment performance;
- tax base.

Rules that suited the discovery period may need revision as the country approaches mature production or depletion.

C33. Annex conclusion

The NRF is one component of national wealth conversion. The stronger case-study test connects petroleum receipts, fund accounting, lawful withdrawals, public investment and service outcomes without collapsing them into one headline.

For a new producer, the operational priority is traceability. Every dollar should have a documented path from petroleum entitlement to the sovereign balance sheet and, where spent, to a measurable public asset or service.

TECHNICAL ANNEX D

Environmental authorization, oil-spill preparedness, financial assurance, biodiversity and climate accounting

Purpose: provide a project-level environmental and emergency-response framework suitable for deepwater petroleum and adaptable to island and coastal states. Source IDs refer to SOURCE_REGISTER.md.

D1. Environmental governance is a chain

An environmental permit is one point in a longer chain: baseline science -> impact assessment -> alternatives -> permit -> monitoring -> inspection -> incident response -> enforcement -> remediation -> closure.

A country should not treat any single link as proof that environmental risk is controlled.

Guyana's Environmental Protection Act predates the offshore boom. [S39] Project-specific environmental permits and EIA records later developed for offshore projects. [S48-S56, S86]

D2. Baseline science

Before drilling or development, the state should possess or have access to baseline data on:

- water quality;
- currents;
- sediment;
- fisheries;
- plankton;
- marine mammals;
- seabirds;
- turtles;
- reefs;
- mangroves;
- wetlands;
- shoreline use;
- protected areas;
- tourism;
- community livelihoods.

The baseline must record seasonality. Sampling only once can miss migration, spawning or wet/dry-season variation.

D3. Data ownership and independence

Environmental studies are often operator-funded. Funding source alone does not invalidate them, but the state should define:

- methodology approval;
- laboratory accreditation;
- raw-data access;
- metadata;
- quality assurance;
- independent review;
- publication rules.

A regulator unable to access raw environmental data is dependent on summaries.

D4. EIA scope

A petroleum EIA should consider:

- alternative locations;
- alternative technology;
- no-project alternative;
- routine discharges;
- flaring;
- greenhouse gases;
- produced water;
- drilling waste;
- vessel traffic;
- noise;
- marine fauna;
- cumulative impacts;
- accident scenarios;
- climate hazards;
- decommissioning.

The EIA should identify uncertain impacts rather than convert uncertainty into "no impact."

D5. Cumulative impact

A single project may have manageable impacts while multiple projects create cumulative pressure.

Cumulative analysis should consider:

- number of FPSOs;
- support vessels;
- helicopters;
- shore bases;
- emissions;
- waste;
- fisheries exclusion;
- lighting;
- noise;
- port activity;
- emergency-response demands.

Guyana's rapid multi-project expansion makes cumulative analysis essential.

D6. Permit register

Each project should have a record containing:

- permit number;
- issue date;
- expiry;
- project;
- operator;
- EIA version;
- monitoring requirements;
- flaring conditions;
- waste conditions;
- spill plan;
- wildlife plan;
- financial assurance;
- reporting frequency;
- audit requirement;
- amendment history;

- enforcement history.

Public indices establish document availability, while compliance requires additional evidence. [S49-S56]

D7. Flaring

The Liza Phase 1 permit modification record identifies additional emissions reporting, time limits and a payment for flaring beyond specified periods. [S55]

A complete flaring analysis needs:

- flare volume;
- duration;
- reason;
- equipment failure;
- planned maintenance;
- emergency event;
- permit limit;
- payment;
- corrective action.

Installed flare capacity is not the same as emissions.

D8. Greenhouse-gas accounting

The paper should separate:

1. direct offshore operational emissions;
2. purchased electricity and shore-base emissions;
3. domestic combustion of petroleum products;
4. exported oil combustion;
5. land-use and forest emissions;
6. carbon-credit issuance.

The boundaries are different and should not be netted casually.

Forest credits do not automatically cancel petroleum emissions. The final paper should avoid economy-wide carbon-neutral claims unless the entire accounting boundary is defined and independently supported.

D9. Forest and petroleum coexistence

Guyana's LCDS and forest-finance framework predate first oil. [S22] Later forest-credit transactions exist alongside petroleum expansion. [S23, S32]

The analytical framework should therefore report:

- forest cover;
- deforestation rate;
- credit issuance;
- credit sale;
- cash receipt;
- community allocation;
- petroleum operational emissions;
- exported fossil carbon.

These are parallel national accounts.

D10. Oil-spill hazard identification

Potential spill sources include:

- well blowout;
- subsea equipment;
- FPSO storage;
- tanker transfer;
- support vessel collision;
- pipeline or riser;
- fuel transfer;
- shore-base handling.

Each hazard has a different probability, volume, response strategy and legal pathway.

D11. Spill consequence map

A spill map should identify:

- likely trajectory ranges;
- seasonal currents;
- fisheries;
- protected habitat;
- neighbouring states;
- coastlines;
- ports;
- response bases;
- hospitals;

- wildlife rehabilitation;
- waste disposal.

The map should show uncertainty rather than a single deterministic path.

D12. Response tiers

Tier 1: operator resources for small local events. Tier 2: national or regional reinforcement. Tier 3: large international response.

The plan should specify when escalation occurs, who declares it, and who pays.

D13. Incident command

The response structure should define:

- lead authority;
- operator role;
- maritime authority;
- environmental agency;
- coast guard/military;
- health authority;
- fisheries;
- local government;
- communications;
- legal unit;
- finance/claims unit.

Exercises should test authority conflict.

D14. Capping and well control

A deepwater blowout may require specialized capping equipment and subsea intervention.

The final paper should identify:

- capping-stack access;
- location;
- mobilization time;
- compatible connectors;
- vessels;
- remotely operated vehicles;
- relief-well capability;

- drilling-rig availability.

Licence obligations should be compared with demonstrated access. [S56]

D15. Containment and recovery

Response resources may include:

- booms;
- skimmers;
- storage;
- dispersants;
- in-situ burning where lawful;
- shoreline protection;
- waste management.

Effectiveness depends on sea state, weather, oil type and response time. Equipment count alone is not readiness.

D16. Dispersants

Dispersant use requires:

- approved products;
- toxicity data;
- decision authority;
- ecological trade-off;
- application method;
- exclusion zones;
- monitoring.

The plan should identify who decides when dispersants create less harm than mechanical recovery.

D17. Wildlife response

A wildlife plan should pre-identify:

- specialist organizations;
- rehabilitation sites;
- transport;
- veterinary care;
- handling protocols;

- protected species;
- carcass documentation.

Wildlife response should form part of exercises.

D18. Public health

Spill response can expose workers and communities to fumes, contaminated seafood and cleanup chemicals.

The plan should define:

- occupational protection;
- air monitoring;
- seafood closure;
- drinking-water monitoring;
- medical surveillance;
- public advice.

D19. Fisheries

Fisheries require both ecological and economic assessment.

The compensation record should distinguish:

- closed fishing area;
- lost catch;
- damaged gear;
- market stigma;
- longer-term stock effects.

Claims need transparent eligibility and evidence.

D20. Tourism and island economies

For an island state, beaches, reefs and cruise or resort infrastructure may dominate the exposure.

The emergency model should include:

- shoreline protection priorities;
- visitor communications;
- port closure;
- airport disruption;
- desalination intake;

- reef monitoring;
- reputational damage.

These risks differ from Guyana's continental forest context.

D21. Transboundary risk

Offshore spills can cross borders.

The response plan should include:

- notification thresholds;
- contact points;
- mutual assistance;
- customs clearance for equipment;
- aircraft and vessel access;
- evidence sharing;
- claims coordination.

A regional drill is stronger evidence than a memorandum alone.

D22. Financial assurance

Environmental liability and financial assurance are different.

Liability defines who owes. Financial assurance defines what resources are immediately available or guaranteed.

Possible layers include:

- operator balance sheet;
- insurance;
- parent guarantee;
- letter of credit;
- escrow;
- industry mutual;
- statutory fund.

The final publication should identify amount, trigger, exclusions, duration and legal enforceability.

D23. Parent-company support

A parent-company guarantee should be read as a legal instrument.

Key questions:

- guarantor entity;
- amount;
- term;
- governing law;
- trigger;
- conditions;
- termination;
- dispute route;
- successor responsibility.

A press statement about a guarantee is not enough for a legal conclusion.

D24. Insurance review

Insurance analysis should examine:

- policy limit;
- deductible;
- exclusions;
- pollution wording;
- control-of-well;
- third-party liability;
- business interruption;
- claims-made/occurrence basis;
- insurer credit quality;
- aggregate limits.

Nominal insurance value may overstate available compensation if exclusions apply.

D25. Financial-assurance litigation

Guyana's 2026 appellate dispute should be analyzed from authenticated judgments and permit text. [\[S20\]](#)

The final paper should distinguish:

- underlying environmental liability;
- regulator's permit discretion;
- security amount;
- High Court order;
- appellate outcome;

- any subsequent proceedings.

The author should not infer political motive or legal quality beyond the documented record.

D26. Oil-pollution statute

The 2025 Oil Pollution Prevention, Preparedness, Response and Responsibility Act expands the statutory architecture. [S53-S54]

The final legal audit should verify:

- commencement;
- regulations;
- competent authority;
- contingency planning;
- reporting;
- liability;
- financial responsibility;
- enforcement;
- transboundary provisions.

An enacted act may require regulations before some procedures operate.

D27. Emergency exercise design

A credible annual exercise should simulate:

- communications outage;
- severe weather;
- equipment delay;
- cross-border trajectory;
- injured personnel;
- fishery closure;
- media demand;
- public claims.

Observers should score response time and decision quality, then publish a corrective-action summary.

D28. Incident-data register

Each incident should record:

- date/time;

- project;
- source;
- estimated volume;
- substance;
- duration;
- affected area;
- response;
- regulator notification;
- investigation;
- enforcement;
- remediation;
- closure.

The register should include near misses when required.

D29. Independent sampling

After a material event, independent sampling protects the credibility of evidence.

Protocols should define:

- chain of custody;
- laboratory;
- duplicate samples;
- sampling locations;
- control sites;
- publication;
- retention.

The operator should not be the sole holder of incident evidence.

D30. Decommissioning environment

End-of-life planning should include:

- well plugging;
- subsea equipment;
- FPSO removal;
- waste;
- seabed disturbance;
- monitoring;

- financial security.

A state should update cost estimates during production.

D31. Climate physical risk

Offshore and shore facilities face:

- storms;
- extreme rainfall;
- sea-level rise;
- heat;
- flooding;
- changing ocean conditions.

The environmental management system should incorporate physical climate risk into design and emergency planning.

D32. Biodiversity hierarchy

The mitigation hierarchy is: avoid -> minimize -> restore -> offset residual impact where appropriate.

The first three should not be replaced by an offset payment when avoidance is feasible.

D33. Community participation

Public participation should occur early enough to influence alternatives.

The record should include:

- notice;
- accessible documents;
- meeting locations;
- language;
- comments;
- agency response;
- appeal route.

Participation should not be measured only by number of meetings.

D34. Indigenous rights

Where Indigenous or customary rights are affected, the process should identify applicable domestic law and international standards.

The final paper should include community and independent evidence rather than rely solely on government programme descriptions. [S62-S63]

D35. Environmental performance dashboard

Publish by project:

- permit status;
- emissions;
- flaring;
- discharges;
- waste;
- spills;
- inspections;
- enforcement;
- drills;
- financial assurance;
- biodiversity monitoring.

This creates longitudinal evidence.

D36. Carbon-credit integrity

For forest carbon:

- methodology;
- baseline;
- monitoring period;
- verification;
- issuance;
- registry transfer;
- buyer claim;
- payment;
- community allocation.

Contract value should not be reported as cash received. [S23, S32]

D37. Blue-carbon caution

Small island states may consider mangrove, seagrass or coastal carbon projects.

They should verify:

- tenure;
- methodology;
- permanence;
- leakage;
- additionality;
- monitoring cost;
- buyer eligibility.

Blue carbon should not be assumed to replicate Guyana's forest programme.

D38. Environmental regulator capacity

Staffing should be assessed by workload:

- EIAs;
- permits;
- audits;
- inspections;
- incident response;
- data review;
- enforcement.

A nominal headcount is less useful than cases per specialist.

D39. Independent technical panel

A small regulator can use an independent panel for complex projects if:

- conflicts are disclosed;
- terms are public;
- raw data are accessible;
- government retains the decision;
- reports are published where lawful.

The panel should supplement, not replace, permanent institutional capacity.

D40. New-producer environmental readiness test

Before first oil, verify:

- baseline data;
- permit system;

- incident command;
- transboundary agreements;
- financial assurance;
- capping access;
- laboratories;
- claims process;
- wildlife capability;
- annual drill.

Readiness should be demonstrated in an exercise.

D41. Annex conclusion

Environmental stewardship in an offshore petroleum state depends on evidence before, during and after production. Guyana's permits, pollution legislation, forest programme and public legal disputes provide a broad record for studying how environmental governance evolves alongside rapid production.

The transferable lesson is a chain of responsibility. A newcomer needs baseline science, project-specific conditions, independent monitoring, emergency capability, enforceable financial resources and transparent correction when controls fail.

TECHNICAL ANNEX E

Local content, workforce, household welfare, regional distribution and social accountability

Purpose: define how to test whether petroleum activity creates durable domestic value and broad social gains. Source IDs refer to SOURCE_REGISTER.md.

E1. Local content is not one number

Local content can refer to legal ownership, procurement, employment, wages, skills, domestic manufacturing, tax residence, retained profit or technology transfer. These are different measures.

Guyana's Local Content Act 2021 created a statutory framework and registration system. [S57-S58] Government mid-year reports describe procurement and employment activity in covered sectors. [S59-S60] The final paper should preserve the distinction between programme reporting and independently verified domestic value added.

E2. Supplier classification

For each supplier, record:

- legal entity;
- beneficial owner;
- Guyanese ownership share;
- employees by residence;
- payroll in Guyana;
- imports;
- domestic purchases;
- local subcontractors;
- profit retained;
- tax paid;
- technical capability;
- safety certification;
- export sales.

A supplier registered locally but importing nearly all value has a different economic effect from a firm manufacturing or engineering locally.

E3. Gross procurement versus domestic value

A procurement invoice may contain:

- imported equipment;
- freight;
- foreign specialist services;
- local labour;
- local materials;
- local margin;
- tax.

Only part of the invoice is domestic value added. The final methodology should estimate each component where data permit.

The formula is: Domestic value added = local wages + local operating surplus + local taxes less subsidies + domestic intermediate value added.

This is conceptually different from gross spend.

E4. Beneficial ownership

Local-content systems should verify beneficial ownership rather than rely solely on nominal shareholding.

The register should identify:

- natural persons with control;
- voting rights;
- related parties;
- management control;
- financing arrangements;
- changes in ownership.

Front-company risk rises when statutory local-content thresholds create economic rents.

E5. Employment quality

Employment analysis should include:

- job count;
- occupation;
- contract status;
- wage;
- benefits;

- training;
- safety;
- duration;
- promotion;
- gender;
- region.

A temporary support job and a certified petroleum engineer are both employment, but their long-run productivity effects differ.

E6. Skills pipeline

The skills strategy should map:

- required occupations;
- current supply;
- training duration;
- accreditation;
- employer demand;
- placement;
- retention.

Training should be evaluated by completion and employment outcome, not enrolment alone.

E7. Public-sector talent competition

Petroleum expansion can draw engineers, lawyers, accountants and IT specialists from regulators and ministries.

The state should monitor:

- vacancy rates;
- turnover;
- wage gap;
- recruitment time;
- training cost;
- contractor secondments.

A successful local-content programme should not unintentionally weaken the regulator.

E8. Immigration and specialist labour

A small workforce may require foreign specialists during rapid scale-up.

Policy choices should distinguish:

- temporary expertise;
- permanent migration;
- knowledge transfer;
- understudy arrangements;
- professional licensing.

The analysis should describe labour-market effects rather than present a political preference.

E9. Safety and competence

Local participation targets should never lower safety requirements.

For safety-critical occupations, track:

- certification;
- hours of supervised practice;
- medical fitness;
- emergency training;
- incident record;
- refresher training.

Competence is a prerequisite to localization.

E10. Supplier development

Supplier policy can focus on:

- finance;
- standards;
- quality systems;
- bidding;
- insurance;
- cybersecurity;
- environmental compliance;
- export readiness.

A grant or registration certificate should not be counted as commercial capability without contracts and performance.

E11. Exportable capability

The strongest domestic value may come from services that can later export to Suriname, Trinidad and Tobago or other markets.

Potential examples include:

- logistics;
- environmental monitoring;
- engineering;
- safety training;
- fabrication;
- marine services;
- data;
- professional services.

The final paper should identify sectors based on evidence rather than assume all petroleum services are scalable.

E12. Household welfare measurement

GDP growth is not a household welfare measure.

Use:

- real household consumption;
- income;
- poverty/deprivation;
- food security;
- housing;
- utilities;
- health;
- education;
- transport;
- digital access.

Guyana's latest cited household survey in the current source base covers 2018/2019. [S61] It should not be presented as a 2026 poverty estimate.

E13. Survey freshness

Every social statistic should show:

- survey year;

- publication year;
- sample;
- regions;
- price base;
- methodology.

Where recent data do not exist, say so.

E14. Distribution

National averages can conceal differences across:

- coast;
- hinterland;
- urban;
- rural;
- Indigenous villages;
- income groups;
- gender;
- age.

A petroleum case study should disaggregate when data permit.

E15. Inflation and real income

Rapid investment can raise nominal wages and prices simultaneously.

The relevant measure is real purchasing power. Track:

- wage growth;
- food prices;
- rent;
- construction cost;
- transport;
- utilities.

A nominal wage increase may not improve welfare if living costs rise faster.

E16. Housing

Resource booms often place pressure on land and housing.

Monitor:

- rent;
- home price;
- building permits;
- serviced lots;
- mortgage cost;
- informal housing;
- regional concentration.

Housing programmes should be tested through completed, occupied, serviced units.

E17. Health

Petroleum revenue may fund health infrastructure, but outcomes depend on staff and operations.

Track:

- facility completion;
- doctors/nurses;
- equipment uptime;
- medicine availability;
- travel time;
- waiting time;
- mortality indicators.

A hospital building is not equivalent to a functioning service.

E18. Education

Track:

- attendance;
- completion;
- learning;
- technical certification;
- tertiary graduation;
- employer placement.

Capital spending and scholarships are inputs.

E19. Regional equity

A regional dashboard should compare service access before and after major petroleum-funded programmes.

Use consistent denominators and document data gaps. Per-capita spending alone does not indicate need or outcome.

E20. Amerindian and hinterland evidence

LCDS materials describe village participation and a 15% community allocation from carbon-credit revenue. [S62-S63]

The independent research test should examine:

- village decision process;
- funds received;
- project selected;
- procurement;
- completion;
- maintenance;
- local satisfaction;
- grievance.

Government-reported participation should be triangulated with community sources where available.

E21. Free, prior and informed participation

Where applicable law or policy requires consultation, the evidence record should include:

- notice;
- information;
- time for deliberation;
- representative authority;
- consent/decision;
- dissent;
- conditions;
- appeal.

Participation should not be inferred from attendance alone.

E22. Grievance system

A grievance register should record:

- date;
- complainant category;

- issue;
- project;
- responsible body;
- acknowledgement;
- investigation;
- remedy;
- closure;
- appeal.

Personal information should be protected.

E23. Social licence

"Social licence" is not a legal permit.

The paper should use the term only to describe stakeholder acceptance or relationship quality and should support claims with surveys, consultation records or documented stakeholder positions.

E24. Gender

Petroleum labour markets can be male dominated.

Track:

- employment by occupation and gender;
- pay;
- training;
- leadership;
- contractor workforce;
- safety and harassment procedures.

The goal is evidence, not a generic equality statement.

E25. Youth

Youth outcomes can be tracked through:

- training;
- apprenticeships;
- first employment;
- school-to-work transition;
- entrepreneurship.

Programme participation should be linked to employment and income where possible.

E26. Small business finance

Petroleum suppliers often need working capital, insurance and equipment finance.

Track:

- loan approvals;
- cost of capital;
- payment terms;
- contract duration;
- default;
- procurement concentration.

A local-content target without finance may privilege already-capitalized firms.

E27. Procurement concentration

Calculate:

- top five supplier share;
- top ten;
- related-party share;
- foreign-import component;
- local SME share.

High concentration is not automatically improper, especially in specialist services, but it deserves explanation.

E28. Wage spillovers

Petroleum wage premiums may increase pay elsewhere or create staff shortages.

Track public service, construction, mining, hospitality and professional wages over time.

E29. Migration

Rapid growth can attract return migrants and foreign workers.

Measure:

- work permits;
- return migration;
- remittances;

- housing demand;
- regional origin.

Migration affects both capacity and service demand.

E30. Social protection

A resource boom can finance transfers, but permanent entitlements should be stress-tested under lower petroleum revenue.

Model:

- beneficiary count;
- unit benefit;
- inflation indexation;
- administrative cost;
- funding source;
- cost under low-price scenario.

E31. Community investment by operators

Corporate social investment should be reported separately from statutory payments and public revenue.

Track:

- donor;
- amount;
- beneficiary;
- project;
- outcome.

Private philanthropy should not be counted as government expenditure.

E32. Social outcome scorecard without ranking politics

The paper can display indicators but should not convert them into an overall political score.

A dashboard can show:

- direction;
- observation year;
- confidence;
- distribution;

- source.

Readers retain the decision about policy interpretation.

E33. Local-content audit sample

Select suppliers across:

- catering;
- transport;
- fabrication;
- engineering;
- environmental services;
- legal/accounting;
- aviation;
- marine;
- construction.

For each, calculate domestic value using invoice, payroll, import and ownership data.

E34. New-producer design principles

A newcomer should:

- set realistic categories;
- verify ownership;
- prioritize capability;
- protect safety;
- measure value added;
- publish results;
- review targets.

Targets should adjust as the supplier base matures.

E35. Annex conclusion

The social case for petroleum cannot be established from GDP, supplier spending or project announcements alone. It requires recent household evidence, verified local value and service outcomes.

Guyana's local-content law, supplier system, household-data gaps and village-level climate-finance programmes provide a useful research setting for separating inputs from outcomes.

TECHNICAL ANNEX F

Macroeconomic transformation, infrastructure, energy security, diversification and statistical control

Purpose: explain how a petroleum boom transmits through a small developing economy and how analysts should distinguish durable transformation from temporary expenditure effects. Source IDs refer to SOURCE_REGISTER.md.

F1. GDP is the starting point

Guyana has experienced exceptional real GDP growth after offshore production. Government and IMF publications document the scale. [S06-S08, S64]

GDP captures production, not distribution, public-service quality or sustainability. A petroleum project can raise GDP significantly while much of the value accrues to capital, imports or external owners.

The final paper should therefore pair GDP with broader indicators.

F2. Oil and non-oil GDP

Separate:

- petroleum extraction;
- construction;
- agriculture;
- mining;
- manufacturing;
- services;
- public administration.

Non-oil growth is important because it indicates whether activity is expanding beyond extraction, but even non-oil sectors can be stimulated by petroleum spending.

F3. Real versus nominal

High commodity prices can raise nominal output without equivalent volume growth.

Every GDP and fiscal series should state:

- real/nominal;
- price base;

- currency;
- calendar/fiscal year.

F4. GDP per capita

GDP per capita can rise rapidly in a small population. It remains an average, not a household income measure.

Use alongside household consumption and distribution.

F5. Gross national income

Where foreign ownership is significant, gross national income may differ from GDP.

The final analysis should examine factor income flowing abroad and retained domestically where reliable data exist.

F6. Balance of payments

Petroleum affects:

- exports;
- imported capital goods;
- services;
- investment income;
- foreign direct investment;
- reserves.

A large export surplus does not automatically mean the domestic economy retains the full amount.

F7. Exchange-rate channels

Resource inflows can create appreciation pressure even under a managed or relatively stable nominal exchange rate.

Real appreciation can occur through domestic inflation.

Track:

- nominal exchange rate;
- CPI;
- trading-partner inflation;
- real effective exchange rate where available.

F8. Dutch-disease mechanism

The mechanism is: resource income -> higher domestic demand -> higher wages/prices in non-tradables -> pressure on non-resource tradable sectors.

Evidence should be tested, not assumed.

Indicators:

- manufacturing exports;
- agriculture exports;
- real wages;
- construction prices;
- import share;
- tourism competitiveness.

F9. Construction boom

Public and private investment can accelerate construction.

Track:

- cement/aggregate demand;
- building permits;
- wages;
- contractor backlog;
- unit cost;
- delays.

Rapid growth with rising unit cost may signal capacity constraints.

F10. Imports

A petroleum boom often raises imports of:

- machinery;
- construction material;
- vehicles;
- food;
- professional services.

Import growth is not automatically negative if it builds productive capacity. The composition matters.

F11. Inflation

Disaggregate:

- food;
- housing;
- transport;
- utilities;
- services;
- construction input prices.

Consumer inflation may understate capital-project inflation.

F12. Labour market

Track:

- employment;
- participation;
- vacancies;
- wages;
- skills;
- migration.

Official statistics should be current enough to capture rapid change.

F13. Public wages

Government competes with industry for technical talent.

Public wage policy should be analyzed alongside retention and productivity, not only fiscal cost.

F14. Credit

Resource booms can expand bank deposits and credit.

Track:

- private credit;
- mortgages;
- construction loans;
- non-performing loans;
- concentration.

Financial stability matters if asset prices rise rapidly.

F15. Public debt

A country receiving oil revenue can still borrow.

Assess:

- debt/GDP;
- debt/non-oil GDP;
- interest/revenue;
- currency;
- maturity;
- guarantees.

Low debt risk does not remove project-selection discipline.

F16. Infrastructure pipeline

The petroleum transformation can finance:

- roads;
- bridges;
- ports;
- drainage;
- power;
- water;
- health;
- education;
- housing;
- digital systems.

Each should be evaluated at project level.

F17. Transport

Measure:

- travel time;
- freight cost;
- reliability;
- maintenance;
- traffic;
- safety.

A new bridge or road should show service performance after commissioning.

F18. Ports

Offshore petroleum increases marine logistics.

Port investment should consider:

- draft;
- berth;
- storage;
- customs;
- spill response;
- congestion;
- fisheries interaction.

A petroleum logistics port may have limited wider value unless connections and services support other sectors.

F19. Aviation

Remote offshore operations require helicopters and fixed-wing support.

Regulatory capacity, safety, weather and emergency response should be included in the infrastructure case.

F20. Electricity demand

Rapid growth can increase electricity demand from households, industry, data centers, construction and services.

The GEA has published power-system plans and demand-side management targets. [S66-S67]

The final paper should compare planned capacity with actual generation, grid reliability and losses.

F21. Gas-to-energy

Domestic gas use may reduce fuel imports and generation cost, but evaluation requires:

- delivered gas price;
- pipeline cost;
- plant cost;
- uptime;
- transmission;

- demand;
- carbon implications;
- backup fuel.

The final paper should distinguish construction status from delivered electricity.

F22. Renewable complement

Solar, hydro and other renewables can diversify supply.

Compare:

- levelized cost;
- firm capacity;
- storage;
- transmission;
- land/ecological impact;
- reliability.

The petroleum boom does not eliminate the value of energy diversification.

F23. Energy subsidies

If government subsidizes electricity or fuel, record:

- fiscal cost;
- beneficiaries;
- price signal;
- distribution.

Subsidy policy is a political choice. The paper should present effects rather than endorse one option.

F24. Agriculture

Petroleum revenue can finance irrigation, drainage, roads and processing, but higher wages can also raise farm costs.

Track:

- output;
- yield;
- exports;
- food imports;
- labour cost;

- processing.

F25. Mining

Guyana already had gold and other mining activity before oil.

The analysis should distinguish petroleum-driven macro changes from pre-existing mining cycles.

F26. Manufacturing

Manufacturing competitiveness depends on:

- energy;
- logistics;
- skills;
- scale;
- exchange rate;
- financing.

Public investment should be evaluated against measurable export or productivity outcomes.

F27. Services

Professional services may expand through legal, accounting, engineering and logistics demand.

Track domestic ownership and export potential.

F28. Tourism

Tourism diversification depends on air access, infrastructure, environment and destination development.

A resource boom can finance tourism while also competing for labour.

F29. Digital economy

Improved connectivity and public systems can support technology and services.

Measure adoption and export revenue rather than infrastructure announcements alone.

F30. Public investment multiplier

A fiscal multiplier measures economic response to spending but varies with import leakage and capacity.

In a small open economy with high imports, a large capital budget may leak abroad through imported machinery and services.

The final paper should avoid applying multipliers from unrelated economies without adjustment.

F31. National wealth accounting

A broader balance sheet includes:

- petroleum;
- forests;
- land;
- financial assets;
- infrastructure;
- human capital;
- debt;
- environmental liabilities.

Growth in GDP can coexist with a decline in total wealth if resource depletion is not replaced.

F32. Forest natural capital

Guyana's forest assets have ecological and climate value alongside carbon-market revenue.

The paper should avoid valuing the entire forest solely at carbon-credit prices.

F33. Climate resilience as capital

Sea defenses, drainage and resilient infrastructure protect existing wealth.

Benefit analysis should include avoided loss.

F34. Productivity

Long-run transformation requires output per worker to rise.

Track productivity in non-oil sectors.

F35. Firm creation

Measure:

- new firms;
- survival;
- employment;

- exports;
- ownership.

A surge in registrations may not mean durable enterprise growth.

F36. Tax base diversification

Petroleum revenue can weaken incentives to collect non-oil tax if poorly designed.

Track non-oil tax revenue relative to non-oil GDP.

A diversified tax base supports post-petroleum fiscal sustainability.

F37. Statistical capacity

Rapid structural change can outpace survey and national-account systems.

The IMF has noted data gaps and modernization needs. [S64, S68]

The final data appendix should record:

- frequency;
- lag;
- revision;
- coverage;
- last survey.

F38. Rebased GDP

When GDP is rebased, historical comparisons may change.

The publication should use consistent series and state the base year.

F39. Nowcasting

Where official data lag, nowcasts can support planning but should be labelled models.

They should not replace official observed data in the historical record.

F40. Scenario architecture

The macro scenario model should combine:

- oil price;
- production;
- government revenue;

- public investment;
- imports;
- inflation;
- exchange-rate pressure;
- non-oil growth;
- fund balance.

It should show uncertainty bands.

F41. Low-price scenario

A sustained price decline affects:

- exports;
- fund deposits;
- government spending;
- contractor activity;
- imports.

The model should distinguish immediate and lagged effects.

F42. Project-delay scenario

A delayed FPSO or gas/power project changes:

- production;
- supplier demand;
- revenue;
- infrastructure timing.

Remove delayed output from the baseline rather than smoothing it over.

F43. Overheating scenario

A high-price/high-spending case may produce:

- inflation;
- labour shortages;
- construction delays;
- import congestion.

The policy response involves trade-offs, which the paper should describe neutrally.

F44. Diversification test

A sector is becoming more durable if:

- productivity rises;
- exports rise;
- domestic value rises;
- private investment persists without petroleum-linked demand.

A sector that grows only by serving petroleum construction is less diversified than headline non-oil GDP suggests.

F45. Post-oil test

The strongest test asks whether the economy could sustain public services if petroleum revenue fell materially.

Model:

- non-oil tax;
- recurrent spending;
- debt service;
- fund return;
- maintenance;
- pensions.

F46. Infrastructure outcome matrix

For each major infrastructure class:

- baseline problem;
- project;
- cost;
- completion;
- utilization;
- maintenance;
- measured effect.

This shifts analysis from announcements to performance.

F47. Small-state adaptation

An island producer faces:

- smaller labour pool;
- higher import dependence;
- concentrated tourism;
- limited land;
- disaster exposure.

Its optimal spending pace may differ from Guyana's.

F48. Statistical dashboard

Minimum quarterly/annual series:

- oil output;
- non-oil GDP;
- CPI;
- wages;
- employment;
- imports;
- non-oil exports;
- public revenue;
- fund assets;
- debt;
- construction cost;
- household welfare.

F49. Annex conclusion

Macroeconomic transformation should be judged by the conversion of petroleum income into productive capacity, resilience and durable non-oil income.

Guyana's growth, infrastructure programme, energy transition and data gaps create a strong case for a multi-indicator approach that goes beyond headline GDP.

TECHNICAL ANNEX G

Comparator implementation dossiers: Norway, Ghana, Timor-Leste, Suriname, Trinidad and Tobago, and small-island states

Purpose: compare institutional mechanisms without ranking political systems or selecting a preferred country model.

G1. Comparator protocol

Every country case uses the same questions:

1. What stage of petroleum development?
2. What is the revenue-management vehicle?
3. How are withdrawals linked to the budget?
4. What is the investment policy?
5. Which institutions audit and publish?
6. How concentrated is petroleum revenue?
7. What is the reserve/depletion profile?
8. What is the non-oil tax base?
9. What environmental exposure is distinctive?
10. Which elements are transferable?

The protocol prevents a large mature producer from being compared to a new island producer through headline fund size alone.

G2. Norway

Norway's Government Pension Fund Global operates within a mature fiscal and institutional system. Official publications describe the transfer of net state petroleum cash to the fund and a fiscal guideline associated with the expected long-run real return. [S25, S74]

Implementation features:

- petroleum revenue is separated from ordinary budget revenue;
- fund assets are invested largely abroad;
- withdrawals enter the national budget;
- performance is benchmarked;
- parliamentary and public reporting is extensive.

Transfer constraints:

- decades of institutional development;
- large non-oil tax base;
- mature capital markets;
- very large accumulated fund;
- strong administrative capacity.

New-producer adaptation: Use budget integration, transparent reporting and external diversification as design references. Do not copy the numerical spending rule without modeling local infrastructure deficits, revenue volatility and absorption.

G3. Ghana

Ghana's Petroleum Revenue Management Act established stabilization and heritage purposes. Bank of Ghana reporting provides repeated fund statements. [S24, S75-S76]

Implementation features:

- statutory fund structure;
- stabilization function;
- heritage function;
- central-bank management;
- recurring reporting.

Transfer constraints:

- different reserve trajectory;
- different fiscal regime;
- different currency and debt environment;
- different production maturity.

New-producer adaptation: Define stabilization and intergenerational purposes separately even if one legal fund is used. Publish allocation rules and exception procedures.

G4. Timor-Leste

Timor-Leste's Petroleum Fund provides a small-state comparison. The central bank publishes monthly, quarterly and annual reporting and describes liquidity and growth portfolios. [S26, S77-S78]

Implementation features:

- transparent central-bank reporting;
- financial asset accumulation;
- formal withdrawal framework;
- investment portfolios linked to liquidity needs.

Core risk: A well-managed financial portfolio does not eliminate fiscal dependence on withdrawals if petroleum production declines and the domestic tax base remains narrow.

New-producer adaptation: Model the post-field budget before first oil. A permanent public salary or subsidy should survive a conservative production and return scenario.

G5. Suriname

Suriname's GranMorgu project had reached final investment decision and remained under construction at the September 2026 source cut-off, with first oil targeted for 2028. [S27, S79]

Implementation relevance:

- regional geology;
- offshore deepwater development;
- pre-first-oil preparation period;
- national oil company role;
- potential regional services and emergency cooperation.

Transfer constraint: Staatsolie gives Suriname a different institutional structure from Guyana. The state-company role changes access to commercial information and internal technical capability.

New-producer adaptation: A country without a national oil company should not assume the same information flows. It needs explicit regulatory data rights and independent expertise.

G6. Trinidad and Tobago

Trinidad and Tobago is a mature Caribbean petroleum and gas producer with a Heritage and Stabilisation Fund reporting framework. [S80]

Implementation relevance:

- long experience with commodity cycles;
- petroleum and gas industry capabilities;
- regional professional services;
- fiscal stabilization;
- industrial downstream history.

Transfer constraint: Its established petrochemical base, infrastructure and long production history are not available to a newcomer.

New-producer adaptation: Study mature-cycle risk, stabilization and industrial capability. Test downstream proposals against market scale and delivered feedstock economics.

G7. Guyana as a comparator in its own right

Guyana's distinctive features include:

- rapid sequence from 2015 discovery to 2019 first oil;
- large discoveries in a small economy;
- production scale-up through multiple FPSOs;
- NRF law before first oil;
- evolving fund rules;
- local-content legislation after production started;
- pre-existing LCDS and forest finance;
- low coastal geography and major hinterland forest estate.

These features make the case rich but limit automatic transfer.

G8. Small-island producer archetype

Assume a hypothetical island with:

- population below one million;
- tourism above 30% of exports;
- limited skilled workforce;
- sensitive reefs;
- hurricane exposure;
- no large sovereign fund;
- narrow tax base;
- one commercial offshore field.

The preferred analysis is not "Which country should it copy?" It is "Which mechanism fits each constraint?"

G9. Fund design for the island archetype

Potential requirements:

- high liquidity reserve for production interruption and disasters;
- external investment to limit domestic overheating;
- conservative recurrent-spending commitments;
- explicit post-field fiscal plan;
- transparent budget integration.

A heritage component may be useful, but the share is a sovereign policy choice.

G10. Environmental design for the island archetype

Priority capabilities:

- reef and fishery baseline;
- tourism-impact plan;
- hurricane-plus-spill exercise;
- transboundary agreements;
- rapid claims process;
- regional equipment access;
- desalination protection.

These differ from Guyana's forest-heavy natural-capital profile.

G11. Local-content design for the island archetype

Avoid setting statutory percentages that exceed workforce and supplier capacity.

Prioritize:

- safety-critical training;
- professional services;
- environmental monitoring;
- marine logistics;
- data;
- transferable technical skills.

Measure domestic value added rather than registrations.

G12. Comparator metrics

Use common metrics:

- petroleum revenue/non-oil GDP;
- fund assets/non-oil GDP;
- withdrawals/fund assets;
- non-oil tax/non-oil GDP;
- debt/non-oil GDP;
- production per capita;
- reserve life;

- capital spending/non-oil GDP;
- current-account balance;
- inflation;
- non-resource exports;
- environmental response time.

Common metrics improve comparison without creating a political ranking.

G13. Institutional maturity dimensions

Describe, do not score:

- legal clarity;
- data availability;
- independent audit;
- budget integration;
- environmental monitoring;
- dispute resolution;
- public participation;
- statistical freshness.

A qualitative narrative is preferable to a single composite grade.

G14. Transfer decision record

For each proposed borrowed policy, document:

- source country;
- policy;
- original stage;
- prerequisite;
- receiving-country gap;
- adaptation;
- cost;
- risk;
- review date.

Example: "External sovereign-fund investment" may transfer from Norway, but the receiving country may need a larger liquid reserve and a lower risk tolerance.

G15. Comparator failure modes

Common errors:

- selecting Norway because of fund size without adjusting for institutional history;
- selecting Ghana because legal language appears familiar without modeling local revenue volatility;
- selecting Timor-Leste without examining depletion dependence;
- treating Suriname's planned first oil as observed success;
- assuming Trinidad's downstream industry is reproducible at smaller scale;
- copying Guyana's forest-carbon arrangements into an island with different ecosystems.

G16. Comparator chapter output

Each country profile in the final PDF should include:

- two-page institutional summary;
- fiscal-flow diagram;
- fund/budget relationship;
- key risks;
- transfer limits;
- dated sources.

The purpose is practical comparison.

G17. Annex conclusion

Comparators are strongest when they reveal prerequisites and failure modes. The new producer should borrow mechanisms selectively, preserve local democratic decision-making and publish why each adaptation was made.

TECHNICAL ANNEX H

Guyana offshore project profiles, production controls and project-level evidence template

Purpose: organize each offshore project as a separate auditable case. Source IDs refer to SOURCE_REGISTER.md.

H1. Why project-level analysis matters

The Stabroek Block is not one homogeneous project. Each development has its own:

- production licence;
- FPSO;
- wells;
- capacity;
- capital programme;
- EIA;
- permit conditions;
- startup;
- production;
- cost recovery;
- spill plan;
- decommissioning exposure.

Aggregating the entire block can hide differences.

H2. Liza Phase 1

Key evidence:

- first project;
- production began December 2019; [S01, S83]
- environmental permit record available; [S48]
- production licence record available. [S86]

Final project profile should include:

- discovery/appraisal wells;
- development approval date;
- production licence;

- EIA;
- permit;
- FPSO;
- nameplate capacity;
- actual monthly output;
- flaring history;
- cost audit period;
- environmental amendments;
- first government cargo;
- decommissioning plan.

H3. Liza Phase 2

Liza Phase 2 began production in February 2022. [S82]

The profile should separate:

- planned ramp;
- actual output;
- licence addendum;
- environmental permit;
- development cost;
- cost-recovery entry;
- operational incidents.

Operator-planned production should not substitute for regulator observations.

H4. Payara

Payara began production in November 2023. [S83]

The final profile should identify:

- Prosperity FPSO;
- permit/EIA record;
- production licence;
- first oil;
- actual monthly output;
- environmental conditions;
- cost recovery;
- government lifts.

H5. Yellowtail

Yellowtail began production in August 2025 and became the fourth operating project. [S84]

The operator stated that combined installed capacity exceeded 900,000 barrels per day. That is a capacity measure, not an annual average.

Project profile requirements:

- ONE GUYANA FPSO;
- licence;
- EIA;
- permit;
- startup;
- ramp;
- actual monthly output;
- government entitlement;
- environmental performance.

H6. Uaru

Uaru was not yet producing at the evidence cut-off used for this research.

Public environmental permit and production licence materials exist. [S50, S56]

The profile should therefore label:

- sanctioned;
- under development;
- no realized production yet at cut-off.

Forecast first oil belongs in a separate field.

H7. Whiptail

Public environmental permit material exists for Whiptail. [S51]

The final profile should identify project status at publication date and avoid filling production cells until observed output exists.

H8. Future projects

Any project approved after Whiptail should use the same template.

The publication should never silently update an old edition. New project status should carry a version date.

H9. Project master table

Columns:

- project;
- FPSO;
- discovery link;
- FID/approval;
- production licence;
- environmental permit;
- first oil;
- capacity;
- 2026 actual average where available;
- peak;
- cost;
- cost-bank status;
- government lifts;
- flaring;
- incidents;
- decommissioning security.

H10. Production measurement

Production data should distinguish:

- gross wellstream;
- stabilized crude;
- water;
- gas;
- reinjection;
- flare;
- fuel use;
- export.

Public dashboards often focus on crude output. Technical regulators need the full material balance.

H11. Metering

Meter governance should cover:

- calibration;
- seals;
- redundancy;
- maintenance;
- data transmission;
- audit trail;
- correction procedure.

A government must be able to reproduce the entitlement calculation.

H12. Downtime

For each project, record:

- planned maintenance;
- unplanned outage;
- equipment failure;
- weather;
- reservoir constraint.

Downtime affects annual output and revenue.

H13. Capacity versus throughput

Nameplate capacity is an engineering design measure.

Actual throughput depends on:

- wells;
- reservoir;
- maintenance;
- debottlenecking;
- safety;
- weather.

A daily record above nameplate for a short period should not be extrapolated to annual production without evidence.

H14. Reservoir performance

The state should receive sufficient data to monitor:

- pressure;
- water cut;

- gas-oil ratio;
- injection;
- decline.

Reservoir uncertainty affects future revenue.

H15. Development cost

Each project should have a versioned cost estimate:

- sanction estimate;
- revised estimate;
- audited eligible cost;
- final cost where known.

Operator headline development cost and recoverable cost are not necessarily identical.

H16. Cost attribution

Costs should be tagged to:

- project;
- shared infrastructure;
- exploration;
- appraisal;
- corporate overhead.

Shared costs require contractual allocation rules.

H17. Environmental project register

For every project:

- EIA date;
- permit date;
- modifications;
- flaring limits;
- discharge rules;
- spill plan;
- wildlife plan;
- monitoring reports;
- audit;
- enforcement.

This register should connect to TECHNICAL ANNEX D.

H18. Incident taxonomy

Classify:

- safety;
- spill;
- flare;
- discharge;
- equipment failure;
- collision;
- regulatory noncompliance.

Not every incident has equal consequence.

H19. Production licence controls

Licence review should extract:

- term;
- area;
- development plan;
- reporting;
- measurement;
- safety;
- spill preparedness;
- abandonment;
- assignment.

The final text should quote sparingly and cite the original.

H20. Project economics template

Inputs:

- capex;
- opex;
- oil price;
- production;
- downtime;
- cost recovery;

- royalty;
- profit share;
- tax;
- decommissioning.

Outputs:

- gross revenue;
- government revenue;
- contractor cash flow;
- cost-bank balance;
- sensitivity.

H21. Government-lift template

For each project/month:

- government entitlement barrels;
- cargo date;
- realized price;
- sale value;
- settlement;
- NRF deposit.

This connects operational and financial records.

H22. Cumulative Stabroek analysis

After building project profiles, aggregate only compatible measures:

- actual barrels;
- government lifts;
- cost recovery;
- fund receipts.

Maintain project detail underneath.

H23. Project closeout

Eventually each project needs:

- cessation plan;
- well abandonment;

- facility removal;
- environmental monitoring;
- decommissioning cost;
- security;
- residual liability.

Start tracking before decline.

H24. New-producer project template

A new country should adopt the project record before sanction.

Mandatory folders:

1. legal;
2. reservoir;
3. economics;
4. environment;
5. safety;
6. metering;
7. fiscal;
8. local content;
9. emergency;
10. closure.

H25. Evidence maturity labels

Use:

- observed;
- audited;
- regulator-approved;
- operator-reported;
- government-reported;
- forecast;
- disputed;
- pending.

This prevents mixed evidence.

H26. Annex conclusion

Project-level evidence is the bridge between national petroleum headlines and actual regulatory performance. Guyana's multi-FPSO expansion makes the project record essential for understanding production, costs, environmental conditions and public revenue.

TECHNICAL ANNEX I

Source-to-claim verification, corrections, uncertainty and publication integrity protocol

Purpose: define the evidentiary rules used in the flagship publication so readers can reproduce major claims and distinguish verified facts from forecasts, interpretations and unresolved questions.

I1. Claim classes

Every material statement should be classified internally as one of:

- legal fact;
- contractual fact;
- observed operational fact;
- audited financial fact;
- official administrative report;
- operator report;
- multilateral analysis;
- stakeholder position;
- forecast;
- model output;
- author synthesis.

The class determines the required source and wording.

I2. Legal claims

Legal claims require the enacted text, official gazette, regulation, licence, permit or authenticated judgment where available.

A ministry press release describing a judgment is secondary evidence for the legal proposition. It may establish the ministry's interpretation but not replace the judgment.

I3. Contract claims

Contract terms should come from the executed agreement or official published contract.

Model agreements belong in a separate category. They should not be applied retroactively to an older executed contract.

I4. Financial claims

Priority order:

1. audited financial statement;
2. central-bank or treasury transaction report;
3. official budget or mid-year account;
4. operator statement where it concerns operator finances;
5. media only when primary records are unavailable.

If an audited account later restates an earlier number, the restated number should be used with a note.

I5. Production claims

Use regulator data for actual production where available.

Operator announcements are acceptable for first-oil dates and milestones, but a stated capacity or daily record should not be described as annual average production.

I6. Social claims

Use recent representative surveys for household outcomes.

Administrative programme counts may support statements about participation or spending, but not household welfare unless linked to outcome evidence.

I7. Environmental claims

A permit establishes authorization and conditions. An inspection establishes regulator observation. Monitoring data establish measured environmental variables. An enforcement order establishes a regulatory finding. None alone proves overall environmental performance.

I8. Carbon-finance claims

Separate:

- methodology approval;
- verified units;
- issuance;
- sale contract;
- transfer;
- cash receipt;
- community allocation;

- expenditure;
- outcome.

The minimum contract value is not the same as cash received.

I9. Evidence confidence

Use three internal labels: **High:** primary authoritative record with clear date and definition. **Moderate:** reliable official or independent analysis with some limitation. **Unresolved:** conflicting sources, missing primary document or material ambiguity.

The public paper should explain uncertainty rather than display a hidden numerical score.

I10. Observation date

Every time-sensitive figure needs:

- observation period;
- publication date;
- evidence cut-off.

This is essential for 2026 production, fund balances and EITI status.

I11. Forecast wording

Use:

- projected;
- targeted;
- planned;
- estimated;
- operator expects;
- government projects.

Do not use:

- will produce;
- will receive;
- completed;

unless observed evidence supports it.

I12. Causality

Avoid causal claims from temporal association alone.

Example: Petroleum revenue and public construction may rise together, but a specific welfare outcome requires a plausible mechanism and evidence.

I13. Political claims

Where a source criticizes or supports a political decision, attribute the view.

The author should describe evidence, institutional process and policy effects without endorsing or opposing a party, official or electoral choice.

I14. Quote control

Use short quotations only when wording matters legally or analytically.

Most source material should be paraphrased with a citation.

I15. Source register fields

Each source record should include:

- ID;
- author/institution;
- title;
- date;
- source type;
- jurisdiction;
- claim supported;
- page/section;
- URL;
- access date;
- limitation;
- superseded status.

I16. Claim register

For each high-impact numerical claim:

- claim ID;
- manuscript location;
- value;
- unit;
- date;

- source ID;
- calculation;
- reviewer;
- status.

This allows automated checking.

I17. Cross-source reconciliation

Where ministry, central bank and IMF figures differ:

1. compare period;
2. compare currency;
3. compare cash/accrual;
4. compare revision date;
5. compare scope;
6. use the most appropriate source and explain the difference.

Do not average incompatible figures.

I18. Contradictory evidence

A contradiction register should preserve:

- source A;
- source B;
- disputed proposition;
- likely reason;
- unresolved status;
- final treatment.

Deleting inconvenient sources reduces credibility.

I19. Corrections

A corrections register should show:

- date identified;
- original text;
- corrected text;
- reason;
- affected pages;
- source;

- version released.

Material corrections should remain visible in the digital archive.

I20. Versioning

Use version numbers:

- research draft;
- review draft;
- publication 1.0;
- corrected 1.1.

Every PDF should carry:

- version;
- evidence cut-off;
- SHA-256;
- publication date.

I21. External review

External review should identify:

- reviewer expertise;
- conflicts;
- scope;
- date;
- comments;
- author response.

A paper should never imply peer review when only internal checking occurred.

I22. Fiscal model review

Review:

- contract interpretation;
- units;
- price;
- production;
- cost recovery;
- tax;

- timing;
- discounting.

A model should be reproducible from disclosed inputs.

I23. Environmental review

Review:

- permit version;
- baseline;
- monitoring;
- spill assumptions;
- liability;
- insurance;
- transboundary risk.

Environmental review should not be limited to climate emissions.

I24. Social review

Review:

- survey year;
- sample;
- regional coverage;
- Indigenous/community evidence;
- distribution;
- programme versus outcome.

I25. Comparator review

Review whether comparator data use common dates and definitions.

A mature producer's current fund balance should not be compared directly with a new producer's first-year cash without context.

I26. Repetition audit

A 500-plus-page publication risks repeated narrative.

The final build should detect repeated long paragraphs and flag identical text. Technical annexes may restate definitions when needed, but repeated prose should be minimized.

I27. Page-substance audit

Page count is not a quality metric by itself.

The final audit should report:

- total pages;
- pages with at least 100 words;
- figures/tables;
- source pages;
- blank pages;
- duplicate paragraphs.

A page with a full analytical table may be substantive despite fewer than 100 words and should be identified separately.

I28. Link audit

Every URL should be checked for:

- response;
- authoritative domain;
- correct document;
- date;
- version.

Archived copies should be used where lawful and appropriate for fragile links.

I29. PDF audit

Check:

- searchable text;
- bookmarks;
- page numbers;
- TOC;
- metadata;
- broken glyphs;
- clipping;
- table overflow;
- citation legibility;
- file checksum.

I30. Publication decision

The paper is ready for a publication edition when:

- all material claims have source IDs;
- unresolved material issues are disclosed;
- models reproduce;
- legal versions are correct;
- final PDF passes visual and structural QA.

External peer review is an additional status, not a prerequisite to honestly publishing independent research if its absence is stated.

I31. Annex conclusion

Publication integrity depends on traceability and correction. The value of the Guyana case study will come from readers being able to distinguish what happened, what was projected, what remains disputed and how the evidence was checked.

TECHNICAL ANNEX J

Adverse scenarios, combined shocks and decision playbooks

Purpose: test whether resource-governance institutions continue functioning when assumptions fail. The scenarios are analytical exercises, not predictions.

J1. Scenario architecture

Every scenario has:

- trigger;
- direct effect;
- fiscal channel;
- operational channel;
- environmental/social channel;
- decision owner;
- immediate actions;
- medium-term actions;
- public disclosure;
- recovery test.

J2. Oil price shock

Assume realized oil price falls 40% for two years.

Test:

- gross revenue;
- royalty;
- profit oil;
- fund deposit;
- withdrawal rule;
- capital programme;
- debt;
- suppliers.

Do not multiply government revenue by -40% mechanically. Cost recovery and production-sharing rules interact.

J3. Production outage

Assume one major FPSO is unavailable for 90 days.

Test:

- lost barrels;
- government entitlement;
- cargo schedule;
- supplier activity;
- budget forecast;
- maintenance cause.

The scenario should identify whether insurance affects government cash.

J4. Delayed new project

Assume first oil slips two years.

Remove the project from:

- production forecast;
- petroleum revenue;
- local-content demand;
- port forecast;
- power demand.

Do not smooth the delay into a lower annual growth rate without showing the missing project.

J5. Cost-bank rebuild

Assume a new sanctioned development materially increases recoverable cost.

Test government profit oil under:

- high cost recovery;
- declining cost recovery;
- later plateau.

This scenario prevents a high-receipt year from being treated as permanent.

J6. Audit dispute

Assume a large cost audit produces a proposed adjustment.

Record:

- proposed amount;
- operator response;
- contractual procedure;
- expected timing;
- no immediate cash recovery unless settled.

The budget should not spend a disputed amount.

J7. Lower fund investment returns

Assume global markets deliver negative returns for a year.

Test:

- fund market value;
- liquidity;
- withdrawal;
- budget reliance.

Petroleum deposits and investment returns should remain separate.

J8. Higher inflation

Assume construction inflation exceeds CPI by 10 percentage points.

Test:

- project cost;
- variation;
- completion;
- maintenance;
- real public-investment volume.

A larger nominal budget may buy less infrastructure.

J9. Labour shortage

Assume technical vacancies double.

Test:

- regulator inspections;
- audit backlog;
- project schedule;
- contractor wages;

- public-service retention.

Potential responses include compensation, training and targeted external recruitment. The paper does not prescribe one.

J10. Currency pressure

Assume domestic prices rise relative to trading partners.

Test:

- non-oil exports;
- imports;
- wages;
- tourism competitiveness.

The scenario focuses on real, not only nominal, exchange conditions.

J11. Major spill

Assume a severe offshore release during adverse weather.

Test:

- notification;
- command;
- capping;
- equipment;
- fisheries closure;
- transboundary notification;
- financial assurance;
- claims;
- independent sampling.

The exercise should use a range of release volumes rather than one false-precision estimate.

J12. Financial-assurance shortfall

Assume immediate cleanup cost exceeds readily available security.

Test:

- operator liquidity;
- insurance;
- parent support;

- state emergency spending;
- recovery rights.

This separates legal liability from cash available in the first weeks.

J13. Hurricane plus spill for an island

Assume:

- airport closure;
- damaged port;
- telecom outage;
- oil release.

Test regional response agreements and redundant logistics.

This combined scenario is more relevant to some islands than Guyana.

J14. Forest-carbon price shock

Assume carbon-credit demand or price weakens.

Test:

- expected climate-finance receipts;
- village allocation;
- LCDS programmes.

Do not transfer the shortfall to petroleum accounts.

J15. EITI or disclosure delay

Assume a required report is late.

Test:

- legal deadline;
- cause;
- interim data;
- stakeholder communication;
- corrective action.

A transparency system should have a late-report protocol.

J16. Procurement failure

Assume a major petroleum-funded infrastructure contract has:

- 30% cost overrun;
- one-year delay;
- weak maintenance plan.

Test whether the public-investment framework pauses similar projects or continues unchanged.

J17. Public-enterprise liability

Assume a state enterprise requires a fiscal transfer.

Add the exposure to the national balance sheet and test fund-withdrawal pressure.

J18. Debt-market shock

Assume borrowing cost rises sharply.

Test:

- refinancing;
- capital programme;
- fund assets;
- debt reduction option.

A resource fund and debt strategy must be analyzed together.

J19. Household cost-of-living shock

Assume food and housing prices rise faster than wages for lower-income households.

Test:

- real consumption;
- transfer policy cost;
- housing response;
- survey evidence.

GDP growth should not be used to dismiss the distributional effect.

J20. Local-content fraud risk

Assume evidence emerges that nominal local firms are controlled by foreign beneficial owners.

Test:

- certification;
- enforcement;
- procurement contracts;

- remediation;
- future verification.

A suspected case should be investigated, not generalized to the entire programme.

J21. Data-system failure

Assume production, fund and budget data disagree.

Freeze publication of a consolidated headline until reconciliation identifies:

- timing;
- currency;
- scope;
- error.

Publish the correction.

J22. Political transition

A change in administration should not break core records.

Test:

- data custody;
- contract archive;
- fund accounts;
- regulator files;
- audit continuity.

The exercise is institutional and does not assume any electoral outcome.

J23. Operator exit

Assume a contractor sells its interest.

Test:

- assignment approval;
- new owner capability;
- decommissioning liability;
- financial assurance;
- tax;
- data transfer.

J24. Decommissioning shock

Assume end-of-life cost doubles.

Test:

- reserve;
- contractor security;
- state exposure.

This scenario should run long before cessation.

J25. Combined fiscal shock

Combine:

- oil at 40;
- production -15%;
- cost recovery at ceiling;
- negative fund return;
- high construction inflation.

This is the severe fiscal case.

J26. Combined environmental-economic shock

Combine:

- spill;
- tourism or fisheries loss;
- cleanup;
- project outage;
- lower fund inflow.

This shows why environmental risk is also fiscal risk.

J27. Recovery indicators

After a shock, monitor:

- production restoration;
- fund reconciliation;
- project reprioritization;
- environmental recovery;
- claims;

- household indicators.

A response is not complete when the press release ends.

J28. Decision log

Each crisis decision should record:

- time;
- authority;
- evidence;
- option;
- reason;
- review date.

The log supports later audit.

J29. Exercise programme

Run annually:

- fiscal shock;
- spill;
- cyber/data failure.

Run every two years:

- combined national exercise.

Publish corrective actions.

J30. Annex conclusion

Institutions are best tested under failure. A newcomer should design stress tests before the first crisis and update them as production, the sovereign fund and public spending scale.

TECHNICAL ANNEX K

Government implementation templates and readiness instruments

Purpose: provide copy-ready institutional templates for a prospective petroleum-producing country. These templates are neutral administrative tools and require adaptation to national law.

K1. Exploration licence decision sheet

Fields:

- application ID;
- applicant;
- beneficial owners;
- acreage;
- work programme;
- financial capability;
- technical capability;
- environmental screening;
- conflicts;
- fiscal terms;
- data obligations;
- relinquishment;
- security;
- decision;
- reasons;
- publication date;
- appeal.

The decision file should preserve rejected alternatives.

K2. Adviser conflict declaration

Every external petroleum adviser should disclose:

- current operator clients;
- former operator employment;
- financial interests;
- family interests;

- competing government engagements;
- confidentiality obligations.

The procuring agency records mitigation or exclusion.

K3. Commercial discovery notice

The state record should identify:

- well;
- test;
- resource classification;
- operator declaration;
- regulator review;
- independent technical review;
- next appraisal step.

A discovery notice should not include unverified public-revenue estimates.

K4. Contract clause crosswalk

Rows:

- royalty;
- cost recovery;
- profit split;
- tax;
- valuation;
- ring-fencing;
- audit;
- stabilization;
- assignment;
- local content;
- environment;
- decommissioning;
- dispute.

Columns:

- executed clause;
- plain-language effect;
- fiscal model treatment;

- audit test;
- later amendment.

K5. Development approval sheet

Require:

- reserves;
- wells;
- facility;
- capex;
- opex;
- schedule;
- first-oil range;
- production range;
- decommissioning;
- EIA;
- spill analysis;
- local-content plan;
- fiscal model;
- independent review.

Record conditions of approval.

K6. Environmental permit summary

Fields:

- permit ID;
- project;
- issue/expiry;
- EIA;
- conditions;
- flaring;
- discharges;
- waste;
- monitoring;
- audit;
- spill plan;

- wildlife plan;
- financial assurance;
- amendment;
- enforcement.

K7. Spill-readiness certificate

The regulator verifies:

- incident command;
- capping access;
- vessels;
- aircraft;
- booms/skimmers;
- dispersant policy;
- lab;
- wildlife;
- claims;
- transboundary contacts;
- exercise date;
- corrective actions.

The certificate records evidence, not a generic "ready" statement.

K8. First-cargo worksheet

Fields:

- project;
- production period;
- entitlement;
- cargo;
- barrels;
- quality;
- price;
- buyer;
- invoice;
- royalty;
- profit oil;

- settlement;
- fund deposit.

Attach source records.

K9. Cost-audit query

Fields:

- cost ID;
- project;
- category;
- amount;
- related party;
- invoice;
- clause;
- audit concern;
- operator response;
- adjustment;
- status.

K10. Sovereign-fund deposit notice

Publish:

- date;
- receipt type;
- amount;
- currency;
- related cargo/royalty period;
- cumulative annual deposits.

Do not include forecasts.

K11. Quarterly fund report

Sections:

1. opening assets;
2. deposits;
3. investment income;
4. withdrawals;

5. closing assets;
6. asset allocation;
7. benchmark;
8. fees;
9. risk;
10. legal notices.

K12. Withdrawal authorization

Record:

- statutory formula;
- maximum lawful amount;
- requested amount;
- parliamentary authority;
- transfer date;
- consolidated-fund receipt.

The authorization does not identify final programme outcomes.

K13. Public investment business case

Required:

- problem;
- options;
- demand;
- benefits;
- capex;
- opex;
- maintenance;
- climate risk;
- procurement;
- implementation capacity;
- beneficiaries;
- KPI;
- independent review.

K14. Procurement outcome record

Publish:

- tender;
- bidders;
- evaluation method;
- winner;
- beneficial owner;
- contract value;
- completion;
- variations.

Protect information only under defined legal grounds.

K15. Project commissioning certificate

Verify:

- physical completion;
- defects;
- utilities;
- staffing;
- maintenance;
- safety;
- operating budget;
- handover.

A completed construction contract is not the same as an operating service.

K16. Local supplier verification

Fields:

- supplier;
- beneficial owner;
- local ownership;
- payroll;
- imports;
- domestic purchases;
- capability;

- standards;
- contract;
- retained value.

K17. Training outcome record

Fields:

- programme;
- enrolment;
- completion;
- certification;
- employment;
- wage;
- retention.

K18. Community consultation record

Fields:

- community;
- issue;
- notice;
- information;
- meeting;
- representative;
- comments;
- agency response;
- decision;
- grievance.

K19. Household data release

Metadata:

- survey year;
- sample;
- regions;
- weights;
- price base;

- fieldwork;
- revision.

This prevents stale data from being presented as current.

K20. EITI/disclosure calendar

List every disclosure with:

- owner;
- legal deadline;
- data period;
- publication date;
- assurance;
- correction.

K21. Annual petroleum report

Minimum chapters:

- exploration;
- reserves/resources;
- production;
- projects;
- government entitlement;
- revenue;
- cost audits;
- local content;
- environment;
- incidents;
- decommissioning;
- next-year risks.

K22. National wealth statement

Report:

- petroleum asset estimate;
- financial fund;
- debt;
- infrastructure;

- human capital indicators;
- forest/natural capital;
- liabilities.

Values should include uncertainty.

K23. Cabinet decision memo

Every material petroleum policy memo should show:

- issue;
- evidence;
- legal authority;
- options;
- fiscal effect;
- environmental effect;
- social effect;
- risks;
- implementation;
- review.

The document supports accountability while preserving lawful cabinet confidentiality where applicable.

K24. Annual institutional review

Ask:

- What rule failed?
- What data were missing?
- Which forecast was wrong?
- Which audit is late?
- Which permit needs revision?
- Which project lacks maintenance?
- Which community grievance remains open?

Publish a non-confidential action plan.

K25. Twenty-four-month pre-first-oil programme

Months 1-3: legal map, data inventory, environmental baseline. Months 4-6: fiscal benchmark, regulator plan, emergency map. Months 7-9: revenue-fund options, audit protocol, ownership register. Months 10-12: draft laws, metering protocol, spill drill. Months 13-15: contract/fiscal publication rules, audit team, review panel. Months 16-18: synthetic revenue transactions, parliamentary test, regional response. Months 19-21: combined shocks and corrections. Months 22-24: independent readiness review.

K26. Five-year production programme

Year 1: reconcile cash. Year 2: verify domestic value. Year 3: verify project outcomes. Year 4: test resilience. Year 5: independent institutional review.

K27. Annex conclusion

Templates convert principles into records. A country is better prepared when each decision leaves an auditable file, each transaction reconciles and each public claim can be tested.

TECHNICAL ANNEX L

Integrated national petroleum risk register

Purpose: show how fiscal, operational, environmental, social and institutional risks interact. The register describes controls without predicting political or market outcomes.

L1. Risk fields

Every risk record should contain:

- ID;
- category;
- event;
- cause;
- affected asset;
- evidence;
- existing control;
- control owner;
- early indicator;
- contingency;
- review date;
- residual uncertainty.

Avoid a single political score.

L2. Geological risk

Event: discovery is smaller or less productive than expected. Control:

- independent resource review;
- staged appraisal;
- conservative public forecasts.

Budget response: Do not create permanent spending before commercial certainty.

L3. Oil-price risk

Event: sustained lower realized price. Control:

- conservative budget assumptions;
- fund buffer;

- stress test;
- flexible capital programme.

L4. Production risk

Event: reservoir or equipment underperformance. Control:

- regulator access;
- maintenance;
- redundant systems;
- conservative forecast.

L5. Project-delay risk

Event: first oil delayed. Control:

- schedule ranges;
- no borrowing based solely on operator date;
- periodic forecast revision.

L6. Cost-recovery risk

Event: cost bank higher than forecast. Control:

- early audit;
- project tagging;
- related-party review;
- scenario model.

L7. Transfer-pricing risk

Event: related-party costs exceed arm's-length value. Control:

- documentation;
- comparables;
- audit;
- contract rights.

L8. Revenue-collection risk

Event: invoice or settlement error. Control:

- cargo ID;

- independent calculation;
- bank reconciliation;
- audit.

L9. Sovereign-fund market risk

Event: financial asset loss. Control:

- diversified allocation;
- benchmark;
- liquidity;
- risk limit.

L10. Withdrawal risk

Event: withdrawals rise faster than durable revenue. Control:

- statutory rule;
- parliament;
- long-run scenario;
- published calculation.

L11. Public-investment inflation risk

Event: construction cost rises. Control:

- sequencing;
- unit-cost database;
- independent estimates.

L12. Maintenance risk

Event: new assets deteriorate. Control:

- lifecycle budget;
- condition monitoring;
- responsible operator.

L13. Debt risk

Event: debt service rises as oil revenue falls. Control:

- debt strategy;

- maturity;
- currency limits;
- asset-liability analysis.

L14. Public-enterprise risk

Event: SOE losses create fiscal transfer. Control:

- audited statements;
- borrowing limits;
- performance contracts;
- guarantee register.

L15. Spill risk

Event: major offshore release. Control:

- prevention;
- capping;
- command;
- response assets;
- financial assurance;
- exercise.

L16. Financial-assurance risk

Event: available security below actual response need. Control:

- periodic severe-case estimate;
- insurance review;
- parent support;
- regulator revision.

L17. Fisheries risk

Event: pollution or exclusion affects fishing. Control:

- baseline;
- consultation;
- closure protocol;
- claims.

L18. Biodiversity risk

Event: cumulative offshore activity harms sensitive species. Control:

- monitoring;
- mitigation hierarchy;
- permit limits;
- independent review.

L19. Climate-transition risk

Event: long-run oil demand or financing changes. Control:

- lower-price scenarios;
- economic diversification;
- no reliance on indefinite high prices.

L20. Carbon-market risk

Event: methodology or buyer eligibility changes. Control:

- separate climate-finance scenarios;
- diversified financing;
- verified issuance.

L21. Local-content inflation risk

Event: mandatory local procurement raises cost where capacity is insufficient. Control:

- capacity study;
- phased targets;
- competition.

L22. Front-company risk

Event: nominal local ownership hides foreign control. Control:

- beneficial ownership;
- audit;
- enforcement.

L23. Skills risk

Event: shortage delays projects and weakens regulator. Control:

- workforce model;
- compensation;
- training;
- targeted recruitment.

L24. Household inequality risk

Event: national income rises while lower-income real consumption lags. Control:

- current survey;
- targeted services;
- regional data.

The paper describes evidence and policy tools without choosing an electoral programme.

L25. Housing risk

Event: rent and land prices rise. Control:

- planning;
- serviced land;
- data;
- infrastructure.

L26. Food-price risk

Event: wage and demand shock raises food cost. Control:

- market monitoring;
- agricultural productivity;
- logistics.

L27. Procurement risk

Event: weak tender or contract management reduces value. Control:

- competition;
- beneficial ownership;
- variation audit;

- outcome review.

L28. Corruption risk

Event: public or private decision is distorted for improper benefit. Control:

- conflict declarations;
- procurement transparency;
- audit;
- investigation;
- enforcement.

Risk description should not imply misconduct without evidence.

L29. EITI participation risk

Event: stakeholder confidence in representation declines. Control:

- transparent selection;
- documented consultation;
- independent process.

Guyana's ongoing 2026 validation provides a documented reason to include the risk. [S69-S70]

L30. Statistical risk

Event: stale data distort policy. Control:

- survey calendar;
- revision policy;
- metadata.

L31. Cyber risk

Event: production or financial data compromised. Control:

- access control;
- backup;
- incident response;
- offline archive.

L32. Records risk

Event: contract or permit history lost. Control:

- versioned digital archive;
- redundant storage;
- retention law.

L33. Litigation risk

Event: permit or fiscal dispute creates uncertainty. Control:

- reasoned decisions;
- legal review;
- complete administrative record.

L34. Political-transition continuity risk

Event: institutional records or procedures change abruptly after an election or cabinet change.

Control:

- statutory roles;
- professional civil service;
- archived data;
- audited accounts.

This is an institutional continuity issue, not a forecast about any election.

L35. Transboundary risk

Event: spill or maritime dispute affects neighbour. Control:

- treaties;
- notification;
- mutual assistance;
- maritime clarity.

L36. Decommissioning risk

Event: operator exits before full closure liability is funded. Control:

- reserve;
- security;
- assignment conditions;
- periodic estimate.

L37. Risk interaction

Risks combine.

Example: price fall + project delay + construction inflation can reduce revenue while increasing project cost.

The integrated scenario should be tested annually.

L38. Early-warning dashboard

Indicators:

- oil price;
- production variance;
- cost-bank variance;
- fund withdrawal ratio;
- construction inflation;
- debt service;
- regulator vacancy;
- audit backlog;
- permit breaches;
- grievances;
- household inflation.

L39. Control assurance

Each control should have:

- design owner;
- operating owner;
- test;
- frequency;
- evidence;
- deficiency;
- remediation.

A policy document without an operating test is not assurance.

L40. Risk review cycle

Quarterly: financial and production risk.

Semi-annual: project delivery, local content, labour.

Annual: environment, sovereign fund, social data, institutional controls.

After material event: special review.

L41. Annex conclusion

A petroleum state needs one integrated risk picture because shocks cross institutional boundaries. The risk register supports disciplined preparation without turning uncertainty into a political judgment.

TECHNICAL ANNEX M

Data dictionary, indicator definitions and reproducible publication tables

Purpose: ensure that major numbers in the flagship edition use stable definitions.

M1. Petroleum production

Unit: barrels of crude oil. Frequency: daily/monthly. Primary source: regulator dashboard where available. Do not mix with nameplate capacity.

M2. Nameplate capacity

Unit: barrels per day. Definition: engineering design throughput. Not an observed annual average.

M3. Government lift

Unit: cargo/barrels. Definition: petroleum entitlement physically lifted for government account.

M4. Profit-oil receipt

Unit: USD or reporting currency. Definition: cash proceeds from government profit-oil sale settled in period.

M5. Royalty

Unit: currency. Definition: contractual royalty paid under the applicable agreement.

M6. Petroleum deposit

Definition: qualifying petroleum cash deposited into the NRF during the period.

Do not include projected future sales.

M7. Fund investment income

Definition: interest, dividend or other investment income under fund accounting.

Keep separate from petroleum receipts.

M8. Fund market value

Definition: end-period fair value of financial assets.

May differ from cash.

M9. Fund withdrawal

Definition: lawful transfer from NRF under the statutory framework.

Not equivalent to final public spending.

M10. Consolidated-fund receipt

Definition: cash received by the government's budget account following withdrawal.

M11. Capital appropriation

Definition: parliament-approved spending authority.

Not equivalent to contract award or completed asset.

M12. Capital expenditure

Definition: amount recorded as spent under public accounts.

Must be distinguished from physical completion.

M13. Project completion

Definition: physical works substantially complete according to defined criteria.

Separate from commissioning.

M14. Commissioning

Definition: asset has passed tests and is available for intended service.

M15. Service outcome

Definition: measured benefit such as travel-time reduction, power reliability or health service.

M16. Gross local procurement

Definition: invoice value purchased from entities counted under local-content reporting.

Not domestic value added.

M17. Domestic value added

Definition: value created domestically through wages, operating surplus and local taxes net of imported intermediate inputs.

M18. Local employment

State whether definition is citizenship, residence or statutory local-content category.

M19. Training participant

Person enrolled.

Separate from completion and employment.

M20. Household poverty

Requires:

- survey year;
- poverty line;
- methodology;
- population coverage.

Never carry an old estimate forward as current.

M21. Real household consumption

Household expenditure or consumption adjusted for prices.

Useful for distributional analysis.

M22. Real GDP

Output volume measured at constant prices.

State base year.

M23. Non-oil GDP

Define excluded petroleum activities and any treatment of petroleum-related services.

M24. Non-oil primary balance

Fiscal balance excluding petroleum revenue and interest, under the adopted methodology.

State denominator.

M25. Public debt

Report gross and, where useful, net of liquid financial assets.

M26. Construction-cost index

Separate from CPI.

Tracks cost pressure in capital programme.

M27. Flaring

Specify gas volume, CO2e or duration.

Do not mix units.

M28. Spill

Specify substance, estimated volume, confidence and affected area.

M29. Financial assurance

Specify instrument, amount, currency, trigger and expiry.

Do not confuse with unlimited legal liability.

M30. Forest-credit issuance

Units issued under the applicable standard.

Separate from sale.

M31. Carbon-credit contract value

Future contractual consideration subject to delivery.

Separate from cash received.

M32. Carbon-credit cash receipt

Actual cash received in period.

M33. Community allocation

Amount allocated or transferred under programme rules.

Separate from outcome.

M34. EITI status

Use official EITI process status and date.

Do not invent a score while validation is ongoing.

M35. Audit adjustment

Specify proposed, agreed or final.

Do not label proposed amount recovered.

M36. Source table convention

Every table footer should include:

- source IDs;
- observation date;
- unit;
- calculation;
- rounding.

M37. Forecast table convention

Forecast tables use:

- scenario name;
- assumptions;
- date prepared;
- uncertainty.

They should never appear visually identical to historical actuals.

M38. Revision triangle

For key annual forecasts, preserve:

- original budget;
- mid-year revision;
- outturn;
- later audited outturn.

This shows forecast quality.

M39. Geographic coding

Use Guyana's administrative regions for public-service data when available.

Offshore projects use block/project coordinates.

M40. Currency

Identify:

- USD;
- GYD;
- exchange rate;
- nominal/real.

Do not compare multi-year nominal GYD without adjustment.

M41. Rounding

Totals may differ because of rounding. Store unrounded calculations.

M42. Missing data

Use blank/NA.

Do not substitute zero.

M43. Estimated data

Mark with "estimate" and source.

M44. Confidential data

State why unavailable and what public proxy is used.

M45. Indicator provenance

Each chart should link to the source register.

M46. Machine-readable supplement

The final publication package should include CSV tables for:

- production;
- fund;
- scenarios;
- comparator metrics;
- project register;
- source register.

M47. Checksums

Publish SHA-256 for PDF and data supplement.

M48. Reproducibility

A user should be able to recalculate displayed tables using the published data and formulas.

M49. Archive date

The online research page should display the evidence cut-off and edition date.

M50. Annex conclusion

Stable definitions protect a long publication from internal contradiction. The data dictionary should be treated as part of the research, not an afterthought.

Scenario tables | Contract sensitivity

Illustrative arithmetic only. Values are not forecasts and do not represent observed government receipts.

Barrels	Price	Cost share	Gross sales	Government	Gov. share
10.00m	\$40	20.0%	\$400.00m	\$168.00m	42.0%
10.00m	\$40	50.0%	\$400.00m	\$108.00m	27.0%
10.00m	\$40	65.0%	\$400.00m	\$78.00m	19.5%
10.00m	\$40	75.0%	\$400.00m	\$58.00m	14.5%
10.00m	\$60	20.0%	\$600.00m	\$252.00m	42.0%
10.00m	\$60	50.0%	\$600.00m	\$162.00m	27.0%
10.00m	\$60	65.0%	\$600.00m	\$117.00m	19.5%
10.00m	\$60	75.0%	\$600.00m	\$87.00m	14.5%
10.00m	\$80	20.0%	\$800.00m	\$336.00m	42.0%
10.00m	\$80	50.0%	\$800.00m	\$216.00m	27.0%
10.00m	\$80	65.0%	\$800.00m	\$156.00m	19.5%
10.00m	\$80	75.0%	\$800.00m	\$116.00m	14.5%
10.00m	\$100	20.0%	\$1.00bn	\$420.00m	42.0%
10.00m	\$100	50.0%	\$1.00bn	\$270.00m	27.0%
10.00m	\$100	65.0%	\$1.00bn	\$195.00m	19.5%
10.00m	\$100	75.0%	\$1.00bn	\$145.00m	14.5%

Scenario tables | Production shocks

Combined volume, price and cost sensitivities. All inputs are illustrative.

Volume factor	Price	Cost share	Gross sales	Government	Gov. share
70.0%	\$40	50.0%	\$280.00m	\$75.60m	27.0%
70.0%	\$40	75.0%	\$280.00m	\$40.60m	14.5%
70.0%	\$60	50.0%	\$420.00m	\$113.40m	27.0%
70.0%	\$60	75.0%	\$420.00m	\$60.90m	14.5%
70.0%	\$80	50.0%	\$560.00m	\$151.20m	27.0%
70.0%	\$80	75.0%	\$560.00m	\$81.20m	14.5%
85.0%	\$40	50.0%	\$340.00m	\$91.80m	27.0%
85.0%	\$40	75.0%	\$340.00m	\$49.30m	14.5%
85.0%	\$60	50.0%	\$510.00m	\$137.70m	27.0%
85.0%	\$60	75.0%	\$510.00m	\$73.95m	14.5%
85.0%	\$80	50.0%	\$680.00m	\$183.60m	27.0%
85.0%	\$80	75.0%	\$680.00m	\$98.60m	14.5%
100.0%	\$40	50.0%	\$400.00m	\$108.00m	27.0%
100.0%	\$40	75.0%	\$400.00m	\$58.00m	14.5%
100.0%	\$60	50.0%	\$600.00m	\$162.00m	27.0%
100.0%	\$60	75.0%	\$600.00m	\$87.00m	14.5%
100.0%	\$80	50.0%	\$800.00m	\$216.00m	27.0%
100.0%	\$80	75.0%	\$800.00m	\$116.00m	14.5%

Scenario tables | Sovereign-fund stress

Synthetic sovereign-fund stress paths. They are decision tests, not projections of the Natural Resource Fund.

Scenario	Year	Opening	Deposits	Withdrawals	Closing
illustrative baseline	1	\$4.00bn	\$4.00bn	\$2.00bn	\$6.16bn
illustrative baseline	2	\$6.16bn	\$4.20bn	\$2.20bn	\$8.41bn
illustrative baseline	3	\$8.41bn	\$4.40bn	\$2.40bn	\$10.74bn
illustrative baseline	4	\$10.74bn	\$4.50bn	\$2.50bn	\$13.17bn
illustrative baseline	5	\$13.17bn	\$4.50bn	\$2.50bn	\$15.70bn
low price and lower deposits	1	\$4.00bn	\$2.40bn	\$2.00bn	\$4.52bn
low price and lower deposits	2	\$4.52bn	\$2.10bn	\$2.00bn	\$4.76bn
low price and lower deposits	3	\$4.76bn	\$2.20bn	\$2.00bn	\$5.10bn
low price and lower deposits	4	\$5.10bn	\$2.50bn	\$2.00bn	\$5.75bn
low price and lower deposits	5	\$5.75bn	\$2.80bn	\$2.00bn	\$6.72bn
higher spending	1	\$4.00bn	\$4.00bn	\$3.00bn	\$5.16bn
higher spending	2	\$5.16bn	\$4.20bn	\$3.30bn	\$6.27bn
higher spending	3	\$6.27bn	\$4.40bn	\$3.50bn	\$7.42bn
higher spending	4	\$7.42bn	\$4.50bn	\$3.70bn	\$8.51bn
higher spending	5	\$8.51bn	\$4.50bn	\$3.90bn	\$9.45bn
market loss then recovery	1	\$4.00bn	\$4.00bn	\$2.00bn	\$5.60bn
market loss then recovery	2	\$5.60bn	\$4.00bn	\$2.00bn	\$7.71bn
market loss then recovery	3	\$7.71bn	\$4.00bn	\$2.00bn	\$10.17bn
market loss then recovery	4	\$10.17bn	\$4.00bn	\$2.00bn	\$12.68bn
market loss then recovery	5	\$12.68bn	\$4.00bn	\$2.00bn	\$15.19bn

Adversarial publication audit

ADVERSARIAL PUBLICATION AUDIT

Guyana's Petroleum Transformation, Edition 1.0

Audit date: 27 September 2026 Scope: content integrity, source traceability, numerical logic, legal status, PDF QA, publication controls and auditability.

Audit conclusion

Edition 1.0 is a substantial research compendium, but it should not be represented as a fully closed, scrutiny-ready final publication yet.

The strongest parts are the documented chronology, separation of observed data from forecasts, the distinction between the 2016 Stabroek terms and later model terms, the Natural Resource Fund accounting framework, the transferability cautions, and the explicit effort to separate government, operator, multilateral and stakeholder positions.

The present release fails its own publication logic in several material areas. The failures are remediable. PR #61 has therefore been returned to draft and issue #60 reopened.

This audit is an internal adversarial review. It is not independent external peer review.

Critical findings

C1. Publication-status contradiction

The PDF cover describes a Flagship Reference Edition 1.0. The build audit marked ``publication_complete: true``.

Yet the executive brief states that the final white paper is not externally reviewed or complete. The final page is titled "Open evidence items" and lists ten checks described as required before final publication. The publication review gates state: "No flagship release until each gate is evidenced." The progress tracker still contains many FOUNDATION and RESEARCHING statuses.

A publication cannot be both complete and incomplete under the same control framework.

Required correction:

- classify the current document as a publication candidate;
- set ``publication_complete: false`` until every mandatory gate is evidenced;
- preserve a separate internal-audit status and an external-review status.

C2. The PDF page-count gate is materially overstated

Edition 1.0 has 518 rendered pages, but the total includes:

- 44 pages of table of contents;
- 3 pages of executive brief;
- 122 pages of integrated flagship manuscript;
- 185 pages of evidence dossiers and workbooks;
- 138 pages of technical annexes;
- 12 pages of scenario tables;
- 3 pages of publication review gates;
- 9 pages of source register;
- 1 page of open evidence items.

The builder's "substantive page" test used ``len(text.split()) >= 100``. Dot leaders and fragmented tokens can inflate this count. A regex word audit found:

- median page density: about 154 words;
- 76 pages below 100 lexical words;
- integrated manuscript: 122 pages and about 25,600 lexical words.

The 518-page total remains a legitimate compendium length, but it is not equivalent to 518 pages of continuous core analysis.

Required correction:

- report total rendered pages and core analytical pages separately;
- exclude table of contents and source register from substantive-page claims;
- use lexical word counts;
- never use page count as a publication-quality proxy.

C3. Architecture drift

The approved architecture contemplated roughly 54 core chapters. The typeset corpus contains 263 unique numbered chapter headings because research dossiers and operating workbooks were promoted into the same chapter hierarchy.

This weakens navigation and makes the work look more fragmented than the underlying research warrants.

Required correction:

- retain the 54-chapter core architecture;
- relabel supporting material as Annex, Evidence Note, Workbook or Checklist;
- keep supporting headings out of the main chapter numbering and primary TOC.

C4. Source traceability is not strong enough for audit-grade publication

The source register contains 86 records. In the rendered PDF, 63 source IDs are cited at least once. In the 122-page integrated manuscript, 81 pages contain no source-ID marker.

A page without a source ID is not automatically unsupported, since some pages contain analytical method or guidance. The density nevertheless makes claim tracing slower than it should be for a publication intended to withstand legal, fiscal and technical challenge.

Required correction:

- create a source-to-claim matrix for every material factual claim;
- add page, section, table, paragraph or statutory provision pinpoints for Tier 1 sources;
- ensure every material number has source, observation date, unit and accounting boundary;
- distinguish primary evidence from attributed interpretation.

C5. Several high-risk legal and regulatory gates remain open

The publication itself identifies these gaps:

- authenticated May 2026 Court of Appeal judgment not reviewed directly;
- commencement/implementing status of the 2025 oil-pollution statute not fully established;
- precise insurance, guarantee and financial-assurance instruments not fully reviewed;
- later cost-audit dispute outcomes remain unsettled.

Current external checks on 27 September 2026 confirm:

- EITI still lists Guyana's 2026 Validation as ongoing, not completed;
- Parliament and the Official Gazette establish enactment/publication of Act 6 of 2025;
- the available public government summary describes the May 2026 appellate result, but this audit has not located an authenticated full judgment;
- public petroleum sources continue to describe unresolved or ongoing cost-audit processes.

Required correction:

- do not characterize an unresolved legal point as settled;
- cite authenticated judgments and commencement instruments when found;
- label absence of a located public instrument as "not established in sources reviewed", not "does not exist".

C6. Scenario tables fail the publication's readability gate

Pages 494-505 use wide CSV tables compressed into narrow portrait columns. Column labels break into fragments such as individual syllables and characters. This fails Gate 12's requirement that tables be readable at normal zoom.

Required correction:

- replace raw CSV dumps with grouped scenario tables;
- use human-readable labels;
- split fiscal, production and fund outputs into separate tables;
- add units, formula notes and scenario interpretation;
- preserve full CSV files as machine-readable supplements.

C7. Accessibility is incomplete

The PDF is searchable and opens cleanly, but the document catalog does not expose a tagged PDF structure tree. The publication therefore does not satisfy a strong accessibility standard for semantic headings, table structure and figure alternative text.

Required correction:

- produce a tagged accessible PDF where feasible;
- ensure reading order, headings and table headers are encoded;
- add captions and text equivalents for material figures;
- run an accessibility check before public release.

C8. External expert review is explicitly outstanding

The publication requires independent review across:

- petroleum fiscal economics;
- Guyana petroleum law and contract interpretation;
- public financial management;
- offshore environmental science and spill response;
- Indigenous/community safeguards;
- macroeconomics;
- data visualization and statistical interpretation.

No internal model review should be described as external review.

Required correction:

- keep external-review status pending;
- publish reviewer scope, conflicts policy and response-to-review log when reviews occur.

Major coverage gaps or underdeveloped areas

M1. Petroleum Activities Act 2023

The 2023 statutory framework should have an explicit chapter-level crosswalk showing what changed relative to the earlier legal framework and how the new Act interacts with legacy Stabroek rights, new licences, decommissioning, data, safety and regulation.

M2. Maritime and geopolitical risk

A newcomer blueprint should address sovereign-boundary and maritime-security risk. Guyana's continuing case before the International Court of Justice concerning the 1899 Arbitral Award and related provisional measures is material contextual evidence. It should be presented neutrally, with no prediction of the merits outcome.

M3. Latest NRF evidence

The Bank of Guyana now publishes audited NRF financial statements for 2025. These should be incorporated into the annual reconciliation before Edition 1.1 is treated as current through September 2026. Monthly 2026 NRF reporting should also be reconciled to the mid-year figures.

M4. Forest-carbon market developments

The climate chapter should explicitly address CORSIA eligibility, corresponding adjustments, Article 6 accounting boundaries and competing views on Indigenous safeguards. Government programme claims, registry evidence and critical stakeholder analysis should remain distinguishable.

M5. Independent social and Indigenous evidence

The current household evidence remains constrained by survey timing. The paper should deepen community-level evidence, including documented critical perspectives, without assuming either programme success or failure.

M6. Local-content outcome measurement

Gross procurement is not domestic value added. Edition 1.1 should add a transparent bridge from gross spend to imports, wages, local ownership, retained profit, supplier capability and exportable capacity.

M7. Project-level environmental performance

Permit and EIA existence is well documented. The harder audit question is performance:

- permit compliance;
- flaring and emissions;
- incident history;
- drill results;
- marine monitoring;
- independent sampling;
- financial-assurance enforceability.

The final publication should state where public data do not permit independent verification.

PDF technical observations

Pass:

- file opens without encryption;
- 518 pages render successfully;
- searchable text is present;
- page numbering is consistent;
- no black-square glyph failures were found in the visual scan;
- fonts used for primary body text are embedded;
- source hyperlinks are present.

Needs correction:

- 44-page TOC is excessive;
- supporting material dominates the main hierarchy;
- scenario tables are unreadable in portrait format;
- PDF is not semantically tagged for accessibility;
- Helvetica is present as an unembedded Type 1 font in the file;
- publication metadata lacks a fuller keyword/version/rights package.

Current evidence-status checks, 27 September 2026

1. EITI Validation: ongoing. EITI's current schedule still lists Guyana among ongoing Validations.
2. NRF 2025: audited statements are publicly available from the Bank of Guyana.
3. Oil Pollution Act 2025: enactment/publication established. Full operational commencement and implementing regulations require authoritative confirmation.
4. Court of Appeal financial-assurance case: government and press summaries are available. Authenticated judgment remains a required source for final legal characterization.
5. Cost audits: public records show material disputed costs and continuing resolution processes. Audit exceptions must not be reported as cash recovered.

Edition 1.1 mandatory remediation plan

Gate A: truth-in-publication controls

- change status from final/complete to publication candidate;
- remove contradictory completion claims;
- keep issue #60 open and PR #61 draft until mandatory gates pass.

Gate B: source-to-claim audit

- produce claim IDs for all material numbers and legal propositions;
- add pinpoint references;
- validate all source URLs;
- add a corrections register.

Gate C: legal hardening

- explicit Petroleum Activities Act 2023 crosswalk;
- authenticated appellate judgment if publicly obtainable;
- pollution-law commencement/regulations check;
- licence/permit version control.

Gate D: financial hardening

- reconcile audited NRF 2020-2025;
- reconcile 2026 monthly reports through the evidence cut-off;
- connect lifts, receipts, deposits, returns and withdrawals without double counting;
- preserve cost-audit dispute status.

Gate E: environmental and social hardening

- project-level compliance evidence;
- spill-response capability and financial assurance;
- independent Indigenous/community evidence;
- local-value-added bridge;
- current-data limitations prominently disclosed.

Gate F: geopolitical and transferability hardening

- add maritime-boundary and territorial-dispute risk module;
- add institutional continuity and political-transition controls;
- keep political and electoral judgments out of the analysis.

Gate G: PDF and data QA

- rebuild TOC around the core 54-chapter architecture;
- demote workbooks and evidence notes;
- redesign scenario tables;
- create accessible/tagged output where practical;
- separate total pages from core analytical pages;

- run link, citation, formula, metadata and visual checks.

Gate H: external review

- independent expert reviews in the disciplines listed above;
- documented responses and unresolved-objection register;
- only then move from publication candidate to externally reviewed edition.

Release rule

Edition 1.1 may be described as internally audited once Gates A-G pass.

It should not be described as independently peer reviewed, externally validated or immune from future correction unless the corresponding external evidence exists.

No publication can honestly be guaranteed to survive every conceivable criticism. The defensible standard is stronger: every material claim should be traceable, versioned, reproducible where quantitative, transparent about uncertainty and open to correction.

Remediation status after Edition 1.1 rebuild

C1 Publication-status contradiction: REMEDIATED. Edition 1.1 is labelled an internally audited publication candidate. The audit record sets `publication_complete=false` and `external_peer_review=pending`.

C2 Page-count overstatement: REMEDIATED IN METRICS. Total rendered pages, lexical-density pages, core integrated pages and core lexical words are now reported separately. Page count is no longer used as proof of research quality.

C3 Architecture drift: REMEDIATED IN PRESENTATION. The ten-part structure remains primary. The 263 granular units are labelled analytical sections rather than chapters and are removed from the primary table of contents.

C4 Source traceability: PARTIALLY REMEDIATED. The source register has expanded from 86 to 97 records and a 45-item high-risk claim-to-source matrix has been added. Full sentence-level pinpoints remain a continuing editorial control.

C5 High-risk legal/regulatory evidence: OPEN. The paper uses narrow language where authenticated appellate text, commencement instruments, financial-assurance instruments or final dispute outcomes have not been established in public sources reviewed.

C6 Scenario-table readability: REMEDIATED. Wide raw CSV dumps have been replaced by six-column human-readable scenario tables while the full CSVs remain machine-readable supplements.

C7 Tagged-PDF accessibility: OPEN. Searchable text and reading order are present, but semantic PDF tagging is not claimed.

C8 External expert review: OPEN. No external peer-review claim will be made until specialist reviews and response logs are completed.

The internally audited candidate is therefore stronger than Edition 1.0, but the external-review and specified open-evidence gates remain visible rather than being masked by a completion label.

High-risk claim audit

C001 Liza-1 well date and 2015 discovery sequence

Sources: S01, S36 | Status: VERIFIED

Retain operator history and official well register separately

C002 Stabroek agreement date and effective/licence dates

Sources: S02, S03, S41 | Status: VERIFIED

Keep signature, effective and licence dates distinct

C003 2016 Stabroek 2% royalty, cost recovery and residual profit-oil structure

Sources: S02, S15 | Status: VERIFIED WITH CONTRACT PINPOINT

Final legal edition should cite exact executed clauses

C004 Later model fiscal terms differ from Stabroek

Sources: S14 | Status: VERIFIED

Never apply later model terms retroactively

C005 Petroleum Activities Act 17 of 2023 supplies a newer statutory framework

Sources: S87 | Status: VERIFIED

Transitional effect on legacy rights requires agreement-specific legal analysis

C006 NRF Act 2019 preceded first oil

Sources: S37, S04 | Status: VERIFIED

Distinguish passage, assent, operation and later replacement

C007 2021 NRF framework replaced earlier legislation and 2024 rules changed withdrawals

Sources: S04, S05 | Status: VERIFIED

Clause-level crosswalk retained in legal annex

C008 First oil occurred in December 2019

Sources: S01, S81 | Status: VERIFIED

Production dashboard is preferred for realized series

C009 Liza Phase 2 started February 2022

Sources: S82 | Status: VERIFIED

Operator startup date, regulator series for realized output

C010 Payara started November 2023

Sources: S83 | Status: VERIFIED

Keep future production estimates separate from actual output

C011 Yellowtail started August 2025

Sources: S84 | Status: VERIFIED

Installed capacity is not annual average production

C012 Daily production reached 900,000 bpd in November 2025

Sources: S85 | Status: VERIFIED AS POINT-IN-TIME

Do not generalize to annual average

C013 Monthly realized production series exists through July 2026

Sources: S81 | Status: VERIFIED

Retain unit/date metadata

C014 H1 2026 profit-oil, royalty, withdrawals and end-June NRF figures

Sources: S07 | Status: VERIFIED AS GOVERNMENT MID-YEAR DATA

Reconcile cash timing to Bank of Guyana reporting

C015 Audited NRF financial statements exist through 2025

Sources: S44, S88 | Status: VERIFIED

Use audited annual baseline before 2026 bridge

- C016** 2026 NRF monthly reports are available through July at cut-off
Sources: S96 | Status: VERIFIED
Reconcile monthly, mid-year and annual accounting bases
- C017** Government cost audits identified disputed items
Sources: S42, S71, S72 | Status: VERIFIED AS AUDIT/DISPUTE STATUS
Audit exception is not recovered cash
- C018** Guyana 2026 EITI Validation began 15 May and remained ongoing at cut-off
Sources: S16, S34, S69, S89 | Status: VERIFIED
No final score or outcome should be stated
- C019** EITI Board recorded concern about stakeholder confidence in civil-society selection
Sources: S16, S34, S69 | Status: VERIFIED AS ATTRIBUTED EITI VIEW
Do not convert into author's political conclusion
- C020** Oil Pollution Act 6 of 2025 was enacted and formally published in the Official Gazette
Sources: S19, S53, S54, S98 | Status: VERIFIED
Gazetting does not establish commencement of every provision
- C021** Full commencement/implementing status of Act 6 of 2025
Sources: S19, S53, S54, S98 | Status: OPEN AFTER TARGETED GAZETTE CHECK
Official Gazette publication located; no relied-upon commencement order or implementing-regulation instrument established in sources reviewed
- C022** May 2026 Court of Appeal reversed the earlier unlimited-guarantee order and distinguished liability from financial assurance
Sources: S20, S101, S105 | Status: VERIFIED AS ATTRIBUTED APPELLATE RESULT
Authenticated Court of Appeal reasons still required for detailed ratio or quotation
- C023** Permits require insurance and parent/affiliate financial-assurance arrangements, while exact executed guarantee/insurance terms remain separate evidence
Sources: S31, S99, S100, S102, S103 | Status: PARTIAL / OPEN
Permit clauses verified; executed guarantee amount, operative wording, policy limits and loss-event liquidity remain unreviewed
- C024** EPA permits and EIAs exist for named projects
Sources: S48-S56, S99-S100 | Status: VERIFIED DOCUMENT AVAILABILITY
Permit existence is not performance evidence
- C025** Project-level environmental compliance, incidents and drill performance
Sources: S48-S56, S94, S99-S100 | Status: PARTIAL / OPEN
Permit terms are better pinned down; independent compliance, incident and response-performance evidence still needs expansion
- C026** Local Content Act 2021 created statutory framework
Sources: S57 | Status: VERIFIED
Separate legal requirements from outcome claims
- C027** Gross local procurement is not equivalent to domestic value added
Sources: S57-S60 | Status: ANALYTICAL CONTROL
Build ownership/import/wage/profit bridge
- C028** 2018/2019 HBS is historical, not a 2026 welfare measure
Sources: S61 | Status: VERIFIED
Do not label old poverty or consumption data as current
- C029** LCDS predates oil production and Norway partnership generated reported forest finance
Sources: S22 | Status: VERIFIED AS OFFICIAL PROGRAMME HISTORY
Outcome evaluation remains separate
- C030** Hess minimum multiyear forest-credit purchase commitment
Sources: S23 | Status: VERIFIED AS CONTRACT/PROGRAMME CLAIM
Contract value is not cash already received

- C031** 2021-vintage ART credits were reported CORSIA eligible
Sources: S95 | Status: VERIFIED AS OFFICIAL PROGRAMME/REGISTRY CLAIM
Preserve corresponding-adjustment and retirement boundaries
- C032** Indigenous safeguard criticisms exist
Sources: S93 | Status: VERIFIED AS ATTRIBUTED CRITICAL VIEW
Triangulate with programme and community evidence
- C033** Village allocations and consultation process are described by LCDS
Sources: S62, S63 | Status: VERIFIED AS OFFICIAL PROGRAMME CLAIM
Independent village-level outcomes remain limited
- C034** Guyana-Venezuela matter remains before the ICJ
Sources: S90, S91 | Status: VERIFIED
No prediction or merits ranking
- C035** CARICOM supports peaceful resolution through the ICJ
Sources: S92 | Status: VERIFIED AS REGIONAL POSITION
Diplomatic statement is not a judicial merits finding
- C036** IMF 2026 staff highlighted absorption, procurement, audit and data issues
Sources: S64 | Status: VERIFIED AS IMF STAFF ASSESSMENT
Not an author verdict
- C037** Norway fiscal/fund framework used as comparator
Sources: S74 | Status: VERIFIED
Mature-state context limits transfer
- C038** Ghana stabilization and heritage fund architecture
Sources: S75, S76 | Status: VERIFIED
Compare mechanisms, not political outcomes
- C039** Timor-Leste Petroleum Fund reporting and portfolio framework
Sources: S77, S78 | Status: VERIFIED
Analyze alongside depletion and withdrawals
- C040** Suriname GranMorgu remained pre-first-oil at cut-off
Sources: S79 | Status: VERIFIED AS OPERATOR UPDATE
Planned 2028 first oil is not observed production
- C041** Trinidad and Tobago HSF is a mature regional comparator
Sources: S80 | Status: VERIFIED
Different industrial history and gas structure
- C042** Strategic environmental assessment identifies cumulative offshore risks
Sources: S94 | Status: VERIFIED AS TOR/RISK SCOPE
TOR is not outcome evidence
- C043** Carbon-credit cash, issuance, transfer and retirement are separate accounting events
Sources: S23, S32, S95 | Status: ANALYTICAL CONTROL
Keep petroleum and carbon ledgers separate
- C044** Public investment allocation is not equivalent to delivered public service
Sources: S64, S65, S73 | Status: ANALYTICAL CONTROL
Project-level completion and maintenance evidence required
- C045** No external peer review has been completed
Sources: Publication records | Status: VERIFIED
Must remain explicit until reviewers and response log exist
- C046** Authenticated CCJ reasons identify the underlying Collins/Whyte litigation, Liza 1 permit condition 14 and Environmental Protection Act section 31
Sources: S103 | Status: VERIFIED WITH PARAGRAPH PINPOINTS
2024 CCJ decision concerns intervention/procedure, not the May 2026 merits appeal

C047 EITI Validation Committee discussed preliminary Guyana mission findings in June 2026

Sources: S104 | Status: VERIFIED AS PROCESS STATUS

Not a final Validation decision

C048 Contemporary reporting says leave was later granted to challenge the May 2026 appellate result before the CCJ

Sources: S106 | Status: ATTRIBUTED SECONDARY / OPEN FOR DOCKET CONFIRMATION

Do not describe as a current CCJ merits decision without official docket evidence

C049 Uaru permit conditions 14.3-14.5 distinguish credible-cost estimation, insurance and parent/affiliate guarantee obligations

Sources: S99 | Status: VERIFIED WITH PERMIT PINPOINTS

Does not establish exact current guarantee amount or insurance limits

C050 Liza Phase 1 permit history confirms EIA, management plan and spill/wildlife-response materials are incorporated into the project authorization record

Sources: S100 | Status: VERIFIED DOCUMENT INCORPORATION

Incorporation does not establish implementation quality or compliance

Publication review gates

Publication Review Gates

No flagship release until each gate is evidenced

Gate 1: Chronology

- Exact dates for discovery, PSA signature, PSA effective date, production licences, NRF statutes, first oil and each project start.
- Earlier exploration and 2009 LCDS clearly separated from post-discovery institutions.
- Every event tagged as observed, enacted, commenced, announced, projected or disputed.

Gate 2: Petroleum contract

- Executed 2016 Stabroek text checked directly.
- Simplified royalty/cost/profit examples reconciled to agreement language.
- 2023 model terms never applied retroactively.
- Audit, tax, valuation, decommissioning and dispute provisions traced.

Gate 3: Production

- Monthly series through latest verified 2026 observation.
- Capacity, actual daily output, monthly output, lifts and receipts stored as separate fields.
- Startup dates and sanctioned projects independently dated.

Gate 4: NRF

- Annual opening balance, receipts, investment income, transfers and closing balance reconciled.
- 2019, 2021 and 2024 legal versions crosswalked.
- Cash balances not mixed with forecasts or accrual estimates.
- Public investment outcomes kept separate from fund accounting.

Gate 5: Environment

- Permit and EIA register per producing project.
- 2025 oil-pollution statute commencement status verified from authoritative instruments.
- Actual insurance, guarantee and liability documents reviewed where public.
- Court decisions read from authentic text before legal characterization.

- Spill-response preparedness assessed with disclosed evidence limits.

Gate 6: Social and local value

- Local-content figures distinguish gross procurement, domestic ownership, wages and retained value.
- Household welfare uses current survey dates.
- No 2019 poverty measure described as a 2026 poverty rate.
- Amerindian and hinterland outcomes include independent or community-level evidence where available.

Gate 7: Climate and natural capital

- Forest-carbon credits, contract values, issued units, receipts and village disbursements reconciled.
- Petroleum emissions and forest-carbon accounting kept separate.
- No claim of economy-wide carbon neutrality without full emissions accounting.
- Carbon-market methodology changes and eligibility status dated.

Gate 8: EITI and governance

- 2026 Validation status checked on publication date.
- EITI Board concerns attributed accurately.
- Contract publication, beneficial ownership, cost audits and procurement disclosures assessed separately.
- No political ranking or judgment.

Gate 9: Comparators

- Ghana, Norway, Timor-Leste and Suriname use current primary legal or fund sources.
- Every comparator includes transfer limits.
- No institutional ranking or selected political winner.

Gate 10: Quantitative models

- All assumptions visible.
- Units tested.
- Sensitivities reproducible.
- Forecast ranges never described as expected fact.
- No confidential company data.

Gate 11: Editorial integrity

- Zero fabricated citations.

- Each material number has a source and observation date.
- Contradictory evidence is preserved.
- External review described as pending until completed.
- Corrections register maintained.

Gate 12: PDF QA

- Searchable text.
- Correct page numbering and bookmarks.
- Tables readable at normal zoom.
- Figures have alt text/captions and source notes.
- No filler or repeated pages.
- Actual substantive page count verified after rendering.
- SHA-256 checksum recorded.
- Website links tested before publication.

Required external reviewers

Independent review should cover:

- petroleum fiscal economics;
- Guyana petroleum law and contract interpretation;
- public financial management;
- offshore environmental science and spill response;
- Indigenous/community safeguards;
- macroeconomics;
- data visualization and statistical interpretation.

External review remains a requirement, not a completed fact.

Dated source register

Source IDs are used throughout the publication. URLs point to the public source used for the stated claim. Limitations remain part of the evidence record.

S01 ExxonMobil Guyana 2023 Annual Report

Early seismic and onshore exploration, May 2015 Liza commercial discovery, 2019 first oil and FPSO dates

Limit: Operator is an interested party; cross-check public licence register

<https://corporate.exxonmobil.com/locations/guyana/publications/2023-annual-report>

S02 Guyana Ministry of Natural Resources, Stabroek petroleum agreement

Official text of the 2016 agreement, contract terms and baseline obligations

Limit: Distinct from future PSAs and public summaries

<https://petroleum.gov.gy/docs-and-pubs/petroleum-agreement-stabroek-block/>

S03 Ministry of Natural Resources, block register

June 27 2016 agreement / October 7 2016 effective date and licence information

Limit: Different contract milestone dates refer to different acts

<https://petroleum.gov.gy/oil-blocks/stabroek-block/>

S04 2024 NRF Annual Report, Ministry of Finance, notes/legal framework

NRF law assented January 23 2019; replaced December 30 2021; withdrawal rule amended February 6 2024

Limit: Full audited report must be checked for changing cash period

https://finance.gov.gy/wp-content/uploads/2025/11/2024_NRF_Annual_Report_Final_Draft-JD0715_GE1507-Final-Print.pdf

S05 Parliament of Guyana, Acts index

Local Content Act 2021 and NRF Act 2021 parliamentary dates

Limit: Parliamentary passing, assenting and commencement dates differ

<https://www.parliament.gov.gy/publications/acts-of-parliament/P80>

S06 Finance, 2026 Budget Speech, January 26 2026, 124-page PDF

January production/revenue/NRF assumptions and policy descriptions

Limit: January projections were superseded by September mid-year update

https://finance.gov.gy/wp-content/uploads/2026/01/Budget_Speech_2026.pdf

S07 Finance, 2026 Mid-Year Report, issued September 2026, printed pp. 21-25 and 104

H1 2026 21 government lifts, USD 1,778.6m profit-oil proceeds, USD 218.4m royalties, USD 4,294.2m end-June fund cash, USD 1,020m withdrawals; revised forecast USD 6,497.6m petroleum deposits

Limit: Government estimates and projections, not audited end-year results

<https://finance.gov.gy/wp-content/uploads/2026/09/Mid-Year-Report-2026.pdf>

S08 IMF, July 31 2026 mission concluding statement

IMF staff views on balance/withdrawal rule, absorption, audit, procurement, AML/CFT, statistical gaps and LCDS

Limit: Staff preliminary findings, not IMF Executive Board Article IV report

<https://www.imf.org/en/news/articles/2026/07/31/mcs073126-guyana-staff-concluding-statement-of-the-2026-article-iv-mission>

S09 IMF, 2025 Article IV report and press

Detailed macro, fiscal and debt context, external baseline

Limit: Historical to 2024; do not use as September 2026 snapshot

<https://www.imf.org/en/publications/cr/issues/2025/05/07/guyana-2025-article-iv-consultation-press-release-staff-report-and-statement-by-the-566712>

S10 World Bank, Guyana country overview

Social indicators, non-oil growth and lack of current household poverty data

Limit: Explicitly labels last cited poverty estimate 2019, not 2026

<https://www.worldbank.org/ext/en/country/guyana>

S11 ExxonMobil, Yellowtail startup, August 8 2025

Fourth FPSO, installed capacity exceeded 900k bopd

Limit: Installed capacity is not sustained actual output

https://corporate.exxonmobil.com/news/news-releases/2025/0808_exxonmobil-guyana-begins-production-at-fourth-offshore-guyana-project

S12 ExxonMobil, November 12 2025

900k bopd daily output milestone

Limit: A single-day threshold is not 2025 annual average

<https://corporate.exxonmobil.com/locations/guyana/news-releases/11122025-daily-oil-production-hits-900000-barrels-in-guyanas-stabroek-block>

S13 Petroleum Management Programme oil-production chart

Government monthly production chart through July 31 2026

Limit: Preserve date and distinguish unit from capacity

<https://petroleum.gov.gy/data-chart/oil-production/>

S14 Government of Guyana, 2023 model fiscal terms

Announced 10% royalty, 65% cost ceiling and 10% corporate tax for future licensing

Limit: Do not apply revised terms retroactively to Stabroek

<https://petroleum.gov.gy/article/govt-announces-wave-of-enhanced-fiscal-terms-for-first-offshore-oil-and-gas-competitive-licensing-round-highlights/>

S15 Guyana Revenue Authority, royalty treatment, 2026

2016 2% royalty added to simplified 12.5% government profit share when cap binds

Limit: 14.5% example is not constant government take across the life of the block

<https://gra.gov.gy/royalty-payments-to-guyana-government/>

S16 EITI, May 12 2026 Board decision

Second validation from May 15; concern over civil-society selection and stakeholder confidence

Limit: No 2026 final validation result has been established in this work

<https://eiti.org/es/node/28602>

S17 EITI, 2026 call for stakeholder views

2022 validation and 2026 validation scope

Limit: Distinguish earlier assessment from later validation

<https://eiti.org/offers/guyana-2026-eiti-validation-call-views-stakeholder-engagement>

S18 Guyana EITI country secretariat

December 2025 report covers fiscal year 2023; validation visits June 2026

Limit: Secretariat claims require triangulation with EITI global board

<https://eiti.gy/about/>

S19 Parliament, Oil Pollution Prevention, Preparedness, Response and Responsibility Act 2025

Enacted Act No. 6 of 2025

Limit: September mid-year report says draft regulations progressing and commencement expected end-2026; enactment is not commencement

<https://parliament.gov.gy/publications/acts-of-parliament/oil-pollution-prevention-preparedness-response-and-responsibility-act-2025-act-6-of-2025>

S20 Government statement about May 7 2026 Court of Appeal decision

Appeal set aside High Court unlimited financial-guarantee order

Limit: Obtain authenticated judgment and independent legal commentary before final jurisprudential analysis

<https://dpi.gov.gy/press-release-ag-hails-coas-ruling-on-exxonmobil-epas-financial-guarantees/>

S21 EPA environmental authorisation services

Environmental authorisation/permit process and regulated-project categories

Limit: Online application portal alone does not prove monitoring quality

<https://eservices.epa.gov.gy/permits>

S22 Guyana LCDS 2030, October 2024 update

2009 framework, Norway USD 227m for 2009-15, 2022 policy and forest-credit programme; 15% community allocation

Limit: LCDS describes government strategy and self-reported expenditure; impact audit separate

<https://lcds.gov.gy/guyanas-low-carbon-development-strategy-lcds-2030-october-2024-update/>

- S23** LCDS, Guyana ART TREES fact sheet, 2022
Hess minimum USD 750m purchase agreement, credit volume and minimum prices
Limit: Contracted value is not money already received
<https://lcds.gov.gy/fact-sheet-on-guyanas-engagement-in-art-trees/>
- S24** Ghana Petroleum Revenue Management Act 2011
Stabilisation Fund, Heritage Fund, publication cadence and public accountability
Limit: Comparator institutional design, not a performance ranking
<https://ghalii.org/akn/gh/act/2011/815/eng%402011-04-15>
- S25** Norway, 2025 Government Pension Fund white paper
Entire net state petroleum cash transferred to GPF; 3% real-return fiscal guideline
Limit: Norway's long accumulation and mature state capacity differ from a new producer
<https://www.regjeringen.no/contentassets/185dca55ef5649d1bc08db2699bb861f/en-gb/pdfs/stm202420250022000engpdfs.pdf>
- S26** Timor-Leste central bank, 2005/2011 Petroleum Fund statutes
Statutory fund structure and amendments
Limit: Cross-check 2026 depletion pressures independently before claims about outcomes
https://www.bancocentral.tl/publication/laws_np/
- S27** TotalEnergies Suriname, September 19 2026
GranMorgu 2024 FID, first oil planned 2028
Limit: Planned start is not operating output
<https://totalenergies.sr/press/suriname-granmorgu-project-takes-shape-major-production-equipment-arrives>
- S28** Guyana pre-oil sovereign wealth fund Green Paper
Documents work beginning in late 2016 with Commonwealth Secretariat support, inter-agency working group in 2017, IMF missions and FARI training before first oil
Limit: Historical government policy paper. Treat forecasts in the paper as superseded by actual later events
<https://petroleum.gov.gy/wp-content/uploads/2024/10/Green-Paper-Final.pdf>
- S29** IMF 2023 Article IV staff report
Explains two-rule system: Stabroek PSA revenue sharing plus NRF withdrawal framework; notes first oil receipts deposited in 2020 and first budget transfer in 2022; records 2023 model-PSA changes and governance priorities
Limit: IMF analysis reflects information available in 2023 and should not substitute for 2026 data
<https://www.elibrary.imf.org/view/journals/002/2023/379/article-A001-en.xml>
- S30** Natural Resource Governance Institute, natural-resource fund governance guidance
Clear objectives, deposit/withdrawal rules, investment rules, independent oversight, broad consensus and budget integration; warns a fund alone does not produce good governance
Limit: Comparative policy guidance, not jurisdiction-specific legal advice
https://resourcegovernance.org/sites/default/files/NRF_Complete_Report_EN.pdf
- S31** NRGI sovereign wealth fund guidance for policymakers
Highlights oversight, disclosure, fiscal-rule fit, dangers of direct domestic investment from funds and the need to fit rules to national conditions
Limit: Older cross-country guidance; current country practices require fresh verification
<https://resourcegovernance.org/articles/sovereign-wealth-funds-guidance-policymakers>
- S32** LCDS homepage, accessed September 2026
Current official payment dashboard lists cumulative Phase II carbon-credit payments of USD 353.5m, including March 2026 USD 57.9m
Limit: Official programme reporting; independent audit of final use and outcomes remains separate
<https://lcds.gov.gy/>
- S33** EITI validation schedule, accessed September 2026
Guyana validation commenced 15 May 2026 and remains listed among ongoing validations
Limit: Do not state a final 2026 validation score until EITI publishes a decision
<https://eiti.org/validation-schedule>
- S34** EITI Board decision 2026-19, 12 May 2026
Full validation launched with external expert; Board expressed concern over continuing lack of full stakeholder

confidence in civil-society representative selection

Limit: Board concern is attributed to EITI, not adopted as the author's independent conclusion

<https://eiti.org/es/node/28602>

S35 Petroleum Management Programme, official exploration history

Records the long pre-2015 exploration sequence, 1938 licence history, 1965 offshore licences, 1980s investment promotion and later exploration interruptions

Limit: Government historical summary; individual dates should be cross-checked against original licences where material

<https://petroleum.gov.gy/history/summary-history-petroleum-exploration-guyana/>

S36 Petroleum Management Programme, Liza-1 well register

Records Liza-1 as an exploration oil well in the Stabroek Block with well date 15 May 2015 and mapped coordinates

Limit: Administrative well register; commerciality and reserve estimates require separate sources

<https://petroleum.gov.gy/well/liza-1/>

S37 Parliament of Guyana, Natural Resources Fund Act 2019 record

Parliament lists passage on 3 January 2019 and publication on 23 January 2019; the official act records presidential assent on 23 January 2019

Limit: Statutory establishment does not establish later implementation quality or unchanged withdrawal rules

<https://parliament.gov.gy/publications/acts-of-parliament/natural-resources-fund-act-2019>

S38 Parliament of Guyana, Petroleum (Exploration and Production) Act 1986

Establishes the 1986 petroleum-law foundation for exploration, exploitation, conservation and management, including maritime areas

Limit: Later legislation and regulations must be version-checked before applying historical provisions to 2026

<https://parliament.gov.gy/publications/acts-of-parliament/petroleum-exploration-and-production-act-1986/>

S39 Parliament of Guyana, Environmental Protection Act 1996

Establishes the pre-oil statutory environmental framework and environmental-impact mandate

Limit: Existence of statutory authority is separate from project-level permitting, enforcement and outcomes

<https://parliament.gov.gy/publications/acts-of-parliament/environmental-protection-act-1996>

S40 Permanent Court of Arbitration, Guyana v. Suriname case record

Documents the 2000 oil-concession incident, 2004 commencement of arbitration and maritime-boundary dispute framework

Limit: Use the final award itself for detailed legal propositions; case page provides procedural history

<https://pca-cpa.org/en/cases/9/>

S41 Petroleum Management Programme, Stabroek Block register

Records June 27 2016 petroleum agreement date, October 7 2016 effective/licence date and June 15 2017 production licence date

Limit: Administrative register; contract clauses must be checked against executed text

<https://petroleum.gov.gy/oil-blocks/stabroek-block/>

S42 Petroleum Management Programme, VHE initial 2018-2020 Stabroek cost-recovery audit page

Government-published audit record covering cost and revenue review and identifying the operator as responsible for cost-recovery accounting

Limit: Initial audit status and later government resolution of exceptions must be distinguished

<https://petroleum.gov.gy/docs-and-pubs/vhes-initial-audit-report-for-the-stabroek-block-cost-recovery-audit-2018-to-2020/>

S43 Petroleum Management Programme, regulatory-news document index

Lists public cost-recovery audits, production licences and environmental permits, including 1999-2017 and 2018-2020 audit records

Limit: Document availability does not establish final settlement of every audit finding

<https://petroleum.gov.gy/documents-categories/regulatory-news/>

S44 Bank of Guyana, NRF audited financial statements index

Provides audited NRF financial statements for 2020 through 2025 in one official series

Limit: Each annual statement must be read on its own accounting basis and checked for restatements

<https://bankofguyana.org.gy/bog3/publications/natural-resource-fund/audited-financial-statements>

S45 Bank of Guyana, NRF audited financial statements 2020

Records 2020 royalties, profit oil, receivables, cash, first-year accounting policies and no government withdrawal during the year

Limit: GYD reporting and accrual receivables must be reconciled to USD cash deposits before cross-year comparisons

https://bankofguyana.org.gy/bog/images/accounts_budgeting/natural_resource_fund/financial_statements/nrf_audited_financial_statements_2020.pdf

S46 Bank of Guyana, NRF audited financial statements 2021

Records 2021 royalties, profit oil, cash flows, no government withdrawal and replacement of the 2019 Act by the 2021 Act

Limit: Contains restated comparative 2020 figures; use audited restatement rather than mixing earlier presentation

https://bankofguyana.org.gy/bog/images/accounts_budgeting/natural_resource_fund/financial_statements/nrf_audited_financial_statements_2021.pdf

S47 Bank of Guyana, NRF Quarterly Report June 2025

Provides quarterly inflows, market value, interest, transfers to the Consolidated Fund and cumulative withdrawal history through June 2025

Limit: Quarterly market value differs from annual audited closing balances and should not be merged without date alignment

https://www.bankofguyana.org.gy/bog/images/accounts_budgeting/natural_resource_fund/quarterly/nrf-june2025-quarterly.pdf

S48 Petroleum Management Programme, Liza Phase 1 Modified Environmental Permit

Records original permit dated 1 June 2017, the approved EIA/ESMP and revised oil-spill/wildlife plans, and modification on 12 May 2021

Limit: Permit existence and conditions do not establish independent compliance; monitoring evidence must be reviewed separately

<https://petroleum.gov.gy/docs-and-pubs/liza-phase-1-modified-environmental-permit/>

S49 Petroleum Management Programme, EPA-EEPGGL project document index

Official index lists Liza Phase 1 permit, Liza Phase 2 amended permit and EIA materials, Payara permit/EIA/OSRP documents and related environmental records

Limit: Index establishes document availability, not project performance; each permit must be reviewed individually

<https://petroleum.gov.gy/projects-initiatives/environmental-protection-agency-esso-exploration-and-production-guyana-limited/>

S50 Petroleum Management Programme, Uaru Environmental Permit

Permit Ref. 2023-023-EEPGGL, effective 27 April 2023 to 26 April 2028; includes EPA powers for sampling, permit modification and annual review linked to audit

Limit: Project-level compliance and audit results require separate evidence

<https://petroleum.gov.gy/wp-content/uploads/2024/10/Uaru-Signed-Permit.pdf>

S51 Petroleum Management Programme, Whiptail Environmental Permit page

Official publication dated 12 April 2024 for the Whiptail environmental permit

Limit: Detailed permit conditions require document-level extraction and monitoring evidence

<https://petroleum.gov.gy/whiptail-environmental-permit/>

S52 Petroleum Management Programme, Yellowtail public notice

Records the 12 May 2021 environmental-authorization application and EPA determination that an EIA was required before approval or rejection

Limit: Application/EIA scoping is not evidence of a final permit or later compliance

<https://petroleum.gov.gy/notice-to-the-public-yellowtail-development-project/>

S53 Parliament of Guyana, Oil Pollution Prevention, Preparedness, Response and Responsibility Act 2025

Parliament records Act 6 of 2025, passed 16 May 2025 and published 30 May 2025

Limit: Enactment/publication must be kept separate from commencement and implementing regulations

<https://parliament.gov.gy/publications/acts-of-parliament/oil-pollution-prevention-preparedness-response-and-responsibility-act-2025-act-6-of-2025>

S54 Official Gazette of Guyana, Act 6 of 2025 publication

Official Gazette archive records the petroleum pollution Act as an extraordinary gazette dated 28 May 2025

Limit: Gazette publication alone does not establish that every provision had commenced by the research cut-off

<https://officialgazette.gov.gy/index.php/publications/2778-extraordinary-gazettes-28th-may-2025-act-no-6-of-2025-the-oil-pollution-prevention-preparedness-response-and-responsibility-bill-2025>

S55 Petroleum Management Programme, Liza 1 flaring permit modification notice

EPA states the modified permit added emissions reporting, flaring time limits and a US\$30/tCO₂e payment for flaring beyond specified timelines

Limit: Agency statement documents the regulatory requirement; actual emissions, invoices and payments need period evidence

<https://petroleum.gov.gy/epa-issues-a-modified-environmental-permit-to-eeptl-for-the-liza-1-project/>

S56 Petroleum Management Programme, Uaru Petroleum Production Licence

Licence requires compliance with environmental law, expert personnel/equipment for spill response and specified capping-stack/dispersant arrangements

Limit: Licence requirements are obligations, not evidence that all readiness conditions were demonstrated in practice

<https://petroleum.gov.gy/index.php/documents/uaru-petroleum-production-licence>

S57 Parliament of Guyana, Local Content Act 2021

Parliament records Act 18 of 2021, passed 29 December 2021 and published 4 January 2022

Limit: Enactment establishes the statutory framework; measured domestic value added requires separate evidence

<https://www.parliament.gov.gy/publications/acts-of-parliament/local-content-act-2021-no.-18-of-2021>

S58 Guyana Local Content Register

Official supplier/employment registration system and Local Content Secretariat portal

Limit: Certification establishes registry status, not independent proof of domestic value added, capability or retained profit

<https://lcregister.petroleum.gov.gy/>

S59 Ministry of Finance, 2024 Mid-Year Report

Government reported H1 2024 procurement and employment information from contractor/subcontractor reports, including reporting scope and projected training/procurement

Limit: Self-reported sector figures require independent ownership, customs, payroll and value-added checks

<https://finance.gov.gy/wp-content/uploads/2024/08/Mid-Year-Report-2024.pdf>

S60 Ministry of Finance, 2025 Mid-Year Report

Describes expansion of local-content reporting to additional sectors/subsectors and ongoing assessment of local capacity

Limit: Government implementation update; not an independent measure of outcome quality

<https://finance.gov.gy/wp-content/uploads/2025/09/Mid-Year-Report-2025.pdf>

S61 Guyana Bureau of Statistics, Household Budget Survey 2018/2019

Official survey page states HBS/Living Condition fieldwork ran from November 2018 through November 2019 across all ten administrative regions

Limit: Historical household evidence must not be represented as current 2026 welfare data

<https://statisticsguyana.gov.gy/surveys/>

S62 LCDS, consultations and land-tenure fact page

Official programme account describes 15% community revenue share, village-plan process and Amerindian Land Titling programme

Limit: Government account requires triangulation with Indigenous/community sources and independent safeguards evidence

<https://lcds.gov.gy/facts-on-consultations-and-land-tenure-guyanas-art-trees-programme/>

S63 LCDS 2030 October 2024 Update

Government reported more than 240 villages receiving carbon-credit revenue and village-led projects under the 15% allocation

Limit: Programme spending/participation figures require independent outcome and consent verification

<https://lcds.gov.gy/guyanas-low-carbon-development-strategy-lcds-2030-october-2024-update/>

S64 IMF, 2026 Article IV mission concluding statement

IMF staff reports 2025 growth, continued H1 2026 expansion, fiscal/absorption issues, procurement/audit priorities, data gaps and diversification observations

Limit: Preliminary staff findings, not an Executive Board Article IV report; policy recommendations are attributed to IMF staff

<https://www.imf.org/en/news/articles/2026/07/31/mcs073126-guyana-staff-concluding-statement-of-the-2026-article-iv-mission>

S65 Ministry of Finance, Budget Speech 2026

Official 2026 budget policy, public-investment allocations and announced energy/infrastructure programme

Limit: Budget allocation and target do not establish project completion or delivered outcome

https://finance.gov.gy/wp-content/uploads/2026/01/Budget_Speech_2026.pdf

S66 Guyana Energy Agency, Demand Side Management Action Plan handover, July 2026

GEA reports a 2030 target of 516 GWh energy savings against projected generation of 2,508 GWh

Limit: Government/agency plan and target; implementation and achieved savings require later measurement

<https://gea.gov.gy/2026/07/01/6793/>

S67 Guyana Energy Agency, Demerara-Berbice Integrated System transition page

Official description of the electricity mix, gas-to-energy concept and planned renewable complement

Limit: Some project dates on the page are historical planning dates and must be updated against 2026 delivery evidence

<https://gea.gov.gy/renewable/demerara-berbice-integrated-system/>

S68 IMF, 2025 Article IV staff report

Describes official-statistics modernization, planned new household survey and macroeconomic context before the 2026 mission

Limit: Historical to 2025 and superseded where 2026 evidence exists

<https://www.elibrary.imf.org/view/journals/002/2025/103/article-A001-en.xml>

S69 EITI Board Decision 2026-19

Board launched a full Guyana Validation from 15 May 2026 and recorded concern about stakeholder confidence in civil-society representative selection

Limit: Validation remained a process at the research cut-off; do not state a final score without a later Board decision

<https://eiti.org/es/node/28602>

S70 EITI 2026 stakeholder consultation

International Secretariat sought views on Guyana implementation from April 2022 to May 2026, including civil-society participation and barriers

Limit: Consultation questions and submissions are inputs to Validation, not a final assessment

<https://eiti.org/offers/guyana-2026-eiti-validation-call-views-stakeholder-engagement>

S71 Petroleum Management Programme, IHS Markit final 1999-2017 cost-recovery audit

Government-published audit states audited cost-recovery submission totaled USD1.6778bn and identified USD214.4m plus overhead adjustments as having grounds for dispute

Limit: Audit recommendation and government dispute are not identical to final cash recovery or settlement

<https://petroleum.gov.gy/documents/ihs-markit-final-audit-report-stabroek-block-cost-recovery-audit-1999-2017>

S72 Ministry of Natural Resources, audit-status update

Government states the 1999-2017 audit moved to the PSA sole-expert process and describes status of multiple cost audits

Limit: Ministry status statement requires later verification of settlement, expert decision or arbitration outcome

<https://petroleum.gov.gy/article/update-on-status-of-stabroek-block-cost-recovery-and-verification-of-profit-oil-audits/>

S73 Audit Office of Guyana, 2024 public-accounts report portal

Supreme audit institution states the 2024 public-accounts/performance report was presented to the National Assembly in September 2025

Limit: Public-account audit observations must be tied to specific entities/projects; do not generalize individual findings across government

<https://www.audit.org.gy/site/>

- S74** Norway Ministry of Finance, Government Pension Fund 2026 white paper
Describes current GPFG framework, 3% expected-real-return fiscal guideline and long-run role of petroleum wealth in public finance
Limit: Norway's mature institutions, large accumulated assets and economic structure differ materially from a new producer
<https://www.regjeringen.no/en/documents/meld.-st.-7-2025-2026/id3155109/>
- S75** Bank of Ghana, Ghana Petroleum Funds
Official fund page describes Stabilisation and Heritage objectives, semi-annual reporting and annual/quarterly audit framework
Limit: Ghana's statutory allocation formula and production history differ from Guyana; compare mechanisms rather than outcomes by headline balance
<https://www.bog.gov.gh/ghana-petroleum-funds/>
- S76** Bank of Ghana, 2026 Ghana Petroleum Funds reporting index
Official reporting index includes 2025 audited statements and 2026 semi-annual report
Limit: Use individual statements for numerical comparisons and preserve reporting dates
<https://www.bog.gov.gh/ghana-petroleum-funds/gpf-financial-statements/>
- S77** Central Bank of Timor-Leste, Petroleum Fund reporting portal
Official portal publishes annual, quarterly and monthly performance reports including 2025 annual reporting and 2026 updates
Limit: Fund size and investment returns must be analysed alongside budget withdrawals, petroleum depletion and domestic fiscal dependence
https://www.bancocentral.tl/publication/pf_rep/
- S78** Central Bank of Timor-Leste, Petroleum Fund management framework
Describes Liquidity and Growth portfolios and publication requirements
Limit: Current portfolio structure reflects Timor-Leste's withdrawal needs and should not be copied without national cash-flow modelling
https://bancocentral.tl/stats/about_pf/
- S79** TotalEnergies EP Suriname, GranMorgu update 19 September 2026
Operator states project FID occurred in October 2024, construction is progressing and first oil remains targeted for 2028
Limit: Operator project update; planned 2028 first oil is not an observed production result
<https://totalenergies.sr/press/suriname-granmorgu-project-takes-shape-major-production-equipment-arrives>
- S80** Trinidad and Tobago Ministry of Finance, Heritage and Stabilisation Fund reporting portal
Official portal publishes annual and quarterly HSF reports through 2026
Limit: Mature producer and dual stabilization/heritage design offer mechanism comparison, not a direct blueprint for Guyana or a new island producer
<https://www.finance.gov.tt/publications/heritage-and-stabilisation-fund/>
- S81** Petroleum Management Programme, Oil Production dashboard
Official production series covers 20 December 2019 through 31 July 2026
Limit: Monthly realized production must remain separate from installed/nameplate capacity
<https://petroleum.gov.gy/data-chart/oil-production/>
- S82** ExxonMobil, Liza Phase 2 first oil, 11 February 2022
Operator records first production from Liza Unity on 11 February 2022 and planned ramp to 220,000 bpd
Limit: Operator statement is suitable for startup date; realized monthly production should use regulator data
https://corporate.exxonmobil.com/news/news-releases/2022/0211_exxonmobil-starts-production-at-guyanas-second-offshore-development
- S83** Petroleum Management Programme, Payara first oil, 14 November 2023
Government records first oil at Prosperity in November 2023; Liza Destiny started December 2019 and Liza Unity February 2022
Limit: Government media release includes future production estimates that must not be treated as observed output
<https://petroleum.gov.gy/article/govt-welcomes-first-oil-at-prosperity-fpso/>

- S84** ExxonMobil, Yellowtail first production, 8 August 2025
Operator records startup of the fourth project and combined installed capacity above 900,000 bpd
Limit: Installed capacity is not annual average production
https://corporate.exxonmobil.com/news/news-releases/2025/0808_exxonmobil-guyana-begins-production-at-fourth-offshore-guyana-project
- S85** ExxonMobil, 900,000 bpd milestone, 12 November 2025
Operator reports a daily production milestone after Yellowtail ramp-up
Limit: One-day/point-in-time output is not a full-year average
<https://corporate.exxonmobil.com/locations/guyana/news-releases/11122025-daily-oil-production-hits-900000-barrels-in-guyanas-stabroek-block>
- S86** Petroleum Management Programme, Liza Phase 1 Production Licence and Phase 2 Addendum
Official production-licence publication provides legal project authorization documents for Liza developments
Limit: Licence conditions require document-level review and do not establish realized performance
<https://petroleum.gov.gy/docs-and-pubs/liza-phase-1-petroleum-production-licence-and-phase-2-addendum/>
- S87** Parliament of Guyana, Petroleum Activities Act 17 of 2023
Establishes the newer statutory petroleum framework for licensing, regulation, data, safety, decommissioning and administration
Limit: Legacy contractual rights and transitional application require agreement-specific legal analysis
<https://parliament.gov.gy/publications/acts-of-parliament/petroleumactivitiesact17of2023>
- S88** Bank of Guyana, audited NRF financial statements 2025
Audited financial statements for the year ended 31 December 2025, including income, financial position, cash-flow and notes
Limit: Must be reconciled to Ministry reporting using consistent currency, timing and accounting basis
https://bankofguyana.org.gy/bog3/images/accounts_budgeting/natural_resource_fund/financial_statements/nrf_audited_financial_statements_2025.pdf.pdf
- S89** EITI Validation schedule, checked 27 September 2026
Lists Guyana among ongoing Validations and notes commencement on 15 May 2026
Limit: Ongoing status is not a final Validation decision or score
<https://eiti.org/validation-schedule>
- S90** International Court of Justice, Guyana v Venezuela case record
Official case docket for the controversy concerning the 1899 Arbitral Award, including jurisdiction, preliminary objection and provisional-measures material
Limit: The merits remain before the Court; this publication must not predict or decide the outcome
<https://icj-cij.org/case/171>
- S91** International Court of Justice, Order of 1 May 2025
Official provisional-measures order in Guyana v Venezuela
Limit: Interim judicial measure, not a final merits judgment
<https://www.icj-cij.org/node/205403>
- S92** CARICOM statement, 28 April 2026
Regional statement supporting peaceful resolution through the ICJ and addressing conduct concerning the Guyana-Venezuela controversy
Limit: Political/diplomatic statement; use for regional context, not for a merits determination
<https://caricom.org/statement-from-the-caribbean-community/>
- S93** Forest Peoples Programme, ART Guyana carbon-credit case study
Documents criticisms concerning ART certification, Indigenous safeguards and complaint mechanisms
Limit: Advocacy research; contested claims require triangulation with ART, government and community evidence
<https://www.forestpeoples.org/pt/publications-resources/reports/article/arts-controversial-certification-of-carbon-credits-to-the-government-of-guyana-a-case-study-on-challenges-for-high-integrity-labels-in-carbon-markets/>
- S94** Office of the President, Guyana Oil & Gas Capacity Building Project SESA terms of reference
Identifies strategic offshore and onshore environmental and social risk categories including fisheries, spill risk, marine habitats and coastal facilities
Limit: Terms of reference identify risks to assess; they are not measured outcome evidence
<https://op.gov.gy/guyana-oil-gas-capacity-building-project-p166730-terms-of-reference-for-strategic-environmental>

S95 LCDS, CORSIA-eligible ART TREES credits, 28 February 2024

Records issuance of 7.14 million 2021-vintage credits and CORSIA eligibility claims, including corresponding-adjustment context

Limit: Programme statement; registry and UNFCCC records should be used for final accounting verification

<https://lcds.gov.gy/guyana-announces-worlds-first-carbon-credits-for-use-in-un-airline-compliance-programme-corsia/>

S96 Ministry of Finance, NRF monthly reports 2026

Official monthly NRF reporting through July 2026 available at the publication cut-off

Limit: Monthly reports need reconciliation to quarterly, annual and audited accounts before aggregation

<https://finance.gov.gy/monthly-reports/>

S97 Stabroek News, Anand Goolsarran analysis of PSA revenue provisions, August 2025

Independent public commentary using NRF audited accounts to discuss profit-sharing, royalty and taxation

Limit: Commentary, not primary law or audit evidence; use only as an attributed critical interpretation

<https://www.stabroeknews.com/2025/08/04/features/accountability-watch/revisiting-the-profit-sharing-royalty-and-taxation-provisions-of-the-petroleum-sharing-agreement-part-ii/>

S98 Official Gazette of Guyana, Extraordinary Gazette, 28 May 2025, Act No. 6 of 2025

Confirms formal publication of the Oil Pollution, Prevention, Preparedness, Response and Responsibility Act 2025 in the Official Gazette

Limit: Publication confirms enactment/gazetting, not necessarily commencement of every provision or issuance of implementing regulations

<https://officialgazette.gov.gy/index.php/publications/2778-extraordinary-gazettes-28th-may-2025-act-no-6-of-2025-the-oil-pollution-prevention-preparedness-response-and-responsibility-bill-2025>

S99 Petroleum Management Programme, Uaru Environmental Permit, financial assurance conditions 14.3-14.5

Permit states financial assurance should be guided by reasonably credible costs, requires liability/environmental-liability insurance, and requires parent/affiliate guarantee agreements to provide resources for environmental obligations after default

Limit: Permit terms establish obligations and architecture; they do not by themselves prove current insurance limits, guarantee amount, liquidity or enforceability in a future loss event

<https://petroleum.gov.gy/wp-content/uploads/2024/10/Uaru-Signed-Permit.pdf>

S100 Petroleum Management Programme, Liza Phase 1 Modified Environmental Permit

Official page confirms permit issued 1 June 2017 and modified 12 May 2021, and incorporates the EIA, environmental/social management plan, revised oil-spill response plan and wildlife response plan

Limit: Document availability does not establish compliance performance or current financial-assurance sufficiency

<https://petroleum.gov.gy/docs-and-pubs/liza-phase-1-modified-environmental-permit/>

S101 Department of Public Information, Attorney General press release on Court of Appeal ruling, 8 May 2026

Government summary states the Court of Appeal on 7 May 2026 overturned the High Court unlimited-guarantee order and held that EPA retains discretion over financial assurance

Limit: Government account of judgment; authenticated Court of Appeal reasons remain preferred for detailed legal characterization

<https://dpi.gov.gy/press-release-ag-hails-coas-ruling-on-exxonmobil-epas-financial-guarantees/>

S102 Department of Public Information, Attorney General press release, 3 May 2023

Government stated EPA and EEPGL had negotiated a US\$2 billion parent guarantee and indemnity arrangement and described the government's position on the High Court ruling

Limit: Party statement and negotiation description; does not substitute for the executed guarantee instrument

<https://dpi.gov.gy/press-release-by-attorney-generals-chambers/>

S103 Caribbean Court of Justice, [2024] CCJ 16 (AJ) GY, 30 July 2024

Authenticated CCJ reasons identify the underlying Collins/Whyte litigation, renewed Liza 1 permit condition 14, the 2023 High Court order, the Attorney General intervention dispute and Environmental Protection Act section 31

Limit: Procedural/intervention judgment, not the May 2026 Court of Appeal merits reasons

https://ccj.org/wp-content/uploads/2024/07/2024_CCJ_16_AJ_GY.pdf

S104 EITI Validation Committee, June 2026 meeting record

EITI committee page records discussion of preliminary mission findings for Guyana on 23 June 2026 while

Validation remained underway

Limit: Committee agenda/minutes context; not a final EITI Validation decision

<https://eiti.org/fr/node/28362>

S105 Demerara Waves, 7 May 2026 Court of Appeal report

Contemporaneous independent reporting states the appellate court distinguished unlimited liability from financial assurance and treated the permit as requiring a definite financial-assurance amount

Limit: Secondary reporting based on lawyers' accounts; use only alongside official summaries and permit text until authenticated reasons are obtained

<https://demerarawaves.com/2026/05/07/court-of-appeal-overturms-high-courts-ruling-on-exxonmobil-epa-s-financial-guarantees/>

S106 Kaieteur News, 25 August 2026, reported leave to appeal to CCJ

Reports that the Court of Appeal granted leave in July 2026 for Collins and Whyte to challenge the May appellate ruling before the CCJ

Limit: Secondary report; current CCJ docket/filing should be preferred when publicly available

<https://kaieteurnewsonline.com/2026/08/25/exxons-unlimited-guarantee-fight-heads-to-ccj/>

S107 World Bank, Guyana Petroleum Exploration Promotion Project Completion Report, Report No. 12485, 16 November 1993

Documents Credit 1208-GUA, objectives to attract exploration investment and strengthen petroleum administration, the Petroleum Unit, negotiated exploration contracts, training and legislative reform, as well as project shortcomings and capacity lessons

Limit: Historical World Bank evaluation reflects the project and institutional context of the 1980s-early 1990s, not later petroleum performance

<https://documents1.worldbank.org/curated/en/508851468252286613/pdf/multi-page.pdf>

S108 Local Content Secretariat, Local Content Half-Yearly Report Submission Guideline Version 4, 13 June 2025

Defines required reporting on employment, salary scales, procurement, supplier participation, training and capacity development and requires comparative reporting against approved plans

Limit: Reporting architecture establishes required data fields; it does not by itself establish compliance quality or domestic value added

<https://petroleum.gov.gy/wp-content/uploads/2025/07/Local-Content-Half-Yearly-Report-Submission-Guideline-Version-4-1.pdf>

S109 Bank of Guyana, Natural Resource Fund Quarterly Report Q4 2025

Reports end-2025 fund market value, quarterly inflows/outflows, investment return, annual transfers and investment mandate; confirms IFRS reporting and Federal Reserve Bank of New York custody

Limit: Quarterly accounting must be reconciled with audited annual financial statements and later 2026 monthly/mid-year records

https://www.bankofguyana.org.gy/bog/images/accounts_budgeting/natural_resource_fund/quarterly/nrf-december-2025-quarterly.pdf

S110 Bank of Guyana, Natural Resource Fund monthly-report portal

Provides monthly NRF statements for 2025 and 2026, enabling month-by-month bridge analysis rather than reliance only on annual summaries

Limit: Individual monthly reports require consistent cash/accrual treatment and should not be summed without reconciliation

<https://bankofguyana.org.gy/bog3/publications/natural-resource-fund/monthly-reports>

S111 Guyana Revenue Authority, response on inaugural Stabroek cost-recovery audit, 5 April 2023

Describes IHS Markit audit scope, Guyanese inter-agency participation, draft/final report process, Annex C audit-response requirements, institutional disagreements and capacity-transfer objective

Limit: Government revenue-authority account; later settlement/dispute status must be verified separately

<https://www.gra.gov.gy/response-to-stabroek-news-articles-of-sunday-april-2-2023-and-monday-april-03-2023/>

S112 Petroleum Management Programme, 2023 draft shallow-water model Production Sharing Agreement

Provides later-generation model clauses on financial guarantees, liability and insurance, local content, cost recovery, taxation, royalty, decommissioning, audits, dispute resolution and stabilization

Limit: Draft/model terms are not the executed 2016 Stabroek terms and must not be applied retroactively

<https://petroleum.gov.gy/wp-content/uploads/2024/10/Guyana-Draft-Shallow-Water-PSA-2023.pdf>

- S113** EITI, Senegal country profile and petroleum-governance disclosures
Records Senegal's 2019 Petroleum Code, 2022 oil-and-gas revenue-management law, contract publication framework, PETROSEN role, local-content reforms and first-oil/first-gas transition
Limit: EITI country profile summarizes multiple legal and institutional sources; project-level fiscal results require primary national records
<https://eiti.org/countries/senegal>
- S114** EITI, Senegal 2026 Final Validation Report
Reviews Senegal's petroleum transition, state participation, contract review, transparency and reconciliation issues after first oil and gas
Limit: EITI institutional assessment, not a substitute for national law or audited project accounts
https://eiti.org/sites/default/files/Senegal%202026_EITI%20Final%20Validation%20Report%20%28EN%29.pdf
- S115** IMF, Mauritania: Improving Administration of Hydrocarbon Tax Revenues, High-Level Summary TA Report 2025/017
Assesses hydrocarbon-revenue administration and capacity development ahead of Greater Tortue Ahmeyim gas production
Limit: Technical-assistance summary; country implementation must be verified separately
<https://www.imf.org/en/publications/high-level-summary-technical-assistance-reports/issues/2025/05/05/islamic-republic-of-mauritania-improving-administration-of-hydrocarbon-tax-revenues-566696>
- S116** EITI, Validation of Mauritania 2024
Reviews contract publication, national oil company, hydrocarbon fund transparency, beneficial ownership and readiness for emerging gas revenues
Limit: Validation reflects evidence available to EITI and predates later operating years
<https://eiti.org/sites/default/files/2024-10/EITI%20Validation%20of%20Mauritania%20%282024%29%20-%20Final%20Validation%20report%20%28August%202024%29.pdf>
- S117** Uganda EITI Report 2022/23, published 2025
Provides pre-production petroleum legal/institutional framework, local content, cost recovery, environmental management, beneficial ownership and petroleum-revenue architecture
Limit: Reporting period precedes commercial production; future operating outcomes remain separate
<https://www.ugeiti.org/wp-content/uploads/2025/09/UGEITI-Report-2022-23.pdf>
- S118** EITI, Uganda country profile
Describes Uganda Petroleum Fund, revenue institutions, beneficial-ownership framework and pre-first-oil extractive governance
Limit: EITI summary; detailed legal operation should use Uganda statutes and fund reports
<https://eiti.org/countries/uganda>
- S119** Mozambique, Law No. 1/2024 establishing the Sovereign Wealth Fund
Establishes objectives, eligible revenues, transfer rules, stabilization mechanism, parliamentary controls and investment principles for Mozambique's oil and gas sovereign fund
Limit: Statutory design does not establish future implementation performance
https://inp.gov.mz/wp-content/uploads/2024/09/0_Lei-1_2024-Fundo-Soberano.pdf
- S120** Banco de Moçambique, Sovereign Wealth Fund operational management
Describes operational-management structure and records initial capital transferred in December 2025
Limit: Central-bank operational source; future investment performance and fiscal effects require later reporting
<https://www.bancomoc.mz/en/sovereign-wealth-fund/operational-management/>
- S121** IMF, Chile: Strengthening Fiscal and Stabilization Funds, Country Report 23/249
Examines Chile's structural-balance rule, Economic and Social Stabilization Fund, debt anchor and countercyclical resource-revenue management
Limit: Copper-based mature economy differs materially from Guyana's petroleum and development context
<https://www.imf.org/-/media/files/publications/cr/2023/english/1chlea2023004.pdf>
- S122** Bank of Botswana, Pula Fund
Official description of the Pula Fund's purpose, investment of excess reserves and intergenerational savings objective
Limit: Botswana's diamond and reserve-management context differs from petroleum-export accounting
<https://www.bankofbotswana.bw/index.php/content/pula-fund>

S123 IMF, Botswana 2024 Article IV Consultation

Documents pressure on fiscal buffers during a diamond-market downturn and discusses fiscal-rule and sovereign-fund reform options

Limit: IMF staff analysis; Botswana's mature diamond economy is a mechanism comparator, not a direct petroleum analogue

<https://www.imf.org/-/media/files/publications/cr/2024/english/1bwaea2024006-print-pdf.pdf>

S124 Department of Public Information, Guyana, arrival of Errea Wittu FPSO, 21 August 2026

Confirms arrival of Guyana's fifth FPSO for Uaru, project cost US\$12.7 billion, 250,000 bpd design scale and first-oil expectation in Q4 2026

Limit: Government forward-looking startup expectation; does not establish that Uaru had produced first oil by the 27 September 2026 evidence cut-off

<https://dpi.gov.gy/press-release-govt-welcomes-arrival-of-fifth-fpso/>

S125 MODEC, arrival of FPSO Errea Wittu in Guyana, 24 August 2026

Confirms vessel arrival, deployment approximately 200 km offshore in 1,690 m water, about 2 million barrels storage and commissioning ahead of first oil

Limit: Contractor statement; startup and realized production require later operating evidence

https://www.modec.com/news/assets/pdf/20260824_pr_Uaru_en.pdf

S126 Maritime Administration Department, Notice to Mariners 21 of 2026

Confirms continuing deep-water development operations across Yellowtail, Whiptail, Liza, Payara and Uaru fields during 2026

Limit: Operational marine notice, not a production or fiscal record

<https://marad.gov.gy/wp-content/uploads/2026/01/Notice-to-Mariners-21-of-2026.pdf>

S127 Department of Public Information, Hammerhead petroleum production licence approval, 22 September 2025

Confirms approval of seventh Stabroek development, US\$6.8 billion project, planned 150,000 bpd capacity, 18 production/injection wells and anticipated 2029 startup

Limit: Approved/planned development data; future production is not observed output

<https://dpi.gov.gy/gog-approves-7th-petroleum-production-license/>

S128 ExxonMobil, Hammerhead final investment decision, 22 September 2025

Operator confirms seventh-development FID, planned 150,000 bpd FPSO capacity and anticipated 2029 startup

Limit: Operator forward-looking project statement; future startup is not an observed result

https://corporate.exxonmobil.com/news/news-releases/2025/0922_exxonmobil-guyana-expands-capacity-with-seventh-offshore-development

Open evidence items

Priority evidence items for future revision and external review

1. Obtain authenticated PDF text of the May 2026 Court of Appeal judgment and any later appeal.
2. Verify whether the 2025 spill act was commenced after the June 2026 reporting cut-off and before 27 September 2026.
3. Reconcile public Bank of Guyana fund statements to Ministry of Finance June 2026 balance, petroleum deposit and interest categories.
4. Determine public status of 2018-2020 and later cost-oil audits and any contested arbitration.
5. Obtain original EPA permit schedules, environmental impact assessments, baseline marine datasets, third-party incident logs and the precise insurance/parent guarantee instruments.
6. Verify 2026 EITI validation report status at press time, not solely government statements.
7. Independent assessment of Indigenous consultation and village-level benefit delivery with free, prior and informed consultation standards relevant to the jurisdiction.
8. Distinguish project expenditure from commissioned/working infrastructure and verify costs, maintenance and beneficiaries.
9. Validate local content supplier ownership and incremental local value added independently of gross spend.
10. Conduct external technical, financial, legal and environmental review before calling this paper peer reviewed.